



☐ Short Delivery	☐ Only for Engg. Activities	Date	29.09.2025
☐ Bought out valves	☐ Others, Please Specify	CORB Reference	DM25066L
Customer / Buyer Name	LARSEN & TOUBRO LIMITED	CEC Quote Number with Date	250702-0400-PA Dtd 28.08.2025
Site / End User Name & Country	Rashtriya Chemicals and Fertilizers Limited, India	Application Engineer SPOC	Karthikeyan Sekar Satheshkumar Eagambaram
Buyer PO Number/Date	LTES/75001-83664/BS Dtd 03.09.2025	Outside Sales Engineer/Manager	Ashokvijay R
Buyer PO/LOI Received Date	12.09.2025	PO Value	INR 9,99,750
Agreed / Approved Payment Terms	Refer PO clause no. 5.0	1)PO Delivery Date 2)Agreed Delivery Date	26.03.2026
Agreed Incoterms /Shipping Terms	CPT, RCF Thal Project Site	Margin	22%
LD Clause	@0.5% of the undelivered portion of Order value	Agent Name & Commission	NA
Ship to Address with GST#:	Rashtriya Chemical and Fertilizers Limited Thal, Thal-Vaishet, Alibag # 402 208 State: Maharashtra, India GSTIN: 27AAACL0140PEZ6	Bill to Address with GST#:	Larsen & Toubro Limited Divison Name: L&T Energy Hydrocarbon 14th Floor, AMN Tower, AMN Tower, Jogeshwari - Vikroli Link Road, Powai Campus, Gate 3, Mumbai Suburban, Maharashtra, 400072

CIRCOR ORDER REVIEW BOARD

SCOPE OF SUPPLY

APPROVALS (TECHNICAL & COMMERCIAL)

PURCHASE ORDER / LOI – BUYER

CIRCOR PROPOSAL/QUOTE

COST SUMMARY SHEET

SPECIAL NOTES & AE CHECK LIST (INSTRUCTION TO CS / ME/ENG/SCM / QA)

INSTRUMENTATION LIST

CIRCOR VALVE / DESUPERHEATER GAD

CIRCOR VALVE / DESUPERHEATER SPECIFICATION & CALCULATION

ENQUIRY / RFQ FROM CUSTOMER

IMPORTANT CORRESPONDENCES/EMAILS



☐ Short Delivery	☐ Only for Engg. Activities	Date	29.09.2025
☐ Bought out valves	☐ Others, Please Specify	CORB Reference	DM25066L
Customer / Buyer Name	LARSEN & TOUBRO LIMITED	CEC Quote Number with Date	250702-0400-PA Dtd 28.08.2025
Site / End User Name & Country	Rashtriya Chemicals and Fertilizers Limited, India	Application Engineer SPOC	Karthikeyan Sekar Satheshkumar Eagambaram
Buyer PO Number/Date	LTES/75001-83664/BS Dtd 03.09.2025	Outside Sales Engineer/Manager	Ashokvijay R
Buyer PO/LOI Received Date	12.09.2025	PO Value	INR 9,99,750
Agreed / Approved Payment Terms	Refer PO clause no. 5.0	1)PO Delivery Date 2)Agreed Delivery Date	26.03.2026
Agreed Incoterms /Shipping Terms	CPT, RCF Thal Project Site	Margin	22%
LD Clause	@0.5% of the undelivered portion of Order value	Agent Name & Commission	NA
Ship to Address with GST#:	Rashtriya Chemical and Fertilizers Limited Thal, Thal-Vaishet, Alibag # 402 208 State: Maharashtra, India GSTIN: 27AAACL0140PEZ6	Bill to Address with GST#:	Larsen & Toubro Limited Divison Name: L&T Energy Hydrocarbon 14th Floor, AMN Tower, AMN Tower, Jogeshwari - Vikroli Link Road, Powai Campus, Gate 3, Mumbai Suburban, Maharashtra, 400072

SCOPE OF SUPPLY



SCOPE OF SUPPLY - DM25066L

PO Line No	Item No / Tag No	Item Description	Circor Internal No As Refected in CORB Package	Product Line	PO Qty	Customer PO Unit Price	Customer PO Total Price
1	NDSSIW6BCBKACBIXZZ	CIR 3210, ANSI CL 300, ASTM A106 Gr B / A105, 6", RF, Venturi Nozzle, with 6 inch steam pipe & 4 inch liner arrangement	DM25066L	110: DSH	1	₹ 3,10,594	₹ 3,10,594
		CIR 3100, ANSI CL 300, A105, 1", RF, , Std. Bonnet, Graphite, Contoured Plug, Class V, Fail to Close, SASR Actuator, With HOD, SIEMENS Poz, SHAVO AFR etc.,		101: CIR / RTK 3100			
2	NDSSIX6ACX3AC7IXZZ	CIR 3220, ANSI CL 150x300, ASTM A106 Gr B / A105, 3"x1", RF, Fixed Nozzle, without Steam pipe arrangement	DM25066L	110: DSH	1	₹ 2,39,731	₹ 2,39,731
		CIR 3100, ANSI CL 300, A105, 1", RF, , Std. Bonnet, Graphite, MZ Plug, Class V, Fail to Close, SASR Actuator, With HOD, SIEMENS Poz, SHAVO AFR etc.,		101: CIR / RTK 3100			
3	IHMGCVMRP01	Mandatory Spares as per MR Philosophy	DM25066L	114: Spare Parts	1	₹ 4,49,425	₹ 4,49,425
						Total	\$ 9,99,750



☐ Short Delivery	☐ Only for Engg. Activities	Date	29.09.2025
☐ Bought out valves	☐ Others, Please Specify	CORB Reference	DM25066L
Customer / Buyer Name	LARSEN & TOUBRO LIMITED	CEC Quote Number with Date	250702-0400-PA Dtd 28.08.2025
Site / End User Name & Country	Rashtriya Chemicals and Fertilizers Limited, India	Application Engineer SPOC	Karthikeyan Sekar Satheshkumar Eagambaram
Buyer PO Number/Date	LTES/75001-83664/BS Dtd 03.09.2025	Outside Sales Engineer/Manager	Ashokvijay R
Buyer PO/LOI Received Date	12.09.2025	PO Value	INR 9,99,750
Agreed / Approved Payment Terms	Refer PO clause no. 5.0	1)PO Delivery Date 2)Agreed Delivery Date	26.03.2026
Agreed Incoterms /Shipping Terms	CPT, RCF Thal Project Site	Margin	22%
LD Clause	@0.5% of the undelivered portion of Order value	Agent Name & Commission	NA
Ship to Address with GST#:	Rashtriya Chemical and Fertilizers Limited Thal, Thal-Vaishet, Alibag # 402 208 State: Maharashtra, India GSTIN: 27AAACL0140PEZ6	Bill to Address with GST#:	Larsen & Toubro Limited Divison Name: L&T Energy Hydrocarbon 14th Floor, AMN Tower, AMN Tower, Jogeshwari - Vikroli Link Road, Powai Campus, Gate 3, Mumbai Suburban, Maharashtra, 400072

APPROVALS (TECHNICAL & COMMERCIAL)



☐ Short Delivery	☐ Only for Engg. Activities	Date	29.09.2025
☐ Bought out valves	☐ Others, Please Specify	CORB Reference	DM25066L
Customer / Buyer Name	LARSEN & TOUBRO LIMITED	CEC Quote Number with Date	250702-0400-PA Dtd 28.08.2025
Site / End User Name & Country	Rashtriya Chemicals and Fertilizers Limited, India	Application Engineer SPOC	Karthikeyan Sekar Satheshkumar Eagambaram
Buyer PO Number/Date	LTES/75001-83664/BS Dtd 03.09.2025	Outside Sales Engineer/Manager	Ashokvijay R
Buyer PO/LOI Received Date	12.09.2025	PO Value	INR 9,99,750
Agreed / Approved Payment Terms	Refer PO clause no. 5.0	1)PO Delivery Date 2)Agreed Delivery Date	26.03.2026
Agreed Incoterms /Shipping Terms	CPT, RCF Thal Project Site	Margin	22%
LD Clause	@0.5% of the undelivered portion of Order value	Agent Name & Commission	NA
Ship to Address with GST#:	Rashtriya Chemical and Fertilizers Limited Thal, Thal-Vaishet, Alibag # 402 208 State: Maharashtra, India GSTIN: 27AAACL0140PEZ6	Bill to Address with GST#:	Larsen & Toubro Limited Divison Name: L&T Energy Hydrocarbon 14th Floor, AMN Tower, AMN Tower, Jogeshwari - Vikroli Link Road, Powai Campus, Gate 3, Mumbai Suburban, Maharashtra, 400072

PURCHASE ORDER

 LARSEN & TOUBRO		LARSEN & TOUBRO LIMITED L&T ENERGY - HYDROCARBON Technology Centre, North Block -II Powai, Gate No. 1, Saki Vihar Road Mumbai 400072 Maharashtra India	Contact Person : BHUSHAN SHETTY Contact Telephone : Contact Email : bhushan.shetty@larsentoubro.com	PO NO. :LTES/75001-83664/BS PO Date : 03.09.2025
To, VENDOR CODE:60572 Circor Flow Technologies India Pvt Ltd SF No. 337/2, No. 15,Naranapuram Village, Thennampalayam to Annur Road Coimbatore Coimbatore 641659 Tamil Nadu India Telephone: - ATTN:- GSTIN NO.: 33AADCC5649M1ZE	YOUR REFERENCE EMAIL/18.07.2025 For other references refer annexure		PRICING & TAXATION: Please refer details in Annexure I. PAYMENT TERMS: Please refer details in Annexure II.	
Please arrange to provide the goods / services described below strictly as per terms and conditions mentioned in this Purchase Order and its enclosure/s listed below.				
GENERAL DESCRIPTION		CURRENCY	TOTAL VALUE (PRICE)	
Design, Engineering, Procurement, Manufacturing, Supply, Packing & Forwarding and Transportation of Desuperheater for New 1200 MTPD (DAP Basis) NPK Fertilizer Plant for Rashtriya Chemicals and Fertilizers Limited (RCF)at Thal, Maharashtra (India).		INR	11,79,706.00 Eleven Lakh Seventy Nine Thousand Seven Hundred Six only.	
		Value (Price) Basis: CPT, RCF-Thal Project Site basis		
Kindly send all your correspondence regarding this order in duplicate. All items and conditions below and on the reverse are to be strictly complied with.				
DISPATCH DETAILS: For details refer enclosures. Materials / Services to be dispatched only after receiving Dispatch Clearance from LARSEN & TOUBRO LIMITED				
Place of Supply / Destination: Maharashtra				
L&T GST REGISTRATION NO.: 27AAACL0140PEZ6			CIN: L99999MH1946PLC004768	
DOCUMENTATION : Please refer details in Annexure II.				
BILLING : For details refer enclosures. Bills submitted against this Purchase Order must contain Purchase Order No. & Date, Purchase Order Item No. and Vendor code.				
PROJECT: RCF-NPK Project 1200 MTPD (DAP Basis)		Release Code: 11 L.C.: 11.09.2025 11:58:05 Printed On : 12.09.2025 11:09:21		For LARSEN & TOUBRO LIMITED Snehal Shetiya JT. GENERAL MANAGER - PROCUREMENT
Registered Office : L&T House, Ballard Estate, Mumbai : 400 001, INDIA *- Refer Annexure for Details				



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - I
Page No:2 of 2

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

Total Quantity 3

Summary:

Total Gross Price	9,99,750.00
Total IN: Integrated GST	1,79,956.00

Total Value (Rounded off to) : INR 11,79,706.00

For LARSEN & TOUBRO LIMITED

Snehal Shetiya
JT. GENERAL MANAGER - PROCUREMENT

Separate signature is not necessary here in case PO
authorization is through verifiable digital signature.



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 1 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

1.0 References/Scope of work:

Scope of Work shall be strictly as per the Material Requisition Ref. No. E24003-RCF-IN-MRQ-042 enclosed with enquiry document, subsequent addenda, correspondence (Technical Queries, Commercial Queries etc.), discussions and MoM's.

Any exclusions, clarifications etc. stands withdrawn unless otherwise mutually agreed, duly recorded and accepted by L&T prior to placement of the Order.

Reference:

1. RFQ Ref No. RFQ/RCFNPK/0038dt. 02nd Jul 2025 issued by L&T vide Enqueue portal including its MR No. E24003-RCF-IN-MRQ-042_Rev 0.
2. Techno-Commercial Offer No. 250702-0400-PA dt. 18th July 2025 submitted by CIRCOR Flow Technologies India Pvt. Ltd.
3. Final Commercial Offer submitted by CIRCOR Flow Technologies India Pvt. Ltd. vide Email dt. 28th Aug 2025.
4. TQ-1 & Deviation closure sheets dt. 03rd Sep 2025 signed and stamped by CIRCOR Flow Technologies India Pvt. Ltd.
5. Letter of Award (LOA) No. LTEH/RCF/NPK/013 dt. 03-Sep-2025 issued by L&T.
6. All correspondences exchanged between L&T Ltd., and CIRCOR Flow Technologies India Pvt. Ltd. till date.

With reference to the above, Larsen & Toubro Limited, India (hereinafter referred to as "L&T/Purchaser") is pleased to release this Purchase Order (PO) on CIRCOR Flow Technologies India Pvt. Ltd. (hereinafter referred to as "CIRCOR/Supplier/Vendor"), for Design, Engineering, Procurement, Manufacturing, Supply, Packing & Forwarding and Transportation of Desuperheater considering Price Basis according to INCOTERMS 2020, as CPT, Project Site, Raigad, Maharashtra basis for New 1200 MTPD (DAP Basis) NPK Fertilizer Plant for Rashtriya Chemicals and Fertilizers Limited (RCF) at Thal, Maharashtra (India) including guarantees for performance, design, material, & workmanship, in line with scope, specifications, terms & conditions agreed upon as per reference documents stated above.

Mandatory Spares shall be as per MR Document PC193/E-2/P-II/10.0.

2.0 Price & Price Basis:

The Total Price for Supply of Desuperheater for New 1200 MTPD (DAP Basis) NPK Fertilizer Plant along with Associated Utilities/Off Sites Facilities for Rashtriya Chemicals and Fertilizers Limited (RCF) at Thal, Maharashtra (India) shall be INR 9,99,750 (INR Nine Lakhs Ninety Nine Thousand Seven Hundred Fifty Only) (Hereinafter referred to as "Contract Value"), considering Price Basis according to INCOTERMS 2020, as CPT, Project Site, Raigad, Maharashtra basis, including Packing & Forwarding and Transportation.

TPI Charges shall be in L&T's Scope.

The above prices are based on vendor design complying with Material Requisition (MR) Ref. No. E24003-RCF-IN-MRQ-042, its Addendum(s) (if any) and are fixed unless there is an impact attributable to change in design data.

Price in full or in part are firm and fixed with no escalation whatsoever.

The prices are exclusive of GST and shall be paid extra @ 18%.

Payment of GST charged in the invoices will be released, after validation of suppliers' GST compliances, on GSTN Portal, which will be 20 days after filing of GSTR 3B return by supplier, or 20 days from the date of release of basic payment, whichever is later.

Variation in prices of raw material or bought-out items shall not be admissible. Any variation in price on account of exchange rate fluctuations, changes in taxes/duties/law, changes in Codes and standard, increase in wages, Commodity rates, labor rates, fuel prices etc. shall not be admissible.

3.0 Terms of delivery:

CIRCOR Flow Technologies India Pvt. Ltd. shall deliver Desuperheater on or before 26th Mar-2026 on CPT, RCF Thal Project Site, Raigad Maharashtra basis.

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 2 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

The delivery date mentioned above is inclusive of time for requisite approvals from L&T/Owner/TPIA and entire responsibility of delivery, including statutory clearances (if any) remains with Circor Flow Technologies India Pvt. Ltd. and is included in the above-mentioned price.

4.0 Shipment Basis:

CPT, RCF Thal Project Site, Raigad Maharashtra basis.

5.0 Payment Terms:

Payment shall be released as per following milestones:

1.95% of the Basic Order value shall be released after shipment (i.e. Lorry Receipt) on prorata basis as per scope of order, after settling L&T's entitlements of Order price discounts and adjustments if any, as mentioned elsewhere and on submission of following documents to L&T:

#Signed Commercial Invoices (in Triplicate)

#Original GST / Buyer's copy of Tax Invoice (2 Copies)

#Copy of GST Registration Certificate (2 Copies)

#Lorry Receipt/ Railway Receipt (2 Copies)

#Packing List / delivery Challan.

#Original Manufacturer Test Certificates duly endorsed by TPIA -4 Copies

#Inspection Release Note by TPIA (2 Copies)

#Dispatch Clearance in writing from L&T

#Guarantee/Warranty Certificate stating that the equipment, spares, accessories supplied against the subject order meet with the guaranteed requirements as per Project Document No:

E24003-RCF-IN-MRQ-042 and relevant details in addendum issued and points mentioned in TQ closure.

#General Compliance Certificates, issued by Supplier, stating that all terms and conditions of the purchase order until CPT shipment have been fully complied with

#Copy of Email intimation sent to L&T for Insurance

#Copy of e-way bill

2.5% of the Basic Order value (Excluding GST) shall be released after technical document folder (in soft & hard) as per RFQ and submission of as built documents, acceptance of the same by L&T.

GST payment Clause:- Payment of GST charged in the invoices will be released, after validation of suppliers' GST compliances, on GSTN Portal, which will be 20 days after filing of GSTR 3B return by supplier, or 20 days from the date of release of basic payment, whichever is later.

Invoicing Details:

Invoice shall be in the name of:

Larsen & Toubro Limited
Division Name: L&T Energy Hydrocarbon
14th Floor, AMN Tower, AMN Tower, Jogeshwari - Vikroli
Link Road, Powai Campus, Gate 3, Mumbai Suburban,
Maharashtra, 400072
GSTIN: 27AAACL0140PEZ6

The Consignee Address shall be as follows:

Larsen & Toubro Limited
Division Name: L&T Energy Hydrocarbon
Rashtriya Chemical and Fertilizers Limited

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 3 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

Thal, Thal-Vaishet, Alibag # 402 208
State: Maharashtra, India
GSTIN: 27AAACL0140PEZ6

Supplier shall mention GST No. and HSN code on original invoice.

Scanned copies of the invoice and supporting documents shall be uploaded on the SAP Ariba Portal.

Upon uploading of the scanned copies, Hard Copies of the Invoice shall be sent to following address:

Mr. Jignesh Shah / Mr. Sarthak Gulati
L&T Energy Hydrocarbon,(AdVENT)
3rd Floor, IndiQube Lake View,
Saki-Vihar Road, Powai, Mumbai-400076
Tel: +91 9820030267 / +91 7828455361
Email:shah.jignesh@larsentoubro.com / sarthak.gulati@larsentoubro.com

Further to the above, Supplier shall also submit one copy of digitally signed invoice with supporting documents to the following e.mail ID:
rcfnpk.invoices@larsentoubro.co m

For queries related to release of payments & TDS certificate, please contact following personnel:
L&T Shared Services Centre (Operations-Baroda):
Mr. Neeraj Gupta
Tel.: 0265 245 4074
Mob.: 8376804504
Email: Neeraj.Gupta2@larsentoubro.com

Soft Copy of Bank Guarantee shall be uploaded on BG Portal and Original copy of the same shall be couriered at below address.

Ms. Vardhana G
Officer - Operations
L&T Ltd - Shared Services Centre
TCTC Building, 2nd Floor
P.B.No. 979, Mount PoonamalleeRoad
Manapakkam, Chennai - 600089
India.
Tel No: +91 44 22528847

6.0 Price Discount:

In case of delay in supply beyond the Contractual Delivery Date (LR date will be reckoned as delivery date), the price will be discounted @0.5% of the undelivered portion of Order value (excluding taxes & duties) per week of delay or part there of subject to a maximum of 5% of undelivered portion of Order value (excluding taxes & duties).

7.0 Performance Bank Guarantee:

Not Applicable.

8.0 Special terms and conditions:

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 4 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

The following Special Terms & Conditions shall supplement and / or amend the Terms & Conditions. Whenever there is a conflict, the provisions herein shall prevail over those in the Terms & Conditions

GTC Clause Ref 2.1 and 2.2: Prices & Price Basis
Charges of Third Party Inspection (TPI) shall be in L&T's Scope.
Price and Price basis shall be as per clause 2.0 of this Purchase Order.

GTC Clause Ref 3.0: Payment Terms
Payment Terms shall be as per clause 5.0 of this Purchase Order.

GTC Clause Ref 4.10: For Delay in submission of documents /drawings
Not Acceptable

GTC Clause Ref 4.20 & 4.30: Price Discount
LD shall be as per clause 6.0 of this Purchase Order.

GTC Clause Ref 5.00: Delivery
Delivery shall be as per clause 3.0 of this Purchase Order.

GTC Clause Ref 8.00: Shipment Basis
Shipment Basis shall be as per clause 4.0 of this Purchase Order.

9.0 General Terms & Conditions:

1.00 Scope of Work / References

1.10 Scope of work shall be strictly as per the Material Requisition (MR) enclosed with enquiry document, subsequent addenda, correspondence (Technical Queries, Commercial Queries etc.), discussions and MoMs. Any exclusions, clarifications etc. stands withdrawn unless otherwise mutually agreed, duly recorded and accepted by L&T prior to placement of the Order.

2.00 Prices & Price Basis

2.10 "Price shall be on Ex-works basis inclusive of packing and loading onto transporter's vehicle. Price in full or in part are firm and fixed with no escalation whatsoever.

Variation in prices of raw material or bought-out items shall not be admissible. Any variation in price on account of exchange rate fluctuations, changes in taxes/duties/law, changes in Codes and standard, increase in wages, Commodity rates, labor rates, fuel prices etc. shall not be admissible.

Prices to be submitted in Indian Rupees (INR) only and shall be valid for minimum 90 days from the date of offer prior to the Purchase Order "

2.20 Appointment of Third-Party Inspection Agency (TPIA) for performing inspections as per approved Inspection Test Plan (ITP), Quality Procedures/Specifications and Material Requisition at Supplier's works & Sub-Supplier works (Requiring Final Inspection Release Note as per ITP/ agreed with L&T) shall be Supplier's scope, Charges for the same shall be borne by Supplier, including coordination with Third Party Inspection Agencies for timely Inspection shall be in Supplier's scope.

"TPIA Inspection and testing at sub-suppliers works shall be in supplier's scope (unless otherwise mutually agreed).

TPIA Inspection and testing charges for any new procedure qualifications and Material testing shall be under supplier's scope.

TPIA chosen by Suppliers for the TPIA inspection under their scope shall be strictly from following approved TPIA's:

- 1) PDIL
- 2) LRQA

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 5 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

- 3) TUV India Pvt. Ltd.
- 4) Bureau Veritas (India) Pvt. Ltd.
- 5) DNV
- 6) SGS
- 7) TUV Sud South Asia Pvt. Ltd.
- 8) Certification Engineers India Ltd (CEIL)
- 9) Engineers India Limited (EIL)

OR other reputed Inspection Agency may be selected with prior approval from L&T/PMC/Owner.

Cost towards repeat tests and inspection due to failures, repairs etc. for reasons attributable to the supplier/sub-supplier shall be borne by the supplier. Supplier shall not obtain any Inspection waiver from their appointed TPIA without prior approval from L&T."

"TPIA Inspection/L&T Inspection/ Inspection waiver does not relieve supplier/manufacturer/sub-suppliers from contractual obligations and their responsibilities to supply the item(s) concerned in full compliance with the requirements of the Purchase Order and project specifications, applicable codes and standards & approved Quality Plan.

The Owner has the right to inspect and test goods to ensure they meet contract specifications. Inspections can occur at the Buyer's or Supplier's premises or at the final destination. The Supplier must provide necessary facilities and assistances for these inspections. Delays in approval or testing are the Supplier's responsibility. The Supplier must promptly provide copies of all test results to the Buyer."

2.30 Prices are inclusive of following charges: -

2.30.1 Transportation saddles, templates/gauge plates, spider beams, transportation Stools etc., as applicable.

2.30.2 Testing / Trial Run / FAT/ IFAT as per RFQ and approved ITP at Supplier's works, sub-Supplier's works & Site

2.30.3 a) Third-Party Inspection charges for inspection of balance items (not covered under clause 2.2) as per Inspection Test Plan (ITP) at Supplier/sub-Supplier's works shall be in Supplier's scope.
b) Raw materials inspection & certification requirement i.e. EN10204 Type 3.2/3.1, as applicable to be performed through project approved TPIA at Mill / Supplier locations. Including any additional check testing at approved laboratory to comply with PO/MR requirements

c) Inspection & Certification required as per Indian Boiler Regulation (IBR), ASME AI Inspection/Code Stamping and National Board of Boiler & Pressure Vessel Inspectors (NBBI) Registration & and National Board Registration (NBR) as applicable.

d) Third-party Inspection charges to perform check testing of raw material in laboratory with TPIA witness in case raw materials (Plate/ Pipe/ Fitting/ Forgings etc.) is procured from Approved Traders / Stockiest & with mill test certificate EN10204 Type 3.2/3.1.

e) Third-party Inspection Charges at Supplier/sub-Supplier works to perform New/ Fresh Welding Procedure/ personnel Welder Performance Qualifications for welding, NDE, Painting, Refractory, Laboratory Testing Charges shall be in Supplier's scope. In case of any failure/rejection in supplied item/material observed at site, cost of failure analysis (including lab testings, re-inspection) shall be borne by Supplier.

f) For indigenous suppliers, for items deviating IBR requirements, inspection shall be carried out by IBR authorities only.

g) In case of any failure/rejection in supplied item/material observed at site, subject item/material will be sent to the L&T's approved labs for failure analysis. The cost of this failure analysis shall be borne by Supplier, if failure/rejection is attributed to Supplier.

2.30.4 Mandatory Spares - as per list specified in MR.

2.30.5 "2 years O&M spares:

Supplier to quote optional price valid till 31-Dec-2028 as specified in the MR. If the list is not furnished in MR, the same shall be provided as per Supplier's Recommendation in SPIR format provided by L&T.

OWNER reserves the right to place the Order for such O&M spares"

2.30.6 "Spares for Start-up / Erection & Commissioning/Performance Guarantees as specified in the RFQ. If the list is not furnished in RFQ, the same shall be provided as per Supplier's Recommendation. If some Spares, which are considered "Not applicable or not considered in correct quantity as per philosophy" by Supplier becomes applicable directly or becomes applicable by having equivalent function the quantity is corrected based on spares philosophy at the time of order execution, Supplier shall supply them in line with Mandatory Spares Philosophy at no extra cost and / or time implications to L&T."

2.30.7 Special Tools & Tackles required for Erection & Commissioning and performance guarantees as specified in the RFQ. If the list is not furnished in RFQ, the same shall be provided as per Supplier's Recommendation.

2.30.8 "Training Charges (If applicable) for the complete package, individual components as mentioned in MR.

Continued..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 6 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

These shall include charges for training at Supplier/Sub-supplier Works and / or at Project site, training manual, etc.

Training charges shall be quoted as Lump sum or Per Diem basis."

2.30.9 Charges for documentation (hard & soft copy) for approval & as-built etc. in requisite no. of sets as per RFQ. In case the design of any equipment / ordered items is in L&T scope, all such design documents shall be shared to Supplier and all such documents shall be included as part of Final Documentation (soft and hard copies) without any cost implication to L&T.

2.30.10 List along with quantities for all first-fills for all equipment i.e. Lubricating oils, Grease, Silicon Oil, chemicals, consumables etc. in addition to commissioning and start-up requirement for first fill + six months normal operation after commissioning. Supply shall be by L&T

2.30.11 NA

2.30.12 Supplier confirms acceptance of following as MINIMUM across the Table Meetings at L&T / Detail Engineering Consultant / PMC / Owner Office with presence of competent team member :-

a) Kick Off Meeting (2 of days)

b) 1st Design / Document Review Meeting (3 of days)

c) 2nd Design / Document Review Meeting (3 of days)

d) Model Review Meetings (30%, 60%, 90% - 2 day each)

e) Hazop / SIL Review Meeting (2 day each)

Visit Schedule shall be as decided by L&T and conveyed to Supplier.

Supplier to depute competent team member for all above across the table meetings. All costs associated for attending such across the table meetings at L & T Mumbai are included in base price submitted by Supplier.

2.30.13 Providing information and performing required activities in POMS for progress reporting, drawings and document submission & review cycle, inspection calls & TPIA co-ordination, logistics co-ordination, etc as per the project requirements. Performing required activities in SAP Ariba for P.O. Acknowledgement, Shipping notification, Invoicing, etc.

2.40 Taxes and Duties

2.40.1 Raw Materials / Components imported by Indigenous Supplier to their works.

For all raw materials / components sourced by the Supplier/Sub supplier, Supplier/Sub Supplier shall not be entitled for any concessional import duty (whether in India or any other location) for components / raw materials being imported by the Supplier and sent to their works / their sub-supplier works. Any such imported components shall be customs cleared by the Supplier after paying the applicable customs duties. All import duty and other taxes & duties on such raw material / components shall be in the scope of the Supplier and paid by the Supplier. Statutory variation in taxes & duties including FTA, CEPA etc. on raw material / components shall be to Supplier's account at all times.

L&T/OWNER will not bear any liability towards payment of Safeguard Duty, Anti-Dumping Duty, Protective Duty and applicable IGST on same Or Countervailing Duty on subsidized articles or any other such duties of Customs imposed by the Government under Customs Tariff Act, 1975 applicable on import of any materials to India for the subject work. Therefore, the Suppliers are expected to source such materials required for the execution of work only from domestic producers in India.

To make Price competitive, Supplier are requested to consider free trade agreement/CEPA etc. for import of Raw material, wherever applicable. Supplier to note that all responsibility of these import and procedures shall be responsibility of Supplier.

Imported Raw Material / Finished product shall not be despatched directly to Site. These will be sent to supplier works and then despatched as local supply to Site

2.40.2 Finished Goods shipped from Supplier Works:

For all raw materials / components sourced by the Supplier, all import duty and any other taxes & duties on such raw material / components shall be in the scope of the Supplier and paid by the Supplier.

Statutory variation in taxes & duties on inputs (raw material / components), shall be to Supplier's account at all times.

The Supplier shall adhere to the Billing Instructions issued by L&T (to be issued before despatch).

Applicable Goods and Services tax (SGST/UTGST, CGST, IGST) on Supplier's Goods and/or Services shall be paid by L&T, against documentary evidence provided by Supplier subject to following terms

#

- Supplier is registered under GST and have valid GST registration at all the time.
- Supplier shall submit GST tax invoice as per applicable SGST / UTGST / CGST / IGST Act, 2017 and corresponding GST rules
- Supplier shall discharge the applicable CGST & SGST/UTGST / IGST liability, as the case may be, and file GST Returns within the stipulated time.
- Applicable GST Amount & rate to be indicated in tax invoice as per relevant GST Act.
- Credit and Debit note to be raised as per the relevant provision of GST Act and rules made thereunder.
- Any increase in GST Rate over and above the applicable rate, whether arising out of statutory changes or otherwise beyond the "Contractual Delivery Date" (CDD) shall be to Supplier's account.
- Any increase in the GST Liability or any other duties, taxes and levies due to wrong classification of HSN (Harmonized System of Nomenclature) / SAC (Services Accounting Code) by the Supplier, within Contractual Delivery Date, shall be to Supplier's account.
- In the event the Supplier does not discharge the applicable GST in time as per the GST payment rules, L&T reserves the right to retain the said amount from Supplier's invoice and release the same

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 7 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

only after evidence of tax payment by the Supplier is furnished.

Supplier shall furnish the details of returns filed against invoice issued to L&T as evidence of their tax liability discharge.

Statutory Variation in Taxes and Duties:

Statutory variation in GST mentioned above and any new taxes and duties, within the Contractual Delivery period on finished Goods and/or Services and Goods directly despatched to site from Supplier works shall be to L&T's account.

Statutory variation in GST mentioned above and any new taxes and duties, beyond the Contractual Delivery period on finished Goods and/or Services and Goods directly despatched to site from Suppliers works shall be to Supplier's account. Any other statutory variations shall be to Supplier's account.

Statutory variation in taxes & duties on raw material / components shall be to Supplier's account at all times.

Income tax / GST TDS shall be deducted from the Supplier Invoices / Claims and deposited with respective Govt. as and when applicable.

Statutory Approvals:

Unless otherwise specified, it shall be the Supplier's sole responsibility to obtain all approvals from any authority (except for CRZ clearance) required under any statute, rule or regulation of the Central or State Government concerned with the performance of the Order and/or the contractual Work. The application on behalf of the L&T for submission to relevant authorities along with copies of required certificates shall be prepared and submitted by the Supplier well ahead of time so that the actual construction/commissioning of the works is not delayed for want of the approval/inspection by the concerned authorities. The SUPPLIER shall arrange for the inspection of the works by the authorities and will undertake necessary coordination and liaison required and shall not be entitled to any extension of time for any delay in obtaining such approvals.

Any Goods and Services Tax (GST) liability arising in India on account of place of supply of goods and / or services deemed to be within India (as per GST provisions) shall be on account of the Supplier. If required, the Supplier will obtain GST registration in India and comply with all GST related requirements in India, including but not limited to provide GST registration certificate, issue of proper tax invoices, uploading of tax returns, and transfer of input tax credit to L&T etc.

Please refer Annexure I "GST Check List" to be ensured for generation of Invoice and Packing list in compliance with GST Act. However, it is sole responsibility of Supplier to ensure total compliance to applicable SGST / UTGST/ CGST Act, 2017 and corresponding GST rules.

Statutory approval (if any) including any type of permission for movement of ODC consignments from Supplier's works to Project site, shall be in Scope of Supplier. Supplier shall arrange such applicable approvals in advance so that actual movement of goods is not delayed for want of the approval/inspection by the concerned authorities.

2.40.3 Supplier shall comply to Tax collected at source (TCS) on Sale of Goods u/s 206C(1H) of Finance Act, 2020 and subsequent amendments. TCS shall form part of the invoice for sale of goods

3.00 Payment Terms

3.10 "10% of the Basic Order value (excluding taxes and duties) shall be released on fulfillment of below conditions:

- Submission of unconditional Purchase Order acceptance.

- Approval of Critical drawings / documents viz. PR, P&IDs and GA drawings by L&T / PMC in Code-2.

- Submission of Advance Bank Guarantee (@ no cost to BUYER) for advance of equivalent amount "

3.20 "10% of the Basic Order value (excluding taxes and duties) shall be released on fulfilment of below conditions:

- Receipt of major raw materials / Bought out Components at Supplier's Works

- Submission of Advance Bank Guarantee (@ no cost to BUYER) for advance of equivalent amount

- Approval of Billing Schedule

- L&T / OWNER / PMC Approval of FAT AND / OR SAT Procedures for establishing Guaranteed Parameters at Supplier Shop AND / OR Project Site

- Delivery of Foundation Templates as applicable

- Approval of Spares List (2 Years O&M Spares - SPIR Format, Erection & Commissioning Spares, Mandatory Spares). Format shall be provided by L&T.

Supplier shall submit the copy of the TPIA Material Identification & Inspection clearance reports as a backup confirming receipt and acceptance of materials"

"All Advance Milestone Payments shall be released on submission of Advance Bank Guarantee for advance for equivalent amount valid until the end of the Contractual Delivery Date Plus 3 months Plus 3 months claim period. Advance Bank Guarantee (ABG) shall be furnished as per L&T's format and from L&T approved banks.

In case delivery date is extended due to whatsoever reason, then Supplier shall extend the validity of ABG without any cost implication to L&T till expected delivery date plus claim period of 3 months.

In case partial delivery is done, the ABG value may be reduced by the Supplier after obtaining specific approval in writing from L&T"

3.30 65% of the Basic Order value shall be released after shipment (i.e. Lorry Receipt) on prorata basis as per scope of order, after settling L&T's entitlements of Order price discounts and adjustments if any, as mentioned elsewhere and on submission of following documents to L&T:

Continued..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 8 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

- # Signed Commercial Invoices (in Triplicate).
- # Original GST / Buyer's copy of Tax Invoice (2 Copies)
- # Copy of GST Registration Certificate (2 Copies)
- # Lorry Receipt/ Railway Receipt (2 Copies)
- # Packing List / delivery Challan.
- # Original Manufacturer Test Certificates duly endorsed by TPIA -4 Copies
- # Inspection Release Note by TPIA (2 Copies)
- # Dispatch Clearance in writing from L&T
- # Guarantee/Warranty Certificate stating that the equipment#s + spares + accessories supplied against the subject order are meeting the guarantee requirements of the P.O. no. XXXX and PR No. XXXX
- Rev XX (Latest revision of the approved PR to be mentioned)
- # General Compliance Certificates, issued by Supplier, stating that all terms and conditions of the purchase order until Ex Works shipment have been fully complied with
- # U-1 / U-2 form as per ASME and As-built drawings signed by AI
- # Copy of Email intimation sent to L&T for Insurance
- # Copy of e-way bill

In case Performance Bank Guarantee as per L&T's format & from the approved banks as specified by L&T is not issued, 10% of the value of items shipped shall be kept on hold till PBG is received

3.40 Not used

3.50 10% of Basic Order value shall be released on submission of Technical document folder (in soft & hard) as per RFQ, acceptance of the same by L&T, and PBG extension for delay / extended delivery beyond original CDD as applicable.

3.60 Not used

3.70 5% of the Order value shall be released after Project commissioning and Performance Test Certificate of the Project. For the purpose of this payment, the date for Project Commissioning and Performance Test shall be reckoned as the date certified by L&T Site-in-charge in the Commissioning and Performance Test Certificate which shall be issued upon completion of all jobs, receipt of Final Bill from the Supplier and L&T receives Final Certificate from Owner/PMC, or six (6) months from the date of delivery of material, whichever occurs earlier.

3.80 All Advance and Retention Payments shall be made to Supplier within 30 Days from receipt of Original Invoice along with all the correct & complete documents specified above by Wire Transfer.

3.90 Despatch Payments shall be made to the Supplier within 90 days from receipt of Original invoice along with all the correct & complete documents specified above by Wire Transfer.

3.100 In case partial delivery is done, the ABG value may be reduced by the Supplier after obtaining specific approval in writing from L&T.

4.00 Price Discount:

4.10 "For delay in submission of documents/ drawings: In case of delay in submission of drawings/ documents (agreed as milestones), the Order value shall be discounted @ 0.5% for non-achievement of each milestone (irrespective of the extent of delay). Milestones shall be as follows :

PR inputs - XX Weeks from LOI

GAD - XX Weeks from LOI

P&ID - XX Weeks from LOI

Equipment Datasheet Curves - XX Weeks from LOI

Motor / VFD Datasheet & Curves - XX Weeks from LOI

ITP - XX Weeks from LOI

In order to be considered as "submission" for the purpose of this provision, drawings and data submitted by Supplier must meet the Purchase Order requirements sufficiently to receive Buyers review "Code 1" (proceed) or "Code 2" (proceed, subject to acceptance & incorporation of comments) . Documents submitted by Supplier but which do not meet Buyer's minimum requirements will be returned with Buyer's "Code 3" (rejected / do not proceed) review code and are considered not received until revised and resubmitted in acceptable form to Buyer. "

4.20 For delay in delivery of material: In case of delay in supply beyond the Contractual Delivery Date, the price will be discounted @1% of the Order value (excluding taxes & duties) per week of delay or part thereof subject to a maximum of 10% of Order value (excluding taxes & duties)

4.30 "The total Price discount due to clause 4.1 and 4.2 above shall be limited to 10% of Order value.

It is further agreed above clause(s) shall not constitute a waiver of any rights of Buyer to damages or other remedies of Buyer under this Purchase Order or otherwise for Supplier's improper performance or default in performance of any other aspect of this Purchase Order.

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 9 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

It is agreed by the Parties that the amount corresponding to Price Discount, pursuant to Article 4.30, is genuine pre-estimate of the likely loss to Buyer as a result of Supplier's failure to complete the Work as per the Subcontract/PO Schedule and, as such, should not be considered to constitute a penalty under any applicable laws.

Should the Supplier fail to give effect to the imposition of Price Discount pursuant to Article 4.30, the same shall be payable by the Supplier on demand by Buyer. In the event of such failure by the Supplier, the Buyer reserves the right to recover / set off an amount equivalent to the applicable Price Discount out of the sums payable to the Supplier or from the Guarantees or Deposits furnished by the Supplier or the Retention Money retained from the Invoices of the Supplier.

Any Extra ordinary delay (beyond 10 weeks from CDD) not attributable to L&T will be considered very seriously and Supplier may be kept under Holiday list for any subsequent enquiries/ RFQ for a period of not less than three years."

4.40 Non-fulfilment of Utility Guarantee parameters (As Applicable):

Without prejudice to any other reduction, discount or adjustment in price to which the L&T is entitled to, if the actual consumption of Power /Cooling Water/ MP Steam/LP Steam/ Fuel Gas for the Equipment/Packages exceeds and / or actual generation of MP Steam by Equipment/packages is lower than the Stipulated guaranteed figures of the respective utilities furnished by the SUPPLIER in its offer for the Equipment/Packages, L&T shall be entitled to an adjustment in the lump sum price by way of discount as follows. No benefit shall be passed on to the supplier in case the utility consumption demonstrated by the supplier during guarantee test/s is/are less than the guaranteed figures quoted in the bid. The reduction in price shall be in addition to any price reduction mentioned elsewhere in this document / PO.

A. Reduction in price with respect to power consumption (As Applicable)

All items mentioned above shall be performance tested at Supplier works and performance demonstrated. Further, for all the other power consuming equipment /packages as defined in this section, Supplier is required to establish the performance of the equipment / packages during PGTR at site. In case the total KW consumed by the equipment / packages exceeds the stipulated guaranteed value by more than 4%, the subject equipment/Package shall be rejected.

However for excess power (kW) consumed over and above the guaranteed/quoted KW to a maximum of four (4%) percent, reduction in overall prices shall be applicable for excess power consumption as per the following formula on extra Power consumption of the equipment & Packages as identified in this section.

"Reduction in price for Excess Power Consumption CPC= (KWREA- KWREQ) x CP x 8000 x DF

KWREQ = Guaranteed Power (kW) of the identified Equipment quoted by the Supplier

KWREA = Actual Power Consumption of the identified equipment/package (KW) as observed during Performance Guarantee test run (PGTR)

DF (Discount Factor)= 4.02

CP = Cost of Electric Power INR 7.35 per kWh

8000= Number of operating hours per year."

Power consumption shall be measured at motor terminal.

B. Reduction in price for Steam Consumption (As Applicable)

During performance tests MP & LP Steam Net Consumption (MT/Hr) by the Equipment/package exceeds the stipulated guaranteed value by more than 5%, the subject equipment/package shall be rejected.

However for excess steam consumption over and above the guaranteed Net steam consumption up to a maximum of 5% (five percent), reduction shall be applicable for excess total steam consumption as per the following formula:

"CSteam con = CMPSteam con + CLPSteam con

i. CMPSteam con = (SMEPAC -SMEPQC) x CSMP x 8000 x DF

SMEPAC = Actual MP steam Consumption of the identified equipment/packages (MT/hr) as observed during Performance Guarantee test run.

SMEPQC = Guaranteed MP Steam Consumption (MT/Hr) quoted by Supplier.

CSMP = Price of MP Steam INR 2143 per MT

DF (Discount Factor)=4.02

8000= Number of operating hours per year."

"CLPSteam con = (SLEPAC -SLEPQC) x CSLP x 8000 x DF

SLEPAC = Actual LP steam Consumption in the identified equipment/packages (MT/hr) as observed during Performance Guarantee test run.

SLEPQC = Guaranteed LP Steam Consumption (MT/Hr) quoted by Supplier.

CSLP = Price of LP Steam INR 1990 per MT

DF (Discount Factor) = 4.02

8000 = Number of operating hours per year.

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 10 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

Steam consumption shall be measured individually for each equipment."

C. Reduction in price for excess Cooling Water Consumption (As Applicable)

For all the Cooling Water Consumers (as specified in the RFQ), in case during performance tests the Cooling Water consumed (M3/Hr) by the equipment/package exceeds the stipulated guaranteed value by more than 5%, the subject equipment /systems/Packages shall be rejected.

However for excess Cooling Water Consumption over and above the guaranteed Cooling Water consumption up to a maximum of 5% (five percent), reduction shall be applicable for excess Cooling water consumption as per the following formula.

"Excess Cooling water consumption $CC_{Cooling\ water} = (ECCWA - CCWQ) \times CCW \times 8000 \times DF$

$ECCWQ$ = Total Cooling Water Consumption (M3/Hr) of the Equipment/package as quoted by the Supplier

$ECCWA$ = Actual Cooling Water Consumption (M3/Hr) of the Equipment/Package observed during Performance Guarantee test run

8000 = Number of operating hours per year.

CCW = Cost of Cooling water is INR 1.25 per M3.

DF (Discount Factor) = 4.02 "

Cooling water shall be measured individually for each applicable Cooling water users.

In the event of deviation with respect to guarantees as specified above, Supplier is required to rectify the same without any commercial/technical implication. In case of Supplier is not able to rectify the same, equipment(s) not meeting the guarantees shall be rejected and required to be replaced by Supplier without any time and cost implication to L&T.

D. Additional Guarantees with respect to Equipment Performance

Supplier during the performance of contract is required to carry out hydraulics check and required to guarantee the hydraulics for meeting the throughput & Specification requirement as defined in the MR to meet the hydraulic guarantee of the supplied Equipment / package for both Design Case-1 & Case-2 (including SOR & EOR) as specified in MR.

E. Repair / Rectification / Modification

In case the shortfall in the guaranteed parameters, Supplier may carry out necessary repair / rectification / modification / replacement at his own cost and through his own agency on urgent basis and complete the repair / rectification / modification / replacement within a maximum period of three months to improve the supplied systems to prove guaranteed parameters in the final performance test.

Should the systems continue to underperform, reduction in price shall be leviable after this as specified for shortfall in guaranteed parameters above.

However, if the shortfall in guaranteed parameters is beyond the rejection parameters, Supplier and L&T will mutually agree on the method and possibilities of rectification and if the plant is still not capable of performing within the acceptance limits, it will be L&T's decision with respect to rejection or penalties, which shall be final and binding on the Supplier.

F. Test Run Requirements

Detailed procedure for PGTR shall be discussed with Supplier and finalized mutually, however for field tested equipment/Packages the individual utility consumptions shall be recorded jointly for a period of 72 hours continuous run of the unit. After the completion of successful performance test, average consumption of individual utilities will be worked out. These shall be considered as actual consumption for computation of price adjustment.

All instruments required, if any, to establish the performance are within Supplier's scope of work / supply at no extra cost to OWNER.

In an event that unit operates in a mode / feed case different than specified above or at a throughput lower than design throughput, the utility numbers shall be prorated for reduced Capacity for the purpose of guarantee in mutual discussion with PMC/OWNER.

4.50 Back-charges for Demurrages / Cancellation charges :

ODC/OWC consignments: Supplier shall inform the date for shipment (complete / partial) along with transportation drawings and packing list (Weights and overall dimensions for ODC/OWC) at least six months before the shipment date in order for L&T to finalize the shipping contract, take statutory approvals for movement of ODC consignments from supplier's works to Project site, construction of necessary by-passes, etc. Supplier shall then confirm the firm date of shipment one month in advance for L&T to book the ship / barge and the axles, as applicable and for completing all procedures for further movement to site.

For all other consignments: Supplier shall confirm the firm date of shipment one month in advance for L&T to make the necessary transportation arrangements.

In case the cargo is not ready for shipment as confirmed, any demurrages / detention / cancellation charges, approval cost, etc shall be back-charged to the Supplier at actuals and shall be adjusted from the dispatch payment milestone.

5.00 Delivery:

5.10 Timely delivery is essence of this Purchase Order.

5.20 "Contractual Delivery Date (CDD) is on Ex-works duly packed and loaded onto transporters vehicle, (INCOTERMS 2020) basis.

Delivery is inclusive of time required for acceptance of equipment / item and Release/ Issue of IRN after completion of Inspection activities by Inspection by Third Party Inspection Agency and/or L&T and/or PMC and/or OWNER as per L&T/OWNER approved Inspection Test Plan (ITP)."

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 11 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

- 5.30 "The date of Lorry Receipt shall be reckoned as the date on which the delivery has been effected for delivery on Ex-works basis. The date recorded as "Site Receipt Date" (by the Stores In-charge of L&T at Site) shall be reckoned as the date on which the delivery has been effected for delivery of DAP / DDP Project site basis"
- 5.40 Effective Date of the Order shall be the date of the Order by L&T by way of LOI or P.O., (whichever is earlier), unless specified otherwise.
- 5.50 The number of despatchable units with dimensions & weight (after packing) shall be clearly specified in the offer. Maximum pre-fabrication shall be carried out at Supplier / Supplier's sub-Supplier's works to minimize Site work. The extent of Site fabrication to complete the equipment shall be specified in the offer. All category of spares (i.e. Erection & commissioning, Mandatory, Critical, Capital, 2 years operation, as applicable) shall be packed separately with item wise packing list.
- 5.60 If the SUPPLIER fails to deliver the GOODS within the Contractual Delivery Date due to reasons attributable to the SUPPLIER, the L&T shall have the right to cause SUPPLIER to ship the GOODS by the most expeditious means of transportation, such as air freight, at sole risk, cost and expense of the SUPPLIER.
- 5.70 Supplier shall submit draft Packing List showing all despatchable units with weight & dimensions before 12 weeks# ex-works readiness. Supplier to provide packing list for mandatory spares separately. Box for spares should be separate during dispatch. Any partial supplies of spares is not acceptable
- 5.80 List of inputs required by Supplier to meet the Contractual obligations along with the "Time Schedule" of requirement will be submitted during the Kick-off meeting / within 2 weeks of receipt of LOI/PO whichever is earlier.
- 5.90 Detailed "Delivery Schedule" (Bar Chart) showing Engineering, Procurement, Manufacture, Shop Inspection, Transportation, etc. will be furnished within 2 weeks from the LOI/PO mentioned in PR (Code-I approved by PMC/OWNER). The cost of all such visits by Supplier is included in their price. No separate per diem rate shall be paid to Supplier for such visit.
- 5.11 "For all items except spares - L&T shall, within 16 Weeks of receipt of material at Site, revert regarding any short Supplies w.r.t. Packing List.
For Spares: - L&T shall revert regarding any short supplies during handover of spares to customer.
Supplier shall immediately supply the short supplied items on DDP Site basis with no Price and time implication to L&T."
- 5.12 Supplier shall depute their representative to Site, at least once on a mutually agreed date for reconciliation of all materials dispatched, in order to ensure completeness of the items as per scope mentioned in PR (Code-I approved by PMC/OWNER). The cost of all such visits by Supplier is included in their price. No separate per diem rate shall be paid to Supplier for such visit.
- 5.13 Delay and Recovery Plan (Schedule Recovery)
- 5.13.1 If, in Buyer's sole opinion, the Supplier's performance or progress is inadequate, deficient, or unsatisfactory or the Supplier may fail to achieve any of the milestone dates, Buyer may (but is not obliged to) notify the Supplier of such inadequacy, deficiency, or inability ("Delay Notice").
- 5.13.2 Upon receipt of a Delay Notice, the Supplier must, within seven (7) days or any other period stated in the Delay Notice, submit to Buyer a recovery plan.
- 5.13.3 "The Supplier shall prepare and submit a detailed plan ("Recovery Plan") at its own risk and cost setting out the various measures that it intends to implement in order to recover the delay to the works and milestone dates and eliminate the causes and events which are resulting in such delay, inadequacy, deficiency, or failure, and shall include, but not limited to the following:
- (a) mobilization of additional equipment and personnel;
 - (b) working overtime, increasing the number of shifts or work days or both;
 - (c) expediting delivery of materials by airlifting;
 - (d) enhancing the manufacturing and fabrication capacity;
 - (e) substitution of higher-grade materials or specifications (with prior written authorization of Buyer)."
- 5.13.4 The Recovery Plan shall be subject to Buyer's approval and acceptance and shall be implemented at the Supplier's sole cost.
- 5.13.5 In the event the Supplier fails to submit the Recovery Plan or the submitted Recovery Plan is not acceptable to the Buyer, then the Supplier must, at his sole risk, cost, and expense, implement any remedial measures Buyer may require or instruct.
- 5.13.6 If, in L&T's sole opinion, Supplier's performance or progress is inadequate, deficient, and/or unsatisfactory and the Supplier may fail to achieve any or all of the milestone dates, L&T may depute its own resources at Supplier's Group for expediting purpose. It shall be carried out in consultation with Supplier. Any and all cost incurred by L&T in deputing its resources shall be to Supplier's account and shall be paid by Supplier to L&T.
- 5.13.7 At any time, if the Buyer is of the opinion that the Recovery Plan or any other remedial measures as instructed by the Buyer are not being implemented or not being followed by the Supplier to the satisfaction of the Buyer, the Buyer is free to take all such actions or avail all such rights and remedies available to him under the Purchase Order including but not limited to Step-in, or getting the Work done by third parties Termination for default as per Clause 11.10.2 etc.
- 6.00 Performance Bank Guarantee:
- 6.10 Performance Bank Guarantee for 10% of the basic Order value (excluding taxes & duties) (@ no cost to BUYER) securing the performance of the Supplier as per Order terms & conditions, including but not limited to execution of the Order, delivery, design, material, quality, workmanship, commissioning and satisfactory performance of the Equipment / Package throughout the Defect Liability period (DLP) shall be submitted.
- 6.20 The PBG shall be furnished within one month from the date of LOI or PO, whichever is earlier
- 6.30 Performance Bank Guarantee (PBG) shall be furnished as per L&T's format (Annexure B) and from the Banks approved by L&T. The Performance Bank Guarantee shall be valid until the completion

Continues..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 12 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

of Defect liability period i.e. 31-Dec-2028 or 24 months from the date of delivery whichever is later with a further claim period of 3 Months as specified in the Format of PBG
6.40 The Bank Guarantee(s) shall be extended by supplier at no cost implication to L&T by further period(s) as the L&T may require until performance of all the Supplier's obligations under the PO/Contract.
6.50 "ABGs and PBGs shall be issued as per L&Ts format as attached at Annexure-B and Annexure-C. BGs shall be issued from Indian Branch of one of the Banks listed in L&Ts format. BGs issued by other Banks will not be accepted.

Bidders are requested to ensure that the BG confirmation is transmitted by their bank only in IFN 760 format through SFMS (Structured Financial Messaging system) to facilitate faster processing of the BGs. The procedure / requirements is enlisted herein below for reference:

- a) While making the BG application, Bidder to inform their Bank to send BG confirmation only through IFN 760 (Guarantee) vide SFMS to our Bank.
- b) IFN 760 (Guarantee) should contain the following details: BG No, BG Amount, BG Type, date of issuance, Expiry date, Name of BG Applicant & Beneficiary's name and address.
- c) Bidder should obtain a copy of IFN 760 sent through SFMS by their bank and attach the same with the Original hardcopy of BG and submit the same to L&T.

The details of our bank to which SFMS confirmation has to be sent is as under:

Name of Beneficiary # Larsen & Toubro Limited
Bank Name: ICICI BANK LIMITED
IFSC Code: ICIC0000009
Bank Address: 110, First Floor, Nungambakkam High Road, Nungambakkam, Chennai # 600034"

7.00 "Insurance:
Insurance of the items up to loading on Transporter Vehicle as per delivery basis shall be in Supplier's scope.
Transit Insurance of items from dispatch point up to Project Site shall be in L&T's scope as per below:

Clause 7.0
POLICY NO.:5001004424P121101475 (SCE) / 5001002124P121101476 (MCE)
Policy issued by UNITED INDIA INSURANCE COMPANY LIMITED

7.10 "L&T / OWNER shall arrange the Marine / Transit Insurance for the Project Supplies. Supplier shall ensure that the Policy number is clearly written on the dispatch documents. Supplier shall intimate L&T following details of the dispatch within 24 hours of dispatch.

- Purchase Order No. and Date
- Invoice No. & Date
- LR No. and Date / BL No. and Date/AWB No. and Date
- Name of Transporter / Freight Forwarder
- Place of Works
- Value of shipment
- Number of Packaged cases and gross weight & Net weight, Gross Volume
- Shipment from & to"

In case the Supplier fails to communicate the above information, Supplier will be liable to replace/repair the goods at his cost, in case of damage during transit. Supplier shall provide all the support in case of a claim under the insurance.

Supplier shall provide all the support in case of a claim under the insurance.

8.00 Shipment Basis:

8.10 Shipment basis shall be on Ex-works duly packed and loaded onto transporters vehicle, (INCOTERMS 2020) basis.

8.20 Transfer of Property shall take place on exchange of negotiable documents required at the time of dispatch

9.00 General Conditions of Purchase Order

9.10 Conformance to Engineering Specifications / "Supplier-List" / PTR:

9.10.1 All Materials / items / equipment shall be supplied strictly as per MR, except for mutually agreed deviations duly recorded & accepted by L&T prior to placement of Purchase Order. All other deviations, clarifications etc. stand withdrawn.

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 13 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

9.10.2 All Materials / sub components shall be procured only from Suppliers as per OWNER / PMC "Project Specific Supplier list" and sub-Supplier list agreed pre-order, if any.

9.10.3 Any Material Sourced from a Supplier Outside OWNER/PMC issued "Project Specific Approved Supplier-List" and sub-Supplier List agreed pre-order, if any, shall stand rejected without any commercial implication to L&T.

Suppliers on Owner / PMC "Holiday List" shall not be considered for ordering. Names of such Suppliers who are put on Owner / PMC "Holiday List" subsequent to Supplier approval at bidding stage shall be informed to the Supplier at any stage of the project. Supplier shall comply with this requirement without any time and cost implication to the L&T / PMC / Owner.

"For items not covered in the OWNER / PMC issued "Project Specific Approved Supplier list", the proposed Supplier list shall be approved by Purchaser / Owner / PMC prior to placement of order by Supplier. Supplier shall submit following with necessary documents for such identified items for Purchaser/ Owner / PMC review / approval

a) Complete List of such items / components along with proposed Suppliers

b) PTR of sub Suppliers for supplies made to OWNER in last 5 years. PTR to include details as item, MOC, project, End user name, year of supply and capacity/ dimensions, etc.

c) Any other data / document as requested by Purchase / Owner / PMC.

Above data to be submitted by Supplier within 2 weeks from the date of L&T LOI / PO (whichever is earlier). "

Non-acceptance of any proposed Supplier due to any reasons whatsoever shall not be a cause of schedule and cost implication. Purchaser/ Owner / PMC shall be the final authority for the approval of proposed Suppliers. PMC / Owner / L&T decision shall be accepted by Supplier without any time and cost implication to L&T.

9.10.4 Proven Track Record (PTR) / Supplier Qualification Criteria (VQC) approval (as applicable) from OWNER shall be the responsibility of the Supplier without any Price and / or Delivery implications to L&T. L&T shall submit the PTR/VQC to OWNER. Supplier's competent representative shall attend across the-table discussions with OWNER for approval. In case of non-acceptance by OWNER, L&T reserves the right to cancel the Order without any Price implications to L&T.

9.10.5 Valid Enlistment of the applicable items as per this order with OWNER / PMC shall be the responsibility of the Supplier. In case of non-availability of valid enlistment from OWNER/PMC, L&T reserves the right to cancel the Order without any Price implications to L&T.

9.20 Drawings/Documents:

9.20.1 Supplier shall submit all drawings/documents for approval in soft copy through L&T's POMS as per the time schedule specified in Supplier Document Requirement (SDR) of MR. Draft DCI in required format to be submitted at least 5 days in advance from KOM date. In case KOM is not applicable, Draft DCI in required format to be submitted within 10 days from LOI/PO whichever is earlier. The SDR list shall be populated in the SDR format provided as per RFQ. L&T / PMC / OWNER will approve / comment on the critical drawings/documents within Three weeks from the receipt. For the remaining drawings/ documents, the duration for submission and review / approval shall be mutually agreed during the KO meeting. Any Change in technical specification (for supplier designed equipment / package) shall not be permitted without approval of OWNER / PMC / Licensor / L&T.

9.20.2 Supplier shall get the designs / drawings reviewed & approved by L&T/ PMC / OWNER. The comments shall be incorporated and implemented without any Price and / or Delivery implications to L&T. Supplier shall be responsible for any delay in approval (for reasons attributable to Supplier), and shall not be eligible for any Delivery extension on this account.

9.20.3 Supplier shall ensure re-submission of drawings/ documents after incorporation of L&T's / PMC / OWNER's comments within 10 days from receipt of comments (for first re-submission) and within 7 days for subsequent re-submissions. Supplier shall also ensure that the above documents/ drawings are complete in all respects. In case presence of Supplier is necessary for approval of drawings / documents / clarifications by L&T/ PMC / OWNER, Supplier shall deploy competent personnel and facilitate early approvals/clarifications as required by L&T/ PMC / OWNER at no extra Delivery and / or Price implications to L&T.

Supplier shall submit soft copy of final QC dossier documents within two (2) weeks of dispatch of an equipment. QC dossier shall contain all stage and final inspection reports including Approved QA-QC procedures, Stage inspections and Final inspection reports, Material Certifications, NDT Records, Welding Procedures, Weld Maps, Manufacturing Reports, Factory Test Records (Equipment/ Piping), Statutory Certificate and Drawings etc

9.20.4 Final Documentation as per SDR in RFQ will be submitted in requisite number of copies (both Hard & Soft) failing which the Order will be deemed to be incomplete. Index for final documentation shall be submitted 4 weeks prior to dispatch for approval. Comments from L&T / PMC / OWNER shall be incorporated and re-submitted within 1 week from receipt of comments.

9.20.5 Soft copy of final documents (based on approved index) shall be submitted within 1 month from dispatch for approval. After approval of the soft copy the final documents in soft copies and hard copies as mentioned in MR /RFQ shall be couriered to the address informed by L&T.

9.30 Inspection, Testing and Quality Control:

9.30.1 Supplier confirms acceptance to Indicative Inspection Test Plan (IITP) sent along with the RFQ. In case L&T does not furnish an IITP, Supplier shall furnish ITP along with the offer, as per Inspection requirements specified in the MR

9.30.2 After award of Order, detailed Inspection Test Plan (ITP)/ Quality Control Plan (QCP) shall be, submitted for approval to L&T within three weeks from the date of LOI/PO. L&T / PMC / OWNER shall approve / give comments within three weeks of receipt. Supplier to re submit all quality documents after incorporation of L&T's comments within 10 days. Supplier shall submit all ITP/Procedures for approval through L&T's web based portal (POMS) as per the agreed schedule specified in Supplier Document Requirement (VDR) or MR " L&T/PMC/Owner reserves the right for inspection visits and Quality Audits at Supplier/ Sub Suppliers place/ works wherever felt necessary. Supplier to extend necessary support.

9.30.3 Obtaining approval of ITP from L&T / PMC / OWNER shall be Supplier's responsibility during execution without any price and/or delivery implications to L&T. All comments shall be incorporated

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 14 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

and implemented without any price and / or delivery implications to L&T.

9.30.4 Inspection of items shall be by TPIA / L&T / PMC / OWNER as per approved QCP/ITP. Co-ordination for timely inspection by TPIA shall be Supplier's responsibility. Failure to do so, shall not be a reason for delivery extension.

9.30.5 Failure to inspect goods or payment for goods or acceptance of goods shall in no way impair L&T's right to reject non-conforming goods or to avail of any other remedies to which L&T may be entitled. Any inspection and acceptance thereof, shall not relieve the Supplier of their contractual obligations.

Supplier confirms acceptance to L&T's Supplier's Shop Inspection Co-ordination Procedure. (Annexure M).

9.30.6 TPIA charges for re-inspection due to rejection, incomplete offering, non-readiness etc. for the reasons attributable to Supplier Group shall be to Supplier's account.

9.30.7 Supplier to offer sufficient sizable quantity for Inspection by TPIA / L&T / PMC / Owner (to optimize inspection visit), as applicable as per approved ITP.

Supplier shall raise Inspection Notification through L&T's web based portal (POMS) in line with advance notification period as specified in Supplier Shop Inspection Procedure or as mutually agreed.

In case of abortive visit of TPIA/ PMC/ OWNER due to non-readiness of item for inspection/ cancellation of Inspection Notification without sufficient time as per agreed contract, charges for the same shall be back charged to vendor.

"For any witness and Hold activity as per approved ITP, Supplier cannot proceed for further activity without inspection clearance from LTEH and LTEH TPIA.

TPIA will be available for inspection in normal working day/working hour. Any inspection call/ Visit on weekends/ holidays or extra working hours shall be based on the availability of inspector. Any delays caused due to non-availability of TPIA for aforementioned reasons cannot be claimed for delivery extension.

Supplier to ensure availability of L&T/ PMC/ OWNER approved prerequisites documents for inspection at point of inspection, including sub Supplier shop.

"Participate in inspection activities through VC/ use of Remote Inspection Technique in lieu of physical presence at Supplier/ Sub-Supplier premises shall be at the sole discretion of L&T/ PMC / OWNER.

Participation in remote inspection by L&T/ PMC/ OWNER will be irrespective of TPIA's physical presence at Supplier/ Sub-#Suppliers works.

Supplier / Sub-Supplier should be equipped with IT infrastructure for conducting such remote inspection"

9.30.8 Vendor Inspection Documents

Vendor shall upload MTC and necessary backup document duly endorsed by L&T TPIA in L&T's web portal (POMS) against inspection notification after completion of inspection activities.

Vendor located within India shall submit original hardcopies of MTCs with necessary backup (IRN wise) to L&T QA QC within 1 Week after getting Quality Clearance.

9.40 Billing Schedule / Billing Breakup:

9.40.1 For package items involving multiple dispatches / multiple locations Billing Schedule / Billing Break-up indicating the place of dispatch shall be submitted by the Supplier along with LOI acceptance for approval by L&T and generation of the Purchase Order. Final billing break-up for dispatches shall be clearly mention individual dispatch items with clear unit of measure & quantities. . The generic terms like "Lot", "Set" as unit of measure shall not be acceptable. This shall be subject to L&T's approval and Final Billing break-up to be submitted at least three (3) months prior to commencement of dispatch.

9.50 Progress Reports:

9.50.1 Supplier shall submit, within 2 weeks from the date of LOI/PO, bar chart/S-curve showing start and finish dates of various activities. Updated progress reports shall be submitted on weekly and monthly basis as per L&T's format. After start of manufacturing / production progress report reflecting the updated manufacturing status to be submitted on weekly basis.

9.50.2 Supplier shall provide and/or upload required information and/or pictures and/or data by accessing POMS. The information and/or pictures and/or data (like planned/forecast/actual dates, manufacturing/inspection stages, unpriced Purchase orders issued by Supplier, certain information for logistics arrangement, so on and so forth) to be provided by filling pre-defined and/or mutually agreed MS Excel based templates/formats for facilitating Monitoring, Progress calculation & monitoring, related report generation from POMS for information.

L&T / PMC / OWNER's representative reserves the right to visit Supplier's and/or Supplier Group's premises and/or facilities and/or works to review, assess physical progress, expedite etc. Supplier shall ensure and provide free access during normal working hours (and extended working hours, if required) at Supplier's and/or Supplier Group's premises and/or facilities and/or works. Such expediting shall, in no way, relieve the Supplier of their delivery obligations under the terms of the Order. Supplier shall also provide unpriced copies of Purchase Orders, for major parts & components placed by Supplier and/or Supplier Group to support progress calculated & reported as part of progress report. Cut-off date and submission date for monthly and/or weekly reports shall be defined in the Kickoff meeting.

"Supplier shall convey all applicable Purchase Order requirements to sub-Suppliers. All sub-orders shall show the following statement:

"THIS ORDER MAY BE SUBJECT TO INSPECTION, EXPEDITING AND/OR AUDITING BY BUYER AND/OR OWNER WHOSE REPRESENTATIVES SHALL BE ENTITLED TO ENTER THE ADMINISTRATIVE AND PRODUCTION FACILITIES OF SUB-SUPPLIER'S PREMISES of any tier ""."

Upon request copies of sub-order(s) shall be submitted to L&T. Copies may be unpriced but should clearly show the description of material or equipment involved, agreed delivery date(s) and evidence of

Continued..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 15 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

acceptance of the sub-order by the sub-Supplier.

9.50.3 Kickoff Meeting shall be carried out at L&T's offices. Supplier's Engineering, Quality & Project Manager and sales personal involved in bidding stage shall be present (as may be required by L&T/OWNER). Supplier to ensure presence of key representatives of their Critical / KEY sub-Suppliers as may be required for items falling on critical path for order execution.

9.50.4 Detailed organization chart with contact details (E-mail, Hand-phone numbers & direct numbers) of all key personnel viz. Project Manager, Engineering Manager, Quality Manager & Production / Manufacturing manager and Service Team for Site activities to be submitted by Supplier in KOM.

9.50.5 Supplier shall provide CV of the Project manager who shall be responsible for execution of the order. L&T reserve the right for interaction with the proposed project manager and will reserve the right to accept or reject the CV. In case of rejection, Supplier shall propose an alternate candidate for L&T approval.

9.50.6 All Review meetings (Design, document review and approval, progress review, etc.) shall be held at L&T / PMC / OWNER's office as may be required by L&T/ OWNER. Supplier and their representative including sub-Suppliers shall attend all such meetings without any cost implication to L&T / PMC /Owner.

9.50.7 Project sponsors (members of management that sponsor the Project) from "L&T" and "Supplier" shall attend the meetings once in two months as minimum or frequency as felt necessary.

9.50.8 Progress measurement weightages & Milestones shall be discussed with supplier during KOM and accordingly vendor to submit the progress reports

9.60 Dispatch clearance:

9.60.1 "Supplier shall submit copy of IRN issued by TPIA, Inspection reports and other QA documents duly reviewed / approved by TPIA & any other inspection agencies as per approved ITP for dispatch clearance by L&T. Goods / individual items shall be dispatched only after issue of Code-I approval (approved with no comments) of PR and drawings/ documents by L&T / PMC / OWNER

Items shall not be dispatched by Supplier without L&T's written dispatch clearance."

9.70 Billing Instructions:

9.70.1 The Billing Instructions (including Formats of Invoice, Packing List, and LR) shall be forwarded to the Supplier along with the Purchase Order or along with dispatch instructions. Supplier shall prepare specimen of these documents TWO weeks prior to dispatch and submit the same to L&T for approval. If the Billing Instructions are not strictly followed, L&T shall withhold the amount payable till such time the corrected documents are submitted.

9.70.2 Advance Dispatch Documents:

"Supplier shall forward to L&T copies of the following documents within 48 hours of dispatch by fax/e-mail: --

- Commercial Invoices
 - GST Invoice
 - Lorry Receipt
 - Packing List duly approved by Supplier.
 - Inspection Release Note issued by TPIA (PMC Endorsed # as applicable)
 - Manufacturer's Certificate of Inspection for Shipment issued by Supplier, Stating that all terms and conditions of the Purchase Order / Contract until Ex works shipment have been fully complied"
- "Supplier shall attach / send following documents along with consignment : GST invoice (Transporter copy)

1) E-way / Road permit bill where necessary

2) Packing list with proper quantity and HSN codes

E-way Bill shall be in Supplier's Scope."

9.80 Transportation:

9.80.1 If transportation is in Supplier's scope, items shall be dispatched on "Door-Delivery and Freight Pre-paid basis". Goods shall be sent on full truckload basis only. Prior permission of L&T shall be taken for dispatch on Part-Load basis. Retail booking or Transshipment is not permitted. Under no circumstances, consignment shall be dispatched on #delivery against consignee copy. Items should not be dispatched in Shipper's / Supplier's own container.

Currently Entry Permits are not required. However if required, L&T shall provide the required Entry Permits for supplies (meant for permanent incorporation into the Project) made to site. Supplier shall dispatch goods only after receipt of the same. Supplier/ transporter shall mention the Form No. on Invoice and LR. The Supplier shall furnish reconciliation of Road Permits issued to them along with final Bill. Supplier shall issue necessary way bill / e-way bill as and when goods are dispatched from Suppliers works / site.

9.90 Damage to items in Transit:

9.90.1 In case the consignment has met with an accident in transit and/or the item(s) are received by L&T in damaged condition at the Site, it shall notify the same to the Supplier and Supplier shall send competent representative immediately to Site / accident Site as required for assessing the damage so that the damaged components / parts can be re-Ordered. Any other documents Support as required by Insurance company / L&T/ OWNER will be furnished by Supplier. Supplier shall arrange to obtain the #Damage Certificate# for the value declared by Buyer and all other documents such as first information report (FIR) etc. from the transporter for claiming / settlement of insurance claims.

10.00 Packing:

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO

LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 16 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

10.10 Supplier shall adhere to Approved Packing & Marking Instructions. (Annexure-D)

The Supplier wherever applicable, shall, after proper painting, pack and crate all materials for shipment in a manner suitable for export to a tropical, humid climate in accordance with internationally accepted export practices and in such a manner so as to protect them from damage and deterioration in transit by road, rail and/or sea and during storage at the Site till the time of erection. Without prejudice to any other liabilities or obligations of the Supplier, the Supplier shall be responsible for all damage(s) to the materials due to improper packing. Depending on their type, goods are to be protected against both mechanical damage (shocks, rupture, breakage, loss) and corrosion (moisture, rain, salty atmosphere, sand, etc.). All the items for this Project shall be shrink wrapped. All packing materials will be considered non-returnable.

Should L&T, OWNER or any of its representatives incur any additional costs in rectification works due to a failure of the Goods, rectification work of damage(s) due to improper packaging or documentation to conform to the requirements of the Purchase Order, L&T shall back charge such costs to Supplier and shall deduct such amounts from unpaid amounts owed to Supplier.

10.20 Packing lists shall be reviewed and endorsed by supplier prior to the dispatch. All short Supplies shall be listed separately & communicated to L&T by supplier immediately upon dispatch along with the dispatch plan of the short supplied items.

10.30 Purchase Order No. and P.O. Sr. No. shall be mentioned in all the commercial documents like Packing lists, Invoices etc. Failure to do so shall result in returning of documents to the Supplier and delay in payments.

10.40 Supplier shall submit lashing/ loading scheme to L&T for approval for ODC consignments, if applicable. L&T's Logistics Engineer reserves the right for inspection prior to dispatch of the consignment at the loading point.

10.50 Supplier to pack the Start up / Erection & Commissioning Spares, 2 years# operating spares and Special Tools & Tackles in separate wooden boxes.

10.60 Supplier shall submit, "Transportation Methodology" which includes, but not limited to the following at least five months prior to Contractual Delivery Date:

- Protection arrangements for Internals to avoid the damage during multimodal transportation up to Project Sites
- Lifting & loading plans at Supplier works for loading the cargo on Axle
- Transportation Sketch showing Overall Length, Width, Height, Weight & Three Dimensional CG Location, Saddles locations, Lifting and lashing Points
- Transportation Loading diagram on Hydraulic Axles & marching plan from Fabrication Area to Jetty
- Supplier to provide required Jetty to LSP free of Cost.
- Special requirements for unloading at on Barge & Roll-off at Jetty and subsequent transportation to Site etc.
- Design and fabrication drawing for Stools (Pedestals) for Saddles for off-loading on stools on barge as well as while off-loading the equipment at site.
- Scope of Stools to be provided while off-loading the equipment at Site.
- Saddle Width and Length to be discussed with the Nominated LSP and Supplier to arrange for the design accordingly to avoid any issue while off-loading the cargo on barge and further movement of the cargo to Project Site.
- Transportation Drawing should also have the details for the point load for individual saddle so that LSP can arrange for the stability calculation on barge and axle.
- Supplier to arrange for the individual Invoice, Packing List and Certificate of Origin (COO) for shipment equipment wise.
- Supplier to arrange for the separate Invoice, Packing list for the cargo such as loose items, spares and accessories which will be shipped or transporter separately.
- Supplier to arrange for the Lifting Diagram for individual packages based on which LSP shall discuss with shipping line for loading pattern for cargo loading on vessel.

Supplier shall design and provide the adequate number of transportation saddles for transportation by Multimodal way - road & barge transportation.

Supplier will check adequacy of lashing Supplier shall submit the above documents at least six months prior to dispatch to L&T & Insurance Company for approval. L&T or Insurance Company reserves the right to comment and the comments will be incorporated without any time & cost to L&T prior to the dispatch of the consignment. L&T reserve the right to inspect the transportation arrangement before dispatch

11.00 Special Conditions of Purchase Order

11.10 Deviations to the RFQ Specifications:

11.10.1 Deviations shall not be acceptable after release of P.O.

11.10.2 If unavoidable, same will be processed as follows:

- Supplier to submit the same to L&T along with all backups for approval (format as per OWNER/ PMC standard).

- Responsibility of processing the request and acceptance thereafter by OWNER as applicable shall rest with L&T. In case the deviation is acceptable there shall be no cost and time implications (in case what is being offered is superior to the bid specifications in terms of cost).

- If rejected by OWNER / PMC, L&T shall in no way be responsible and all terms & conditions including delivery / time of completion and / or price as per the Purchase Order shall remain unaffected and the deviations stand withdrawn by Supplier.

- If the deviation is accepted by OWNER, the Cost benefit sought by OWNER, if any, will be deducted and the Order value shall be discounted accordingly. Any recoveries if due, shall be made from Supplier's subsequent invoices / pending payments if any. This deduction shall be over and above any of the other deductions / limitation of liability indicated elsewhere. Supplier shall provide all the

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 17 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

necessary and sufficient back-ups for justifying the cost benefit applicable for the deviation passed on to L&T.

11.10.3 Deviation(s) taken by Supplier during pre-order stage which is responded by L&T as "Subject to PMC / OWNER acceptance" shall also be processed as per above methodology.

11.20 Liability under the Order:

11.20.1 Supplier's maximum aggregate liability under this Order shall not exceed the Order value. This does not include liability for "Make-Good" and "Risk Purchase" as specified elsewhere in this document.

No such limit shall apply in respect of:

- i. Any liability pursuant to Supplier's indemnity obligations under the PO ; or
- ii. any loss resulting from fraud, intentional or willful misconduct or illegal or unlawful acts or omissions of Supplier, its affiliates or any sub-contractor or any supplier or any of its or their respective officers, directors, employees, servants or agents; or
- iii. any liability to rectify, repair, restore or replace any materials and/or works or deficiencies therein in terms of the PO;
- iv. any liability under this PO because of which Owner is unable to take the possession of the Equipment/system/package/Plant;
- v. In the event of any claim or loss or damage arising out of infringement of Intellectual Property;
- vi. For any damage to any third party, including death or injury of any third party caused by the Supplier or any person or firm acting on behalf of the Supplier in executing the works
- vii. Supplier's Liability for Price Adjustment for utility consumption as PO
- viii. Supplier's Liability for Process Guarantees (if any) or other Guarantees as mentioned in the Technical Part/PO.
- ix. Any liability arises out due to perform of any work or part thereof under this PO by L&T/Owner at risk, cost and expense of Supplier.

11.20.2 The deductions on account/ Recovery of any of the reasons at Sr. No. 4 & 11 of this document shall be made from the pending bills of the Supplier under this Purchase Order or any other Purchase Order and/or the Bank Guarantees and/ or submitted by the Supplier. It is clarified that the Buyer reserves the right to initiate appropriate proceedings against the Supplier and for the recovery of the shortfall amounts if any.

11.20.3 Levy of Price discount shall not relieve the Supplier from any of their contractual obligations and shall not limit the Buyer's rights in any manner.

11.20.4 Neither Party shall be liable to the other for any indirect or consequential loss or damage howsoever arising, including without limitation loss of product or equipment or facility down-time or loss of profit or loss of opportunity or revenue or increased costs of finance or like items of loss or damage, without regard to the causes thereof including, without limitation, willful misconduct or fraud, the fault, negligence, breach of statutory or contractual duty or obligation or otherwise of either Supplier or OWNER provided that (a) no liability or indemnity under this clause shall be considered as a liability to pay or to indemnify a Party for indirect or consequential loss or damage and (b) the provisions of this Clause shall not limit the Supplier's obligations to pay Price Discount under this Order. This Article shall survive termination of the Purchase Order.

11.30 Guarantee/Warranty

11.30.1 All materials / Equipment supplied, (irrespective of whether engineering, design data or other information has been furnished, reviewed or approved by L&T/PMC/OWNER) shall be free from faulty design, material & workmanship, and shall be of sufficient size and capacity and of proper materials so as to fulfill in all respects all operating conditions, specified in the Order.

11.30.2 Engineering Guarantee:

11.30.2.1 It shall be the prime responsibility of the Supplier to carry out design, engineering and supply in accordance with the RFQ and such design shall also be in accordance with good and sound engineering practices. Supplier must ensure that all equipment and items supplied by him shall comply with the technical specifications to achieve the guarantees and warranties specified under this Purchase Order.

11.30.2.2 In case any errors or omissions in such design or engineering, attributable to the Supplier, which require re-engineering resulting in any new requirements for the subject item / materials / packages, the same shall be supplied and / or carried out by the Supplier within the scope of work and/or supply and within the contractual delivery date without any extra price and / or delivery implications to L&T.

11.30.2.3 Supplier shall check and satisfy themselves with Material Requisition, Equipment datasheets and other attachments for its accuracy, correctness and completeness to meet the agreed guarantees and in case of deficiency, Supplier shall inform L&T the modifications proposed to be carried out to overcome the deficiency and rectify the same at no extra cost.

Supplier, if required, shall carry out corrective technical studies and engineering as may be required without any extra cost to the L&T. Before proceeding with the re-work, L&T/Owner/PMC approval, if applicable, shall be obtained.

11.30.3 Material/ Workmanship Guarantee:

11.30.3.1 Workmanship for manufacture and construction shall be carried out in accordance with the stipulations of the MR/RFQ.

11.30.3.2 Supplier shall guarantee that the subject Package (i.e. Equipment and/ or Materials including spare parts, chemicals, lubricants and consumables, if any) supplied under this Order shall be new and unused (not reconditioned) and of recent manufacture, are of the best quality & workmanship, free from defects in design, material & workmanship, are suitable for the operating conditions specified in the Order and that the performance of the Equipment / Material etc. supplied shall be as per stipulated parameters.

11.30.3.3 Any rectification of any defective material supplied or work done shall be made good without any Price implications or entitlement of extension of time to L&T or if so considered by L&T/OWNER, shall be replaced by new equipment within the scope of work and/or supply on DDP Site basis.

Continued..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO

LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 18 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

11.30.3.4 "The Supplier guarantees and accepts full, responsibility for all materials supplied / installed / erected (including all components/parts) with regard to:

- Selection of material (if not specified by L&T)
- Material specification and metallurgy as per Specification
- Fabrication and Manufacturing workmanship
- Work specification
- Engagement of highly experienced, reliable and qualified sub-Suppliers/sub-contractors, engineers, workmen etc."

11.30.3.5 The above shall exclude damages due to normal wear and tear, application for other than the intended purpose, improper handling or inadequate maintenance by L&T/OWNER.

11.30.4 Performance Guarantee :

11.30.4.1 The items designed, supplied and installed / erected shall operate to the desired capacity to produce the desired and guaranteed duty conditions.

11.30.4.2 This guarantee shall cover the guaranteed mechanical performance for each and every related equipment, their efficiencies and their performance as laid down in the specifications including guarantee for utilities(power / Steam / Cooling Water/ Fuel Gas) consumption/generation as indicated by the Supplier in their Offer / agreed in PO.

The Supplier shall select the equipment considering the economy of utilities consumption/generation within the guaranteed figures provided and minimum effluent or pollution discharge satisfying the specifications/requirements specified in the MR.

11.30.4.3 Equipment / items which do not perform to the guarantees shall either be replaced or altered, repaired or replaced in parts or components (including dismantling, transportation, erection, hook-up, commissioning etc.) at no Price implication to L&T/OWNER and without entitlement of extension of delivery date to the Supplier.

11.30.4.4 Supplier shall carry out all activities for collecting the required data during guarantee runs to identify problems of non-performance for further analysis and modifications required to meet process performance parameters.

11.30.4.5 Supplier shall be solely responsible for supply of all instruments required for establishing performance of the subject item at no extra Price to L&T.

11.30.4.6 Supplier shall provide a list of laboratory test procedures and frequencies thereof required for validating the performance guarantees.

11.30.4.7 Performance tests shall be started when the machine/equipment is stabilized under design conditions. The equipment / machine / package shall be operated and controlled in accordance with procedures set up beforehand. The performance shall be measured on the basis of the average of data obtained during 72 hours of performance test runs under continuous operation of the machine.

11.30.4.8 If on any testing any material or equipment or any individual Unit of the equipment does not meet the design, rated or guaranteed performance relative thereto, the Supplier shall forthwith, within the Supplier's scope of work and at no additional Price to L&T, undertake such additional tests and/ or operations that are necessary to identify the cause of such failure in conjunction with L&T.

11.30.4.9 If as a result of tests and / or operation it is determined that the design/ rated/guaranteed outputs/capacities have not been met because of defect(s) in any material, service or workmanship, the Supplier shall forthwith arrange to repair / replace part or whole of the item supplied with no Price implications to L&T/OWNER. The Supplier shall, for this purpose, forthwith establish a "Time Schedule" acceptable to L&T / OWNER for such replacement / rectification bearing in mind the exigencies of the Project requirements. Should the Supplier fail to establish such "Time Schedule", L&T shall establish the "Time Schedule" and the "Time Schedule" so established shall be binding on the Supplier. Should the Supplier thereafter fail to adhere to the "Time Schedule" so established for the replacement / rectification, L&T shall (but without obligation to do so), take-over in whole or part such replacement / rectification at the Risk & Cost of the Supplier. In so doing, L&T shall be entitled to identify and employ the quickest available resources of supply and / or work on its own and/ or from third parties. L&T reserves the right to charge Supplier overheads and / or administrative charges for any such supply and / or work.

11.40 Guarantee / Defect Liability Period:

11.40.1 Notwithstanding anything stated to the contrary elsewhere in the documents comprising the RFQ, but subject to 11.4.2 & 11.4.3) below, the Guarantee / Defect Liability Period (DLP) of the Equipment / Material / Services for design, material, workmanship and performance shall be till 31-Dec-2028 or 24 months from the date of delivery whichever is later. In case the delivery of the order is delayed beyond the original Contractual Delivery Date (CDD), the DLP shall be extended by the period equivalent to the delay.

11.40.2 Should any defect and / or under performance in the Equipment / Material supplied be noticed during the guarantee / defect liability period, the Supplier shall, at no Price to L&T / OWNER make the defect good at Site promptly, by way of repair / modification or replacement and re-installment of the Equipment / Material.

11.40.3 Any Equipment / Materials repaired / modified or replaced / re-installed during the above Defect Liability Period shall be subject to defect free guarantee / defect liability period for a further period of twelve (12) months from the date of correction, repair, rectification, replacement and/ or re-installation with the intent that the provisions of other clauses under "Guarantee / Defect Liability Period" shall apply thereto in respect of all defects discovered during a period of twelve (12) months from the date of completion of the relative correction, repair, rectification, replacement and/ or re-installation as the case may be, in the same manner and to the same extent as apply in respect of the original works.

11.40.4 In case of replacement of goods Supplier shall not be entitled to return of the defective goods.

11.40.5 In case of any claim arising out of this guarantee within guarantee / defect liability period, L&T shall notify the Supplier in writing and upon receipt of the same, immediately Supplier shall, start repair/ replace/ work for the defective goods without any Price to L&T / OWNER. The delivery schedule for the same shall be mutually discussed & agreed upon.

11.40.6 The Supplier shall provide at his own cost and initiative all labor, material, equipment, crafts and inputs, transportation, taxes, duties etc. whatsoever required in and relative to the correction, repair, rectification, replacement, erection, installation and / or re-performance of the relative work(s) and / or systems(s) within the guarantee / defect liability period.

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 19 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

11.40.7 Should the Supplier fail to fulfill his obligations during Guarantee / Defect liability period, all costs / charges / taxes & duties / expenses related to engineering / supply / construction / erection / testing / commissioning etc. for the said equipment / material etc. incurred by L&T/OWNER on this account shall be back-charged to the Supplier. However, this shall not relieve the Supplier from any of the contractual obligations including guarantees.

11.40.8 In case Supplier's guarantee period / defect liability period as per clause above expires and L&T requires an extension in the defect liability period, then Supplier shall extend the defect liability period, including extension in validity of PBG and implications thereof if any, will be discussed at that point of time.

11.40.9 At the sole discretion of L&T/OWNER, Supplier shall supply New, re-manufactured, Buyer approved repaired part or assembly components required to correct the defect(s) observed during Defect Liability Period. Rejected items/defected items under this warranty shall remain property of Buyer.

11.50 Technical Co-operation # Joint Responsibility :

11.50.1 Should the supply be complementary to materials or equipment supplied by third parties, the SUPPLIER undertakes to co-operate with said other SUPPLIERS, in order to co-ordinate with them the planning, manufacture, erection and/or setting in operation of whatever is pertaining to his supply.

11.50.2 Any cost arising from a Supplier's wrong technical information to other Suppliers shall be at Supplier's charge.

11.50.3 The Supplier shall deliver to the BUYER, a copy of all pertinent documentation.

11.50.4 The Supplier and such other Supplier/s shall be jointly liable for the good running of groups, plants, equipment, etc., supplied to BUYER. Should any inconvenience occur due to erroneous exchange of technical information, the Supplier/s involved shall directly co-operate with each other, in order to eliminate such inconveniences as soon as possible according to their technical responsibilities, at no cost to the BUYER.

11.60 "Non-Waiver:
Failure by BUYER to insist upon strict performance of any terms or conditions of the Order, or failure or delay to exercise any rights or remedies provided therein or by law, or failure to properly notify Supplier in the event of breach, or the acceptance of or payment for any goods or services thereunder, or the review or failure to review designs/ documents or inspection of material by Buyer / OWNER / PMC / Buyer nominated TPIA shall not release Supplier from any of the warranties or obligations of the Order and shall not be deemed a waiver of any right of BUYER to insist upon strict performance thereof or any of its rights or remedies as to any prior or subsequent default thereunder nor shall any termination of work under the Order by BUYER operate as a waiver of any of the terms thereof."

11.70 "Non Disclosure agreement:
The Buyer and the OWNER (if any) reserve the sole right to disclose and/or publicize in any manner (i.e., for instance, through press release, speeches, photographs, videos, etc.), directly or indirectly (i.e., for instance, authorizing someone to make such disclosure), any kind of data, document and/or information concerning the Supply (materials and/or services) or any part thereof. The Supplier shall not make any such disclosure and/or publicity without the prior written consent of the Buyer and the OWNER.
Supplier shall limit its disclosure of any information supplied by Buyer/Company in connection with this PO only to those of its personnel, Sub-Vendors, Sub-Contractors who require it for the performance of the Work. If Company/ Buyer designates any other person who may receive such information, Buyer shall obtain from each such person a written confidentiality and non-disclosure agreement obligating them with respect to such information and to the same extent that Supplier is obligated under this PO.
Except as specifically permitted by the terms of this PO or otherwise authorized in writing by Buyer/Company, the Supplier and members of Supplier Group shall not duplicate, use, or disclose any information supplied by Buyer or its agents during the course of the Work so long as, and to the extent that, the information has not become part of the public domain. To the extent Company / Buyer approves any disclosure pursuant to this clause, the Supplier shall ensure that any third party to whom the Supplier discloses such information, executes a confidentiality agreement containing provisions protecting such disclosed information from further disclosure that are no less stringent than those contained herein, and which grant both Company and Buyer the right to directly enforce such provisions. Supplier shall provide copies of such confidentiality agreements to Buyer upon its request."

11.80 "Code of Conduct
Supplier shall adhere to the L&T Code of Conduct.
Supplier shall ensure that all members of Supplier/s Group comply with the requirements of this clause 11.80 of the PO. Any violation of the provisions of this clause 11.80 shall constitute a material breach of this PO which, without prejudice to Buyer/s right to enforce any other remedy provided by law or under this PO, shall allow Buyer to terminate this PO for default and recover damages including, but not limited to, any increased costs incurred by Buyer as a result of such breach."

11.90 "HSE Requirements
The Supplier shall abide by all the health, safety and environmental requirements in compliance with the applicable local statutory rules & regulations and L&T HSE Policy and shall ensure that all such requirements are communicated to all the personnel involved in executing this Purchase Order"

11.10 Termination / Cancellation and Suspension:
11.10.1 L&T may, at any time terminate the Order in whole or part by giving written notice thereof and without assigning any reasons. Supplier, shall on receipt of notice of termination discontinue all further activities connected with the execution of the Order and inform L&T of details of work carried out as on date, together with financial implication. In the event of such termination, L&T shall discuss with Supplier to ascertain extent of work carried out by Supplier and to reach an agreement on cost mutually acceptable to L&T and Supplier.

11.10.2 In case specified delivery schedule is not adhered to or the progress of manufacture of the goods is not satisfactory or not in accordance with the Delivery schedule, L&T shall inform the Supplier

Continues..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 20 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

about the delay. The Supplier shall take necessary actions to catch-up with the agreed delivery schedule that result in Supplier meeting the agreed schedule. In the event that the Supplier takes no action or even after the Supplier taking necessary actions, as mentioned above, the anticipated delay is in excess of 4 weeks, L&T reserves the right to cancel the Purchase Order in whole or in part without liability for damages/ compensation. In such event, L&T may procure from elsewhere goods, at Supplier's Risk & Cost, which in L&T's opinion would meet the same purpose as the goods for which Purchase Order is cancelled and Supplier shall be liable for reimbursing L&T cost of such substitution. L&T reserves the right to charge Supplier overheads and / or administrative charges for any such supply and / or work.

11.10.3 In the event of rejection of non-conforming goods, Supplier shall be allowed, without any extension of delivery time, to rectify the non-conformities. Should, however, the Supplier fail to do so within the stipulated time, L&T may cancel the Order as to the non-conforming goods and retain the same right with respect to substitution as are set out in preceding paragraph and in addition recover actual expenses incurred by L&T, in installing and removing the non-conformity to be rectified at Supplier's Risk and Cost. L&T reserves the right to charge Supplier overheads and / or administrative charges for any such supply and / or work.

11.10.4 If the Order is suspended (By Written instructions) for whatsoever reasons and for whatever time duration by L&T, the Delivery and price shall be amended based on mutually agreed terms. Buyer shall pay all reasonable and substantiated costs incurred by the Supplier in giving effect to the Suspension, unless the suspension is:

- "a) necessary because of Force Majeure; or
- b) necessary for the proper execution of the Work or any part thereof; or
- c) necessary because of negligence, error or other default on part of the Supplier; or
- d) necessary for the health or safety of Company /Contractor/Supplier Personnel, Third Parties or for the protection of the work, Company Property or environment; or
- e) otherwise provided for in the Purchase order."

11.11 Insolvency and Bankruptcy

11.11.1 "The Supplier shall inform the Buyer in case of occurrence of one of the following events or if there is substantial reason for Buyer to believe that any one of the following events may occur:

- a) Supplier becomes insolvent,
- b) a petition in bankruptcy is filed with respect to or by Supplier,
- c) opening of an official receivership procedure with respect to Supplier,
- d) Supplier's request for obtaining controlled administration,
- e) makes an arrangement or composition with (or assignment in favour of) his creditors,
- f) agrees to carry out the Purchase Order under a committee of inspection of his creditors,
- g) goes into liquidation,
- h) has an execution/restraining order levied on his goods,
- i) important changes on the Supplier's organisational structure or Ownership which can undermine the successful conclusion of the supply,
- j) any authority having jurisdiction takes or institutes any action or proceedings for the dissolution or winding up of Supplier or for the suspension of his operations"

11.11.2 Upon knowledge of any such events from the Supplier or from any other sources, the Buyer, upon notice in writing to Supplier, has the right without prejudice to the procedural consequences, to terminate totally or in part the Purchase Order as per clause no. 11.10.2 and to take all the special and/or particular measures needed to fulfil his obligations towards the Buyer. In case of insolvency no prior notice for cure is required to be issued by the Buyer before issuing such termination notice.

11.11.3 In case of the above events and irrespective to the fact that whether the Purchase Order is terminated by the Buyer or still in subsistence when it has reason to believe that any of such event are likely to occur, the Buyer, at his option, can forthwith enter upon the worksite and take possession of all the material, drawings and designs, work in progress pertaining to the Purchase Order whether issued by the Buyer or otherwise. The Buyer at his option may use and utilise the plant and equipment of the Supplier for completion of the works. At the option of the Buyer, the Supplier shall assign all the purchase orders, service contracts that it had executed pertaining to the Works with its sub-Suppliers in favour of the Buyer and accordingly inform the sub-Suppliers.

11.11.4 The Buyer shall not be liable to pay the Supplier any money under the purchase order until the costs of completion are ascertained.

11.12 Force Majeure

11.12.1 "The term ""Force Majeure"" shall mean any of the following acts, events, or circumstances which renders a Party unable to perform its obligations which in each instance is not reasonably foreseeable, cannot be sufficiently mitigated by reasonable efforts, and the cause of which and the inability to perform is not caused by any act or omission of the Party asserting Force Majeure:

- a) act of Terrorism
- b) riot, war, invasion, act of foreign enemies, rebellion, revolution, insurrection of military or usurped power
- c) Ionizing radiation or contamination, radio activity from any nuclear fuel or from any nuclear waste from the combustion of nuclear fuel, radioactive toxic explosive or other hazardous properties of any explosive assembly or nuclear component.
- d) epidemics(Except Covid 19 or Coronavirus Pandemic), earthquakes, flood, fire, hurricanes, typhoons or other physical natural disaster, but excluding weather conditions regardless of severity
- e) freight embargoes, strikes at national or state-wide level or industrial disputes at a national or state-wide level in any country where Works are performed, and which affect an essential portion of the

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 21 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

Works but excluding any industrial dispute which is specific to the performance of the Works or the Contract"

11.12.2 The following shall not afford the Supplier a ground for extension of time for delivery of materials:

- a. Inclement or unforeseen weather, strike or lock-out, shutdown, third party breach, delay in payment or commercial hardship shutdown or idleness or other impediment in progress or completion of the supply or work due to any reason whatsoever shall not afford the Supplier a ground for extension of time or relieve the Supplier of his/its full obligations under the contract.
- b. Delay in the supply of any material to Supplier due to any whatsoever reasons (including Force Majeure) from Supplier's vendors/sub-suppliers / sub-contractors shall not entitle Supplier for any extension of time for completion or to any claim for costs/loss/remuneration or damages or compensation whatsoever.

11.12.3 If either Party is rendered unable, wholly or in part, by Force Majeure to perform its obligations under this Purchase Order, it is agreed that performance of such obligations by such Party, so far as they are affected by Force Majeure, shall be excused from the inception of any such inability until it is corrected or relieved, but for no longer period. If either Party desires to invoke Force Majeure, it shall notify the other Party in writing of the act, event, or circumstance that constitutes the event of Force Majeure, the date on which such event of Force Majeure commenced, with proper substantiation (duly certified by the local Chamber of Commerce or statutory authorities, relevant to the Supplier). Such notice shall be delivered not later than three (3) days after the date on which such Party first had knowledge of the effect of such event of Force Majeure. Either party shall forthwith inform the effect of the event of Force Majeure on the affected Party's ability to perform its obligations under this Purchase Order. The Party claiming inability to perform must make all reasonable efforts to minimize the delay caused to the Works as a result of the occurrence of an event of Force Majeure and mitigate any other effects of Force Majeure.

11.12.4 Affected parties shall diligently mitigate or remove the effect of Force Majeure. Each party shall upon receipt of the Notice of Force Majeure under the preceding Article shall confer promptly with the affected party and agree upon, within fifteen days, a course of action to be adopted by the affected party to remove or alleviate such effect(s), and shall seek reasonable methods of resuming full performance of affected party's obligations and achieving the objectives under the Order.

11.12.5 In the event of any force majeure cause, either party shall not be liable for delays in performing their obligations under this Order and the delivery dates will be extended without being subject to Price Discount for delayed deliveries, as stated elsewhere.

11.12.6 For delays arising out of Force Majeure, Supplier shall not claim extension in completion date for a period exceeding the period of delay attributable to the causes of Force Majeure and neither Buyer nor Supplier shall be liable to pay extra costs or claim any damages or compensation in addition to or in lieu of extension in completion date. To clarify, during the Force Majeure no party shall be liable to each other for the payment of any hire charges for equipment or manpower, storage charges of material, stand-by charges of equipment etc.

11.12.7 The Supplier shall not be entitled for extension in completion / Delivery Date if the effect of Force Majeure does not exceed 7 days.

11.12.8 Immediately after the end of the event of Force Majeure, the affected Party having invoked such Force Majeure shall notify the other Party in writing that the event of Force Majeure has ended and such Party shall resume performance of its obligations under the Purchase Order.

11.12.9 In case of an overlapping delay that is when more than one event causes a delay and the cause of at least one of those events is attributable to Supplier, then to the extent and period that such delays are concurrent, the Supplier shall not be entitled to seek resort under the Force Majeure clause and seek an extension of time for the period of such delay which is attributable to the Supplier.

11.12.10 The following are specifically excluded as Force Majeure and shall not constitute a basis for claims for schedule extensions, under this Purchase Order:

"# Events or circumstances involving a previous or existing condition on or before the effective date of the Purchase Order; or

Delay of Material by Supplier's Vendor, Subcontractors, Sub suppliers

Late performance by a Supplier and / or his sub-Supplier caused by a shortage of supervisors or labour, inefficiencies, or similar occurrences; or

Late delivery of equipment or materials by any sub-Supplier caused by congestion at a manufacturer's plant or elsewhere, an oversold condition of the market, inefficiencies, or similar occurrences; or

Any inability to obtain visas or renewal of visas for expatriate workers; or

Changes in market conditions

Social distancing norms/ consequences of COVID 19 pandemic/quarantine requirements or the like.

Inclement or unforeseen weather, strike or lock-out, shutdown, third party breach, delay in payment or commercial hardship, shutdown or idleness or other impediment in progress or completion of the supply or work due to any reason whatsoever shall not afford the Supplier a ground for extension of time or relieve the Supplier of his/its full obligations under the PO.

Any other clause mentioned in this order"

11.12.11 Should the Works be delayed for more than thirty (30) consecutive days or sixty (60) days in aggregate as a result of Force Majeure, Buyer at its option may either suspend the Work affected or terminate this Purchase Order or the portion of the Work involved. The provisions governing Cancellation shall apply. However, L&T shall not be liable to pay any Cancellation charges over and above the charges as admissible under Clause of Cancellation.

11.12.12 Failure by Supplier to comply with the notice requirements mentioned hereinabove shall be deemed to conclusively waive Supplier's rights to invoke Force Majeure in respect of the relevant acts, events or circumstances and to any consequent entitlements to extension of time for completion of the Work or any separable portion thereof.

11.12.13 Notwithstanding anything mentioned hereinabove, the Buyer shall have right to invoke Force Majeure if the Buyer is unable to utilize/use the services supplied by Supplier under the Purchase Order due to events mentioned above.

11.13 "Infringement of Intellectual Property Rights etc.

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 22 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

Suppliers shall warrant that the sale or use of goods supplied under the Contract, whether designed and manufactured so as to comply with L&T's specifications or otherwise, shall be free of any claim, whether rightful or otherwise of any third person, by way of infringement of any patent, copyright, trademark or industrial design or the like, and the Supplier shall hold L&T harmless and indemnify L&T at Supplier's own cost, from any and all such claims and legal proceeding. L&T makes no warranty that the production, sale or use of goods designed and manufactured so as to comply with L&T's specification will not give rise to the claim, whether rightful or otherwise, of any third person by way of infringement of any patent, copy-right trademark or industrial design or the like and in no event shall L&T be liable to Supplier for indemnification in the event of any action being brought against Supplier in connection with any such claim. If Buyer's right to use such designs, drawings, calculations, software, or Materials is enjoined or otherwise restrained or precluded as a result of a claim of infringement, Supplier shall either procure for Buyer the right to use the infringing item, replace the infringing item with a substantially compatible and functionally equivalent non-infringing item, or remove the infringing item and compensate Buyer for the costs associated with procuring and installing a suitable substitute.

This Clause shall survive termination of the Order."

11.14 Dispute Resolution:

11.14.1 Any dispute or difference between the Parties arising out of or in connection with this Order which cannot be resolved amicably between the Parties at the level at which it arose within fifteen days thereof shall, in the first instance, be referred to the nominees of Directors of the respective Party for resolution. If the dispute cannot be settled within a further period of thirty days such dispute shall be resolved by arbitration in accordance with The Arbitration and Conciliation Act, India 1996 and any amendments thereof, by one arbitrator to be appointed by agreement between the Parties, failing which the appointment shall be made under the said Act.

11.14.2 The arbitration proceedings shall be conducted in English language. The arbitration venue shall be in Mumbai, India. The Parties agree that the dispute resolution process provided herein shall be the sole remedy and the Parties shall not suspend their obligations under the Order.

11.14.3 This Order shall be subject to the laws in force in India.

11.14.4 This Clause to arbitrate shall survive termination of this Order.

11.14.5 Notwithstanding the existence of any dispute or arbitration in terms hereof or otherwise, unless otherwise directed in writing by L&T, the Supplier shall continue and be bound to continue and perform the works to completion in all respects according to the Order (unless the Order or works be determined by L&T or by the Supplier under the provisions of the Order), and the Supplier shall remain liable and bound in all respects under the Order.

11.14.6 The work under the order shall, however, continue during the Arbitration proceedings and no payment due or payable to the concerned party shall be withheld (except to the extent disputed) on account of initiation, commencement or pendency of such proceedings.

11.14.7 "Local rules and regulations:

Supplier shall ensure that the Goods within the scope of supply for this Purchase Order are in compliance with local codes and regulations. Supplier shall ensure that local rules and regulations and procedures have been complied with in obtaining certificates if and when required. Costs related to compliance with aforementioned are included in the Order Value.

Supplier has to take note of steel policy, Value addition norms, Subcontractor and import restrictions as applicable to order while quoting and execution of the project."

11.14.8 "Harmonized Tariff Codes:

Supplier shall provide Buyer with the applicable Harmonized Tariff (HSN) Codes per line item for this Purchase Order / Packing List. Failure to do so may result in back charges to the Supplier."

11.15 Assignment and Sub-Contracting:

11.15.1 Supplier shall not, except with the prior written consent of L&T, transfer or assign their obligations and interest in supply of materials, or sub-contract whole or part of the work under the Purchase Order in any manner whatsoever.

11.15.2 Grant of any such consent or approval shall not however absolve the Supplier in any manner from all or any of their obligations, duties and responsibilities, including delivery of materials under this Order within the stipulated Delivery schedule.

11.15.3 Moreover, delay, if any, by L&T in approving any Sub-Contractor(s) proposed by the Supplier, or, rejection if any, by the L&T of any Sub-Contractor(s) proposed by the Supplier, shall not entitle the Supplier for any claims for additional compensation or any claim for an extension in the Contractual Date of delivery. Supplier shall ensure that competent and resourceful agencies with proven track record should be proposed for the work to be sub-contracted.

11.15.4 Delay in the supply of any material to Supplier by Supplier's vendors, sub-suppliers or sub-contractors due to whatsoever reasons, shall not entitle Supplier for any extension of time for completion or to any claim for costs, loss, remuneration or damages or compensation.

11.16 Modifications and Amendments:

11.16.1 L&T shall have the right to modify or amend this Purchase Order, by way of a written communication to the Supplier, subject to an adjustment in the price and/or delivery date. Adjustments in price and/or delivery date shall be mutually agreed between L&T and the Supplier.

In case price and/or delivery date adjustment cannot be settled immediately, upon written instruction from Purchaser, Supplier will proceed with the modification and price and/or delivery date adjustment

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 23 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

shall be settled by mutual agreement before order close-out. Any delay in executing the modification after receipt of written instructions shall be to Supplier's account.

11.16.2 Supplier shall forthwith inform regarding any claim for "Change-Order" prior to commencing with the same.

"If there is any alteration or change in the scope of work or extra item and the Supplier or L&T considers that the change requires revision in the Order price and / or delivery, Supplier shall issue / submit the change order proposal (COP) within 7 days of such instructions issued to Supplier.

Supplier shall submit the following minimum information and documents in support thereof -

- (i) a list of affected deliverables
 - (ii) a breakdown of man-hours by deliverable and job title/equipment item
 - (iii) a list of Materials and quantities with associated cost for each individual item or - cost per unit quantity as applicable
 - (iv) copies of all related Sub-contractor quotations for the Work / Materials / equipment as stated in the proposed Change together with any discount or rebate Supplier would receive from Sub-contractor(s), directly or indirectly, for those Work / Materials / equipment
 - (v) a network schedule fragment showing how the proposed Change Order Work would be incorporated into the Work Schedule and any resulting impact on CDD
 - (vi) any other information / documents / clarifications as may be required by the Buyer
- e) In addition to describing the Change, a Change Order shall include:
- (i) any adjustment in the CDD / milestone dates resulting from the Change; and
 - (ii) the lump sum price or other basis (as determined by Buyer) for determining any increase in the PO Price either due to Vendor or credit due to Contractor as a result of such Change, if any as mutually discussed and agreed between the Parties."

"Change-Orders" informed later or at the time of delivery shall not be entertained.

If Supplier proceeds with the additional Work involved in a Change without first obtaining a Change Order signed by both Parties, the Supplier shall not be entitled to any additional compensation for the Work so performed or to any adjustment of the CDD and/or milestone dates as a result of the said Change.

The Supplier specifically waives any and all rights to claim from Buyer for such costs or additional time to perform the Work as a result of compliance by Supplier with any such oral changes."

11.16.3 The Supplier shall carry out such amendments to the Purchase Order and shall be bound by the same terms and conditions incorporated in the original Purchase Order.

11.16.4 "Nothing shall constitute a change order to the extent that additional work / efforts / resource deployment etc. are carried out / required only:

- a) to ensure that the Work is carried out and completed as per the delivery schedule
- b) As a consequence of Supplier's failure to comply with any of its obligations under the Purchase Order. "

"Provided always that the provisions with respect to a Change Order or a Change Order Proposal shall not be applicable if:

- a) the change of supply or services or work is necessary at any time in order for the Supplier to comply with the requirements of the Contract or the Supplier's responsibilities under the Contract,
- b) the additional Work/ services and/or materials are required to make good the Supplier's Warranties and/or Guarantees as set forth in the Contract.
- c) the change relates to the repair, rectification or re-performance of any of the defective Work(s) or materials within the Supplier's scope or supply or services or work."

11.16.5 Any extension of time, if granted by L&T, shall be the sole remedy of the Supplier for any cause or event of delay and the Supplier shall not be entitled in addition to or in lieu of such extension, to claim any damages or compensation whether under the law governing contracts or quasi-contracts or any other relationship, and the Supplier hereby waives and disclaims any and all contrary rights.

11.17 "Right of the Buyer to set-off:

L&T shall be entitled to recover from the Supplier any sum(s) due to L&T on account of any damage(s) or otherwise, whether in respect of Supplies under this Purchase Order or under any of their previous Purchase Orders / Contracts, by deducting such sums from the amount due by them to the Supplier in respect of Supplies made under this Purchase Order or any of their prior or subsequent Purchase Orders / Contracts and/or Bank Guarantees."

11.18 "Confidentiality:

All drawings, specifications and details furnished by L&T to the Supplier shall be kept confidential. The technical information imparted and supplied to the Supplier by L&T shall at all times remain the absolute property of L&T. Supplier shall preserve and cause its employees to preserve the secrecy of any confidential information.

Supplier shall read and un-conditionally accept the terms and conditions of Non-Disclosure Agreement (NDA) as per Annexure E and Countersign (with stamp).

Supplier shall not use or cause to be used any confidential information for any purpose other than the performance of the Work, and as such Supplier shall not a) reproduce, copy or use any confidential information, or b) disclose to, place at the disposal of, or use on behalf of any third party, or enable any third party to peruse, copy or use any confidential information unless with prior written consent of L&T.

This Clause shall survive termination of the Order."

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 24 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

11.19 Post Warranty Comprehensive Annual Maintenance Contract (PWCAMC)

Supplier to submit the quote and confirm acceptance to Terms and Conditions for Post Warranty Comprehensive Annual Maintenance Contract (PWAMC) enclosed as Annexure H of this document (will be provided during Project phase). For Proposal, Supplier to refer RFQ for respective PWCAMC items.

The AMC Charges shall be inclusive of all the charges for Transportation, Lodging, Boarding, all insurances including third party insurance, all Taxes /Duties / Cess /Levies / Fees and all other incidental charges etc. but excluding GST which shall be payable extra at actual on Submission of Invoice as per GST Invoicing rules. Purchase/OWNER will not have any liability, whatsoever, over and above the quoted prices of PWCAMC.

Other Terms & Conditions of PWCAMC:

Payment Term: The payment terms for the AMC shall be on quarterly basis after completion of each quarter against submission of Bills duly certified by the Engineer-in-charge. Payment will be made through Wire transfer within 60 days from certified invoice.

Validity: The order for PWCAMC will be placed before the completion of Project Defect Liability Period PWCAMC Prices offered must be valid for ordering till one year after completion of Project Defect Liability Period. The price of AMC services will remain firm & fixed for the complete execution of AMC contract.

PBG: Within 10 Days of placement of Order for PWCAMC, SUPPLIER shall submit PBG for 5% of the annual PWCAMC amount which shall be valid for the entire AMC period

Logistic Support: Supplier to furnish Logistic support certificate as per Annexure J. The logistic support certificate shall be signed by corporate level person of the Supplier.

OWNER may decide to enter into separate contract with AMC service provider / Supplier at its discretion on the same prices and terms & conditions.

11.20 NA

11.21 NA

11.22 "Supplier's Sales / Commercial Conditions, if any, and/or any other terms & conditions indicated in their offer shall not be applicable and stand withdrawn in lieu of "Terms & Conditions" agreed and recorded between "Supplier" and "L&T" before placement of Order.

Deviations whatsoever after placement of LOI / P.O. (whichever is earlier) are not acceptable & valid.

In case of any contradictions in the Terms & Conditions, the more stringent shall be applicable."

11.23 NA

11.24 STEP-IN RIGHTS:

11.24.1 If, in the opinion of Buyer, the Supplier's financial affairs become unsound or he suspends payment of his commercial debts or if there are reasons for Buyer to believe that Supplier may become insolvent or it has committed the breach of the terms of Purchase Order or there is any delay in performance of the Purchase order, then Buyer may without prejudice to any other rights it may have under the Purchase Order, have the right to enter upon the worksite and without thereby terminating the Purchase Order, or releasing the Supplier of his obligations or liability under the Purchase Order take over the balance Works and complete the same by using Supplier resources including but not limited to yard, equipment, personnel, workmen and/or other sub-Supplier, as Buyer in its absolute discretion may think proper and at the sole risk and cost of Supplier.

11.24.2 If Buyer in exercising any of the rights conferred by this clause incurs any expenditure in excess of the Purchase Order Value, then such excess amount shall be paid by the Supplier/ Sub-Supplier to Buyer, and such amount shall be considered as debt due to the Buyer.

11.24.3 Notwithstanding any provision to the contrary the Buyer while exercising its rights under this clause shall not be liable to pay to the Supplier any money until the cost of completion of the works and all other expenses incurred by Buyer have been ascertained. Supplier shall provide Indemnity Bond as per Annexure N attached with this Terms & Conditions.

11.24.4 The Supplier guarantees that such Buyer's right of taking over of the works will be accepted by its sub-Supplier and will be included in the respective sub-purchase orders.

11.24.5 In case the Buyer decides to make use of its aforesaid right and if so requested by him, the Supplier shall facilitate and assist Buyer. Notwithstanding Buyer's use of such rights, it shall also have a right to pay any sub-Supplier directly to prevent from suspending or terminating performances. The amount so paid will subsequently be set-off from the Purchase Order Value.

11.25 FREE ISSUE MATERIAL

11.25.1 The Buyer can supply, free of charge certain material (hereinafter referred as the Free-Issued Material) (if any) in accordance with the details stated Purchase Order. The Buyer can provide such Free Issued Material either at the Supplier's Site / Yard / Warehouse or any other location stated in the Purchase Order. Supplier shall be liable for all demurrage or other charges in connection with any delay in unloading, hauling, lifting or handling of Free-Issued Material.

11.25.2 As soon as Free-Issued Material is received in Supplier's Site/Yard/Warehouse, it shall mark them with an identification number and Buyer's name, and shall keep them separate from other items. Upon receipt of such Free-Issued Material, the Supplier shall thoroughly inspect them, and shall promptly give notice within three (3) days from the receipt of the Free Issued Material to the Buyer of any shortfall, theft, pilferage, defect, or damage in these Free-Issued Material failing which the Supplier shall be deemed to have accepted and confirmed that the Free Issued Material are free from any shortfall, defects, or damage.

11.25.3 Once the Free-Issued Material is delivered by the Buyer, the Supplier shall be responsible for the care, custody, control, storage, handling, protection, preservation, re-packing return of surplus or

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 25 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

unused material and use of the Free Issued Material. Supplier shall provide Indemnity Bond as per Annexure N attached with this Terms & Conditions.

11.25.4 Supplier must, at all times, segregate the Free Issued Materials from the other material, Construction Equipment, and any other Supplier's property. Supplier must use the Free Issued Materials solely and exclusively for incorporation in and the performance of Works.

11.25.5 The Free Issued Materials shall be always considered to be the property of Buyer and the Supplier shall ensure that it is free from any lien / charge / encumbrances. Supplier shall acknowledge receipt of all materials and/or equipment furnished to Supplier by Buyer for its packaging or assembly into a "complete package". It is specifically understood and agreed that title to all said items furnished by Buyer shall remain with Buyer at all times regardless of whether said items are incorporated into material or equipment being constructed hereunder and whether said items remain separable or not. Said items can at any time be reclaimed and repossessed by Buyer.

11.25.6 If during the term of Purchase Order there is evidence of a lien or a charge created by the Supplier on the Free-Issued Material, the Buyer may retain an amount from Supplier sufficient to indemnify Buyer against such lien or claim. Buyer may keep said amounts until the Supplier pays and discharges the lien or charge. The Buyer at his discretion may pay and discharge the same and deduct the amount paid from amounts due to Supplier under Purchase Order.

11.25.7 The Supplier shall be responsible for any theft, loss or damage to any such Free-Issued Material and any deterioration to any such Free-Issued Materials as a result of the Supplier's care, custody, control, handling and storage and shall indemnify and hold harmless the Buyer from any and all claim in this respect. The Supplier shall execute and sign Indemnity Bond in the format annexed herewith. The Supplier shall not remove Free-Issued Material from the Site/Yard/Warehouse without first obtaining a written release authorizing such removal from Buyer.

11.25.8 Upon the completion of the Works relating to the Free Issued Materials, the Supplier must notify the Buyer of any unused or surplus Free Issued Materials and the Supplier must immediately return the surplus Free Issued Materials as instructed by the Buyer in the same condition as received by the Supplier from the Buyer, except for normal wear and tear.

11.25.9 FIM Insurance, material reconciliation, wastages, scrap and excess material terms and conditions mutually agreed between Buyer and Supplier and covered under Purchase Order shall govern.

11.26 TITLE AND RISK

11.26.1 "Title to the raw material/ Work in Progress (WIP) / finished goods shall pass to Buyer:

- a) on arrival of raw material on Supplier's Site / Yard / Warehouse, or
- b) when payment for raw material is made by Buyer, or
- c) when the Supplier has commenced manufacturing at its/ sub-Supplier's Works, or
- d) when raw materials, WIP, finished goods are dispatched from Supplier's Works whichever occurs earlier"

11.26.2 Notwithstanding any transfer of title, Supplier shall at all times be solely responsible for all costs arising in respect of payment for Supplier supplied materials and for transport and importation of such raw material/ WIP / finished goods including without limitation, shipment costs, customs duties and insurance costs.

11.26.3 Notwithstanding the above, the risk and responsibility for any damage, or loss of the raw material/WIP/finished goods shall remain with the Supplier until the final delivery.

11.26.4 The Supplier shall, at its sole risk and cost, take insurance cover for the risk of loss or damage to the raw material/ WIP / finished goods for its full replacement value until delivery of the finished goods in accordance with the terms of the Purchase Order.

11.26.5 The raw material/ WIP / finished goods shall remain under the care and custody of the Supplier until the date of delivery or achieving completion.

11.26.6 Supplier shall properly store and ensure maintenance, custody and control of any and all goods in accordance with the requirements of this Purchase Order and the best professional practices and shall protect the same against weather hazards, water, humidity, dirt, fire, theft, vandalism and the like. Supplier shall use every means to prevent at all times any defect, corrosion, damage, theft or loss whatsoever to the goods in its custody.

11.26.7 Raw material/ WIP / finished goods pertaining to the Works shall be considered to be the property of Buyer and the Supplier shall always ensure that it is free from any lien / charge / encumbrances. If during the term of Purchase Order there is evidence of a lien or a claim for which Buyer might be or become liable or to which Buyer's property might be or become subject and which is chargeable to Supplier, its sub-Supplier, Buyer may retain an amount from Supplier sufficient to indemnify Buyer against such lien or claim. Buyer may keep said amounts until the Supplier pays and discharges the lien or claim or until Buyer is satisfied that such lien or claim is invalid.

11.26.8 If such lien or claim is valid in Buyer's judgement, Buyer may pay and discharge the same and deduct the amount paid from amounts due to Supplier under Purchase Order.

11.26.9 If a lien or claim remains unsatisfied after termination of Work or completion, Supplier shall promptly refund to Buyer all amounts paid by Buyer in discharging such lien or claim, including costs and reasonable attorney fees.

11.26.10 "Liens.

The Supplier shall take full responsibility for the payment of its Sub-vendors / Sub-contractors under all subcontracts or purchase orders issued by it and for the management of all claims for payment thereunder, including payments to all its Sub-vendors / Sub-contractors of any tier, and shall take all proper precautions to prevent the imposition of any lien or charge against any property of the Buyer/ Company and to indemnify, defend and hold the Buyer / Owner harmless against any and all claims, demands, actions, suits, proceedings and judgments which may be brought or secured against the Buyer / Owner by reason of the Supplier's failure to make payments under any of the sub-contracts / purchase orders to its Sub-vendors / Sub-contractors.

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 26 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

Breach of law

Supplier shall defend, indemnify and hold harmless Buyer / Owner from and against any claims including libels, maritime liens and other liens, claims, charges, encumbrances, suits or penalties which may be #led on or levied upon Buyer / Owner or their associates, employees, due to any breach of any Applicable Law (including but not limited to Applicable Economic Sanctions Laws and Regulations, Applicable Export Control Laws and Regulations, other laws, regulations and requirements relating to personnel, health, safety and environmental matters, international maritime laws, regulations or requirements, or any laws restricting the import, use or possession or firearms, explosives, alcohol, pornography or other controlled or prohibited materials or substances) by any member of Supplier Group in connection with the performance of the Work and provision of Supplies under this PO.

Criminal misappropriation or misapplication

Notwithstanding any other provision of this PO, Suppliers shall release and indemnify and hold harmless Owner and Buyer from any and all claims arising from or related to any criminal misappropriation or misapplication by any personnel of Supplier Group of any property, whether tangible or intangible, occurring during the course of or in connection with the performance of the Work and provision of Supplies under this PO. Supplier shall obtain similar indemnities running to Owner and Buyer from each of its Sub-Vendor and Sub#Contractor."

11.27 "COVID Clause:

Supplier acknowledges that it is aware of change in the working conditions that has arisen due to the ongoing Coronavirus situation (COVID19). Supplier while submitting its bid has analysed all the aspects of the present situation of COVID19 including but not limited to - the quarantine requirements of personnel and vessels, social distancing measures, restrictions on transport, requirements of personal protection equipment, testing requirements, requirements of medical insurance etc. The consideration amount under this Purchase Order is inclusive of all such costs arising out of COVID19."

11.27.1 "Notwithstanding anything contained elsewhere in this Purchase Order, the Buyer shall not bear any additional time and costs impact arising out of COVID19. Supplier shall indemnify the Buyer from:

(i) all costs arising out of COVID19 pertaining to the scope of work of the Supplier and

(ii) all claims arising out of Supplier's employees, agents, subcontractors, personnel etc. including but not limited to their medical tests, hospitalisation, medical insurance, transportation, quarantine requirement, accommodation arrangements, stand-by etc.

(iii) All claims arising out due to different notifications from State/Central authorities."

11.28 "The Supplier agrees at all times to indemnify and keep indemnified L&T/Owner/PMC and its officers, servants, agents from and against any and all losses or damage caused at any time to its own property or personnel. It shall hold harmless and keep indemnified L&T/Owner/PMC against -

any third party claims (including but not limited to property loss and damage, personal accident, injury or death of any person)

any and all losses arising directly or indirectly from acts of Supplier, its affiliates, sub-suppliers or any of their respective personnel during performance of obligations under the PO.

any and all losses, arising directly or indirectly from or incurred by reason of claims or sanctions or penalties imposed by any governmental authorities or others for any actual or asserted failure of Supplier to comply with any applicable law or applicable consent(s), sanction(s) and/or authorization(s) or with any rules or regulations applicable to it

any and all losses, arising directly or indirectly from or incurred by reason of failure of Supplier to pay any taxes or any amounts in respect thereof."

11.29 Other Terms & Conditions

Data Protection and Privacy: The Supplier shall comply with all applicable data protection and privacy laws and regulations. The Supplier shall ensure that any personal data processed in connection with the Purchase Order is handled in accordance with applicable laws and regulations. The Supplier shall indemnify and hold the Buyer harmless from any claims, damages, or penalties arising from the Supplier's breach of data protection and privacy obligations.

Anti-Bribery and Corruption: The Supplier shall comply with all applicable anti-bribery and anti-corruption laws including the Prevention of Corruption Act, 1988, and any other relevant legislations and regulations amended from time to time. The Supplier shall not offer, give, or receive any bribes, gifts, or other inducements in connection with the execution of this Purchase Order. Any breach of this clause shall entitle the Buyer to terminate the Purchase Order with immediate effect and claim damages.

Intellectual Property Rights: The Supplier shall ensure that all goods and services supplied under this Purchase Order do not infringe any third-party intellectual property rights. The Supplier shall indemnify and hold the Buyer harmless from any claims, damages, or expenses arising from any infringement or alleged infringement of intellectual property rights.

All intellectual property rights, including but not limited to patents, copyrights, trademarks, and trade secrets, created or used in the performance of this contract shall remain the property of the respective party. The Supplier grants Buyer (L&T) a non-exclusive, royalty-free license to use any intellectual property necessary for the completion and operation of the project.

Compliance with Laws and Regulations: The Supplier shall comply with all applicable laws, regulations, and standards in the performance of its obligations under this Purchase Order. This includes, but is not limited to, labor laws, tax/custom laws, environmental laws, and safety regulations. The Supplier shall indemnify and hold the Buyer harmless from any claims, damages, or penalties arising from the Supplier's non-compliance with applicable laws. Any breach of this clause shall be considered a material breach of the contract.

Ethical Conduct: The Supplier shall conduct its business in an ethical manner and in accordance with the highest standards of integrity and professionalism. The Supplier shall ensure that its employees, agents, and subcontractors adhere to these standards in the performance of their duties under this Purchase Order. Supplier shall comply with the terms and conditions of Buyer's Supplier Code of Conduct available at Buyer's website.

Sustainability and Environmental Responsibility: The Supplier shall adopt sustainable practices and minimize the environmental impact of its operations. The Supplier shall comply with all applicable

Continues..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO

LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 27 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

environmental laws and regulations and shall take all necessary measures to prevent pollution, reduce waste, and conserve resources.

Audit Rights: Notwithstanding any other provision of this PO, Contractor shall have the right to audit all of Vendor's records for the purpose of verifying compensation paid for terminated Work, including Lump Sum and Work Unit Basis compensation paid. The Supplier shall provide access to all relevant records and cooperate fully with the Buyer's audit team. Any discrepancies or non-compliance identified during the audit shall be rectified by the Supplier at its own cost.

Business Continuity and Disaster Recovery: The Supplier shall have in place a business continuity and disaster recovery plan to ensure the uninterrupted supply of goods and services under this Purchase Order. The Supplier shall provide a copy of this plan to the Buyer upon request and shall notify the Buyer immediately in the event of any disruption to its operations.

Subcontractor Approval: The Supplier shall not subcontract any part of its obligations under this Purchase Order without the prior written consent of the Buyer. The Supplier shall remain fully responsible for the performance of its subcontractors and shall ensure that they comply with all terms and conditions of this Purchase Order.

Prior Agreements: This PO supersedes all previous contracts, correspondence, and understandings between the Parties concerning the subject matter hereof and constitutes their entire agreement and understanding in relation to the performance of Work / provision of Supplies as stated herein. The Parties shall not be bound by, or be liable for, any statement, representation, promise, inducement, or understanding of any kind or nature not recorded in this PO or its subsequent amendments and Change Orders.

No Release The obligations and liabilities of the Supplier under this PO shall not be released, diminished or in any other way affected by any enquiry, direction, inspection, comment, consent, sanction, acknowledgement, confirmation, approval, or advice made or given by or on behalf of Buyer / Company, any Buyer/ Company representative or lenders, a Buyer's financiers (including any of their nominated entities), or any of their respective employees, servants, or agents; or any failure to enquire, direct, inspect, comment, consent, sanction, acknowledge, confirm, approve, or advise, whether or not such act or omission might give rise to an independent liability of any such person.

Invalidity if any one or more of the provisions contained in this PO is held as invalid, illegal or unenforceable in any respect under any Applicable Law, the validity, legality and enforceability of the remaining provisions contained in this PO shall not in any way be affected or impaired.

Independent Contractors The Parties to this PO are independent contractors and have entered into this agreement for the execution of Work / provision of Supplies on a principal-to-principal basis. Neither Supplier nor any member of the Supplier Group shall be deemed to be servants, agents, or employees of the Buyer.

Survival The provisions in relation to Guarantee/Warranty, Patents and Royalties, liabilities, Indemnity, Governing law and Dispute resolution, infringement of IPR and Confidentiality as stated in this Purchase Order shall survive its expiry or early termination.

Payment Of Costs, Expenses And Damages Payment of costs, expenses and damages to the Buyer shall not release the Supplier from its obligation to carry out and complete the work in accordance with the provisions of this Purchase Order and/or to discharge its obligations and responsibilities arising of this Purchase Order, nor prevent the Buyer from exercising its right to terminate the Purchase Order in accordance with Article 11.10.2.

Standards The goods and services must meet the technical specifications# standards. If no standard is mentioned, they should follow the Owner's/manufacturer's/licensor's standards.

Cooperation with other contractors The Supplier must not hinder other contractors and should facilitate their work. The Supplier should cooperate and share technical information with the Buyer/Owner and other contractors. Supplier must comply with all safety measures at the Supplier's expense.

Repeat Order In case of additional requirement, Supplier shall confirm to supply as per Unit Rate mentioned in the price schedule. Delivery schedule of such repeat order shall be prioritised by Supplier to enable L&T to meet project requirement.

11.3 Terms & Conditions for Supplier's Supervision of Erection, Commissioning & Performance Tests and for Training of Owner's Personnel (as applicable)

A The per diem rates shall be inclusive of cost of Boarding & Lodging (at Site or during transit), travel during Mob and Demob, daily to & fro travel to Site, communication cost, overseas & local telephone calls, insurance, pocket money & all other incidental expenses.

Per Diem Rates shall be inclusive of all taxes, duties, levies including the income tax / withholding taxes, which are required to be withheld by LTHE for the services rendered in India.

Per Diem Rate for Year 2025 -- Physical Visits -- INR XXX

Per Diem Rate for Year 2026 -- Physical Visits -- INR XXX

Per Diem Rate for Year 2027 -- Physical Visits -- INR XXX

Per Diem Rate for Year 2028 -- Physical Visits -- INR XXX

Per Diem Rate for Year 2025 -- Remote Visits/Supervisions -- INR XXX

Per Diem Rate for Year 2026 -- Remote Visits/Supervisions -- INR XXX

Per Diem Rate for Year 2027 -- Remote Visits/Supervisions -- INR XXX

Per Diem Rate for Year 2028 -- Remote Visits/Supervisions -- INR XXX

To & Fro Travel (Economy Class Air Fare / II AC Train Fare) to Project Site shall be reimbursed extra at actuals by LTHE.

Above Per-diem rate is also valid for Sub-supplier visit during Erection, Commissioning & Performance test at Project Site.

The Per diem rate for day /Mandays shall not be payable/considered, if Supplier representative(s) are required to be "Quarantined" at site as per the guidelines of the Local Authorities / LTHE / Owner HSE requirement etc.

Continues..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 28 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

B The Per Diem charges shall be applicable from the day the engineer reports to Site-In-Charge to the day the engineer leaves the site.

C Non-work days, such as Sunday or any holiday observed by OWNER / LTHE falling during the period of work shall not be considered for calculation of number of days payable, unless such days are certified as work days by LTHE#s site in-charge.

D The per diem rate mentioned above is considering 10 hours of working per day (excluding lunch hours) for 6 working days in a week. No overtime shall be paid for working beyond 10 hours in a day. Working on Sundays / Holidays shall be paid at normal per diem rate mentioned above. Time consumed during travelling to & fro between hotel / lodging location to site shall not be considered under working hour.

E Supplier will bring along Special apparatus / instrument / measuring devices etc., duly calibrated (if required) to carry out adjustment/setting etc. and to perform various tests as required.

F In the event the duration of stay of Supplier's supervision is solely attributable to the Supplier, (i.e. due to re-work or repair, waiting for material or instructions from head office etc.) then all costs incurred by the Supplier(s) for those number of days shall be to account of the Supplier.

G The supervision expert shall be mobilized within 10 days of notice given to Supplier.

H Supplier shall take all necessary Insurance Policies as applicable to cover Supplier#s owned, leased or hired equipment, tools, vehicles etc. Necessary Insurance such as Workman compensation policy and/or ESI policy and/or Employer#s liability policy as applicable, for the personnel deputed to site by Supplier shall be taken by the Supplier and the charges for the same shall be to Supplier#s account and is included in the per diem rates mentioned above. Third Party Liability Insurance (TPL) to be procured by Supplier shall cover legal liability for bodily injury to loss of life of and/or damage to and loss of properties of the third person party arising out of the performance by the Supplier of the works.

I "Supplier shall submit the following at least 4 weeks prior to deputation of the engineer :-

- Check-list of all activities / resources (manpower, material, machinery) to be conducted / arranged by LTHE before the arrival of engineer at project site.
- Detailed Bar Chart highlighting all the activities with duration for Erection / Installation / Testing of the Equipment.
- List of Tools & Tackles, measuring Instruments and manpower resources required for Erection / Installation / Testing which are to be arranged by LTHE."

J Payment for Supervision of erection & commissioning shall be effected by Wire Transfer against time sheets duly certified by LTHE#s site in-charge within 60 days of submission of monthly Invoice with following correct supporting documents like:

- # Invoice of Flight & Boarding Pass or Train Ticket.
- # Time sheet duly certified by site in-charge.
- # Copy of MOM made at site.
- # Copy of Insurance Policy for Personnel and/or equipment/tools as applicable.

The Supplier shall adhere to the Billing Instructions issued by LTHE while submitting Service Invoice.

Applicable Goods and Services tax (SGST/UTGST, CGTS, IGST) on Services shall be paid by LTHE, subject to following terms #

- Suppliers are registered under GST and have valid GST registration at all the time.
- Supplier shall submit GST tax invoice as per applicable SGST / UTGST / CGST / IGST Act, 2017 and corresponding GST rules
- Supplier shall discharge the applicable CGST & SGST/UTGST / IGST liability, as the case may be, and file GST Returns within the stipulated time.
- Applicable GST value & rate to be indicated in tax invoice, as per relevant GST Act / Rules.
- Credit and Debit note to be raised as per the relevant provision of GST Act and rules made thereunder.
- Any increase in GST Rate over and above the applicable rate, whether arising out of statutory changes or otherwise beyond the "Contractual Delivery Date" (CDD) shall be to Supplier#s account.
- Any increase in the GST Liability or any other duties, taxes and levies due to wrong classification of SAC (Services Accounting Code) by the Supplier shall be to Supplier#s account.
- In the event the Supplier does not discharge the applicable GST in time as per the GST payment rules and input tax credit is not available to LTHE as on the date of payment to Supplier, the Company (LTHE) reserves the right to retain such GST amounts from the Supplier#s Payments and release the same only after evidence of tax payment by the Supplier is furnished.
- Supplier shall furnish the details of returns filed against invoice issued to LTHE as evidence of his tax liability discharge.

K Health, Safety & Environmental Protection Procedures (HSE)

The Purchaser and the Supplier agree to abide to all HSE norms as established by LTHE, PMC, Owner and laws of India.

L Other Terms

The Supplier shall carry out the Technical Field Services using sufficient persons having adequate knowledge of the operations to be carried out (including the methods and techniques required, the hazards likely to be encountered and methods of preventing accidents) for the satisfactory and safe performance of the Technical Field Services.

The Supplier shall employ (or cause to be employed) only persons who are careful and appropriately qualified, skilled and experienced in their respective trades or occupations in connection with the works. LTHE/ PMC / OWNER may remove (or cause to be removed) from the site or from involvement in the works, any employee, who in the opinion of LTHE site in-charge/PMC/Owner :

- a. persists in any misconduct
- b. is incompetent or negligent in the performance of his duties

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 29 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

- c. fails to comply with any provisions in the P.O. deemed important to LTHE/OWNER, acting reasonably for
d. persists in any conduct which is prejudicial to safety, health, or the protection of the environment.
If appropriate, the Supplier shall then appoint (or cause to be appointed) a suitable replacement person at its own cost.

Definitions / Abbreviations

"OWNER/

COMPANY" Rashtriya Chemicals & Fertilisers Limited (RCF)

Licensor INCRO S.A. (Spain)

Project 1200 MTPD (DAP Basis) NPK Project and Associated Facilities

Order "Shall mean the totality of Contract between L&T and Supplier and shall include the following in the listed order of precedence:

- (a) The Purchase Order and any Amendments thereof
(b) Letter of Award / Letter of Intent
(c) Agreed Terms and Conditions of Purchase, and any mutually agreed amendments to the Agreed Terms and Conditions
(d) Correspondences exchanged between the Parties i.e.(Technical Queries, Commercial Queries and responses thereof by the Contractor).
(e) Approved ITP
(f) MATERIAL REQUISITION / Technical Specifications / Drawings / Applicable Code & Standard.
(g). Description of Bill of Quantity
(h) RFQ & its Amendments / Addendums

Any exclusions, clarifications etc. exchanged between the parties shall stand withdrawn unless otherwise mutually agreed, duly recorded and accepted by Buyer in writing.

"

Mechanical Completion "Shall mean installation of all Supplies by Supplier thereafter completion of all activities as per approved specification and drawings/acceptable International Engineering practices, to be performed including pre-commissioning and start-up activities, completion of all punch-list items, as provided by OWNER/PMC and/or Process Licensor and/or statutory bodies like IBR, OISD, CCE, TAC, Factory Inspector, Pollution Control Authorities etc.

Mechanical Completion shall be certified to Supplier after L&T receives Mechanical Completion Certificate from Owner/PMC"

Commissioning Shall mean pressing into service of the plant(s), Equipment(s), Vessel(s), Pipeline(s), Machinery (ies) and systems & sub-systems comprised within the Unit(s) in accordance with the procedures as approved in the applicable Operating Manual and as per the requirements of OWNER/PMC and/ or Licensor after successful testing, pre-commissioning and trial run of the Unit(s).

Performance Test Shall mean the test or series of tests to be carried out by the SUPPLIER/CONTRACTOR to prove the contractual guarantees as per the requirement of OWNER/ PMC/ Licensor / L&T. (PGT/ PGTR).

Code-1 Shall mean final approval of drawings/documents incorporating all comments of the OWNER / Licensor / PMC / L&T.

Code-2 Shall mean tentative approval of drawings / documents by OWNER / Licensor / PMC / L&T with comments to be incorporated by Supplier, with the intent that subject to incorporation of and/or rectification in accordance with such comments, construction / manufacture can proceed based on commented drawings / documents pending their re-submission and grant of Code-1.

Code-3 Shall mean that on review of the drawings/documents by the OWNER / Licensor / PMC / L&T the same have not been found in accordance with the applicable codes, standards, design basis and contractual stipulations and must therefore be re-submitted for review and approval after appropriate correction and/or re-preparation.

Code# R Shall mean Document is retained for records. Proceed with manufacture / fabrication.

Parties Shall mean the Parties executing this Contract, viz. L&T and the Supplier.

Party Shall mean one of the Parties.

AI Authorised Inspector (ASME)

TPIA Third Party Inspection Agency

LOI Letter of Intent

PO Purchase Order

DLP Defect Liability Period

ITP Inspection Test Plan

Continue..

L.C.: 11.09.2025 11:58:05



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - II
Page No: 30 of 30

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

SDR Supplier Data Requirement
PTR Proven Track Record
VQC / SQC Vendor / Supplier Qualification Criteria
MR Material Requisition
PR Purchase Requisition
LSP Logistic Service Provider
PBG Performance Bank Guarantee
KOM Kick Off Meeting
HAZOP Hazard and Operability Study
SIL Safety Integrity Level
Supplier Shall mean the Supplier of Goods and/or Services on whom the Purchase Order is released
Sub-Supplier Shall mean Supplier#s Supplier of any tier
Supplier Group Shall mean Supplier, its Sub-Supplier of any tier, and their respective representatives, but shall not include any member of L&T
POMS Shall mean "Post Order Management System" a digital (web based) portal developed by L&T for information capturing and/or data sharing and/or performing various required activities by Supplier subsequent to LOI/PO issuance.
Free Issued material Shall mean any equipment and/or materials and/or Goods provided by L&T/OWNER.
Package Supplies under this Contract.
Supplies Shall mean the totality of Goods and/or Services by expression or implication required to be supplied by the Supplier within the Scope of Work as per Order.
RFQ Request for Quotation including amendments issued to Supplier and covers instruction to bidders, Terms & conditions, Material requisition and all other documents.
L&T/Buyer Larsen & Toubro Limited
FAT / IFAT Factory Acceptance test / Integrated Factory Acceptance Test to be carried out at Supplier / sub-Supplier facilities / factory as per agreed scope
SAT Site Acceptance Test to be carried out at Project Site as per agreed scope
Day Shall mean Gregorian calendar day unless specified otherwise
QC Quality Control



LARSEN & TOUBRO LIMITED
L&T ENERGY - HYDROCARBON
Technology Centre, North Block -II

PURCHASE ORDER

ANNEXURE - III
Page No: 1 of 1

PO NO. :LTES/75001-83664/BS
PO Date : 03.09.2025

Standard data confidentiality & protection related clauses

1. The Vendor shall not use Larsen & Toubro Limited / Subsidiaries / Associates (Hereby referred as **L&T**) confidential & business sensitive Data for any **purpose** other than the provision of the Supplies or Services and shall return any L&T confidential & business sensitive Data to L&T immediately upon request at any time providing such return does not prevent the Vendor from fulfilling its obligations under this Contract.
2. The Vendor shall **retain** L&T confidential & business sensitive Data **no longer** than is necessary for the provision of the Supplies or Services, in accordance with the relevant Data Protection Legislation and such instructions as L&T may provide from time to time. Upon expiry or termination of this Contract for whatever reason, the Vendor shall immediately destroy (if in soft form) / return to L&T (if any hard copies) all confidential & business sensitive Data and certify that no copies have been made or retained by the Vendor or any third party acting on its behalf.
3. The Vendor shall **Process** L&T confidential & business sensitive Data only on the instructions of L&T and to the extent necessary for the performance of this Contract.
4. The Vendor shall not **modify, amend, or alter** the contents of the L&T confidential & business sensitive Data except as required or permitted by this contract or with L&T's prior formal **consent**.
5. The Vendor will transparently share locations of data storage with Customer. In case, Customer has concerns about data storage in specific geographies/locations, The Vendor will take appropriate steps to resolve customer's concerns. The Vendor will take adequate measures to protect customer data **shared with supplier** against information security threats and would bear the cost associated with addressing a data leak/loss of data integrity.
6. The Vendor shall implement the appropriate **technical and organizational measures** to protect L&T confidential & business sensitive Data against accidental or unlawful destruction or accidental loss, alteration, unauthorized disclosure or access, and against all other unlawful forms of processing, and provide to L&T a written description of the measures taken when requested by L&T.
7. The vendor shall comply with all relevant provisions of any L&T codes of practice notified to the Vendor from time to time and the **Data Protection requirements from applicable regulations**.
8. The vendor shall keep all L&T confidential & business sensitive Data secure and confidential, act only on L&T's instructions with respect to it, and comply with such further reasonable requirements from time to time of L&T for the security of it.
9. The vendor shall ensure that, of the Vendor's staff, only those of the Contract Personnel who need to have **access** to the L&T confidential & business sensitive Data are granted access to the data only for the purposes of the performance of this Contract and the Contract Personnel are informed of the confidential nature of the L&T confidential & business sensitive Data and comply with the obligations set out in this Condition.
10. The vendor shall notify L&T in the event that it receives **a request or notice** from any data subject to have access to that person's L&T confidential & business sensitive Data held by it and will provide L&T with full co-operation and assistance in relation to any complaint or request including providing L&T with any relevant L&T confidential & business sensitive Data it holds within the timescales provided by the request or notice or as otherwise required by L&T.
11. The vendor shall abstain the sharing of L&T confidential & business sensitive data at the initial onset however if required the vendor will need to obtain L&T's prior formal consent before transferring L&T confidential & business sensitive Data to any subcontractors in connection with the provision of the Supplies or Services.
12. The vendor shall in respect of L&T confidential & business sensitive Data transferred by L&T to the vendor or acquired by the vendor from L&T's systems to a country outside of India shall be on the basis of the Legislation issued by the Indian Government, or such other data **protection model contract terms** as may be agreed between the Parties from time to time, except where the relevant Data Protection Legislation provides for a derogation from this requirement. Any breach of this Condition by the Vendor shall be deemed to be a material breach of the **Contract and the Vendor shall indemnify** L&T against any costs, losses, damages, proceedings, claims, expenses or demands incurred or suffered by L&T which arise as a result of such breach.
13. The Vendor shall not **disclose** or allow access to any L&T confidential & business sensitive data to any person placed by the Vendor for managing performance of contract i.e. person who is not involved in operational activities.
14. The Vendor shall, upon L&T giving reasonable **notice**, allow L&T or its nominated representatives such access to its premises, Information and records and those of its agents subsidiaries and sub-contractors, as may be reasonably required by L&T from time to time to assess the Vendor's and/or Contract Personnel's compliance with this Condition.
15. During the Term of the Agreement, L&T shall reserve the **right to audit** for the Vendors to check the reasonable security practices followed within the organization to achieve satisfactory assurance that the associate will appropriately safeguard the information.
16. The vendor should maintain system/application logs and accesses being held for at least period of three months for audit or investigation purposes.
17. For all personnel to whom access is granted to L&T Systems having L&T confidential & business sensitive data, information or network, the Vendor shall:
 - o Ensure each individual has a unique user identification and password known only to such individual for his/her sole use
 - o Allow Access using a secure login process
 - o Grant access on least privilege basis
 - o Ensure that audit logs are generated to record user activity and security-relevant events and securely managed and retained with nil ability on the part of the Vendor to allow any un-authorized access or amendment to the audit logs. Such audit logs must be maintained for future reference for a period of at least three months
18. Vendor & it's subcontractors shall comply with L&T Third party information security policy requirements including the above ones.

L.C.: 11.09.2025 11:58:05



☐ Short Delivery ☐ Only for Engg. Activities
☐ Bought out valves ☐ Others, Please Specify

		Date	29.09.2025
		CORB Reference	DM25066L
Customer / Buyer Name	LARSEN & TOUBRO LIMITED	CEC Quote Number with Date	250702-0400-PA Dtd 28.08.2025
Site / End User Name & Country	Rashtriya Chemicals and Fertilizers Limited, India	Application Engineer SPOC	Karthikeyan Sekar Satheshkumar Eagambaram
Buyer PO Number/Date	LTES/75001-83664/BS Dtd 03.09.2025	Outside Sales Engineer/Manager	Ashokvijay R
Buyer PO/LOI Received Date	12.09.2025	PO Value	INR 9,99,750
Agreed / Approved Payment Terms	Refer PO clause no. 5.0	1)PO Delivery Date 2)Agreed Delivery Date	26.03.2026
Agreed Incoterms /Shipping Terms	CPT, RCF Thal Project Site	Margin	22%
LD Clause	@0.5% of the undelivered portion of Order value	Agent Name & Commission	NA
Ship to Address with GST#:	Rashtriya Chemical and Fertilizers Limited Thal, Thal-Vaishet, Alibag # 402 208 State: Maharashtra, India GSTIN: 27AAACL0140PEZ6	Bill to Address with GST#:	Larsen & Toubro Limited Divison Name: L&T Energy Hydrocarbon 14th Floor, AMN Tower, AMN Tower, Jogeshwari - Vikroli Link Road, Powai Campus, Gate 3, Mumbai Suburban, Maharashtra, 400072

CIRCOR PROPOSAL/QUOTE

Price Schedule



Project	1200 MTPD (DAP Basis) NPK Project
Customer	L&T Energy Hydrcarbon
End User	Rashtriya Chemicals andFertilizers Limited
Destination	INDIA
Our Quote Ref	250702-0400-PA
Date	28-Aug-25
Rev	3

Technical Details																	Price Details (INR)			
S.No	Tag Number	Valve/Item Type	Model Name	Body Size	Body Rating	End Conn	Body Mat	Trim Type	Cage / Plug Type	Trim Char	Travel (mm)	Actuator Type	Fail Action	Handwheel	Others	Accessories	Qty	Unit Price (INR)	Total Price (INR)	
1	D5-9801	DSH	CIR 3230	3"	300#	RF	A106/A105	SL Nozzle	-	-	-	None	-	-	-	N.A	1	₹	84,023	
2	TCV-A1-9802	Globe	CIR 3100	1"	300#	RF	A105	Unbalanced	Contour Plug	Equal%	20	Spring & Diaphragm	Close	Yes - Side	-	Refer Spec Sheet	1	₹	2,26,571	
3	D5-9805	DSH	CIR 3220	3"	150#	RF	A106/A105	Fixed Nozzle	-	-	-	None	-	-	-	N.A	1	₹	32,763	
4	TCV-A1-9806	Globe	CIR 3100	1"	300#	RF	A105	Unbalanced	MF Plug	Modified	25	Spring & Diaphragm	Close	Yes - Side	-	Refer Spec Sheet	1	₹	2,06,968	
5	Start-up & commissioning spares																Refer Spec Sheet	1	₹	15,829
6	Mandatory spares (Consumption within 2 years)															-	Refer Spec Sheet	1	₹	4,33,596
																	6	Total	₹	9,99,750

Remarks:
1. Scope: Valve + Actuator and DSH along with accessories as mentioned in CIRCOR Datasheet.

For CIRCOR Flow Technologies India Pvt Ltd

Karthikeyan Sekar
Application Engineering

No.	T&C Clause	CIRCOR Confirmation
	Advanced Bank Guarantee (ABG's) and Performance Bank Guarantee (PBG's) shall be issued as per L&T's format attached in this tender. BG's shall be issued from Indian Branch of one of the Banks listed in L&Ts format. BG's issued by other Banks will not be accepted	Not Applicable
1.00	Scope of Work / References	
1.10	Scope of work shall be strictly as per the Material Requisition (MR) enclosed with enquiry document, subsequent addenda, correspondence (Technical Queries, Commercial Queries etc.), discussions and MoMs. Any exclusions, clarifications etc. stands withdrawn unless otherwise mutually agreed, duly recorded and accepted by L&T prior to placement of the Order.	Agreed
2.00	Prices & Price Basis	
2.10	Price shall be on Ex-works basis inclusive of packing and loading onto transporter's vehicle. Price in full or in part are firm and fixed with no escalation whatsoever. Variation in prices of raw material or bought-out items shall not be admissible. Any variation in price on account of exchange rate fluctuations, changes in taxes/duties/law, changes in Codes and standard, increase in wages, Commodity rates, labor rates, fuel prices etc. shall not be admissible. Prices to be submitted in Indian Rupees (INR) only and shall be valid for minimum 90 days from the date of offer prior to the Purchase Order	Price basis agreed for CPT site basis
2.20	Appointment of Third-Party Inspection Agency (TPIA) for performing inspections as per approved Inspection Test Plan (ITP), Quality Procedures/Specifications and Material Requisition at Supplier's works & Sub-Supplier works (Requiring Final Inspection Release Note as per ITP/ agreed with L&T) shall be Supplier's scope. Charges for the same shall be borne by Supplier, including coordination with Third Party Inspection Agencies for timely Inspection shall be in Supplier's scope. TPIA Inspection and testing at sub-suppliers works shall be in supplier's scope (unless otherwise mutually agreed). TPIA Inspection and testing charges for any new procedure qualifications and Material testing shall be under supplier's scope. TPIA chosen by Suppliers for the TPIA inspection under their scope shall be strictly from following approved TPIA's: 1) PDIL 2) LRQA 3) TUV India Pvt. Ltd. 4) Bureau Veritas (India) Pvt. Ltd. 5) DNV 6) SGS 7) TUV Sud South Asia Pvt. Ltd 8) Certification Engineers India Ltd (CEIL) 9) Engineers India Limited (EIL) OR other reputed Inspection Agency may be selected with prior approval from L&T/PMC/Owner. Cost towards repeat tests and inspection due to failures, repairs etc. for reasons attributable to the supplier/sub-supplier shall be borne by the supplier. Supplier TPIA Inspection/L&T Inspection/ Inspection waiver does not relieve supplier/manufacturer/sub-suppliers from contractual obligations and their responsibilities to supply the item(s) concerned in full compliance with the requirements of the Purchase Order and project specifications, applicable codes and standards & approved Quality Plan. The Owner has the right to inspect and test goods to ensure they meet contract specifications. Inspections can occur at the Buyer's or Supplier's premises or at the final destination. The Supplier must provide necessary facilities and assistances for these inspections. Delays in approval or testing are the Supplier's responsibility. The Supplier must promptly provide copies of all test results to the Buyer	Agreed
2.30	Prices are inclusive of following charges:-	
2.30.1	Transportation saddles, templates/gauge plates, spider beams, transportation Stools etc., as applicable.	Agreed
2.30.2	Testing / Trial Run / FAT/ IFAT as per RFQ and approved ITP at Supplier's works, sub-Supplier's works & Site	Agreed
2.30.3	a) Third-Party Inspection charges for inspection of balance items (not covered under clause 2.2) as per Inspection Test Plan (ITP) at Supplier/sub-Supplier's works shall be in Supplier's scope. b) Raw materials inspection & certification requirement i.e. EN10204 Type 3.2/3.1, as applicable to be performed through project approved TPIA at Mill / Supplier locations. Including any additional check testing at approved laboratory to comply with PO/MR requirements	Agreed

No.	T&C Clause	CIRCOR Confirmation
	c) Inspection & Certification required as per Indian Boiler Regulation (IBR), ASME AI Inspection/Code Stamping and National Board of Boiler & Pressure Vessel Inspectors (NBBI) Registration & and National Board Registration (NBR) as applicable. d) Third-party inspection charges to perform check testing of raw material in laboratory with TPIA witness in case raw materials (Plate/ Pipe/ Fitting/ Forgings etc.) is procured from Approved Traders / Stockiest & with mill test certificate EN10204 Type 3.2/3.1. e) Third-party Inspection Charges at Supplier/sub-Supplier works to perform New/ Fresh Welding Procedure/ personnel Welder Performance Qualifications for welding, NDE, Painting, Refractory, Laboratory Testing Charges shall be in Supplier's scope. In case of any failure/rejection in supplied item/material observed at site, cost of failure analysis (including lab testings, re-inspection) shall be borne by Supplier. f) For indigenous suppliers, for items deviating IBR requirements, inspection shall be carried out by IBR authorities only.	

No.	T&C Clause	CIRCOR Confirmation
	g) In case of any failure/rejection in supplied item/material observed at site, subject item/material will be sent to the L&T's approved labs for failure analysis. The cost of this failure analysis shall be borne by Supplier, if failure/rejection is attributed to Supplier.	
2.30.4	Mandatory Spares - as per list specified in MR.	Agreed
2.30.5	2 years O&M spares: Supplier to quote optional price valid till 30-Jun-2027 as specified in the MR. If the list is not furnished in MR, the same shall be provided as per Supplier's Recommendation in SPIR format provided by L&T. OWNER reserves the right to place the Order for such O&M spares	Agreed
2.30.6	Spares for Start-up / Erection & Commissioning/Performance Guarantees as specified in the RFQ. If the list is not furnished in RFQ, the same shall be provided as per Supplier's Recommendation. If some Spares, which are considered "Not applicable or not considered in correct quantity as per philosophy" by Supplier becomes applicable directly or becomes applicable by having equivalent function the quantity is corrected based on spares philosophy at the time of order execution, Supplier shall supply them in line with Mandatory Spares Philosophy at no extra cost and / or time implications to L&T.	Agreed
2.30.7	Special Tools & Tackles required for Erection & Commissioning and performance guarantees as specified in the RFQ. If the list is not furnished in RFQ, the same shall be provided as per Supplier's Recommendation.	Not Applicable
2.30.8	Training Charges (if applicable) for the complete package, individual components as mentioned in MR. These shall include charges for training at Supplier/Sub-supplier Works and / or at Project site, training manual, etc. Training charges shall be quoted as Lump sum or Per Diem basis.	Not Applicable
2.30.9	Charges for documentation (hard & soft copy) for approval & as-built etc. in requisite no. of sets as per RFQ. In case the design of any equipment / ordered items is in L&T scope, all such design documents shall be shared to Supplier and all such documents shall be included as part of Final Documentation (soft and hard copies) without any cost implication to L&T.	Agreed
2.30.10	List along with quantities for all first-fills for all equipment i.e. Lubricating oils, Grease, Silicon Oil, chemicals, consumables etc. in addition to commissioning and start-up requirement for first fill + six months normal operation after commissioning. Supply shall be by L&T	Not Applicable
2.30.11	NA	
2.30.12	Supplier confirms acceptance of following as MINIMUM across the Table Meetings at L&T / Detail Engineering Consultant / PMC / Owner Office with presence of competent team member :- a) Kick Off Meeting (2 of days) b) 1st Design / Document Review Meeting (3 of days) c) 2nd Design / Document Review Meeting (3 of days) d) Model Review Meetings (30%, 60%, 90% - 2 day each) e) Hazop / SIL Review Meeting (2 day each) Visit Schedule shall be as decided by L&T and conveyed to Supplier. Supplier to depute competent team member for all above across the table meetings. All costs associated for attending such across the table meetings at L & T Mumbai are included in base price submitted by Supplier.	Not Applicable
2.30.13	Providing information and performing required activities in POMS for progress reporting, drawings and document submission & review cycle, inspection calls & TPIA co-ordination, logistics co-ordination, etc as per the project requirements. Performing required activities in SAP Ariba for P.O. Acknowledgement, Shipping notification, Invoicing, etc.	Agreed

No.	T&C Clause	CIRCOR Confirmation
2.40	Taxes and Duties	
2.40.1	Raw Materials / Components imported by Indigenous Supplier to their works. For all raw materials / components sourced by the Supplier/Sub supplier, Supplier/Sub Supplier shall not be entitled for any concessional import duty (whether in India or any other location) for components / raw materials being imported by the Supplier and sent to their works / their sub-supplier works. Any such imported components shall be customs cleared by the Supplier after paying the applicable customs duties All import duty and other taxes & duties on such raw material / components shall be in the scope of the Supplier and paid by the Supplier. Statutory variation in taxes & duties including FTA, CEPA etc. on raw material / components shall be to Supplier's account at all times. L&T/OWNER will not bear any liability towards payment of Safeguard Duty, Anti-Dumping Duty, Protective Duty and applicable IGST on same Or Countervailing Duty on subsidized articles or any other such duties of Customs imposed by the Government under Customs Tariff Act, 1975 applicable on import of any materials to India for the subject work. Therefore, the Suppliers are expected to source such materials required for the execution of work only from domestic producers in India. To make Price competitive, Supplier are requested to consider free trade agreement/CEPA etc. for import of Raw material, wherever applicable. Supplier to note that all responsibility of these import and procedures shall be responsibility of Supplier. Imported Raw Material / Finished product shall not be despatched directly to Site. These will be sent to supplier works and then despatched as local supply to Site	Agreed
2.40.2	Finished Goods shipped from Supplier Works: For all raw materials / components sourced by the Supplier, all import duty and any other taxes & duties on such raw material / components shall be in the scope of the Supplier and paid by the Supplier. Statutory variation in taxes & duties on inputs (raw material / components), shall be to Supplier's account at all times. The Supplier shall adhere to the Billing Instructions issued by L&T (to be issued before despatch). Applicable Goods and Services tax (SGST/UTGST,CGST,IGST) on Supplier's Goods and/or Services shall be paid by L&T, against documentary evidence provided by Supplier subject to following terms – - Supplier is registered under GST and have valid GST registration at all the time. - Supplier shall submit GST tax invoice as per applicable SGST / UTGST/ CGST / IGST Act, 2017 and corresponding GST rules - Supplier shall discharge the applicable CGST & SGST/UTGST / IGST liability, as the case may be, and file GST Returns within the stipulated time. - Applicable GST Amount & rate to be indicated in tax invoice.as per relevant GST Act. - Credit and Debit note to be raised as per the relevant provision of GST Act and rules made thereunder. - Any increase in GST Rate over and above the applicable rate, whether arising out of statutory changes or otherwise beyond the "Contractual Delivery Date" (CDD) shall be to Supplier's account. - Any increase in the GST Liability or any other duties, taxes and levies due to wrong classification of HSN (Harmonized System of Nomenclature) / SAC (Services Accounting Code) by the Supplier, within Contractual Delivery Date, shall be to Supplier's account. - In the event the Supplier does not discharge the applicable GST in time as per the GST payment rules, L&T reserves the right to retain the said amount from Supplier's invoice and release the same only after evidence of tax payment by the Supplier is furnished. - Supplier shall furnish the details of returns filed against invoice issued to L&T as evidence of their tax liability discharge. Statutory Variation in Taxes and Duties: Statutory variation in GST mentioned above and any new taxes and duties, within the Contractual Delivery period on finished Goods and/or Services and Goods directly despatched to site from Supplier works shall be to L&T's account. Statutory variation in GST mentioned above and any new taxes and duties, beyond the Contractual Delivery period on finished Goods and/or Services and Goods directly despatched to site from Suppliers works shall be to Supplier's account. Any other statutory variations shall be to Supplier's account. Statutory variation in taxes & duties on raw material / components shall be to Supplier's account at all times. Income tax / GST TDS shall be deducted from the Supplier Invoices / Claims and deposited with respective Govt. as and when applicable.	Agreed

No.	T&C Clause	CIRCOR Confirmation
	Statutory Approvals: Unless otherwise specified, it shall be the Supplier's sole responsibility to obtain all approvals from any authority (except for CRZ clearance) required under any statute, rule or regulation of the Central or State Government concerned with the performance of the Order and/or the contractual Work. The application on behalf of the L&T for submission to relevant authorities along with copies of required certificates shall be prepared and submitted by the Supplier well ahead of time so that the actual construction/commissioning of the works is not delayed for want of the approval/inspection by the concerned authorities. The SUPPLIER shall arrange for the inspection of the works by the authorities and will undertake necessary coordination and liaison required and shall not be entitled to any extension of time for any delay in obtaining such approvals. Any Goods and Services Tax (GST) liability arising in India on account of place of supply of goods and / or services deem to be within India (as per GST provisions) shall be on account of the Supplier. If required, the Supplier will obtain GST registration in India and comply with all GST related requirements in India, including but not limited to provide GST registration certificate, issue of proper tax invoices, uploading of tax returns, and transfer of input tax credit to L&T etc. : Please refer Annexure I "GST Check List" to be ensured for generation of Invoice and Packing list in compliance with GST Act. However, it is sole responsibility of Supplier to ensure total compliance to applicable SGST / UTGST/ CGST Act, 2017 and corresponding GST rules. Statutory approval (if any) including any type of permission for movement of ODC consignments from Supplier's works to Project site, shall be in Scope of Supplier. Supplier shall arrange such applicable approvals in advance so that actual movement of goods is not delayed for want of the approval/inspection by the concerned authorities.	
2.40.3	Supplier shall comply to Tax collected at source (TCS) on Sale of Goods u/s 206C(1H) of Finance Act, 2020 and subsequent amendments. TCS shall form part of the invoice for sale of goods	
3.00	Payment Terms	
3.10	10% of the Basic Order value (excluding taxes and duties) shall be released on fulfillment of below conditions: - Submission of unconditional Purchase Order acceptance. - Approval of Critical drawings / documents viz. PR, P&IDs and GA drawings by L&T / PMC in Code-2. - Submission of Advance Bank Guarantee (@ no cost to BUYER) for advance of equivalent amount	Same as previous PO# LTES/75001-58231/AS
3.20	10% of the Basic Order value (excluding taxes and duties) shall be released on fulfillment of below conditions: - Receipt of major raw materials / Bought out Components at Supplier's Works - Submission of Advance Bank Guarantee (@ no cost to BUYER) for advance of equivalent amount - Approval of Billing Schedule - L&T / OWNER / PMC Approval of FAT AND / OR SAT Procedures for establishing Guaranteed Parameters at Supplier Shop AND / OR Project Site - Delivery of Foundation Templates as applicable - Approval of Spares List (2 Years O&M Spares - SPIR Format, Erection & Commissioning Spares, Mandatory Spares). Format shall be provided by L&T. Supplier shall submit the copy of the TPIA Material Identification & Inspection clearance reports as a backup confirming receipt and acceptance of materials	1) 95% of the Order value shall be released upon receipt of Desuperheaters at project site as per scope of order, after settling L&T#s entitlements of Order price discounts and adjustments if any, as mentioned elsewhere and on submission of following documents to L&T: - Signed Commercial Invoices (in Triplicate). - Original GST / Buyer#s copy of Tax Invoice (2 Copies). - Copy of GST Registration Certificate (2 Copies). - Lorry Receipt/ Railway Receipt (2 Copies). - Packing List / delivery Challan. - Original Manufacturer Test Certificates duly endorsed by TPIA -4 Copies. - Inspection Release Note by TPIA -2 Copies. - Dispatch Clearance in writing from L&T. - Guarantee/Warranty Certificate stating that the equipment#s + spares + accessories supplied against the subject order meet with the guarantee requirements of MR Ref. No. E23001-COM-IN-MRQ-5056, Rev. 0 and relevant points mentioned in TQ closure. - Copy of Email intimation sent to L&T for Insurance. - Copy of e-way bill. - Certification from site of receipt of material in good condition. - Submission of Performance Bank Guarantee (i.e. within 30 days from the date of PO) equivalent to 10% of basic order value to remain valid until Defect Liability Period, i.e., until 04-Feb-2028 plus claim period of 3 months. Performance Bank Guarantee (PBG) shall be furnished as per L&T#s format and from L&T approved banks. This Payment shall be made to the Supplier within 60 days from the date

No.	T&C Clause	CIRCOR Confirmation
	All Advance Milestone Payments shall be released on submission of Advance Bank Guarantee for advance for equivalent amount valid until the end of the Contractual Delivery Date Plus 3 months Plus 3 months claim period. Advance Bank Guarantee (ABG) shall be furnished as per L&T's format and from L&T approved banks. In case delivery date is extended due to whatsoever reason, then Supplier shall extend the validity of ABG without any cost implication to L&T till expected delivery date plus claim period of 3 months. In case partial delivery is done, the ABG value may be reduced by the Supplier after obtaining specific approval in writing from L&T.	
3.30	65% of the Basic Order value shall be released after shipment (i.e. Lorry Receipt) on prorata basis as per scope of order, after settling L&T's entitlements of Order price discounts and adjustments if any, as mentioned elsewhere and on submission of following documents to L&T: - Signed Commercial Invoices (in Triplicate). - Original GST / Buyer's copy of Tax Invoice (2 Copies) - Copy of GST Registration Certificate (2 Copies) - Lorry Receipt/ Railway Receipt (2 Copies) - Packing List / delivery Challan. - Original Manufacturer Test Certificates duly endorsed by TPIA -4 Copies - Inspection Release Note by TPIA (2 Copies) - Dispatch Clearance in writing from L&T - Guarantee/Warranty Certificate stating that the equipment's + spares + accessories supplied against the subject order are meeting the guarantee requirements of the P.O. no. XXXX and PR No. XXXX Rev XX (Latest revision of the approved PR to be mentioned) - General Compliance Certificates, issued by Supplier, stating that all terms and conditions of the purchase order until Ex Works shipment have been fully complied with - U-1 / U-2 form as per ASME and As-built drawings signed by AI - Copy of Email intimation sent to L&T for Insurance - Copy of e-way bill In case Performance Bank Guarantee as per L&T's format & from the approved banks as specified by L&T is not issued, 10% of the value of items shipped shall be kept on hold till PBG is received	
3.40	Not used	
3.50	10% of Basic Order value shall be released on submission of Technical document folder (in soft & hard) as per RFQ, acceptance of the same by L&T, and PBG extension for delay / extended delivery beyond original CDD as applicable.	
3.60	Not used	
3.70	5% of the Order value shall be released after Project commissioning and Performance Test Certificate of the Project. For the purpose of this payment, the date for Project Commissioning and Performance Test shall be reckoned as the date certified by L&T Site-in-charge in the Commissioning and Performance Test Certificate which shall be issued upon completion of all jobs, receipt of Final Bill from the Supplier and L&T receives Final Certificate from Owner/PMC, or six (6) months from the date of delivery of material, whichever occurs earlier.	
3.80	All Advance and Retention Payments shall be made to Supplier within 30 Days from receipt of Original Invoice along with all the correct & complete documents specified above by Wire Transfer.	
3.90	Despatch Payments shall be made to the Supplier within 90 days from receipt of Original invoice along with all the correct & complete documents specified above by Wire Transfer.	
3.100	In case partial delivery is done, the ABG value may be reduced by the Supplier after obtaining specific approval in writing from L&T.	

No.	T&C Clause	CIRCOR Confirmation
4.00	Price Discount:	
4.10	For delay in submission of documents/ drawings: In case of delay in submission of drawings/ documents (agreed as milestones), the Order value shall be discounted @ 0.5% for non-achievement of each milestone (irrespective of the extent of delay). Milestones shall be as follows : PR inputs - XX Weeks from LOI GAD - XX Weeks from LOI P&ID - XX Weeks from LOI Equipment Datasheet Curves - XX Weeks from LOI Motor / VFD Datasheet & Curves - XX Weeks from LOI ITP - XX Weeks from LOI In order to be considered as "submission" for the purpose of this provision, drawings and data submitted by Supplier must meet the Purchase Order requirements sufficiently to receive Buyers review "Code 1" (proceed) or "Code 2" (proceed, subject to acceptance & incorporation of comments) . Documents submitted by Supplier but which do not meet Buyer's minimum requirements will be returned with Buyer's "Code 3" (rejected / do not proceed) review code and are considered not received until revised and resubmitted in acceptable form to Buyer	Not accepted
4.20	For delay in delivery of material: In case of delay in supply beyond the Contractual Delivery Date, the price will be discounted @1% of the Order value (excluding taxes & duties) per week of delay or part thereof subject to a maximum of 10% of Order value (excluding taxes & duties)	LD is accepted for delay in delivery beyond agreed delivery term at 0.5% of the basic order value per week subject to maximum of 5% of the basic order value.
4.30	The total Price discount due to clause 4.1 and 4.2 above shall be limited to 10% of Order value. It is further agreed above clause(s) shall not constitute a waiver of any rights of Buyer to damages or other remedies of Buyer under this Purchase Order or otherwise for Supplier's improper performance or default in performance of any other aspect of this Purchase Order. It is agreed by the Parties that the amount corresponding to Price Discount, pursuant to Article 4.30, is genuine pre-estimate of the likely loss to Buyer as a result of Supplier's failure to complete the Work as per the Subcontract/PO Schedule and, as such, should not be considered to constitute a penalty under any applicable laws. Should the Supplier fail to give effect to the imposition of Price Discount pursuant to Article 4.30, the same shall be payable by the Supplier on demand by Buyer. In the event of such failure by the Supplier , the Buyer reserves the right to recover / set off an amount equivalent to the applicable Price Discount out of the sums payable to the Supplier or from the Guarantees or Deposits furnished by the Supplier or the Retention Money retained from the Invoices of the Supplier. Any Extra ordinary delay (beyond 10 weeks from CDD) not attributable to L&T will be considered very seriously and Supplier may be kept under Holiday list for any subsequent enquiries/ RFQ for a period of not less than three years.	Total LD shall be 5% of the basic order value.
4.40	Non-fulfilment of Utility Guarantee parameters (As Applicable): Without prejudice to any other reduction, discount or adjustment in price to which the L&T is entitled to, if the actual consumption of Power /Cooling Water/ MP Steam/LP Steam/ Fuel Gas for the Equipment/Packages exceeds and / or actual generation of MP Steam by Equipment/packages is lower than the Stipulated guaranteed figures of the respective utilities furnished by the SUPPLIER in its offer for the Equipment/Packages, L&T shall be entitled to an adjustment in the lump sum price by way of discount as follows. No benefit shall be passed on to the supplier in case the utility consumption demonstrated by the supplier during guarantee test/s is/are less than the guaranteed figures quoted in the bid. The reduction in price shall be in addition to any price reduction mentioned elsewhere in this document / PO.	Not Applicable

No.	T&C Clause	CIRCOR Confirmation
	A. Reduction in price with respect to power consumption (As Applicable) All items mentioned above shall be performance tested at Supplier works and performance demonstrated. Further, for all the other power consuming equipment /packages as defined in this section, Supplier is required to establish the performance of the equipment / packages during PGTR at site. In case the total KW consumed by the equipment / packages exceeds the stipulated guaranteed value by more than 4%, the subject equipment/Package shall be rejected. However for excess power (kW) consumed over and above the guaranteed/quoted KW to a maximum of four (4%) percent, reduction in overall prices shall be applicable for excess power consumption as per the following formula on extra Power consumption of the equipment & Packages as identified in this section. Reduction in price for Excess Power Consumption $C_{PC} = (KW_{REA} - KW_{REQ}) \times C_P \times 8000 \times D_F$ KW_{REQ} = Guaranteed Power (KW) of the Identified Equipment quoted by the Supplier KW_{REA} = Actual Power Consumption of the identified equipment/package (KW) as observed during Performance Guarantee test run (PGTR) D_F (Discount Factor)= 4.02 C_P = Cost of Electric Power INR 7.35 per kWh 8000= Number of operating hours per year. Power consumption shall be measured at motor terminal. B. Reduction in price for Steam Consumption (As Applicable) During performance tests MP & LP Steam Net Consumption (MT/Hr) by the Equipment/package exceeds the stipulated guaranteed value by more than 5%, the subject equipment/package shall be rejected. However for excess steam consumption over and above the guaranteed Net steam consumption up to a maximum of 5% (five percent), reduction shall be applicable for excess total steam consumption as per the following formula: $C_{Steam\ con} = C_{MPSteam\ con} + C_{LPSteam\ con}$ i. $C_{MPSteam\ con} = (SMEPAC - SMEPQC) \times CSMP \times 8000 \times D_F$ $SMEPAC$ = Actual MP steam Consumption of the identified equipment/packages (MT/hr) as observed during Performance Guarantee test run. $SMEPQC$ = Guaranteed MP Steam Consumption (MT/Hr) quoted by Supplier. $CSMP$ = Price of MP Steam INR 2143 per MT D_F (Discount Factor)=4.02 8000= Number of operating hours per year. $C_{LPSteam\ con} = (SLEPAC - SLEPQC) \times CSLP \times 8000 \times D_F$ $SLEPAC$ = Actual LP steam Consumption in the identified equipment/packages (MT/hr) as observed during Performance Guarantee test run. $SLEPQC$ = Guaranteed LP Steam Consumption (MT/Hr) quoted by Supplier. $CSLP$ = Price of LP Steam INR 1990 per MT D_F (Discount Factor) = 4.02 8000 = Number of operating hours per year. Steam consumption shall be measured individually for each equipment. C. Reduction in price for excess Cooling Water Consumption (As Applicable) For all the Cooling Water Consumers (as specified in the RFQ), in case during performance tests the Cooling Water consumed (M³/Hr) by the equipment/package exceeds the stipulated guaranteed value by more than 5%, the subject equipment /systems/Packages shall be rejected. However for excess Cooling Water Consumption over and above the guaranteed Cooling Water consumption up to a maximum of 5% (five percent), reduction shall be applicable for excess Cooling water consumption as per the following formula. Excess Cooling water consumption $C_{Cooling\ water} = (EC_{CWA} - C_{CWQ}) \times C_{CW} \times 8000 \times D_F$ EC_{CWQ} = Total Cooling Water Consumption (M3/Hr) of the Equipment/package as quoted by the Supplier $ECCWA$ = Actual Cooling Water Consumption(M3/Hr) of the Equipment/Package observed during Performance Guarantee test run 8000 = Number of operating hours per year. C_{CW} = Cost of Cooling water is INR 1.25 per M3. D_F (Discount Factor)=4.02 Cooling water shall be measured individually for each applicable Cooling water users.	

No.	T&C Clause	CIRCOR Confirmation
	In the event of deviation with respect to guarantees as specified above, Supplier is required to rectify the same without any commercial/technical implication. In case of Supplier is not able to rectify the same, equipment(s) not meeting the guarantees shall be rejected and required to be replaced by Supplier without any time and cost implication to L&T. D. Additional Guarantees with respect to Equipment Performance Supplier during the performance of contract is required to carry out hydraulics check and required to guarantee the hydraulics for meeting the throughput & Specification requirement as defined in the MR to meet the hydraulic guarantee of the supplied Equipment / package for both Design Case-1 & Case-2 (including SOR & EOR) as specified in MR. E. Repair / Rectification / Modification In case the shortfall in the guaranteed parameters, Supplier may carry out necessary repair / rectification / modification / replacement at his own cost and through his own agency on urgent basis and complete the repair / rectification / modification / replacement within a maximum period of three months to improve the supplied systems to prove guaranteed parameters in the final performance test. Should the systems continue to underperform, reduction in price shall be leviable after this as specified for shortfall in guaranteed parameters above. However, if the shortfall in guaranteed parameters is beyond the rejection parameters, Supplier and L&T will mutually agree on the method and possibilities of rectification and if the plant is still not capable of performing within the acceptance limits, it will be L&T's decision with respect to rejection or penalties, which shall be final and binding on the Supplier. F. Test Run Requirements Detailed procedure for PGTR shall be discussed with Supplier and finalized mutually, however for field tested equipment/Packages the individual utility consumption shall be recorded jointly for a period of 72 hours continuous run of the unit. After the completion of successful performance test, average consumption of individual utilities will be worked out. These shall be considered as actual consumption for computation of price adjustment. All instruments required, if any, to establish the performance are within Supplier's scope of work / supply at no extra cost to OWNER. In an event that unit operates in a mode / feed case different than specified above or at a throughput lower than design throughput, the utility numbers shall be prorated for reduced Capacity for the purpose of guarantee in mutual discussion with PMC/OWNER.	

No.	T&C Clause	CIRCOR Confirmation
4.50	Back-charges for Demurrages / Cancellation charges :	
	<u>ODC/OWC consignments:</u> Supplier shall inform the date for shipment (complete / partial) along with transportation drawings and packing list (Weights and overall dimensions for ODC/OWC) at least six months before the shipment date in order for L&T to finalize the shipping contract, take statutory approvals for movement of ODC consignments from supplier's works to Project site, construction of necessary by-passes, etc. Supplier shall then confirm the firm date of shipment one month in advance for L&T to book the ship / barge and the axles, as applicable and for completing all procedures for further movement to site.	Not applicable as not ODC.
	<u>For all other consignments:</u> Supplier shall confirm the firm date of shipment one month in advance for L&T to make the necessary transportation arrangements.	Not applicable as transportation is in CIRCOR scope.
	In case the cargo is not ready for shipment as confirmed, any demurrages / detention / cancellation charges, approval cost, etc shall be back-charged to the Supplier at actuals and shall be adjusted from the dispatch payment milestone.	Not applicable as transportation is in CIRCOR scope.
5.00	Delivery:	
5.10	Timely delivery is essence of this Purchase Order.	
5.20	Contractual Delivery Date (CDD) is on Ex-works duly packed and loaded onto transporters vehicle, (INCOTERMS 2020) basis. Delivery is inclusive of time required for acceptance of equipment / item and Release/ Issue of IRN after completion of Inspection activities by Inspection by Third Party Inspection Agency and/or L&T and/or PMC and/or OWNER as per L&T/OWNER approved Inspection Test Plan (ITP).	Agreed.
5.30	The date of Lorry Receipt shall be reckoned as the date on which the delivery has been effected for delivery on Ex-works basis. The date recorded as "Site Receipt Date" (by the Stores In-charge of L&T at Site) shall be reckoned as the date on which the delivery has been effected for delivery of DAP / DDP Project site basis	Agreed.
5.40	Effective Date of the Order shall be the date of the Order by L&T by way of LOI or P.O., (whichever is earlier), unless specified otherwise.	Agreed, subject to PO is being released not later than 4 weeks from LOI/LOA.
5.50	The number of despatchable units with dimensions & weight (after packing) shall be clearly specified the offer. Maximum pre-fabrication shall be carried out at Supplier / Supplier's sub-Supplier's works to minimize Site work. The extent of Site fabrication to complete the equipment shall be specified in the offer. All category of spares (i.e. Erection & commissioning, Mandatory, Critical, Capital, 2 years operation, as applicable) shall be packed separately with item wise packing list.	Agreed.
5.60	If the SUPPLIER fails to deliver the GOODS within the Contractual Delivery Date due to reasons attributable to the SUPPLIER, the L&T shall have the right to cause SUPPLIER to ship the GOODS by the most expeditious means of transportation, such as air freight, at sole risk, cost and expense of the SUPPLIER.	Agreed, subject to agreement by the supplier that the reason of the delay is completely attributable to the supplier.
5.70	Supplier shall submit draft Packing List showing all despatchable units with weight & dimensions before 12 weeks' ex-works readiness. Supplier to provide packing list for mandatory spares separately. Box for spares should be separate during dispatch. Any partial supplies of spares is not acceptable	Agreed.
5.80	List of inputs required by Supplier to meet the Contractual obligations along with the "Time Schedule" of requirement will be submitted during the Kick-off meeting / within 2 weeks of receipt of LOI/PO whichever is earlier.	Agreed.
5.90	Detailed "Delivery Schedule" (Bar Chart) showing Engineering, Procurement, Manufacture, Shop Inspection, Transportation, etc. will be furnished within 2 weeks from the LOI/PO	Agreed, subject to PO is being released not later than 4 weeks from LOI/LOA.
5.100	Supplier shall confirm after the last despatch that all materials required for completeness of the items as per PR have been despatched.	Agreed.
5.11	For all items expect spares - L&T shall, within 16 Weeks of receipt of material at Site, revert regarding any short Supplies w.r.t. Packing List. For Spares:- L&T shall revert regarding any short supplies during handover of spares to customer. <u>Supplier shall immediately supply the short supplied items on DDP Site basis with no Price and time implication to L&T.</u>	Intimation regarding any short supply needs to be communicated within 4 weeks from receipt of material at site.
5.12	Supplier shall depute their representative to Site, at least once on a mutually agreed date for reconciliation of all materials dispatched, in order to ensure completeness of the items as per scope mentioned in PR (Code-1 approved by PMC/OWNER). The cost of all such visits by Supplier is included in their price. No separate per diem rate shall be paid to Supplier for such visit.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
5.13	Delay and Recovery Plan (Schedule Recovery)	
5.13.1	If, in Buyer's sole opinion, the Supplier's performance or progress is inadequate, deficient, or unsatisfactory or the Supplier may fail to achieve any of the milestone dates, Buyer may (but is not obliged to) notify the Supplier of such inadequacy, deficiency, or inability ("Delay Notice").	Agreed.
5.13.2	Upon receipt of a Delay Notice, the Supplier must, within seven (7) days or any other period stated in the Delay Notice, submit to Buyer a recovery plan.	Agreed.
5.13.3	The Supplier shall prepare and submit a detailed plan ("Recovery Plan") at its own risk and cost setting out the various measures that it intends to implement in order to recover the delay to the works and milestone dates and eliminate the causes and events which are resulting in such delay, inadequacy, deficiency, or failure, and shall include, but not limited to the following: (a) mobilization of additional equipment and personnel; (b) working overtime, increasing the number of shifts or work days or both; (c) expediting delivery of materials by airlifting; (d) enhancing the manufacturing and fabrication capacity; (e) substitution of higher grade materials or specifications (with prior written authorization of Buyer).	Agreed.
5.13.4	The Recovery Plan shall be subject to Buyer's approval and acceptance and shall be implemented at the Supplier's sole cost.	Agreed.
5.13.5	In the event the Supplier fails to submit the Recovery Plan or the submitted Recovery Plan is not acceptable to the Buyer, then the Supplier must, at his sole risk, cost, and expense, implement any remedial measures Buyer may require or instruct.	Agreed.
5.13.6	If, in L&T's sole opinion, Supplier's performance or progress is inadequate, deficient, and/or unsatisfactory and the Supplier may fail to achieve any or all of the milestone dates, L&T may depute its own resources at Supplier's Group for expediting purpose. It shall be carried out in consultation with Supplier. Any and all cost incurred by L&T in deputing it's resources shall be to Supplier's account and shall be paid by Supplier to L&T.	After mutual agreement between both the parties if Supplier's performance or progress is inadequate, deficient, and/or unsatisfactory and the Supplier may fail to achieve any or all of the milestone dates, L&T may depute its own resources at Supplier's Group for expediting purpose. It shall be carried out in consultation with Supplier. Any and all cost incurred by L&T in deputing it's resources shall be to Supplier's account and shall be paid by Supplier to L&T.
5.13.7	At any time, if the Buyer is of the opinion that the Recovery Plan or any other remedial measures as instructed by the Buyer are not being implemented or not being followed by the Supplier to the satisfaction of the Buyer, the Buyer is free to take all such actions or avail all such rights and remedies available to him under the Purchase Order including but not limited to Step-in, or getting the Work done by third parties Termination for default as per Clause 11.10.2 etc.	Agreed
6.00	Performance Bank Guarantee:	
6.10	Performance Bank Guarantee for 10% of the basic Order value (excluding taxes & duties) (@ no cost to BUYER) securing the performance of the Supplier as per Order terms & conditions, including but not limited to execution of the Order, delivery, design, material, quality, workmanship, commissioning and satisfactory performance of the Equipment / Package throughout the Defect Liability period (DLP) shall be submitted.	Noted.
6.20	The PBG shall be furnished within one month from the date of LOI or PO, whichever is earlier	PBG can be issued only after dispatch of materials.
6.30	Performance Bank Guarantee (PBG) shall be furnished as per L&T's format (Annexure B) and from the Banks approved by L&T. The Performance Bank Guarantee shall be valid until the completion of Defect liability period i.e. 31-Dec-2028 or 24 months from the date of delivery whichever is later with a further claim period of 3 Months as specified in the Format of PBG	Agreed, subject to the delay is solely attributing to supplier.
6.40	The Bank Guarantee(s) shall be extended by supplier at no cost implication to L&T by further period(s) as the L&T may require until performance of all the Supplier's obligations under the PO/Contract.	

No.	T&C Clause	CIRCOR Confirmation
6.50	ABGs and PBGs shall be issued as per L&Ts format as attached at Annexure-B and Annexure-C. BGs shall be issued from Indian Branch of one of the Banks listed in L&Ts format. BGs issued by other Banks will not be accepted. Bidders are requested to ensure that the BG confirmation is transmitted by their bank only in IFN 760 format through SFMS (Structured Financial Messaging system) to facilitate faster processing of the BG's. The procedure / requirements is enlisted herein below for reference: a) While making the BG application, Bidder to inform their Bank to send BG confirmation only through IFN 760 (Guarantee) vide SFMS to our Bank. b) IFN 760 (Guarantee) should contain the following details: BG No, BG Amount, BG Type, date of issuance, Expiry date, Name of BG Applicant & Beneficiary's name and address. c) Bidder should obtain a copy of IFN 760 sent through SFMS by their bank and attach the same with the Original hardcopy of BG and submit the same to L&T. The details of our bank to which SFMS confirmation has to be sent is as under: Name of Beneficiary – Larsen & Toubro Limited Bank Name: ICICI BANK LIMITED IFSC Code: ICIC0000009 Bank Address: 110, First Floor, Nungambakkam High Road, Nungambakkam, Chennai – 600034	Not Applicable
7.00	Insurance: Insurance of the items up to loading on Transporter Vehicle as per delivery basis shall be in Supplier's scope. <u>Transit Insurance of items from dispatch point up to Project Site shall be in L&T's scope.</u>	Agreed.
7.10	L&T / OWNER shall arrange the Marine / Transit Insurance for the Project Supplies. Supplier shall ensure that the Policy number is clearly written on the dispatch documents. Supplier shall intimate L&T following details of the dispatch within 24 hours of dispatch. - Purchase Order No. and Date - Invoice No. & Date - LR No. and Date / BL No. and Date/AWB No. and Date - Name of Transporter / Freight Forwarder - Place of Works - Value of shipment - Number of Packaged cases and gross weight & Net weight, Gross Volume <u>Shipment from L&T to</u>	Agreed.
	In case the Supplier fails to communicate the above information, Supplier will be liable to replace/repair the goods at his cost, in case of damage during transit.	Agreed.
	Supplier shall provide all the support in case of a claim under the insurance.	Agreed.
	Supplier shall provide all the support in case of a claim under the insurance.	Agreed.
8.00	Shipment Basis:	
8.10	Shipment basis shall be on Ex-works duly packed and loaded onto transporters vehicle, (INCOTERMS 2020) basis.	Agreed.
8.20	Transfer of Property shall take place on exchange of negotiable documents required at the time of dispatch	Agreed.
9.00	General Conditions of Purchase Order	
9.10	Conformance to Engineering Specifications / "Supplier-List" / PTR:	
9.10.1	All Materials / Items / equipment shall be supplied strictly as per MR, except for mutually agreed deviations duly recorded & accepted by L&T prior to placement of Purchase Order. All other deviations, clarifications etc. stand withdrawn.	Agreed.
9.10.2	All Materials / sub components shall be procured only from Suppliers as per OWNER / PMC "Project Specific Supplier list" and sub-Supplier list agreed pre-order, if any.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
9.10.3	Any Material Sourced from a Supplier Outside OWNER/PMC issued "Project Specific Approved Supplier-List" and sub-Supplier List agreed pre-order, if any, shall stand rejected without any commercial implication to L&T. Suppliers on Owner / PMC "Holiday List" shall not be considered for ordering. Names of such Suppliers who are put on Owner / PMC "Holiday List" subsequent to Supplier approval at bidding stage shall be informed to the Supplier at any stage of the project. Supplier shall comply with this requirement without any time and cost implication to the L&T / PMC / Owner. For items not covered in the OWNER / PMC issued "Project Specific Approved Supplier list", the proposed Supplier list shall be approved by Purchaser / Owner / PMC prior to placement of order by Supplier. Supplier shall submit following with necessary documents for such identified items for Purchaser/ Owner / PMC review / approval a) Complete List of such items / components along with proposed Suppliers b) PTR of sub Suppliers for supplies made to OWNER in last 5 years. PTR to include details as item, MOC, project, End user name, year of supply and capacity/ dimensions, etc. c) Any other data / document as requested by Purchase / Owner / PMC. Above data to be submitted by Supplier within 2 weeks from the date of L&T LOI / PO (whichever is earlier). Non-acceptance of any proposed Supplier due to any reasons whatsoever shall not be a cause of schedule and cost implication. Purchaser/ Owner / PMC shall be the final authority for the approval of proposed Suppliers. PMC / Owner / L&T decision shall be accepted by Supplier without any time and cost implication to L&T.	Agreed.
9.10.4	Proven Track Record (PTR) / Supplier Qualification Criteria (VQC) approval (as applicable) from OWNER shall be the responsibility of the Supplier without any Price and / or Delivery implications to L&T. L&T shall submit the PTR/VQC to OWNER. Supplier's competent representative shall attend across the-table discussions with OWNER for approval. In case of non-acceptance by OWNER, L&T reserves the right to cancel the Order without any Price implications to L&T.	Agreed.
9.10.5	Valid Enlistment of the applicable items as per this order with OWNER / PMC shall be the responsibility of the Supplier. In case of non-availability of valid enlistment from OWNER/PMC, L&T reserves the right to cancel the Order without any Price implications to L&T.	Agreed.
9.20	Drawings/Documents:	
9.20.1	Supplier shall submit all drawings/documents for approval in soft copy through L&T's POMS as per the time schedule specified in Supplier Document Requirement (SDR) of MR. Draft DCI in required format to be submitted at least 5 days in advance from KOM date. In case KOM is not applicable, Draft DCI in required format to be submitted within 10 days from LOI/PO whichever is earlier. The SDR list shall be populated in the SDR format provided as per RFQ. L&T / PMC / OWNER will approve / comment on the critical drawings/documents within Three weeks from the receipt. For the remaining drawings/ documents, the duration for submission and review / approval shall be mutually agreed during the KO meeting. Any Change in technical specification (for supplier designed equipment / package) shall not be permitted without approval of OWNER / PMC / Licensor / L&T.	Agreed.
9.20.2	Supplier shall get the designs / drawings reviewed & approved by L&T/ PMC / OWNER. The comments shall be incorporated and implemented without any Price and / or Delivery implications to L&T. Supplier shall be responsible for any delay in approval (for reasons attributable to Supplier), and shall not be eligible for any Delivery extension on this account.	Agreed.
9.20.3	Supplier shall ensure re-submission of drawings/ documents after incorporation of L&T's / PMC / OWNER's comments within 10 days from receipt of comments (for first re-submission) and within 7 days for subsequent re-submissions. Supplier shall also ensure that the above documents/ drawings are complete in all respects. In case presence of Supplier is necessary for approval of drawings / documents / clarifications by L&T/ PMC / OWNER, Supplier shall deploy competent personnel and facilitate early approvals/clarifications as required by L&T/ PMC / OWNER at no extra Delivery and / or Price implications to L&T.	Agreed.
	Supplier shall submit soft copy of final QC dossier documents within two (2) weeks of dispatch of an equipment. QC dossier shall contain all stage and final inspection reports including Approved QA-QC procedures, Stage inspections and Final inspection reports, Material Certifications, NDT Records, Welding Procedures, Weld Maps, Manufacturing Reports, Factory Test Records (Equipment/ Piping), Statutory Certificate and Drawings etc	Agreed.
9.20.4	Final Documentation as per SDR in RFQ will be submitted in requisite number of copies (both Hard & Soft) failing which the Order will be deemed to be incomplete. Index for final documentation shall be submitted 4 weeks prior to dispatch for approval. Comments from L&T / PMC / OWNER shall be incorporated and re-submitted within 1 week from receipt of comments.	Agreed.
9.20.5	Soft copy of final documents (based on approved index) shall be submitted within 1 month from dispatch for approval. After approval of the soft copy the final documents in soft copies and hard copies as mentioned in MR/RFQ shall be couriered to the address informed by L&T.	Agreed.
9.30	Inspection, Testing and Quality Control:	
9.30.1	Supplier confirms acceptance to Indicative Inspection Test Plan (IITP) sent along with the RFQ. In case L&T does not furnish an IITP, Supplier shall furnish ITP along with the offer, as per Inspection requirements specified in the MR	Agreed.
9.30.2	After award of Order, detailed Inspection Test Plan (ITP)/ Quality Control Plan (QCP) shall be, submitted for approval to L&T within three weeks from the date of LOI/PO. L&T / PMC / OWNER shall approve / give comments within three weeks of receipt. Supplier to re submit all quality documents after incorporation of L&T's comments within 10 days. Supplier shall submit all ITP/ Procedures for approval through L&T's web based portal (POMS) as per the agreed schedule specified in Supplier Document Requirement (VDR) or MR"	Agreed.

No.	T&C Clause	CIRCOR Confirmation
	L&T/PMC/Owner reserves the right for inspection visits and Quality Audits at Supplier/ Sub Suppliers place/ works wherever felt necessary. Supplier to extend necessary support.	
9.30.3	Obtaining approval of ITP from L&T / PMC / OWNER shall be Supplier's responsibility during execution without any price and/or delivery implications to L&T. All comments shall be incorporated and implemented without any price and / or delivery implications to L&T.	Agreed.
9.30.4	Inspection of items shall be by TPIA / L&T / PMC / OWNER as per approved QCP/ITP. Co-ordination for timely inspection by TPIA shall be Supplier's responsibility. Failure to do so, shall not be a reason for delivery extension.	Agreed.
9.30.5	Failure to inspect goods or payment for goods or acceptance of goods shall in no way impair L&T's right to reject non-conforming goods or to avail of any other remedies to which L&T may be entitled. Any inspection and acceptance thereof, shall not relieve the Supplier of their contractual obligations Supplier confirms acceptance to L&T's Supplier's Shop Inspection Co-ordination Procedure. (Annexure M).	Agreed.
9.30.6	TPIA charges for re-inspection due to rejection, incomplete offering, non-readiness etc. for the reasons attributable to Supplier Group shall be to Supplier's account.	Agreed.
9.30.7	Supplier to offer sufficient sizable quantity for inspection by TPIA / L&T / PMC / Owner (to optimize inspection visit), as applicable as per approved ITP. Supplier shall raise Inspection Notification through L&Ts web based portal (POMS) in line with advance notification period as specified in Supplier Shop Inspection Procedure or as mutually agreed. In case of abortive visit of TPIA/ PMC/ OWNER due to non-readiness of item for inspection/ cancellation of Inspection Notification without sufficient time as per agreed contract, charges for the same shall be back charged to vendor For any witness and Hold activity as per approved ITP, Supplier cannot proceed for further activity without inspection clearance from LTEH and LTEH TPIA. TPIA will be available for inspection in normal working day/working hour. Any inspection call/ Visit on weekends/ holidays or extra working hours shall be based on the availability of inspector. Any delays caused due to non-availability of TPIA for aforementioned reasons cannot be claimed for delivery extension. Supplier to ensure availability of L&T/ PMC/ OWNER approved prerequisites documents for inspection at point of inspection, including sub Supplier shop. Participate in inspection activities through VC/ use of Remote Inspection Technique in lieu of physical presence at Supplier/ Sub-Supplier premises shall be at the sole discretion of L&T/ PMC / OWNER. Participation in remote inspection by L&T/ PMC/ OWNER will be irrespective of TPIAs physical presence at Supplier/ Sub- Suppliers works. <u>Supplier / Sub-Supplier should be equipped with IT infrastructure for conducting such remote inspection</u>	Agreed.
9.30.8	Vendor Inspection Documents Vendor shall upload MTC and necessary backup document duly endorsed by L&T TPIA in L&Ts web portal (POMS) against inspection notification after completion of inspection activities. Vendor located within India shall submit original hardcopies of MTCs with necessary backup (IRN wise) to L&T QA QC within 1 Week after getting Quality Clearance.	Agreed.
9.40	Billing Schedule / Billing Breakup:	
9.40.1	For package items involving multiple dispatches / multiple locations Billing Schedule / Billing Break-up indicating the place of dispatch shall be submitted by the Supplier along with LOI acceptance for approval by L&T and generation of the Purchase Order. Final billing break-up for dispatches shall be clearly mention individual dispatch items with clear unit of measure & quantities. . The generic terms like "Lot", "Set" as unit of measure shall not be acceptable. This shall be subject to L&T's approval and Final Billing break-up to be submitted at least three (3) months prior to commencement of dispatch.	Agreed.
9.50	Progress Reports:	
9.50.1	Supplier shall submit, within 2 weeks from the date of LOI/PO, bar chart/S-curve showing start and finish dates of various activities. Updated progress reports shall be submitted on weekly and monthly basis as per L&Ts format. After start of manufacturing / production progress report reflecting the updated manufacturing status to be submitted on weekly basis.	Agreed.
9.50.2	Supplier shall provide and/or upload required information and/or pictures and/or data by accessing POMS. The information and/or pictures and/or data (like planned/forecast/actual dates, manufacturing/inspection stages, unpriced Purchase orders issued by Supplier, certain information for logistics arrangement, so on and so forth) to be provided by filling pre-defined and/or mutually agreed MS Excel based templates/formats for facilitating Monitoring, Progress calculation & monitoring, related report generation from POMS for information.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
	L&T / PMC / OWNER's representative reserves the right to visit Supplier's and/or Supplier Group's premises and/or facilities and/or works to review, assess physical progress, expedite etc. Supplier shall ensure and provide free access during normal working hours (and extended working hours, if required) at Supplier's and/or Supplier Group's premises and/or facilities and/or works. Such expediting shall, in no way, relieve the Supplier of their delivery obligations under the terms of the Order. Supplier shall also provide unpriced copies of Purchase Orders, for major parts & components placed by Supplier and/or Supplier Group to support progress calculated & reported as part of progress report. Cut-off date and submission date for monthly and/or weekly reports shall be defined in the Kickoff meeting. Supplier shall convey all applicable Purchase Order requirements to sub-Suppliers. All sub-orders shall show the following statement: "THIS ORDER MAY BE SUBJECT TO INSPECTION, EXPEDITING AND/OR AUDITING BY BUYER AND/OR OWNER WHOSE REPRESENTATIVES SHALL BE ENTITLED TO ENTER THE ADMINISTRATIVE AND PRODUCTION FACILITIES OF SUB-SUPPLIER'S PREMISES of any tier ". Upon request copies of sub-order(s) shall be submitted to L&T. Copies may be unpriced but should clearly show the description of material or equipment involved, agreed delivery date(s) and evidence of acceptance of the sub-order by the sub-Supplier.	
9.50.3	Kickoff Meeting shall be carried out at L&T's offices. Supplier's Engineering, Quality & Project Manager and sales personal involved in bidding stage shall be present (as may be required by L&T/OWNER). Supplier to ensure presence of key representatives of their Critical / KEY sub-Suppliers as may be required for items falling on critical path for order execution.	Agreed.
9.50.4	Detailed organization chart with contact details (E-mail, Hand-phone numbers & direct numbers) of all key personnel viz. Project Manager, Engineering Manager, Quality Manager & Production / Manufacturing manager and Service Team for Site activities to be submitted by Supplier in KOM.	Agreed.
9.50.5	Supplier shall provide CV of the Project manager who shall be responsible for execution of the order. L&T reserve the right for interaction with the proposed project manager and will reserve the right to accept or reject the CV. In case of rejection, Supplier shall propose an alternate candidate for L&T approval.	Agreed.
9.50.6	All Review meetings (Design, document review and approval, progress review, etc.) shall be held at L&T / PMC / OWNER's office as may be required by L&T/ OWNER. Supplier and their representative including sub-Suppliers shall attend all such meetings without any cost implication to L&T / PMC / Owner.	Agreed.
9.50.7	Project sponsors (members of management that sponsor the Project) from "L&T" and "Supplier" shall attend the meetings once in two months as minimum or frequency as felt necessary.	Agreed.
9.50.8	Progress measurement weightages & Milestones shall be discussed with supplier during KOM and accordingly vendor to submit the progress reports	Agreed.
9.60	Dispatch clearance: Supplier shall submit copy of IRN issued by TPIA, Inspection reports and other QA documents duly reviewed / approved by TPIA & any other inspection agencies as per approved ITP for dispatch clearance by L&T. Goods / individual items shall be dispatched only after issue of Code-1 approval (approved with no comments) of PR and drawings/ documents by L&T / PMC / OWNER	
9.60.1	Items shall not be dispatched by Supplier without L&T's written dispatch clearance.	Agreed.
9.70	Billing Instructions:	
9.70.1	The Billing Instructions (including Formats of Invoice, Packing List, and LR) shall be forwarded to the Supplier along with the Purchase Order or along with dispatch instructions. Supplier shall prepare specimen of these documents TWO weeks prior to dispatch and submit the same to L&T for approval. If the Billing Instructions are not strictly followed, L&T shall withhold the amount payable till such time the corrected documents are submitted.	Agreed.
9.70.2	Advance Dispatch Documents: Supplier shall forward to L&T copies of the following documents within 48 hours of dispatch by fax/e-mail: -- - Commercial Invoices - GST Invoice - Lorry Receipt - Packing List duly approved by Supplier. - Inspection Release Note issued by TPIA (PMC Endorsed – as applicable) - Manufacturer's Certificate of Inspection for Shipment issued by Supplier, Stating that all terms and conditions of the Purchase Order / Contract until Ex works shipment have been fully complied Supplier shall attach / send following documents along with consignment :GST invoice (Transporter copy) 1) E-way / Road permit bill where necessary 2) Packing list with proper quantity and HSN codes E-way Bill shall be in Supplier's Scope.	Agreed.
9.80	Transportation:	
9.80.1	If transportation is in Supplier's scope, items shall be dispatched on "Door-Delivery and Freight Pre-paid basis". Goods shall be sent on full truckload basis only. Prior permission of L&T shall be taken for dispatch on Part-Load basis. Retail booking or Transshipment is not permitted. Under no circumstances, consignment shall be dispatched on 'delivery against consignee copy. Items should not be dispatched in Shipper's / Supplier's own container.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
	Currently Entry Permits are not required. However if required, L&T shall provide the required Entry Permits for supplies (meant for permanent incorporation into the Project) made to site. Supplier shall dispatch goods only after receipt of the same. Supplier/ transporter shall mention the Form No. on Invoice and LR. The Supplier shall furnish reconciliation of Road Permits issued to them along with final Bill. Supplier shall issue necessary way bill / e-way bill as and when goods are dispatched from Suppliers works / site.	
9.90	Damage to Items in Transit:	
9.90.1	In case the consignment has met with an accident in transit and /or the item(s) are received by L&T in damaged condition at the Site, it shall notify the same to the Supplier and Supplier shall send competent representative immediately to Site / accident Site as required for assessing the damage so that the damaged components / parts can be re-Ordered. Any other documents Support as required by Insurance company / L&T/ OWNER will be furnished by Supplier. Supplier shall arrange to obtain the 'Damage Certificate' for the value declared by Buyer and all other documents such as first information report (FIR) etc. from the transporter for claiming / settlement of insurance claims.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
10.00	Packing:	
10.10	Supplier shall adhere to Approved Packing & Marking Instructions. (Annexure-D) The Supplier wherever applicable, shall, after proper painting, pack and crate all materials for shipment in a manner suitable for export to a tropical, humid climate in accordance with internationally accepted export practices and in such a manner so as to protect them from damage and deterioration in transit by road, rail and/or sea and during storage at the Site till the time of erection. Without prejudice to any other liabilities or obligations of the Supplier, the Supplier shall be responsible for all damage(s) to the materials due to improper packing. Depending on their type, goods are to be protected against both mechanical damage (shocks, rupture, breakage, loss) and corrosion (moisture, rain, salty atmosphere, sand, etc.). All the items for this Project shall be shrink wrapped. All packing materials will be considered non-returnable. Should L&T, OWNER or any of its representatives incur any additional costs in rectification works due to a failure of the Goods, rectification work of damage(s) due to improper packaging or documentation to conform to the requirements of the Purchase Order, L&T shall back charge such costs to Supplier and shall deduct such amounts from unpaid amounts owed to Supplier.	Agreed.
10.20	Packing lists shall be reviewed and endorsed by supplier prior to the dispatch. All short Supplies shall be listed separately & communicated to L&T by supplier immediately upon dispatch along with the dispatch plan of the short supplied items.	Agreed.
10.30	Purchase Order No. and P.O. Sr. No. shall be mentioned in all the commercial documents like Packing lists, Invoices etc. Failure to do so shall result in returning of documents to the Supplier and delay in payments.	Agreed.
10.40	Supplier shall submit lashing/ loading scheme to L&T for approval for ODC consignments, if applicable. L&T's Logistics Engineer reserves the right for inspection prior to dispatch of the consignment at the loading point.	Not applicable.
10.50	Supplier to pack the Start up / Erection & Commissioning Spares, 2 years' operating spares and Special Tools & Tackles in separate wooden boxes.	Agreed.
10.60	Supplier shall submit, "Transportation Methodology" which includes, but not limited to the following at least five months prior to Contractual Delivery Date: - Protection arrangements for Internals to avoid the damage during multimodal transportation up to Project Sites - Lifting & loading plans at Supplier works for loading the cargo on Axle - Transportation Sketch showing Overall Length, Width, Height, Weight & Three Dimensional CG Location, Saddles locations, Lifting and lashing Points - Transportation Loading diagram on Hydraulic Axles & marching plan from Fabrication Area to Jetty - Supplier to provide required Jetty to LSP free of Cost. - Special requirements for unloading at on Barge & Roll-off at Jetty and subsequent transportation to Site etc. - Design and fabrication drawing for Stools (Pedestals) for Saddles for off-loading on stools on barge as well as while off-loading the equipment at site. - Scope of Stools to be provided while off-loading the equipment at Site. - Saddle Width and Length to be discussed with the Nominated LSP and Supplier to arrange for the design accordingly to avoid any issue while off-loading the cargo on barge and further movement of the cargo to Project Site. - Transportation Drawing should also have the details for the point load for individual saddle so that LSP can arrange for the stability calculation on barge and axle. - Supplier to arrange for the individual Invoice, Packing List and Certificate of Origin (COO) for shipment equipment wise. - Supplier to arrange for the separate Invoice, Packing list for the cargo such as loose items, spares and accessories which will be shipped or transporter separately. - Supplier to arrange for the Lifting Diagram for individual packages based on which LSP shall discuss with shipping line for loading pattern for cargo loading on vessel. Supplier shall design and provide the adequate number of transportation saddles for transportation by Multimodal way - road & barge transportation.	Agreed.
	Supplier will check adequacy of lashing Supplier shall submit the above documents at least six months prior to dispatch to L&T & Insurance Company for approval. L&T or Insurance Company reserves the right to comment and the comments will be incorporated without any time & cost to L&T prior to the dispatch of the consignment. L&T reserve the right to inspect the transportation arrangement before dispatch	
11.00	Special Conditions of Purchase Order	
11.10	Deviations to the RFQ Specifications:	
11.10.1	Deviations shall not be acceptable after release of P.O.	Agreed.
11.10.2	If unavoidable, same will be processed as follows: - Supplier to submit the same to L&T along with all backups for approval (format as per OWNER/ PMC standard). - Responsibility of processing the request and acceptance thereafter by OWNER as applicable shall rest with L&T. In case the deviation is acceptable there shall be no cost and time implications (in case what is being offered is superior to the bid specifications in terms of cost).	Agreed.

No.	T&C Clause	CIRCOR Confirmation
	- If rejected by OWNER / PMC, L&T shall in no way be responsible and all terms & conditions including delivery / time of completion and / or price as per the Purchase Order shall remain unaffected and the deviations stand withdrawn by Supplier. - If the deviation is accepted by OWNER, the Cost benefit sought by OWNER, if any, will be deducted and the Order value shall be discounted accordingly. Any recoveries if due, shall be made from Supplier's subsequent invoices / pending payments if any. This deduction shall be over and above any of the other deductions / limitation of liability indicated elsewhere. Supplier shall provide all the necessary and sufficient back-ups for justifying the cost benefit applicable for the deviation passed on to L&T.	
11.10.3	Deviation(s) taken by Supplier during pre-order stage which is responded by L&T as "Subject to PMC / OWNER acceptance" shall also be processed as per above methodology.	Agreed.
11.20	Liability under the Order:	
11.20.1	Supplier's maximum aggregate liability under this Order shall not exceed the Order value. This does not include liability for "Make-Good" and "Risk Purchase" as specified elsewhere in this document. No such limit shall apply in respect of: i. Any liability pursuant to Supplier's indemnity obligations under the PO ; or ii. any loss resulting from fraud, intentional or willful misconduct or illegal or unlawful acts or omissions of Supplier, its affiliates or any sub-contractor or any supplier or any of its or their respective officers, directors, employees, servants or agents; or iii. any liability to rectify, repair, restore or replace any materials and/or works or deficiencies therein in terms of the PO; iv. any liability under this PO because of which Owner is unable to take the possession of the Equipment/system/package/Plant; v. In the event of any claim or loss or damage arising out of infringement of Intellectual Property; vi. For any damage to any third party, including death or injury of any third party caused by the Supplier or any person or firm acting on behalf of the Supplier in executing the works vii. Supplier's Liability for Price Adjustment for utility consumption as PO viii. Supplier's Liability for Process Guarantees (if any) or other Guarantees as mentioned in the Technical Part/PO. ix. Any liability arises out due to perform of any work or part thereof under this PO by L&T/Owner at risk, cost and expense of Supplier.	Agreed.
11.20.2	The deductions on account/ Recovery of any of the reasons at Sr. No. 4 & 11 of this document shall be made from the pending bills of the Supplier under this Purchase Order or any other Purchase Order and/or the Bank Guarantees and/ or submitted by the Supplier. It is clarified that the Buyer reserves the right to initiate appropriate proceedings against the Supplier and for the recovery of the shortfall amounts if any.	Agreed.
11.20.3	Levy of Price discount shall not relieve the Supplier from any of their contractual obligations and shall not limit the Buyer's rights in any manner.	Agreed.
11.20.4	Neither Party shall be liable to the other for any indirect or consequential loss or damage howsoever arising, including without limitation loss of product or equipment or facility down-time or loss of profit or loss of opportunity or revenue or increased costs of finance or like items of loss or damage, without regard to the causes thereof including, without limitation to, willful misconduct or fraud, the fault, negligence, breach of statutory or contractual duty or obligation or otherwise of either Supplier or OWNER provided that (a) no liability or indemnity under this clause shall be considered as a liability to pay or to indemnify a Party for indirect or consequential loss or damage and (b) the provisions of this Clause shall not limit the Supplier's obligations to pay Price Discount under this Order. This Article shall survive termination of the Purchase Order.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
11.30	Guarantee/Warranty	
11.30.1	All materials / Equipment supplied, (irrespective of whether engineering, design data or other information has been furnished, reviewed or approved by L&T/PMC/OWNER) shall be free from faulty design, material & workmanship, and shall be of sufficient size and capacity and of proper materials so as to fulfill in all respects all operating conditions, specified in the Order.	Agreed.
11.30.2	Engineering Guarantee:	
11.30.2.1	It shall be the prime responsibility of the Supplier to carry out design, engineering and supply in accordance with the RFQ and such design shall also be in accordance with good and sound engineering practices. Supplier must ensure that all equipment and items supplied by him shall comply with the technical specifications to achieve the guarantees and warranties specified under this Purchase Order.	Agreed.
11.30.2.2	In case any errors or omissions in such design or engineering, attributable to the Supplier, which require re-engineering resulting in any new requirements for the subject item / materials / packages, the same shall be supplied and / or carried out by the Supplier within the scope of work and/or supply and within the contractual delivery date without any extra price and / or delivery implications to L&T.	Agreed.
11.30.2.3	Supplier shall check and satisfy themselves with Material Requisition, Equipment datasheets and other attachments for its accuracy, correctness and completeness to meet the agreed guarantees and in case of deficiency, Supplier shall inform L&T the modifications proposed to be carried out to overcome the deficiency and rectify the same at no extra cost.	Agreed.
	Supplier, if required, shall carry out corrective technical studies and engineering as may be required without any extra cost to the L&T. Before proceeding with the re-work, L&T/Owner/PMC approval, if applicable, shall be obtained.	Agreed.
11.30.3	Material/ Workmanship Guarantee:	
11.30.3.1	Workmanship for manufacture and construction shall be carried out in accordance with the stipulations of the MR/RFQ.	
11.30.3.2	Supplier shall guarantee that the subject Package (i.e. Equipment and/ or Materials including spare parts, chemicals, lubricants and consumables, if any) supplied under this Order shall be new and unused (not reconditioned) and of recent manufacture. , are of the best quality & workmanship, free from defects in design, material & workmanship, are suitable for the operating conditions specified in the Order and that the performance of the Equipment / Material etc. supplied shall be as per stipulated parameters.	Agreed.
11.30.3.3	Any rectification of any defective material supplied or work done shall be made good without any Price implications or entitlement of extension of time to L&T or if so considered by L&T/OWNER, shall be replaced by new equipment within the scope of work and/or supply on DDP Site basis.	Agreed.
11.30.3.4	The Supplier guarantees and accepts full, responsibility for all materials supplied / installed / erected (including all components/parts) with regard to: - Selection of material (if not specified by L&T) - Material specification and metallurgy as per Specification - Fabrication and Manufacturing workmanship - Work specification - Engagement of highly experienced, reliable and qualified sub-Suppliers/sub-contractors, engineers, workmen etc.	Agreed.
11.30.3.5	The above shall exclude damages due to normal wear and tear, application for other than the intended purpose, improper handling or inadequate maintenance by L&T/OWNER.	Agreed.
11.30.4	Performance Guarantee :	
11.30.4.1	The items designed, supplied and installed / erected shall operate to the desired capacity to produce the desired and guaranteed duty conditions.	Agreed.
11.30.4.2	This guarantee shall cover the guaranteed mechanical performance for each and every related equipment, their efficiencies and their performance as laid down in the specifications including guarantee for utilities(power / Steam / Cooling Water/ Fuel Gas) consumption/generation as indicated by the Supplier in their Offer / agreed in PO.	Agreed.
	The Supplier shall select the equipment considering the economy of utilities consumption/generation within the guaranteed figures provided and minimum effluent or pollution discharge satisfying the specifications/requirements specified in the MR.	Agreed.
11.30.4.3	Equipment / items which do not perform to the guarantees shall either be replaced or altered, repaired or replaced in parts or components (including dismantling, transportation, erection, hook-up, commissioning etc.) at no Price implication to L&T/OWNER and without entitlement of extension of delivery date to the Supplier.	Agreed.
11.30.4.4	Supplier shall carry out all activities for collecting the required data during guarantee runs to identify problems of non-performance for further analysis and modifications required to meet process performance parameters.	Agreed.
11.30.4.5	Supplier shall be solely responsible for supply of all instruments required for establishing performance of the subject item at no extra Price to L&T.	Agreed.
11.30.4.6	Supplier shall provide a list of laboratory test procedures and frequencies thereof required for validating the performance guarantees.	Not applicable.
11.30.4.7	Performance tests shall be started when the machine/equipment is stabilized under design conditions. The equipment / machine / package shall be operated and controlled in accordance with procedures set up beforehand. The performance shall be measured on the basis of the average of data obtained during 72 hours of performance test runs under continuous operation of the machine.	Agreed.
11.30.4.8	If on any testing any material or equipment or any individual Unit of the equipment does not meet the design, rated or guaranteed performance relative thereto, the Supplier shall forthwith, within the Supplier's scope of work and at no additional Price to L&T, undertake such additional tests and/ or operations that are necessary to identify the cause of such failure in conjunction with L&T.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
11.30.4.9	If as a result of tests and / or operation it is determined that the design/ rated/guaranteed outputs/capacities have not been met because of defect(s) in any material, service or workmanship, the Supplier shall forthwith arrange to repair / replace part or whole of the item supplied with no Price implications to L&T/OWNER. The Supplier shall, for this purpose, forthwith establish a "Time Schedule" acceptable to L&T / OWNER for such replacement / rectification bearing in mind the exigencies of the Project requirements. Should the Supplier fail to establish such "Time Schedule", L&T shall establish the "Time Schedule" and the "Time Schedule" so established shall be binding on the Supplier. Should the Supplier thereafter fail to adhere to the "Time Schedule" so established for the replacement / rectification, L&T shall (but without obligation to do so), take-over in whole or part such replacement / rectification at the Risk & Cost of the Supplier. In so doing, L&T shall be entitled to identify and employ the quickest available resources of supply and / or work on its own and/or from third parties. L&T reserves the right to charge Supplier overheads and / or administrative charges for any such supply and / or work.	Agreed.
11.40	Guarantee / Defect Liability Period:	
11.40.1	Notwithstanding anything stated to the contrary elsewhere in the documents comprising the RFQ, but subject to 11.4.2 & 11.4.3) below, the Guarantee / Defect Liability Period (DLP) of the Equipment / Material / Services for design, material, workmanship and performance shall be till 31-Dec-2028 or 24 months from the date of delivery whichever is later. In case the delivery of the order is delayed beyond the original Contractual Delivery Date (CDD), the DLP shall be extended by the period equivalent to the delay.	31-Dec-2028 or 30 months from the date of delivery, whichever is earlier
11.40.2	Should any defect and / or under performance in the Equipment / Material supplied be noticed during the guarantee / defect liability period, the Supplier shall, at no Price to L&T / OWNER make the defect good at Site promptly, by way of repair / modification or replacement and re-installment of the Equipment / Material.	Agreed.
11.40.3	Any Equipment / Materials repaired / modified or replaced / re-installed during the above Defect Liability Period shall be subject to defect free guarantee / defect liability period for a further period of twelve (12) months from the date of correction, repair, rectification, replacement and/or re-installation with the intent that the provisions of other clauses under "Guarantee / Defect Liability Period" shall apply thereto in respect of all defects discovered during a period of twelve (12) months from the date of completion of the relative correction, repair, rectification, replacement and/or re-installation as the case may be, in the same manner and to the same extent as apply in respect of the original works.	Not applicable.
11.40.4	In case of replacement of goods Supplier shall not be entitled to return of the defective goods.	Not applicable.
11.40.5	In case of any claim arising out of this guarantee within guarantee / defect liability period, L&T shall notify the Supplier in writing and upon receipt of the same, immediately Supplier shall, start repair/ replace/ work for the defective goods without any Price to L&T / OWNER. The delivery schedule for the same shall be mutually discussed & agreed upon.	Agreed.
11.40.6	The Supplier shall provide at his own cost and initiative all labor, material, equipment, crafts and inputs, transportation, taxes, duties etc. whatsoever required in and relative to the correction, repair, rectification, replacement, erection, installation and / or re-performance of the relative work(s) and / or systems(s) within the guarantee / defect liability period.	Agreed.
11.40.7	Should the Supplier fail to fulfill his obligations during Guarantee / Defect liability period, all costs / charges / taxes & duties / expenses related to engineering / supply / construction / erection / testing / commissioning etc. for the said equipment / material etc. incurred by L&T/OWNER on this account shall be back-charged to the Supplier. However, this shall not relieve the Supplier from any of the contractual obligations including guarantees.	Agreed.
11.40.8	In case Supplier's guarantee period / defect liability period as per clause above expires and L&T requires an extension in the defect liability period, then Supplier shall extend the defect liability period, including extension in validity of PBG and implications thereof if any, will be discussed at that point of time.	Not applicable.
11.40.9	At the sole discretion of L&T/OWNER, Supplier shall supply New, re-manufactured, Buyer approved repaired part or assembly components required to correct the defect(s) observed during Defect Liability Period. Rejected items/defected items under this warranty shall remain property of Buyer.	Not applicable.
11.50	Technical Co-operation – Joint Responsibility :	
11.50.1	Should the supply be complementary to materials or equipment supplied by third parties, the SUPPLIER undertakes to co-operate with said other SUPPLIERS, in order to co-ordinate with them the planning, manufacture, erection and/or setting in operation of whatever is pertaining to his supply.	Agreed.
11.50.2	Any cost arising from a Supplier's wrong technical information to other Suppliers shall be at Supplier's charge.	Agreed.
11.50.3	The Supplier shall deliver to the BUYER, a copy of all pertinent documentation.	Agreed.
11.50.4	The Supplier and such other Supplier's shall be jointly liable for the good running of groups, plants, equipment, etc., supplied to BUYER. Should any inconvenience occur due to erroneous exchange of technical information, the Supplier's involved shall directly co-operate with each other, in order to eliminate such inconveniences as soon as possible according to their technical responsibilities, at no cost to the BUYER.	Agreed.
11.60	Non-Waiver: Failure by BUYER to insist upon strict performance of any terms or conditions of the Order, or failure or delay to exercise any rights or remedies provided therein or by law, or failure to properly notify Supplier in the event of breach, or the acceptance of or payment for any goods or services thereunder, or the review or failure to review designs/ documents or inspection of material by Buyer / OWNER / PMC / Buyer nominated TPIA shall not release Supplier from any of the warranties or obligations of the Order and shall not be deemed a waiver of any right of BUYER to insist upon strict performance thereof or any of its rights or remedies as to any prior or subsequent default thereunder nor shall any termination of work under the Order by BUYER operate as a waiver of any of the terms thereof.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
11.70	Non Disclosure agreement: The Buyer and the OWNER (if any) reserve the sole right to disclose and/or publicize in any manner (i.e., for instance, through press release, speeches, photographs, videos, etc.), directly or indirectly (i.e., for instance, authorizing someone to make such disclosure), any kind of data, document and/or information concerning the Supply (materials and/or services) or any part thereof. The Supplier shall not make any such disclosure and/or publicity without the prior written consent of the Buyer and the OWNER. Supplier shall limit its disclosure of any information supplied by Buyer/Company in connection with this PO only to those of its personnel, Sub-Vendors, Sub-Contractors who require it for the performance of the Work. If Company/ Buyer designates any other person who may receive such information, Buyer shall obtain from each such person a written confidentiality and non-disclosure agreement obligating them with respect to such information and to the same extent that Supplier is obligated under this PO. Except as specifically permitted by the terms of this PO or otherwise authorized in writing by Buyer/Company, the Supplier and members of Supplier Group shall not duplicate, use, or disclose any information supplied by Buyer or its agents during the course of the Work so long as, and to the extent that, the information has not become part of the public domain. To the extent Company / Buyer approves any disclosure pursuant to this clause, the Supplier shall ensure that any third party to whom the Supplier discloses such information, executes a confidentiality agreement containing provisions protecting such disclosed information from further disclosure that are no less stringent than those contained herein, and which grant both Company and Buyer the right to directly enforce such provisions. Supplier shall provide copies of such confidentiality agreements to Buyer upon its request.	Agreed.
11.80	Code of Conduct Supplier shall adhere to the L&T Code of Conduct. Supplier shall ensure that all members of Supplier's Group comply with the requirements of this clause 11.80 of the PO. Any violation of the provisions of this clause 11.80 shall constitute a material breach of this PO which, without prejudice to Buyer's right to enforce any other remedy provided by law or under this PO, shall allow Buyer to terminate this PO for default and recover damages including, but not limited to, any increased costs incurred by Buyer as a result of such breach.	Agreed.
11.90	HSE Requirements The Supplier shall abide by all the health, safety and environmental requirements in compliance with the applicable local statutory rules & regulations and L&T HSE Policy and shall ensure that all such requirements are communicated to all the personnel involved in executing this Purchase Order	Agreed.

No.	T&C Clause	CIRCOR Confirmation
11.10	Termination / Cancellation and Suspension:	
11.10.1	L&T may, at any time terminate the Order in whole or part by giving written notice thereof and without assigning any reasons. Supplier, shall on receipt of notice of termination discontinue all further activities connected with the execution of the Order and inform L&T of details of work carried out as on date, together with financial implication. In the event of such termination, L&T shall discuss with Supplier to ascertain extent of work carried out by Supplier and to reach an agreement on cost mutually acceptable to L&T and Supplier.	Agreed.
11.10.2	In case specified delivery schedule is not adhered to or the progress of manufacture of the goods is not satisfactory or not in accordance with the Delivery schedule, L&T shall inform the Supplier about the delay. The Supplier shall take necessary actions to catch-up with the agreed delivery schedule that result in Supplier meeting the agreed schedule. In the event that the Supplier takes no action or even after the Supplier taking necessary actions, as mentioned above, the anticipated delay is in excess of 4 weeks, L&T reserves the right to cancel the Purchase Order in whole or in part without liability for damages/ compensation. In such event, L&T may procure from elsewhere goods, at Supplier's Risk & Cost, which in L&T's opinion would meet the same purpose as the goods for which Purchase Order is cancelled and Supplier shall be liable for reimbursing L&T cost of such substitution. L&T reserves the right to charge Supplier overheads and / or administrative charges for any such supply and / or work.	L&T reserves the right to cancel the Purchase Order in whole or in part without liability for damages/ compensation provided the delay is solely attributed to supplier and agreed mutually.
11.10.3	In the event of rejection of non-conforming goods, Supplier shall be allowed, without any extension of delivery time, to rectify the non-conformities. Should, however, the Supplier fail to do so within the stipulated time, L&T may cancel the Order as to the non-conforming goods and retain the same right with respect to substitution as are set out in preceding paragraph and in addition recover actual expenses incurred by L&T, in installing and removing the non-conformity to be rectified at Supplier's Risk and Cost. L&T reserves the right to charge Supplier overheads and / or administrative charges for any such supply and / or work.	Agreed.
11.10.4	If the Order is suspended (By Written instructions) for whatsoever reasons and for whatever time duration by L&T, the Delivery and price shall be amended based on mutually agreed terms. Buyer shall pay all reasonable and substantiated costs incurred by the Supplier in giving effect to the Suspension, unless the suspension is:	Agreed.
	a) necessary because of Force Majeure; or b) necessary for the proper execution of the Work or any part thereof; or c) necessary because of negligence, error or other default on part of the Supplier; or d) necessary for the health or safety of Company /Contractor/Supplier Personnel, Third Parties or for the protection of the work, Company Property or environment; or e) otherwise provided for in the Purchase order.	
11.11	Insolvency and Bankruptcy	
11.11.1	The Supplier shall inform the Buyer in case of occurrence of one of the following events or if there is substantial reason for Buyer to believe that any one of the following events may occur: a) Supplier becomes insolvent, b) a petition in bankruptcy is filed with respect to or by Supplier, c) opening of an official receivership procedure with respect to Supplier, d) Supplier's request for obtaining controlled administration, e) makes an arrangement or composition with (or assignment in favour of) his creditors, f) agrees to carry out the Purchase Order under a committee of inspection of his creditors, g) goes into liquidation, h) has an execution/restraining order levied on his goods, i) important changes on the Supplier's organisational structure or Ownership which can undermine the successful conclusion of the supply, j) any authority having jurisdiction takes or institutes any action or proceedings for the dissolution or winding up of Supplier or for the suspension of his operations	Agreed.
11.11.2	Upon knowledge of any such events from the Supplier or from any other sources, the Buyer, upon notice in writing to Supplier, has the right without prejudice to the procedural consequences, to terminate totally or in part the Purchase Order as per clause no. 11.10.2 and to take all the special and/or particular measures needed to fulfil his obligations towards the Buyer. In case of insolvency no prior notice for cure is required to be issued by the Buyer before issuing such termination notice.	Agreed.
11.11.3	In case of the above events and irrespective to the fact that whether the Purchase Order is terminated by the Buyer or still in subsistence when it has reason to believe that any of such event are likely to occur, the Buyer, at his option, can forthwith enter upon the worksite and take possession of all the material, drawings and designs, work in progress pertaining to the Purchase Order whether issued by the Buyer or otherwise. The Buyer at his option may use and utilise the plant and equipment of the Supplier for completion of the works. At the option of the Buyer, the Supplier shall assign all the purchase orders, service contracts that it had executed pertaining to the Works with its sub- Suppliers in favour of the Buyer and accordingly inform the sub-Suppliers.	Agreed.
11.11.4	The Buyer shall not be liable to pay the Supplier any money under the purchase order until the costs of completion are ascertained.	Agreed.
11.12	Force Majeure	

No.	T&C Clause	CIRCOR Confirmation
11.12.1	The term "Force Majeure" shall mean any of the following acts, events, or circumstances which renders a Party unable to perform its obligations which in each instance is not reasonably foreseeable, cannot be sufficiently mitigated by reasonable efforts, and the cause of which and the inability to perform is not caused by any act or omission of the Party asserting Force Majeure: a) act of Terrorism b) riot, war, invasion, act of foreign enemies, rebellion, revolution, insurrection of military or usurped power c) Ionizing radiation or contamination, radio activity from any nuclear fuel or from any nuclear waste from the combustion of nuclear fuel, radioactive toxic explosive or other hazardous properties of any explosive assembly or nuclear component. d) epidemics (Except Covid 19 or Coronavirus Pandemic),, earthquakes, flood, fire, hurricanes, typhoons or other physical natural disaster, but excluding weather conditions regardless of severity e) freight embargoes, strikes at national or state-wide level or industrial disputes at a national or state-wide level in any country where Works are performed, and which affect an essential portion of the Works but excluding any industrial dispute which is specific to the performance of the Works or the Contract	Agreed.
11.12.2	The following shall not afford the Supplier a ground for extension of time for delivery of materials:	Not accepted.
	a. Inclement or unforeseen weather, strike or lock-out, shutdown, third party breach, delay in payment or commercial hardship shutdown or idleness or other impediment in progress or completion of the supply or work due to any reason whatsoever shall not afford the Supplier a ground for extension of time or relieve the Supplier of his/its full obligations under the contract.	
	b. Delay in the supply of any material to Supplier due to any whatsoever reasons (including Force Majeure) from Supplier's vendors/sub-suppliers / sub-contractors shall not entitle Supplier for any extension of time for completion or to any claim for costs/loss/remuneration or damages or compensation whatsoever.	
11.12.3	If either Party is rendered unable, wholly or in part, by Force Majeure to perform its obligations under this Purchase Order, it is agreed that performance of such obligations by such Party, so far as they are affected by Force Majeure, shall be excused from the inception of any such inability until it is corrected or relieved, but for no longer period. If either Party desires to invoke Force Majeure, it shall notify the other Party in writing of the act, event, or circumstance that constitutes the event of Force Majeure, the date on which such event of Force Majeure commenced, with proper substantiation (duly certified by the local Chamber of Commerce or statutory authorities, relevant to the Supplier). Such notice shall be delivered not later than three (3) days after the date on which such Party first had knowledge of the effect of such event of Force Majeure. Either party shall forthwith inform the effect of the event of Force Majeure on the affected Party's ability to perform its obligations under this Purchase Order. The Party claiming inability to perform must make all reasonable efforts to minimize the delay caused to the Works as a result of the occurrence of an event of Force Majeure and mitigate any other effects of Force Majeure.	Agreed.
11.12.4	Affected parties shall diligently mitigate or remove the effect of Force Majeure. Each party shall upon receipt of the Notice of Force Majeure under the preceding Article shall confer promptly with the affected party and agree upon, within fifteen days, a course of action to be adopted by the affected party to remove or alleviate such effect(s), and shall seek reasonable methods of resuming full performance of affected party's obligations and achieving the objectives under the Order.	Agreed.
11.12.5	In the event of any force majeure cause, either party shall not be liable for delays in performing their obligations under this Order and the delivery dates will be extended without being subject to Price Discount for delayed deliveries, as stated elsewhere.	Agreed.
11.12.6	For delays arising out of Force Majeure, Supplier shall not claim extension in completion date for a period exceeding the period of delay attributable to the causes of Force Majeure and neither Buyer nor Supplier shall be liable to pay extra costs or claim any damages or compensation in addition to or in lieu of extension in completion date. To clarify, during the Force Majeure no party shall be liable to each other for the payment of any hire charges for equipment or manpower, storage charges of material, stand-by charges of equipment etc.	Agreed.
11.12.7	The Supplier shall not be entitled for extension in completion / Delivery Date if the effect of Force Majeure does not exceed 7 days.	Agreed.
11.12.8	Immediately after the end of the event of Force Majeure, the affected Party having invoked such Force Majeure shall notify the other Party in writing that the event of Force Majeure has ended and such Party shall resume performance of its obligations under the Purchase Order.	Agreed.
11.12.9	In case of an overlapping delay that is when more than one event causes a delay and the cause of at least one of those events is attributable to Supplier, then to the extent and period that such delays are concurrent, the Supplier shall not be entitled to seek resort under the Force Majeure clause and seek an extension of time for the period of such delay which is attributable to the Supplier.	Agreed.
11.12.10	The following are specifically excluded as Force Majeure and shall not constitute a basis for claims for schedule extensions, under this Purchase Order:	

No.	T&C Clause	CIRCOR Confirmation
	- Events or circumstances involving a previous or existing condition on or before the effective date of the Purchase Order; or - Delay of Material by Supplier's Vendor, Subcontractors, Sub suppliers - Late performance by a Supplier and / or his sub-Supplier caused by a shortage of supervisors or labour, inefficiencies, or similar occurrences; or - Late delivery of equipment or materials by any sub-Supplier caused by congestion at a manufacturer's plant or elsewhere, an oversold condition of the market, inefficiencies, or similar occurrences; or - Any inability to obtain visas or renewal of visas for expatriate workers; or - Changes in market conditions - Social distancing norms/ consequences of COVID 19 pandemic/quarantine requirements or the like. - Inclement or unforeseen weather, strike or lock-out, shutdown, third party breach, delay in payment or commercial hardship, shutdown or idleness or other impediment in progress or completion of the supply or work due to any reason whatsoever shall not afford the Supplier a ground for extension of time or relieve the Supplier of his/its full obligations under the PO. - Any other clause mentioned in this order	Agreed excluding pandemic clause.
11.12.11	Should the Works be delayed for more than thirty (30) consecutive days or sixty (60) days in aggregate as a result of Force Majeure, Buyer at its option may either suspend the Work affected or terminate this Purchase Order or the portion of the Work involved. The provisions governing Cancellation shall apply. However, L&T shall not be liable to pay any Cancellation charges over and above the charges as admissible under Clause of Cancellation.	Agreed.
11.12.12	Failure by Supplier to comply with the notice requirements mentioned hereinabove shall be deemed to conclusively waive Supplier's rights to invoke Force Majeure in respect of the relevant acts, events or circumstances and to any consequent entitlements to extension of time for completion of the Work or any separable portion thereof.	Agreed.
11.12.13	Notwithstanding anything mentioned hereinabove, the Buyer shall have right to invoke Force Majeure if the Buyer is unable to utilize/use the services supplied by Supplier under the Purchase Order due to events mentioned above.	Agreed.
11.13	Infringement of Intellectual Property Rights etc. Suppliers shall warrant that the sale or use of goods supplied under the Contract, whether designed and manufactured so as to comply with L&T's specifications or otherwise, shall be free of any claim, whether rightful or otherwise of any third person, by way of infringement of any patent, copyright, trademark or industrial design or the like, and the Supplier shall hold L&T harmless and indemnify L&T at Supplier's own cost, from any and all such claims and legal proceeding. L&T makes no warranty that the production, sale or use of goods designed and manufactured so as to comply with L&T's specification will not give rise to the claim, whether rightful or otherwise, of any third person by way of infringement of any patent, copy-right trademark or industrial design or the like and in no event shall L&T be liable to Supplier for indemnification in the event of any action being brought against Supplier in connection with any such claim. If Buyer's right to use such designs, drawings, calculations, software, or Materials is enjoined or otherwise restrained or precluded as a result of a claim of infringement, Supplier shall either procure for Buyer the right to use the infringing item, replace the infringing item with a substantially compatible and functionally equivalent non-infringing item, or remove the infringing item and compensate Buyer for the costs associated with procuring and installing a suitable substitute. This Clause shall survive termination of the Order.	Agreed.
11.14	Dispute Resolution:	
11.14.1	Any dispute or difference between the Parties arising out of or in connection with this Order which cannot be resolved amicably between the Parties at the level at which it arose within fifteen days thereof shall, in the first instance, be referred to the nominees of Directors of the respective Party for resolution. If the dispute cannot be settled within a further period of thirty days such dispute shall be resolved by arbitration in accordance with The Arbitration and Conciliation Act, India 1996 and any amendments thereof, by one arbitrator to be appointed by agreement between the Parties, failing which the appointment shall be made under the said Act.	Agreed.
11.14.2	The arbitration proceedings shall be conducted in English language. The arbitration venue shall be in Mumbai, India. The Parties agree that the dispute resolution process provided herein shall be the sole remedy and the Parties shall not suspend their obligations under the Order.	Agreed.
11.14.3	This Order shall be subject to the laws in force in India.	Agreed.
11.14.4	This Clause to arbitrate shall survive termination of this Order.	Agreed.
11.14.5	Notwithstanding the existence of any dispute or arbitration in terms hereof or otherwise, unless otherwise directed in writing by L&T, the Supplier shall continue and be bound to continue and perform the works to completion in all respects according to the Order (unless the Order or works be determined by L&T or by the Supplier under the provisions of the Order), and the Supplier shall remain liable and bound in all respects under the Order.	Agreed.
11.14.6	The work under the order shall, however, continue during the Arbitration proceedings and no payment due or payable to the concerned party shall be withheld (except to the extent disputed) on account of initiation, commencement or pendency of such proceedings.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
11.14.7	Local rules and regulations: Supplier shall ensure that the Goods within the scope of supply for this Purchase Order are in compliance with local codes and regulations. Supplier shall ensure that local rules and regulations and procedures have been complied with in obtaining certificates if and when required. Costs related to compliance with aforementioned are included in the -Order Value. Supplier has to take note of steel policy, Value addition norms, Subcontractor and import restrictions as applicable to order while quoting and execution of the project.	Agreed.
11.14.8	Harmonized Tariff Codes: Supplier shall provide Buyer with the applicable Harmonized Tariff (HSN) Codes per line item for this Purchase Order / Packing List. Failure to do so may result in back charges to the Supplier.	Agreed.
11.15	Assignment and Sub-Contracting:	
11.15.1	Supplier shall not, except with the prior written consent of L&T, transfer or assign their obligations and interest in supply of materials, or sub-contract whole or part of the work under the Purchase Order in any manner whatsoever.	Agreed.
11.15.2	Grant of any such consent or approval shall not however absolve the Supplier in any manner from all or any of their obligations, duties and responsibilities, including delivery of materials under this Order within the stipulated Delivery schedule.	Agreed.
11.15.3	Moreover, delay, if any, by L&T in approving any Sub-Contractor(s) proposed by the Supplier, or, rejection if any, by the L&T of any Sub-Contractor(s) proposed by the Supplier, shall not entitle the Supplier for any claims for additional compensation or any claim for an extension in the Contractual Date of delivery. Supplier shall ensure that competent and resourceful agencies with proven track record should be proposed for the work to be sub-contracted.	Agreed.
11.15.4	Delay in the supply of any material to Supplier by Supplier's vendors, sub-suppliers or sub-contractors due to whatsoever reasons, shall not entitle Supplier for any extension of time for completion or to any claim for costs, loss, remuneration or damages or compensation.	Agreed.
11.16	Modifications and Amendments:	
11.16.1	L&T shall have the right to modify or amend this Purchase Order, by way of a written communication to the Supplier, subject to an adjustment in the price and/or delivery date. Adjustments in price and/or delivery date shall be mutually agreed between L&T and the Supplier.	Agreed.
	In case price and/or delivery date adjustment cannot be settled immediately, upon written instruction from Purchaser, Supplier will proceed with the modification and price and/or delivery date adjustment shall be settled by mutual agreement before order close-out. Any delay in executing the modification after receipt of written instructions shall be to Supplier's account.	
11.16.2	Supplier shall forthwith inform regarding any claim for "Change-Order" prior to commencing with the same. If there is any alteration or change in the scope of work or extra item and the Supplier or L&T considers that the change requires revision in the Order price and / or delivery, Supplier shall issue / submit the change order proposal (COP) within 7 days of such instructions issued to Supplier. Supplier shall submit the following minimum information and documents in support thereof - (i) a list of affected deliverables (ii) a breakdown of man-hours by deliverable and job title/equipment item (iii) a list of Materials and quantities with associated cost for each individual item or - cost per unit quantity as applicable (iv) copies of all related Sub-contractor quotations for the Work / Materials / equipment as stated in the proposed Change together with any discount or rebate Supplier would receive from Sub-contractor(s), directly or indirectly, for those Work / Materials /equipment (v) a network schedule fragment showing how the proposed Change Order Work would be incorporated into the Work Schedule and any resulting impact on CDD (vi) any other information / documents / clarifications as may be required by the Buyer e) In addition to describing the Change, a Change Order shall include: (i) any adjustment in the CDD / milestone dates resulting from the Change; and (ii) the lump sum price of or other basis (as determined by Buyer) for determining any increase in the PO Price either due to Vendor or credit due to Contractor as a result of such Change, if any as mutually discussed and agreed between the Parties. "Change-Orders" informed later or at the time of delivery shall not be entertained. If Supplier proceeds with the additional Work involved in a Change without first obtaining a Change Order signed by both Parties, the Supplier shall not be entitled to any additional compensation for the Work so performed or to any adjustment of the CDD and/or milestone dates as a result of the said Change. The Supplier specifically waives any and all rights to claim from Buyer for such costs or additional time to perform the Work as a result of compliance by Supplier with any such oral changes.	Agreed.
11.16.3	The Supplier shall carry out such amendments to the Purchase Order and shall be bound by the same terms and conditions incorporated in the original Purchase Order.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
11.16.4	Nothing shall constitute a change order to the extent that additional work / efforts / resource deployment etc. are carried out / required only: a) to ensure that the Work is carried out and completed as per the delivery schedule b) As a consequence of Supplier's failure to comply with any of its obligations under the Purchase Order. Provided always that the provisions with respect to a Change Order or a Change Order Proposal shall not be applicable if: a) the change of supply or services or work is necessary at any time in order for the Supplier to comply with the requirements of the Contract or the Supplier's responsibilities under the Contract, b) the additional Work/ services and/or materials are required to make good the Supplier's Warranties and/or Guarantees as set forth in the Contract. c) the change relates to the repair, rectification or re-performance of any of the defective Work(s) or materials within the Supplier's scope or supply or services or work.	Agreed.
11.16.5	Any extension of time, if granted by L&T, shall be the sole remedy of the Supplier for any cause or event of delay and the Supplier shall not be entitled in addition to or in lieu of such extension, to claim any damages or compensation whether under the law governing contracts or quasi-contracts or any other relationship, and the Supplier hereby waives and disclaims any and all contrary rights.	Agreed.
11.17	Right of the Buyer to set-off: L&T shall be entitled to recover from the Supplier any sum(s) due to L&T on account of any damage(s) or otherwise, whether in respect of Supplies under this Purchase Order or under any of their previous Purchase Orders / Contracts, by deducting such sums from the amount due by them to the Supplier in respect of Supplies made under this Purchase Order or any of their prior or subsequent Purchase Orders / Contracts and/or Bank Guarantees.	Agreed.
11.18	Confidentiality: All drawings, specifications and details furnished by L&T to the Supplier shall be kept confidential. The technical information imparted and supplied to the Supplier by L&T shall at all times remain the absolute property of L&T. Supplier shall preserve and cause its employees to preserve the secrecy of any confidential information. Supplier shall read and un-conditionally accept the terms and conditions of Non-Disclosure Agreement (NDA) as per Annexure E and Countersign (with stamp). Supplier shall not use or cause to be used any confidential information for any purpose other than the performance of the Work, and as such Supplier shall not a) reproduce, copy or use any confidential information, or b) disclose to, place at the disposal of, or use on behalf of any third party, or enable any third party to peruse, copy or use any confidential information unless with prior written consent of L&T. This Clause shall survive termination of the Order.	Agreed.
11.19	Post Warranty Comprehensive Annual Maintenance Contract (PWCAMC) Supplier to submit the quote and confirm acceptance to Terms and Conditions for Post Warranty Comprehensive Annual Maintenance Contract (PWCAMC) enclosed as Annexure H of this document (will be provided during Project phase). For Proposal, Supplier to refer RFQ for respective PWCAMC items. The AMC Charges shall be inclusive of all the charges for Transportation, Lodging, Boarding, all insurances including third party insurance, all Taxes /Duties / Cess /Levies / Fees and all other incidental charges etc. but excluding GST which shall be payable extra at actual on Submission of Invoice as per GST Invoicing rules. Purchase/OWNER will not have any liability, whatsoever, over and above the quoted prices of PWCAMC.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
	Other Terms & Conditions of PWCAMC: Payment Term: The payment terms for the AMC shall be on quarterly basis after completion of each quarter against submission of Bills duly certified by the Engineer-in-charge. Payment will be made through Wire transfer within 60 days from certified invoice. Validity: The order for PWCAMC will be placed before the completion of Project Defect Liability Period PWCAMC Prices offered must be valid for ordering till one year after completion of Project Defect Liability Period The price of AMC services will remain firm & fixed for the complete execution of AMC contract. PBG: Within 10 Days of placement of Order for PWCAMC, SUPPLIER shall submit PBG for 5% of the annual PWCAMC amount which shall be valid for the entire AMC period Logistic Support: Supplier to furnish Logistic support certificate as per Annexure J. The logistic support certificate shall be signed by corporate level person of the Supplier. OWNER may decide to enter into separate contract with AMC service provider / Supplier at its discretion on the same prices and terms & conditions.	
11.20	NA	
11.21	NA	
11.22	Supplier's Sales / Commercial Conditions, if any, and/or any other terms & conditions indicated in their offer shall not be applicable and stand withdrawn in lieu of "Terms & Conditions" agreed and recorded between "Supplier" and "L&T" before placement of Order. Deviations whatsoever after placement of LOI / P.O. (whichever is earlier) are not acceptable & valid. <i>In case of any contradictions in the Terms & Conditions, the more stringent shall be applicable.</i>	Agreed.
11.23	NA	
11.24	STEP-IN RIGHTS:	Agreed.
11.24.1	If, in the opinion of Buyer, the Supplier's financial affairs become unsound or he suspends payment of his commercial debts or if there are reasons for Buyer to believe that Supplier may become insolvent or it has committed the breach of the terms of Purchase Order or there is any delay in performance of the Purchase order, then Buyer may without prejudice to any other rights it may have under the Purchase Order, have the right to enter upon the worksite and without thereby terminating the Purchase Order, or releasing the Supplier of his obligations or liability under the Purchase Order take over the balance Works and complete the same by using Supplier resources including but not limited to yard, equipment, personnel, workmen and/or other sub-Supplier, as Buyer in its absolute discretion may think proper and at the sole risk and cost of Supplier.	Agreed.
11.24.2	If Buyer in exercising any of the rights conferred by this clause incurs any expenditure in excess of the Purchase Order Value, then such excess amount shall be paid by the Supplier/ Sub-Supplier to Buyer, and such amount shall be considered as debt due to the Buyer.	Agreed.
11.24.3	Notwithstanding any provision to the contrary the Buyer while exercising its rights under this clause shall not be liable to pay to the Supplier any money until the cost of completion of the works and all other expenses incurred by Buyer have been ascertained. Supplier shall provide Indemnity Bond as per Annexure N attached with this Terms & Conditions.	Agreed.
11.24.4	The Supplier guarantees that such Buyer's right of taking over of the works will be accepted by its sub-Supplier and will be included in the respective sub-purchase orders.	Agreed.
11.24.5	In case the Buyer decides to make use of its aforesaid right and if so requested by him, the Supplier shall facilitate and assist Buyer. Notwithstanding Buyer's use of such rights, it shall also have a right to pay any sub-Supplier directly to prevent from suspending or terminating performances. The amount so paid will subsequently be set-off from the Purchase Order Value.	
11.25	FREE ISSUE MATERIAL	Agreed.
11.25.1	The Buyer can supply, free of charge certain material (hereinafter referred as the Free-Issued Material) (if any) in accordance with the details stated Purchase Order. The Buyer can provide such Free Issued Material either at the Supplier's Site / Yard / Warehouse or any other location stated in the Purchase Order. Supplier shall be liable for all demurrage or other charges in connection with any delay in unloading, hauling, lifting or handling of Free-Issued Material.	Agreed.
11.25.2	As soon as Free-Issued Material is received in Supplier's Site/Yard/Warehouse, it shall mark them with an identification number and Buyer's name, and shall keep them separate from other items. Upon receipt of such Free-Issued Material, the Supplier shall thoroughly inspect them, and shall promptly give notice within three (3) days from the receipt of the Free Issued Material to the Buyer of any shortfall, theft, pilferage, defect, or damage in these Free-Issued Material failing which the Supplier shall be deemed to have accepted and confirmed that the Free Issued Material are free from any shortfall, defects, or damage.	Agreed.
11.25.3	Once the Free-Issued Material is delivered by the Buyer, the Supplier shall be responsible for the care, custody, control, storage, handling, protection, preservation, re-packing return of surplus or unused material and use of the Free Issued Material. Supplier shall provide Indemnity Bond as per Annexure N attached with this Terms & Conditions.	Agreed.
11.25.4	Supplier must, at all times, segregate the Free Issued Materials from the other material, Construction Equipment, and any other Supplier's property. Supplier must use the Free Issued Materials solely and exclusively for incorporation in and the performance of Works.	
	The Free Issued Materials shall be always considered to be the property of Buyer and the Supplier shall ensure that it is free from any lien / charge / encumbrances.	

No.	T&C Clause	CIRCOR Confirmation
11.25.5	Supplier shall acknowledge receipt of all materials and/or equipment furnished to Supplier by Buyer for its packaging or assembly into a "complete package". It is specifically understood and agreed that title to all said items furnished by Buyer shall remain with Buyer at all times regardless of whether said items are incorporated into material or equipment being constructed hereunder and whether said items remain separable or not. Said items can at any time be reclaimed.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
11.25.6	If during the term of Purchase Order there is evidence of a lien or a charge created by the Supplier on the Free-Issued Material, the Buyer may retain an amount from Supplier sufficient to indemnify Buyer against such lien or claim. Buyer may keep said amounts until the Supplier pays and discharges the lien or charge. The Buyer at his discretion may pay and discharge the same and deduct the amount paid from amounts due to Supplier under Purchase Order.	Agreed.
11.25.7	The Supplier shall be responsible for any theft, loss or damage to any such Free-Issued Material and any deterioration to any such Free-Issued Materials as a result of the Supplier's care, custody, control, handling and storage and shall indemnify and hold harmless the Buyer from any and all claim in this respect. The Supplier shall execute and sign Indemnity Bond in the format annexed herewith. The Supplier shall not remove Free-Issued Material from the Site/Yard/Warehouse without first obtaining a written release authorizing such removal from Buyer.	Agreed.
11.25.8	Upon the completion of the Works relating to the Free Issued Materials, the Supplier must notify the Buyer of any unused or surplus Free Issued Materials and the Supplier must immediately return the surplus Free Issued Materials as instructed by the Buyer in the same condition as received by the Supplier from the Buyer, except for normal wear and tear.	Agreed.
11.25.9	FIM Insurance, material reconciliation, wastages, scrap and excess material terms and conditions mutually agreed between Buyer and Supplier and covered under Purchase Order shall govern.	Agreed.
11.26	TITLE AND RISK	Agreed.
11.26.1	Title to the raw material/ Work in Progress (WIP) / finished goods shall pass to Buyer: a) on arrival of raw material on Supplier's Site / Yard / Warehouse, or b) when payment for raw material is made by Buyer, or c) when the Supplier's has commenced manufacturing at its/ sub-Supplier's Works, or d) when raw materials, WIP, finished goods are dispatched from Supplier's Works whichever occurs earlier	Agreed.
11.26.2	Notwithstanding any transfer of title, Supplier shall at all times be solely responsible for all costs arising in respect of payment for Supplier supplied materials and for transport and importation of such raw material/ WIP / finished goods including without limitation, shipment costs, customs duties and insurance costs.	Agreed.
11.26.3	Notwithstanding the above, the risk and responsibility for any damage, or loss of the raw material/WIP/finished goods shall remain with the Supplier until the final delivery.	Agreed.
11.26.4	The Supplier shall, at its sole risk and cost, take insurance cover for the risk of loss or damage to the raw material/ WIP / finished goods for its full replacement value until delivery of the finished goods in accordance with the terms of the Purchase Order.	Agreed.
11.26.5	The raw material/ WIP / finished goods shall remain under the care and custody of the Supplier until the date of delivery or achieving completion.	Agreed.
11.26.6	Supplier shall properly store and ensure maintenance, custody and control of any and all goods in accordance with the requirements of this Purchase Order and the best professional practices and shall protect the same against weather hazards, water, humidity, dirt, fire, theft, vandalism and the like. Supplier shall use every means to prevent at all times any defect, corrosion, damage, theft or loss whatsoever to the goods in its custody.	Agreed.
11.26.7	Raw material/ WIP / finished goods pertaining to the Works shall be considered to be the property of Buyer and the Supplier shall always ensure that it is free from any lien / charge / encumbrances. If during the term of Purchase Order there is evidence of a lien or a claim for which Buyer might be or become liable or to which Buyer's property might be or become subject and which is chargeable to Supplier, its sub-Supplier, Buyer may retain an amount from Supplier sufficient to indemnify Buyer against such lien or claim. Buyer may keep said amounts until the Supplier pays and discharges the lien or claim or until Buyer is satisfied that such lien or claim is invalid.	Agreed.
11.26.8	If such lien or claim is valid in Buyer's judgement, Buyer may pay and discharge the same and deduct the amount paid from amounts due to Supplier under Purchase Order.	Agreed.
11.26.9	If a lien or claim remains unsatisfied after termination of Work or completion, Supplier shall promptly refund to Buyer all amounts paid by Buyer in discharging such lien or claim, including costs and reasonable attorney fees.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
11.26.10	Liens. The Supplier shall take full responsibility for the payment of its Sub-vendors / Sub-contractors under all subcontracts or purchase orders issued by it and for the management of all claims for payment thereunder, including payments to all its Sub-vendors / Sub-contractors of any tier, and shall take all proper precautions to prevent the imposition of any lien or charge against any property of the Buyer/ Company and to indemnify, defend and hold the Buyer / Owner harmless against any and all claims, demands, actions, suits, proceedings and judgments which may be brought or secured against the Buyer / Owner by reason of the Supplier's failure to make payments under any of the sub-contracts / purchase orders to its Sub-vendors / Sub-contractors. Breach of law Supplier shall defend, indemnify and hold harmless Buyer / Owner from and against any claims including libels, maritime liens and other liens, claims, charges, encumbrances, suits or penalties which may be filed on or levied upon Buyer / Owner or their associates, employees, due to any breach of any Applicable Law (including but not limited to Applicable Economic Sanctions Laws and Regulations, Applicable Export Control Laws and Regulations, other laws, regulations and requirements relating to personnel, health, safety and environmental matters, international maritime laws, regulations or requirements, or any laws restricting the import, use or possession or firearms, explosives, alcohol, pornography or other controlled or prohibited materials or substances) by any member of Supplier Group in connection with the performance of the Work and provision of Supplies under this PO. Criminal misappropriation or misapplication Notwithstanding any other provision of this PO, Supplier shall release and indemnify and hold harmless Owner and Buyer from any and all claims arising from or related to any criminal misappropriation or misapplication by any personnel of Supplier Group of any property, whether tangible or intangible, occurring during the course of or in connection with the performance of the Work and provision of Supplies under this PC. Supplier shall obtain similar indemnities running to Owner	Agreed.
11.27	COVID Clause: Supplier acknowledges that it is aware of change in the working conditions that has arisen due to the ongoing Coronavirus situation (COVID19). Supplier while submitting its bid has analysed all the aspects of the present situation of COVID19 including but not limited to - the quarantine requirements of personnel and vessels, social distancing measures, restrictions on transport, requirements of personal protection equipment, testing requirements, requirements of medical insurance etc. The consideration amount under this Purchase Order is inclusive of all such costs arising out of COVID19.	NA
11.27.1	Notwithstanding anything contained elsewhere in this Purchase Order, the Buyer shall not bear any additional time and costs impact arising out of COVID19. Supplier shall indemnify the Buyer from: (i) all costs arising out of COVID19 pertaining to the scope of work of the Supplier and (ii) all claims arising out of Supplier's employees, agents, subcontractors, personnel etc. including but not limited to their medical tests, hospitalisation, medical insurance, transportation, quarantine requirement, accommodation arrangements, stand-by etc. <i>/(iii) All claims arising out due to different notifications from State/Central authorities.</i>	NA
11.28	The Supplier agrees at all times to indemnify and keep indemnified L&T/Owner/PMC and its officers, servants, agents from and against any and all losses or damage caused at any time to its own property or personnel. It shall hold harmless and keep indemnified L&T/Owner/PMC against - • any third party claims (including but not limited to property loss and damage, personal accident, injury or death of any person) • any and all losses arising directly or indirectly from acts of Supplier, its affiliates, sub-suppliers or any of their respective personnel during performance of obligations under the PO. • any and all losses, arising directly or indirectly from or incurred by reason of claims or sanctions or penalties imposed by any governmental authorities or others for any actual or asserted failure of Supplier to comply with any applicable law or applicable consent(s), sanction(s) and/or authorization(s) or with any rules or regulations applicable to it • any and all losses, arising directly or indirectly from or incurred by reason of failure of Supplier to pay any taxes or any amounts in respect thereof.	Agreed.
11.29	Other Terms & Conditions	
Data Protection and Privacy:	The Supplier shall comply with all applicable data protection and privacy laws and regulations. The Supplier shall ensure that any personal data processed in connection with the Purchase Order is handled in accordance with applicable laws and regulations. The Supplier shall indemnify and hold the Buyer harmless from any claims, damages, or penalties arising from the Supplier's breach of data protection and privacy obligations.	Agreed.
Anti-Bribery and Corruption:	The Supplier shall comply with all applicable anti-bribery and anti-corruption laws including the Prevention of Corruption Act, 1988, and any other relevant legislations and regulations amended from time to time. The Supplier shall not offer, give, or receive any bribes, gifts, or other inducements in connection with the execution of this Purchase Order. Any breach of this clause shall entitle the Buyer to terminate the Purchase Order with immediate effect and claim damages.	Agreed.
Intellectual Property Rights:	The Supplier shall ensure that all goods and services supplied under this Purchase Order do not infringe any third-party intellectual property rights. The Supplier shall indemnify and hold the Buyer harmless from any claims, damages, or expenses arising from any infringement or alleged infringement of intellectual property rights.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
	All intellectual property rights, including but not limited to patents, copyrights, trademarks, and trade secrets, created or used in the performance of this contract shall remain the property of the respective party. The Supplier grants Buyer (L&T) a non-exclusive, royalty-free license to use any intellectual property necessary for the completion and operation of the project.	Agreed.
Compliance with Laws and Regulations:	The Supplier shall comply with all applicable laws, regulations, and standards in the performance of its obligations under this Purchase Order. This includes, but is not limited to, labor laws, tax/custom laws, environmental laws, and safety regulations. The Supplier shall indemnify and hold the Buyer harmless from any claims, damages, or penalties arising from the Supplier's non-compliance with applicable laws. Any breach of this clause shall be considered a material breach of the contract.	Agreed.
Ethical Conduct:	The Supplier shall conduct its business in an ethical manner and in accordance with the highest standards of integrity and professionalism. The Supplier shall ensure that its employees, agents, and subcontractors adhere to these standards in the performance of their duties under this Purchase Order. Supplier shall comply with the terms and conditions of Buyer's Supplier Code of Conduct available at Buyer's website.	Agreed.
Sustainability and Environmental Responsibility:	The Supplier shall adopt sustainable practices and minimize the environmental impact of its operations. The Supplier shall comply with all applicable environmental laws and regulations and shall take all necessary measures to prevent pollution, reduce waste, and conserve resources.	Agreed.
Audit Rights:	Notwithstanding any other provision of this PO, Contractor shall have the right to audit all of Vendor's records for the purpose of verifying compensation paid for terminated Work, including Lump Sum and Work Unit Basis compensation paid. The Supplier shall provide access to all relevant records and cooperate fully with the Buyer's audit team. Any discrepancies or non-compliance identified during the audit shall be rectified by the Supplier at its own cost.	Agreed.
Business Continuity and Disaster Recovery:	The Supplier shall have in place a business continuity and disaster recovery plan to ensure the uninterrupted supply of goods and services under this Purchase Order. The Supplier shall provide a copy of this plan to the Buyer upon request and shall notify the Buyer immediately in the event of any disruption to its operations.	Agreed.
Subcontractor Approval:	The Supplier shall not subcontract any part of its obligations under this Purchase Order without the prior written consent of the Buyer. The Supplier shall remain fully responsible for the performance of its subcontractors and shall ensure that they comply with all terms and conditions of this Purchase Order.	Agreed.
Prior Agreements:	This PO supersedes all previous contracts, correspondence, and understandings between the Parties concerning the subject matter hereof and constitutes their entire agreement and understanding in relation to the performance of Work / provision of Supplies as stated herein. The Parties shall not be bound by, or be liable for, any statement, representation, promise, inducement, or understanding of any kind or nature not recorded in this PO or its subsequent amendments and Change Orders.	Agreed.
No Release	The obligations and liabilities of the Supplier under this PO shall not be released, diminished or in any other way affected by any enquiry, direction, inspection, comment, consent, sanction, acknowledgement, confirmation, approval, or advice made or given by or on behalf of Buyer / Company, any Buyer/ Company representative or lenders, a Buyer's financiers (including any of their nominated entities), or any of their respective employees, servants, or agents; or any failure to enquire, direct, inspect, comment, consent, sanction, acknowledge, confirm, approve, or advise, whether or not such act or omission might give rise to an independent liability of any such person.	Agreed.

No.	T&C Clause	CIRCOR Confirmation
Invalidity	if any one or more of the provisions contained in this PO is held as invalid, illegal or unenforceable in any respect under any Applicable Law, the validity, legality and enforceability of the remaining provisions contained in this PO shall not in any way be affected or impaired.	Agreed.
Independent Contractors	The Parties to this PO are independent contractors and have entered into this agreement for the execution of Work / provision of Supplies on a principal-to-principal basis. Neither Supplier nor any member of the Supplier Group shall be deemed to be servants, agents, or employees of the Buyer.	Agreed.
Survival	The provisions in relation to Guarantee/Warranty, Patents and Royalties, liabilities, Indemnity, Governing law and Dispute resolution, infringement of IPR and Confidentiality as stated in this Purchase Order shall survive its expiry or early termination.	Agreed.
Payment Of Costs, Expenses And Damages	Payment of costs, expenses and damages to the Buyer shall not release the Supplier from its obligation to carry out and complete the work in accordance with the provisions of this Purchase Order and/or to discharge its obligations and responsibilities arising of this Purchase Order, nor prevent the Buyer from exercising its right to terminate the Purchase Order in accordance with Article 11.10.2 .	Agreed.
Standards	The goods and services must meet the technical specifications' standards. If no standard is mentioned, they should follow the Owner's/manufacturer's/licensor's standards.	Agreed.
Cooperation with other contractors	The Supplier must not hinder other contractors and should facilitate their work. The Supplier should cooperate and share technical information with the Buyer/Owner and other contractors. Supplier must comply with all safety measures at the Supplier's expense.	Agreed.
Repeat Order	In case of additional requirement, Supplier shall confirm to supply as per Unit Rate mentioned in the price schedule. Delivery schedule of such repeat order shall be prioritised by Supplier to enable L&T to meet project requirement.	Price validity shall be applicable till delivery of project
11.3	Terms & Conditions for Supplier's Supervision of Erection, Commissioning & Performance Tests and for Training of Owner's Personnel (as applicable)	
A	The per diem rates shall be inclusive of cost of Boarding & Lodging (at Site or during transit), travel during Mob and Demob, daily to & fro travel to Site, communication cost, overseas & local telephone calls, insurance, pocket money & all other incidental expenses. Per Diem Rates shall be inclusive of all taxes, duties, levies including the income tax / withholding taxes, which are required to be withheld by LTHE for the services rendered in India. Per Diem Rate for Year 2025 -- Physical Visits -- INR XXX Per Diem Rate for Year 2026 -- Physical Visits -- INR XXX Per Diem Rate for Year 2027 -- Physical Visits -- INR XXX Per Diem Rate for Year 2028 -- Physical Visits -- INR XXX Per Diem Rate for Year 2025 -- Remote Visits/Supervisions -- INR XXX Per Diem Rate for Year 2026 -- Remote Visits/Supervisions -- INR XXX Per Diem Rate for Year 2027 -- Remote Visits/Supervisions -- INR XXX	To be mutually agreed on applicable charges.

No.	T&C Clause	CIRCOR Confirmation
	The Supplier shall carry out the Technical Field Services using sufficient persons having adequate knowledge of the operations to be carried out (including the methods and techniques required, the hazards likely to be encountered and methods of preventing accidents) for the satisfactory and safe performance of the Technical Field Services. The Supplier shall employ (or cause to be employed) only persons who are careful and appropriately qualified, skilled and experienced in their respective trades or occupations in connection with the works. LTHE/ PMC /OWNER may remove (or cause to be removed) from the site or from involvement in the works, any employee, who in the opinion of LTHE site in-charge/PMC/Owner : - persists in any misconduct - is incompetent or negligent in the performance of his duties - fails to comply with any provisions in the P.O. deemed important to LTHE/OWNER, acting reasonably for - persists in any conduct which is prejudicial to safety, health, or the protection of the environment. If appropriate, the Supplier shall then appoint (or cause to be appointed) a suitable replacement person at its own cost.	Agreed.
Definitions / Abbreviations		
OWNER/ COMPANY	Rashtriya Chemicals & Fertilisers Limited (RCF)	
Licensor	INCRO S.A. (Spain)	
Project	1200 MTPD (DAP Basis) NPK Project and Associated Facilities	
Order	Shall mean the terms of contract between L&T and Supplier and shall include the following in the listed order of precedence: - The Purchase Order and any Amendments thereof - Letter of Award / Letter of Intent - Agreed Terms and Conditions of Purchase, and any mutually agreed amendments to the Agreed Terms and Conditions - Correspondences exchanged between the Parties i.e.(Technical Queries, Commercial Queries and responses thereof by the Contractor). - Approved ITP - MATERIAL REQUISITION / Technical Specifications / Drawings / Applicable Code & Standard. - Description of Bill of Quantity - RFQ & its Amendments / Addendums Any exclusions, clarifications etc. exchanged between the parties shall stand withdrawn unless otherwise mutually agreed, duly recorded and accepted by Buyer in writing.	
Mechanical Completion	Shall mean installation of all Supplies by Supplier thereafter completion of all activities as per approved specification and drawings/acceptable International Engineering practices, to be performed including pre-commissioning and start-up activities, completion of all punch-list items, as provided by OWNER/PMC and/or Process Licensor and/or statutory bodies like IBR, OISD, CCE, TAC, Factory Inspector, Pollution Control Authorities etc. Mechanical Completion shall be certified to Supplier after L&T receives Mechanical Completion Certificate from Owner/PMC	
Commissioning	Shall mean pressing into service of the plant(s), Equipment(s), Vessel(s), Pipeline(s), Machinery (ies) and systems & sub-systems comprised within the Unit(s) in accordance with the procedures as approved in the applicable Operating Manual and as per the requirements of OWNER/PMC and/ or Licensor after success-ful testing, pre-commissioning and trial run of the Unit(s).	
Performance Test	Shall mean the test or series of tests to be carried out by the SUPPLIER/CONTRACTOR to prove the contractual guarantees as per the requirement of OWNER/PMC/ Licensor / L&T. (PGT/ PGTR).	
Code-1	Shall mean final approval of drawings/documents incorporating all comments of the OWNER / Licensor / PMC / L&T.	
Code-2	Shall mean tentative approval of drawings / documents by OWNER / Licensor / PMC / L&T with comments to be incorporated by Supplier, with the intent that subject to incorporation of and/or rectification in accordance with such comments, construction / manufacture can proceed based on commented drawings / documents pending their re-submission and grant of Code-1.	
Code-3	Shall mean that on review of the drawings/documents by the OWNER / Licensor / PMC / L&T the same have not been found in accordance with the applicable codes, standards, design basis and contractual stipulations and must therefore be re-submitted for review and approval after appropriate correction and/or re-preparation.	
Code- R	Shall mean Document is retained for records. Proceed with manufacture / fabrication.	
Parties	Shall mean the Parties executing this Contract, viz. L&T and the Supplier.	
Party	Shall mean one of the Parties.	
AI	Authorised Inspector (ASME)	
TPIA	Third Party Inspection Agency	
LOI	Letter of Intent	

No.	T&C Clause	CIRCOR Confirmation
PO	Purchase Order	
DLP	Defect Liability Period	
ITP	Inspection Test Plan	
SDR	Supplier Data Requirement	
PTR	Proven Track Record	
VQC / SQC	Vendor / Supplier Qualification Criteria	
MR	Material Requisition	
PR	Purchase Requisition	
LSP	Logistic Service Provider	
PBG	Performance Bank Guarantee	
KOM	Kick Off Meeting	
HAZOP	Hazard and Operability Study	
SIL	Safety Integrity Level	
Supplier	Shall mean the Supplier of Goods and/or Services on whom the Purchase Order is released	
Sub-Supplier	Shall mean Supplier's Supplier of any tier	
Supplier Group	Shall mean Supplier, its Sub-Supplier of any tier, and their respective representatives, but shall not include any member of L&T	
POMS	Shall mean "Post Order Management System" a digital (web based) portal developed by L&T for information capturing and/or data sharing and/or performing various required activities by Supplier subsequent to LOI/PO issuance.	
Free Issued material	Shall mean any equipment and/or materials and/or Goods provided by L&T/OWNER.	
Package	Supplies under this Contract.	
Supplies	Shall mean the totality of Goods and/or Services by expression or implication required to be supplied by the Supplier within the Scope of Work as per Order.	
RFQ	Request for Quotation including amendments issued to Supplier and covers instruction to bidders, Terms & conditions, Material requisition and all other documents.	
L&T/Buyer	Larsen & Toubro Limited	
FAT / IFAT	Factory Acceptance test / Integrated Factory Acceptance Test to be carried out at Supplier / sub-Supplier facilities / factory as per agreed scope	
SAT	Site Acceptance Test to be carried out at Project Site as per agreed scope	
Day	Shall mean Gregorian calendar day unless specified otherwise	
QC	Quality Control	

			SPECIFICATION DE-SUPERHEATER CIRCOR Select v15.3.0			Quote No./Rev: 250702-0400-PA / 0 Item Desc.: DSH-Line Item1 Item Rev. / Date: 2 / 19-Aug-2025 Customer: Larsen & Toubro Limited Cust. RFQ / Rev: E24003-RCF-IN-DST-042 Project Name: 1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India) End User: Rashtriya Chemicals and Fertilizers Limited			
Plant Name:			Application: De-Superheater						
Unit No.:			Quantity : 1						
Line/PID No.:			Primary Line:						
Plant Site: -			Tag No.: DS-9801						
1		Fluid : Steam		Service :			ShutOff :		
		Sizing Properties Steam:		Units	Max	Norm	Min	Cond4	Cond5
2	SERVICE / CONDITIONS	Inlet Flow Rate		kg/hr	7258	4698	1410		
3		Pressure, P1		kg/cm2(G)	8	8	8		
4		Inlet Temp., T1		°C	416	416	416		
5		Outlet Flow Rate		kg/hr	8878	5747	1725		
6		Outlet Temp., T2		°C	180	180	180		
		Sizing Properties Spray Water (SW):							
7		Flow Rate		kg/hr	1620	1049	315		
8		Pressure		kg/cm2(G)	16	16	16		
9		Temperature		°C	115	115	115		
10	LINE	Steam Size & Sch. Inlet	DN 150 SCH STD	40	 Reference Sketch Probe Style - SL Nozzle Data Plate Stainless Steel				
11		Steam Size & Sch. Outlet	DN 150 SCH STD	41					
12		Steam Pipe Mat'l In	ASTM A106 Gr B	42					
13		Steam Pipe Mat'l Outlet	ASTM A106 Gr B	43					
14		Liner Size & Sch.	4 inch SCH STD	44					
15		SW Size & Sch. Inlet	DN 25, SCH 160	46					
16		SW Pipe Mat'l Inlet	ASTM A106 Gr B	47					
17	DSH BODY	Type	Spring Loaded - Probe Style	48	POSITIONER				
18		Stub Size	3 Inch	49					
19		Pressure Class	300#	50					
20		Steam Dsgn Press/Temp	18.15 / FV / 420 °C	51					
21		SW Dsgn Press/Temp	18 / FV / 155	52					
22		Mfr. / Model	CIRCOR	53					
23		Model No.	CIR 3230	54					
24		SW Flange Size	1 Inch	55					
25	MATERIAL	SW Flange Rating	300#	56	AIR SET				
26		Rated Cv	SL40B	57					
27		Characteristic		58					
28		End Connection	RF Flanged	59					
29	ACC.	Body/Bonnet Mat'l	ASTM A106 Gr B / A105	60	SOL				
30		Stub Material		61					
31		Trim Material		62					
32		Nozzle Material	A565 Gr 616 / Alloy 422	63					
33	LOC.	Trim Type		64	SWITCH				
34		Certification	IBR	65					
35		Volume Booster		66					
36		Volume Tank		67					
37	TEST	Equip. Location	Outdoor	68	TEST	Seat Leakage Other See Factory Notes / QA Plan			
38		Amb. Temp. Min/Max		69					
39		Area Classification	Non-Hazardous	70					
		Min Required Straight Length	6m	71	Press/Temp Tables: ASME B16.34				
		Min Req. Temp Sensors Dist.	12m	72					
		6 inch Steam Pipe A106 Gr. B / Sch Std / WNRF ends with 4 inch Sch40 A106 Grb Liner offered to improve the velocity in minimum flow condition.		73					
		Corrosion Allowance 2 mm		74					
		Steam Pipe PMS 3A1BC		75					
		Spray water Pipe PMS 3A1BC		76					
		5 Deg C more than saturation level considered at DSH outlet to avoid condensate formation							
Recommended the strainer in the water line (Customer Scope)				Submittal by: Karthikeyan S Company: CIRCOR Flow Technologies India Pvt Ltd Contact Info:					

	DESUPERHEATER ANALYSIS SHEET CIRCOR Select v15.3.0			Quote No./Rev: 250702-0400-PA / 0		
Plant Name: Unit No.: Line/PID No.:	Application:	Attemperator Spray Water			Item Desc.: DSH-Line Item1	
	Quantity :	1			Item Rev. / Date: 2 / 19-Aug-2025	
	Primary Line:				Customer: Larsen & Toubro Limited	
Plant Site: -	Tag No.:	DS-9801			Cust. RFQ / Rev: E24003-RCF-IN-DST-042	
					Project Name: 1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)	
					End User: Rashtriya Chemicals and Fertilizers Limited	
Fluid: Steam						
Sizing Properties Steam:	Units	Max	Norm	Min	Cond4	Cond5
Inlet Flow Rate	kg/hr	7258	4698	1410		
Pressure, P1	kg/cm2(G)	8	8	8		
Inlet Temp., T1	°C	416	416	416		
Saturated Temp	°C	174.69	174.69	174.69		
Inlet Enthalpy	KJ/Kg	3300.17	3300.17	3300.17		
Inlet Specific Volume	m3/kg	0.394	0.394	0.394		
Outlet Flow Rate	kg/hr	8878	5747	1725		
Outlet Temp., T2	°C	180	180	180		
Outlet Enthalpy	KJ/Kg	2786.21	2786.21	2786.21		
Outlet Specific Volume	m3/kg	0.247	0.247	0.247		
Sizing Properties Spray Water (SW):						
Flow Rate	kg/hr	1620	1049	314.73		
Pressure	kg/cm2(G)	16	16	16		
Temperature	°C	115	115	115		
Enthalpy	KJ/Kg	483.63	483.63	483.63		
Rel. Density	Kg/m3	0.95	0.95	0.95		
Steam Inlet Velocity	m/s[Mach No.]	38.44[0.06]	24.88[0.04]	7.47[0.01]		
Steam Outlet Velocity	m/s[Mach No.]	23.99[0.05]	15.53[0.03]	4.66[0.01]		
SW Velocity	m/s					
Velocities		Normal	Normal	Normal		
Required Cv						
Flow Condition		Normal	Normal	Normal		
Shut-Off Pressure						
Steam Dsgn Temp In/Out :						
Minimum Degree of Superheat Required:			5 °C (41 °F)			
Min Required Straight Length (From point of water injection):			6m			
Min Required Temp Sensors Distance (From point of water injection):			12m			
Turn Down Ratio: NA						
Spray Water Quantity Required:						
$Q_w = \frac{H_1 - H_2}{H_2 - H_w} \times Q_s$						
H1 = Enthalpy of Steam at Inlet in KJ/Kg						
H2 = Enthalpy of Steam at Outlet in KJ/Kg						
Hw = Enthalpy of Water in KJ/Kg						
Qs = Steam Flow Rate in Kg/Hr						
$Q_w = \frac{3300.17 - 2786.21}{2786.21 - 1620.08} \times 7258$						
Qw = 1620 kg/hr						
*Invalid/Incomplete.			Submittal by:			
			Company:			
			Contact Info:			

				SPECIFICATION ON-OFF / CONTROL VALVE CIRCOR Select v23.1.0				Quote No./Rev: 250702-0400-PA / 2					
Plant Name: Unit No.: Line/PID No.: Plant Site: Thal, Maharashtra,INDIA				Application: Attemperator Spray Water Quantity : 1 Primary Line: - Tag No.: TCV-A1-9802				Item Desc.: CV-Line Item2					
								Item Rev. / Date: 2 / 19-Aug-2025					
								Customer: L&T Energy Hydrcarbon					
								Cust. RFQ / Rev: 0					
								Project Name: 1200 MTPD (DAP Basis) NPK Project					
								Rashtriya Chemicals and Fertilizers Limited					
								End User:					
1	Fluid : Water			Critical Pressure : 224.52 kg/cm²									
	Sizing Properties		Units	Case1	Case2	Case3	Case4	..	Shut-Off				
2	SERVICE / CONDITIONS	Flow Rate	kg/hr	1620	1049	315			---				
3		Inlet Pressure	kg/cm²[g]	16	16	16			18 kg/cm²				
4		Outlet Pressure	kg/cm²[g]	10.2	10.1	10			---				
5		Choke Drop	kg/cm²	12.904	13.304	13.338			---				
6		Inlet Temperature	°C	115	115	115			115				
7		Vapor Pressure	kg/cm²[g]	0.69	0.69	0.69			---				
8		Rel. Density		0.948	0.948	0.948			---				
9		Absolute Viscosity	cP	0.257	0.257	0.257			---				
10		Required Cv		0.81	0.52	0.15			---				
11		Travel	%	64.5	53	17			---				
12		Allow. / Pred. SPL, IEC	dBA	85 / 39	85 / 35	85 / 21			---				
13		F _L	5	0.91	0.93	0.93			---				
14	LINE	Pipe Size & Sch. In	DN 25 Sch. 160		53	ACTUATOR	Type	Pneumatic Spring & Diaphragm					
15		Pipe Size & Sch. Out	DN 25 Sch. 160		54		Mfr. / Model	CIRCOR / STD					
16		Pipe Insulation	None		55		Size/Eff. Area						
17	VALVE / BODY / BONNET	Type	Cast-Globe		56		Service	Modulating					
18		Body Size	DN25		57		Sprg Actn/Air Fail	Closed					
19		Pressure Class	300 ANSI		58		Spring No.						
20		Dsgn Press	18 kg/cm²[g]	Dsgn 115 °C	59		Max.Allow. Cyl. Press.	6 bar[g]					
21		Mfr. / Model	CIRCOR / 3100		60		Min. Req'd Press.	4 bar[g]					
22		Model No.			61		Avail. Air Supply						
23		Body/Bonnet Mat'l	A105		62		Max / Min Press	5.5 bar[g] / 4 bar[g]					
24		End Connection	DN25 Raised Face (RF) Flange		63		Bench Range						
25		Flg. Face Finish	Std. 125 - 250 rms		64		Orientation	Vertical					
26		End Ext. Mat'l			65		HOD/Override Type	Yes - Side Mounted					
27		Flow Direction	Flow Under Plug		66	Lock Up Valve	No						
28		Type of Bonnet	Standard-UTP		67	Set Press.							
29		Packing Mat'l	PTFE		68	Tubing/Fittings	3/8" NPT SS316						
30	TRIM	Type	Unbalanced - Top Guided		69	Data Plate	Stainless Steel						
31		Cage Type	Contour Plug		70	Mfr. / Model	Metso / YTC / Siemens						
32		Characteristic	Equal Percent		71	Input Signal	4-20 mA DC						
33		Travel / Stroke	20 mm		72	On Incr. Signal O/p	Increase						
34		Rated Cv	3.5		73	Type/Protocol	Smart / HART (IP67)						
35		Plug Mat'l	CF8M / SS 316 + Stellite		74	Gauges/By-Pass	Yes / No						
36		Seat Ring Material	CF8M / SS 316 + Stellite		75	Cam Characteristic	Linear						
37		Cage Mat'l	CF8M / SS 316 Cr. Plated		76	Position Feedback	Yes						
38		Stem Mat'l	A479 TY 316		77	Mfr. / Model	SHAVO (Al. Enclosure)						
39	SPECIAL ACCESSORIES	Cable Glands offered 1/2" NPT Adapter offered at AFR inlet w.r.t process connection			78	Range	0-10 Kg/cm2g						
40					Filter / Gauge	Yes (5 Micron) / Yes							
41					79	Set Pressure	4 Barg						
42					Mfr. / Model								
43								Quantity/Type					
44											Voltage		
45													
46					Contacts/Rating								
47								Type					
48											Actuation Points		
49	Hydro Test Press.												
50				77.57 bar[g]									
51							Seat Leakage						
52										ANSI/FCI 70-2 Class V			
Other													
			See Factory Notes / QA Plan										
						Req Stem Thrust:							
									Press/Temp Tables: ASME/ANSI B16.34				
Submittal by: Karthikeyan Sekar													
			Company: CIRCOR Flow Technologies India Pvt Ltd										
						Contact Info:							

			SPECIFICATION DE-SUPERHEATER CIRCOR Select v15.3.0			Quote No./Rev: 250702-0400-PA / 0 Item Desc.: DSH-Line Item2 Item Rev. / Date: 0 / 16-Jul-2025 Customer: Larsen & Toubro Limited Cust. RFQ / Rev: E24003-RCF-IN-DST-042 Project Name: 1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India) End User: Rashtriya Chemicals and Fertilizers Limited			
Plant Name: Unit No.: Line/PID No.: Plant Site: -			Application: De-Superheater Quantity : 1 Primary Line: Tag No.: DS-9805						
1		Fluid : Steam		Service :		ShutOff :			
		Sizing Properties Steam:		Units	Max	Norm	Min	Cond4	Cond5
2	SERVICE / CONDITIONS	Inlet Flow Rate		kg/hr	5428	4737	1421		
3		Pressure, P1		kg/cm2(G)	3.5	3.5	3.5		
4		Inlet Temp., T1		°C	161.5	161.5	161.5		
5		Outlet Flow Rate		kg/hr	5474	4777	1433		
6		Outlet Temp., T2		°C	153	153	153		
7		Sizing Properties Spray Water (SW):							
8		Flow Rate		kg/hr	46	40	12		
9		Pressure		kg/cm2(G)	16	16	16		
9		Temperature		°C	115	115	115		
10	LINE	Steam Size & Sch. Inlet	DN 150 SCH 40	40	Reference Sketch  Fixed Nozzle Desuperheater				
11		Steam Size & Sch. Outlet	DN 150 SCH 40	41					
12		Steam Pipe Mat'l In	ASTM A106 Gr B	42					
13		Steam Pipe Mat'l Outlet	ASTM A106 Gr B	43					
14		Liner Size & Sch.	N/A N/A	44					
15		SW Size &		45					
16		Sch. Inlet	DN 25, SCH 160	46					
16		SW	N/A	47					
16		Pipe Mat'l Inlet							
17	DSH BODY	Type	Fixed Nozzle - Probe Style	48	Data Plate Stainless Steel				
18		Stub Size	3 Inch	49					
19		Pressure Class	150#	50					
20		Steam Dsgn Press/Temp	18.15 / FV / 420 °C	51					
21		SW Dsgn Press/Temp	18 / FV / 155	52					
22		Mfr. / Model	CIRCOR	53					
23		Model No.	CIR 3210	54					
24		SW Flange Size	1 Inch	55					
25	SW Flange Rating	300#	56	POSITIONER	Mfr. / Model				
26	Rated Cv	1.2 (VN4 X 4)	57		Input Signal				
27	Characteristic		58		On Incr. Signal O/p				
28	End Connection	RF Flanged	59		Type/Protocol				
29	MATERIAL	Body/Bonnet Mat'l	ASTM A106 Gr B / A105	60	Gauges/By-Pass				
30		Stub Material		61	Cam Characteristic				
31		Trim Material		62	Position Feedback				
32		Nozzle Material	SS 316 Stellite	63	AIR SET	Mfr. / Model			
33		Trim Type		64		Range			
34	Certification	None	65	Filter / Gauge					
35	ACC.	Volume Booster		66	Set Pressure				
36		Volume Tank		67	SOL	Mfr. / Model			
37	Equip. Location	Outdoor	68	Quantity/Type					
38	Amb. Temp. Min/Max		69	Voltage					
39	LOC.	Area Classification	Non-Hazardous	70	SWITCH	Mfr. / Model			
	Min Required Straight Length	6m	71	Contacts/Rating					
	Min Req. Temp Sensors Dist.	12m	72	Type					
		Corrosion Allowance 2 mm		73	Actuation Points				
		Steam Pipe PMS 3A1BC		74	TEST	Seat Leakage			
		Spray water Pipe PMS 3A1BC		75		Other	See Factory Notes / QA Plan		
		5 Deg C more than saturation level considered at DSH outlet to avoid condensate formation		76					
					Press/Temp Tables: ASME B16.34				
Recommended the strainer in the water line (Customer Scope)					Submittal by: Prabakaran A Company: CIRCOR Flow Technologies India Pvt Ltd Contact Info: 9150004743				

	DESUPERHEATER ANALYSIS SHEET CIRCOR Select v15.3.0			Quote No./Rev: 250702-0400-PA / 0		
Plant Name: Unit No.: Line/PID No.:	Application: Attemperator Spray Water			Item Desc.: DSH-Line Item2		
	Quantity : 1			Item Rev. / Date: 0 / 16-Jul-2025		
	Primary Line:			Customer: Larsen & Toubro Limited		
Plant Site: -	Tag No.: DS-9805			Cust. RFQ / Rev: E24003-RCF-IN-DST-042		
				Project Name: 1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)		
				End User: Rashtriya Chemicals and Fertilizers Limited		
Fluid: Steam						
Sizing Properties Steam:		Units	Max	Norm	Min	Cond4
Inlet Flow Rate		kg/hr	5428	4737	1421	Cond5
Pressure, P1		kg/cm2(G)	3.5	3.5	3.5	
Inlet Temp., T1		°C	161.5	161.5	161.5	
Saturated Temp		°C	147.46	147.46	147.46	
Inlet Enthalpy		KJ/Kg	2775.14	2775.14	2775.14	
Inlet Specific Volume		m3/kg	0.001	0.001	0.001	
Outlet Flow Rate		kg/hr	5474	4777	1433	
Outlet Temp., T2		°C	153	153	153	
Outlet Enthalpy		KJ/Kg	2755.80	2755.80	2755.80	
Outlet Specific Volume		m3/kg	0.001	0.001	0.001	
Sizing Properties Spray Water (SW):						
Flow Rate		kg/hr	46	40	12.10	
Pressure		kg/cm2(G)	16	16	16	
Temperature		°C	115	115	115	
Enthalpy		KJ/Kg	483.63	483.63	483.63	
Rel. Density		Kg/m3	0.95	0.95	0.95	
Steam Inlet Velocity		m/s[Mach No.]	35.25[0.02]	30.76[0.02]	9.23[0.01]	
Steam Outlet Velocity		m/s[Mach No.]	34.43[0.02]	30.04[0.02]	9.01[0.01]	
SW Velocity		m/s				
Velocities			Normal	Normal	Normal	
Required Cv						
Flow Condition			Normal	Normal	Normal	
Shut-Off Pressure						
Steam Dsgn Temp In/Out :						
Minimum Degree of Superheat Required:			5 °C (41 °F)			
Min Required Straight Length (From point of water injection):			6m			
Min Required Temp Sensors Distance (From point of water injection):			12m			
Contingency:			0 %			
Turn Down Ratio:			NA			
Spray Water Quantity Required:						
$Q_w = \frac{H_1 - H_2}{H_2 - H_w} \times Q_s$						
H1 = Enthalpy of Steam at Inlet in KJ/Kg						
H2 = Enthalpy of Steam at Outlet in KJ/Kg						
Hw = Enthalpy of Water in KJ/Kg						
Qs = Steam Flow Rate in Kg/Hr						
$Q_w = \frac{2775.14 - 2755.8}{2755.8 - 46.21} \times 5428$						
Qw = 46 kg/hr						
*Invalid/Incomplete.			Submittal by:			
			Company:			
			Contact Info:			

				SPECIFICATION ON-OFF / CONTROL VALVE CIRCOR Select v23.1.0				Quote No./Rev: 250702-0400-PA / 2		
Plant Name: Unit No.: Line/PID No.: Plant Site: Thal, Maharashtra,INDIA				Application: Attemperator Spray Water Quantity : 1 Primary Line: - Tag No.: TCV-A1-9806				Item Desc.: CV-Line Item4		
								Item Rev. / Date: 2 / 19-Aug-2025		
								Customer: L&T Energy Hydrcarbon		
								Cust. RFQ / Rev: 0		
								Project Name: 1200 MTPD (DAP Basis) NPK Project		
								Rashtriya Chemicals and Fertilizers Limited		
								End User:		
1	Fluid : Water			Critical Pressure : 224.52 kg/cm²						
	Sizing Properties		Units	Case1	Case2	Case3	Case4	..	Shut-Off	
2	SERVICE / CONDITIONS	Flow Rate	kg/hr	46.21	40.33	12.1			---	
3		Inlet Pressure	kg/cm²[g]	16	16	16			18 kg/cm²	
4		Outlet Pressure	kg/cm²[g]	5.26	4.84	3.62			---	
5		Choke Drop	kg/cm²	15.114	15.114	15.114			---	
6		Inlet Temperature	°C	115	115	115			155	
7		Vapor Pressure	kg/cm²[g]	0.69	0.69	0.69			---	
8		Rel. Density		0.948	0.948	0.948			---	
9		Absolute Viscosity	cP	0.257	0.257	0.257			---	
10		Required Cv		0.02	0.02	0.004			---	
11		Travel	%	65	50	10			---	
12		Allow. / Pred. SPL, IEC	dBA	85 / 2	85 / 1	85 / 17			---	
13		F _L	5	0.99	0.99	0.99			---	
14	LINE	Pipe Size & Sch. In	DN 25 Sch. 160		53	Type	Pneumatic Spring & Diaphragm			
15		Pipe Size & Sch. Out	DN 25 Sch. 160		54	Mfr. / Model	CIRCOR / STD			
16		Pipe Insulation	None		55	Size/Eff. Area				
17	VALVE / BODY / BONNET	Type	Cast-Globe		56	Service	Modulating			
18		Body Size	DN25		57	Sprg Actn/Air Fail	Closed			
19		Pressure Class	300 ANSI		58	Spring No.				
20		Dsgn Press	18 kg/cm²[g]	Dsgn 155 °C	59	Max.Allow. Cyl. Press.	6 bar[g]			
21		Mfr. / Model	CIRCOR / 3100		60	Min. Req'd Press.	4 bar[g]			
22		Model No.			61	Avail. Air Supply				
23		Body/Bonnet Mat'l	A105		62	Max / Min Press	5.5 bar[g] / 4 bar[g]			
24		End Connection	DN25 Raised Face (RF) Flange		63	Bench Range				
25		Flg. Face Finish	Std. 125 - 250 rms		64	Orientation	Vertical			
26		End Ext. Mat'l			65	HOD/Override Type	Yes - Side Mounted			
27		Flow Direction	Flow Under Plug		66	Lock Up Valve	No			
28		Type of Bonnet	Standard-UTP		67	Set Press.				
29		Packing Mat'l	PTFE		68	Tubing/Fittings	3/8" NPT SS316			
30	TRIM	Type	Unbalanced - Top Guided		69	Data Plate	Stainless Steel			
31		Cage Type	MF Plug		70	Mfr. / Model	Metso / YTC / Siemens			
32		Characteristic	Modified		71	Input Signal	4-20 mA DC			
33		Travel / Stroke	25 mm		72	On Incr. Signal O/p	Increase			
34		Rated Cv	0.04		73	Type/Protocol	Smart / HART (IP67)			
35		Plug Mat'l	SS 440C Hardened		74	Gauges/By-Pass	Yes / No			
36		Seat Ring Material	SS 440C Hardened		75	Cam Characteristic	Linear			
37		Cage Mat'l	Integral with seat		76	Position Feedback	Yes			
38		Stem Mat'l	A479 TY 316		77	Mfr. / Model	SHAVO (Al. Enclosure)			
39	SPECIAL ACCESSORIES	Cable Glands offered 1/2" NPT Adapter offered at AFR inlet w.r.t process connection			78	Range	0-10 Kg/cm2g			
40					Filter / Gauge	Yes (5 Micron) / Yes				
41					79	Set Pressure	4 Barg			
42					80	Mfr. / Model				
43					81	Quantity/Type				
44					82	Voltage				
45					83	Mfr. / Model				
46					84	Contacts/Rating				
47					85	Type				
48					86	Actuation Points				
49	LOC.	Equip. Location	Outdoor		87	Hydro Test Press.	77.57 bar[g]			
50		Amb. Temp. Min/Max			88	Seat Leakage	ANSI/FCI 70-2 Class V			
51		Area Classification	Non-Hazardous		89	Other	See Factory Notes / QA Plan			
52		Compliance	IBR		Req Stem Thrust:					
					Press/Temp Tables: ASME/ANSI B16.34					
*Equivalent material forms may be used for the trims.					Submittal by: Karthikeyan Sekar					
					Company: CIRCOR Flow Technologies India Pvt Ltd					
					Contact Info:					



1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)



ANNEXURE - 1 - DEVIATIONS / CLARIFICATIONS / EXCEPTIONS

Project : 1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)
Owner : Rashtriya Chemicals and Fertilizers Limited
Job Code : E24003
Requisition No. : E24003-RCF-IN-MRIQ-042
Item: Desuperheater
Rev: 0

SR. NO.	REFERENCE/ DOC. NO.	SECTION / CLAUSE NO.	SPECIFIED REQUIREMENT	COMMENTS / DEVIATIONS OFFERED BY SUPPLIER	REASON	PURCHASER DECISION	DEVIATION CATEGORY	REMARK	FINAL STATUS (OPEN / CLOSE)
1	Doc. No.: E24003-RCF-IN-DST-042	Page 9	MR Requirement: Outlet temperature - 175.4 deg C Inlet steam pressure to DSH - 14 Kg/cm2g	Offered: Outlet temperature - 179.69 Deg C Inlet steam pressure to DSH - 8 Kg/cm2g	We recommend to keep Min 5 Deg C Superheat from saturation level to avoid condensate formation, hence we have offered with Sat + 5 Deg C. Regarding the pressure, inlet pressure shall be 8 Kg/cm2g since pressure drop across DSH is not possible & customer to take the pressure drop from 14 to 8 Kg/cm2g in PCV.				Open
2	Doc. No.: E24003-RCF-IN-DST-042	Page 16	MR Requirement: Outlet temperature - 147.9 deg C Inlet steam pressure to DSH - 8 Kg/cm2g	Offered: Outlet temperature - 152.5 Deg C Inlet steam pressure to DSH - 3.5 Kg/cm2g	We recommend to keep Min 5 Deg C Superheat from saturation level to avoid condensate formation, hence we have offered with Sat + 5 Deg C. Regarding the pressure, inlet pressure shall be 8 Kg/cm2g since pressure drop across DSH is not possible & customer to take the pressure drop from 14 to 8 Kg/cm2g in PCV.				Open
3	E24003-RCF-IN-DST-042	Page 18	MR Requirement: Trim MOC: SS316 stellite	Offered: SS440C Hardened	We recommend to offer SS440C Hardened MOC for trim as the same is for micro flow requirement. Hence for such small size (Dia) SS440C is recommended over SS316				Open
4	E24003-RCF-IN-DST-042	General	MR Requirement: AFR with IP67	Offered: Not Applicable	IP 67 is not applicable for AFR as the same is not an electrical instruments				Open
5	General	TPIA	MR Requirement: All major bought out items shall be inspected and tested at sub-vendor's works by Supplier/ Supplier appointed TPIA.	Not Applicable	TPIA shall be conducted in CIRCOR factory as for applicable items (Assembled final product)				Open
6									Open
7									Open
8									Open
9									Open
10									Open
11									Open
12									Open
13									Open
14									Open
15									Open

Note:

- 1. Supplier shall submit this form indicating technical deviation if any, indicating each clause no. If there is no technical deviation, Supplier shall submit this form indicating "NO DEVIATION"
- 2. Deviation Listed elsewhere other than the above format shall not be considered for offer evaluation.
- 3. Deviations proposed by finalized vendor will be taken up with PMC/ Owner in post order stage. However, if the deviation is rejected or any other remedial suggestion/ different compensatory requirement is asked by PMC/ OWNER then vendor has to withdraw the deviation or incorporate the suggestion from PMC/ Owner without any cost and time implication to LTEH. Also, if any cost benefit is asked by PMC/OWNER , then same will be borne by supplier.
- 4. Supplier shall very clearly categorize all Clarification, Exception and Deviation under following categories:
Category A: Deviations as per Manufacturer's standards design and cannot be withdrawn in any circumstances including additional payment or time extension.
Category B: Deviations which have been proposed as technical alternative by Supplier. If not accepted by Owner, Supplier shall follow the requisition requirement / specifications without any cost and time implications to Purchaser
Category C: Deviations having either 'time' or 'cost' or 'time & cost' implications. If not accepted by Owner, Supplier shall inform implications for compliance to requisition requirement / specifications.
Category D: Not a deviation against any requisition requirements. Supplier's clarification or basis for the offer.

			
1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)	List of Commissioning Spares		
	Annexure-4A	Rev.0	19-Aug-25

ANNXURE-4A: List of Commissioning Spares

Item Name: Control valve

Item No.: - TCV-A1-9802 & TCV-A1-9806

Sr. No.	Description	Quantity
1	Gland Packing	1 Set Each
2	Gasket	1 Nos Each

			
1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)	List of Mandatory Spares		
	Annexure-4B	Rev.0	19-Aug-25

ANNXURE-4B: List of Mandatory Spares

Item Name: Control valve

Item No.: - TCV-A1-9802 & TCV-A1-9806

Sr. No.	Description	Project Requirement	Quantity Offered
1	Actuator Seal Kit	10% of each type of valve subject to minimum 1 no. of each type	1 Set Each valve
2	Trim Set	Trim set consisting of seat ring / seal ring, plug with stem, cage (wherever applicable), packing material for each make, type, size, Pressure rating valve to be provided as spare	1 Set Each valve
3	Actuator Set	10% of each type or minimum 1 complete Actuator for each type and size	1 no. for each valve
4	Soft Parts (Gaskets & Gland packing)	100% for each valve. i.e. one set for each tag	1 Set Each valve
5	SMART Positioners	10% of each type of instruments, subject to minimum 2 nos. of each type	2 no's offered
6	Other accessories: Quick Exhaust relay, Volume Boosters, Air Filter regulators, position Transmitters, changeover relay, NRV, Pilot valves, Trip valve, etc.	10% of each type of instruments, subject to minimum 2 nos. of each type	2 no' of AFR offered

			
1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)	List of 2 years Op. Spares		
	Annexure-4C	Rev.0	19-Aug-25

ANNXURE-4C: List of 2 years Op. Spares

Item Name: Control valve

Item No.: - TCV-A1-9802 & TCV-A1-9806

Sr. No.	Description	Project Requirement	Quantity Offered
1	Actuator Seal Kit	10% of each type of valve subject to minimum 1 no. of each type	1 Set Each valve
2	Soft Parts (Gaskets & Gland packing)	100% for each valve. i.e. one set for each tag	1 Set Each valve



☐ Short Delivery	☐ Only for Engg. Activities	Date	29.09.2025
☐ Bought out valves	☐ Others, Please Specify	CORB Reference	DM25066L
Customer / Buyer Name	LARSEN & TOUBRO LIMITED	CEC Quote Number with Date	250702-0400-PA Dtd 28.08.2025
Site / End User Name & Country	Rashtriya Chemicals and Fertilizers Limited, India	Application Engineer SPOC	Karthikeyan Sekar Satheshkumar Eagambaram
Buyer PO Number/Date	LTES/75001-83664/BS Dtd 03.09.2025	Outside Sales Engineer/Manager	Ashokvijay R
Buyer PO/LOI Received Date	12.09.2025	PO Value	INR 9,99,750
Agreed / Approved Payment Terms	Refer PO clause no. 5.0	1)PO Delivery Date 2)Agreed Delivery Date	26.03.2026
Agreed Incoterms /Shipping Terms	CPT, RCF Thal Project Site	Margin	22%
LD Clause	@0.5% of the undelivered portion of Order value	Agent Name & Commission	NA
Ship to Address with GST#:	Rashtriya Chemical and Fertilizers Limited Thal, Thal-Vaishet, Alibag # 402 208 State: Maharashtra, India GSTIN: 27AAACL0140PEZ6	Bill to Address with GST#:	Larsen & Toubro Limited Divison Name: L&T Energy Hydrocarbon 14th Floor, AMN Tower, AMN Tower, Jogeshwari - Vikroli Link Road, Powai Campus, Gate 3, Mumbai Suburban, Maharashtra, 400072

COST SUMMARY SHEET

Price Schedule



Project	S20-0770 (Gas Based New Project)
Customer	S&T Energy Hydrocarbon
End User	Hydrocarbon Processing and Petrochemicals
Destination	INDIA
Our Quote Ref	INDIA0061
Date	25-Feb-23
Rev	1

Technical Details																	Material Cost (M)						Conversion Cost (L5)		Total Cost Details		Total Cost (INR)		Gross Margin	Price Details (INR)						
S.No	Tag Number	Valve/Item Type	Model Name	Body Size	Body Rating	End Conn	Body Mat	Trim Type	Cage / Plug Type	Trim Char	Travel (mm)	Actuation Type	Fail Action	Handwheel	Others	Accessories	Qty	Complete Body L/A	Actuator	Air Filter Regulator	Positioner	Material Unit Material (MM)	Unit Material (MM)	Value Unit Conversion (MM)	Unit Conversion (MM)	Unit Cost (INR)	Total Cost (INR)	Unit Cost (INR)	Total Cost (INR)	Margin	Unit Price (INR)	Total Price (INR)				
1	S2-0802	OS&H	CIR 3120	3"	350#	RF	A350/A150	St Nozzle				None	-	-	-	-	N/A	1	₹ 33,622	₹ -	₹ -	₹ -	₹ 675	₹ 38,297	₹ 16,956	₹ 18,956	₹ 51,254	₹ 51,254	₹ 51,254	39.40%	₹ 84,579	₹ 86,579				
2	TCV-A2-0802	Globe	CIR 3120	2"	350#	RF	A350	Unbalanced	Control Plug	Equilib	20	Spring & Diaphragm	None	Close	Top-Side	Refer Spec Sheet	1	₹ 5,300	₹ -	₹ 5,000	₹ 15,000	₹ 450	₹ 5,450	₹ 6,424	₹ 3,170	₹ 3,170	₹ 2,573	₹ 2,573	39.36%	₹ 2,565	₹ 2,565					
3	S2-0802	OS&H	CIR 3120	3"	350#	RF	A350/A150	Fixed Nozzle				None	-	-	-	-	N/A	1	₹ 9,180	₹ -	₹ -	₹ -	₹ 720	₹ 9,900	₹ 4,514	₹ 4,514	₹ 13,414	₹ 13,414	₹ 13,414	39.32%	₹ 17,405	₹ 17,405				
4	TCV-A2-0802	Globe	CIR 3120	2"	350#	RF	A150	Unbalanced	MF Plug	Modified	25	Spring & Diaphragm	Close	Top-Side	-	Refer Spec Sheet	1	₹ 5,300	₹ 15,000	₹ 500	₹ 1,50,400	₹ 6,570	₹ 6,570	₹ 1,25,648	₹ 1,25,648	₹ 1,25,648	₹ 1,25,648	45.49%	₹ 2,22,520	₹ 2,22,520						
5																Refer Spec Sheet	2	₹ 2,267	₹ -	₹ -	₹ -	₹ -	₹ 2,267	₹ -	₹ 2,267	₹ 2,267	₹ 2,267	₹ 2,267	45.04%	₹ 3,598	₹ 3,598					
6																Refer Spec Sheet	1	₹ 2,47,267	₹ -	₹ -	₹ -	₹ -	₹ 2,47,267	₹ 6,674	₹ 6,674	₹ 2,53,941	₹ 2,53,941	₹ 2,53,941	43.04%	₹ 4,45,831	₹ 4,45,831					
																		6														₹ 6,79,726				
																		4														₹ 5,79,726				
																																₹ 2,53,941	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,045			
																																₹ 1,41,045	₹ 1,41,0			

Remarks:
1. Scope: Valve + Actuator and OS&H along with accessories as mentioned in CIRCOR Datasheet.

For CIRCOR Flow Technologies India Pvt Ltd

Southwestern India
Application Engineering

Keyboard Shortcut for Price - CDR-18	
Margin incl. SG&A	22%
Margin incl. SG&A	38%
SG&A	16.0%
BS	0.0%
Tax Warrant	1.0%
Excl. Tax	12.0%
For Use	1.000
INR	
30 months from the date of supply	



☐ Short Delivery
 ☐ Only for Engg. Activities
☐ Bought out valves
 ☐ Others, Please Specify

		Date	29.09.2025
		CORB Reference	DM25066L
Customer / Buyer Name	LARSEN & TOUBRO LIMITED	CEC Quote Number with Date	250702-0400-PA Dtd 28.08.2025
Site / End User Name & Country	Rashtriya Chemicals and Fertilizers Limited, India	Application Engineer SPOC	Karthikeyan Sekar Satheshkumar Eagambaram
Buyer PO Number/Date	LTES/75001-83664/BS Dtd 03.09.2025	Outside Sales Engineer/Manager	Ashokvijay R
Buyer PO/LOI Received Date	12.09.2025	PO Value	INR 9,99,750
Agreed / Approved Payment Terms	Refer PO clause no. 5.0	1)PO Delivery Date 2)Agreed Delivery Date	26.03.2026
Agreed Incoterms /Shipping Terms	CPT, RCF Thal Project Site	Margin	22%
LD Clause	@0.5% of the undelivered portion of Order value	Agent Name & Commission	NA
Ship to Address with GST#:	Rashtriya Chemical and Fertilizers Limited Thal, Thal-Vaishet, Alibag # 402 208 State: Maharashtra, India GSTIN: 27AAACL0140PEZ6	Bill to Address with GST#:	Larsen & Toubro Limited Divison Name: L&T Energy Hydrocarbon 14th Floor, AMN Tower, AMN Tower, Jogeshwari - Vikroli Link Road, Powai Campus, Gate 3, Mumbai Suburban, Maharashtra, 400072

SPECIAL NOTES & CHECK LIST FOR APPLICATION ENGINEERING

(INSTRUCTION TO CS/QA/ME/ENG/SCM)

CIRCOR Flow Technologies India Private Limited

No.15 Naranapuram Village | Thennampalayam to Annur Road | Ponnandampalayam | Coimbatore – 641659 | India

Telephone: - +91 421 232 1600 | www.circor.com



Order reviewing check list - Application Engineering & Special Notes for CS / SCM / QA / Production / Mfg. Engg / Finance

Order # :	DM25066L	Date :	29-09-2025
Customer :	L&T	Doc Rev :	1
Application Engineer :	Karthikeyan Sekar		
SI #	Description	Status	Remarks
For Customer Service / Sales			
1	Checking ABG / PBG status for the order	N.A	
2	Checking for spares & mating parts requirements	Yes	Mandatory & Commissioning spares list attached
3	Checking the payment terms as per PO	Yes	
4	Confirming on LD clause for the order	Yes	
5	Confirming Freight terms for the order & ensure delivery is as per quote	Yes	
6	Confirming on customer inspection	Yes	L&T Scope
7	Attaching credit rating for the customer	N.A	
8	Agent commission applicable or not. If yes, please provide agent name & % of commission.	N.A	
9	Others if any - Special notes	None	
For Engineering			
10	IBR status for the items	Yes	
11	Confirming Pressure & temperature gauge selection	N.A	
12	Confirming the Selection of Temperature transmitter for inlet design temperature	N.A	
13	Confirming the Selection of Pressure transmitter for Outlet design temperature	N.A	
14	Checking Body & Trim Material mismatch between / Spec sheet - Complete spec sheet data validation	Yes	
15	Checking Stem travel from design	N.A	
16	Checking NDT requirements for the order	Yes	
17	Providing category for PED valves.	N.A	
18	Actuator model no# verification in spec sheet	Yes	
19	Checking Actuator handwheel mounting sequence between / Spec sheet (Side mounted / top mounted / With or with out gear box)	Yes	
20	AFR model no# verification between Ins. List / Spec sheet	Yes	
21	Air lock relay model no# verification between Ins. List / Spec sheet	N.A	
22	Positioner model no# verification between Ins. List / Spec sheet	Yes	
23	Limit switch model no# verification between Ins. List / Spec sheet	N.A	
24	Volume booster model no# verification between Ins. List / Spec sheet	N.A	
25	Solenoid valve model no# verification between Ins. List / Spec sheet	N.A	
26	Solenoid valve voltage verification between Ins. List / Spec sheet	N.A	
27	Engineering to provided relevant inputs to Manufacturing asap	Yes	
28	Spec sheet attached for design Consideration	Yes	
29	Any deviation in spec after detailed engineering should be communicated to AE to take approval from customer	Yes	
For Manufacturing			
30	Need of QAP as per customer requirement	Yes	
31	Whether / QAP / Spec sheet approval is needed from customer or not.	Yes	
32	Checking Tubing & fitting nomenclature	N.A	
33	Special paint requirements like RAL no# / No# of coats / Dry film thickness	N.A	
34	Tag no# mismatch between PO / / Spec sheet / Ins. List	Yes	
35	Confirming leakage class between Spec sheet & customer requirement.	Yes	
36	Checking packing requirements for the order	N.A	
37	Checking Welding requirements for the order	Yes	
38	Ensure ABB positioner is available in case of availability volume booster	N.A	
39	Ensure Susin I-tork positioner is replacing CIRCOR positioner	N.A	
40	When Port size is $\geq 3/4"$, ensure the usage of ROTEX & not ASCO for solenoid valves	N.A	
41	Ensure that the standard tubings & fitting size is 3/8".	N.A	
42	Ensure that the PRDS has the fixing length for Pressure transmitter & temperature transmitter	N.A	
43	Ensure that the position transmitter make is not YAMATAKE	N.A	
Order Specific Notes			
44	All Pneumatic Vent Ports (on Actuator and Accessories) shall be plugged with SS304 bug screen		
45	SS 304 tag plate shall be provided on each valve & accessories		
46	TPI Applicable		
47	PMI check shall be carried out for all SS/Alloy steels component		
48	Weld Joints shall be 100% RT		
49	3.1 Mill test certificates as per EN 10204 for Valve Body, Bonnet, Plug, Seat, closure member & trim material (Spray Head, Nozzle & Spindle), Bar/ Forging & Fasteners		
50	Commissioning spares to be offered (1 set of Gaskets & 1 set of Gland packing for each CV involved)		
51	FRP Canopy offered for Positioner		
52	AVL not applicable other than Instrument Model no. defined		
53	All Piping shall be designed in accordance with the Process Piping Code, ASME B31.3 (2020)		
54	Mandatory Spares offered		
55	PMS to be followed		
56	1/2" NPT Adapter offered at AFR inlet w.r.t process connection		
57	All bolts for valve - actuator shall be SS316 (Except Body-Bonnet connectors)		
58	Wake frequency calculation for spray nozzle as per PTC 19.3 year 2024 shall be submitted		
59	Non used positioner port to be plugged with SS304 plug		



☐ Short Delivery
 ☐ Only for Engg. Activities
☐ Bought out valves
 ☐ Others, Please Specify

		Date	29.09.2025
		CORB Reference	DM25066L
Customer / Buyer Name	LARSEN & TOUBRO LIMITED	CEC Quote Number with Date	250702-0400-PA Dtd 28.08.2025
Site / End User Name & Country	Rashtriya Chemicals and Fertilizers Limited, India	Application Engineer SPOC	Karthikeyan Sekar Satheshkumar Eagambaram
Buyer PO Number/Date	LTES/75001-83664/BS Dtd 03.09.2025	Outside Sales Engineer/Manager	Ashokvijay R
Buyer PO/LOI Received Date	12.09.2025	PO Value	INR 9,99,750
Agreed / Approved Payment Terms	Refer PO clause no. 5.0	1)PO Delivery Date 2)Agreed Delivery Date	26.03.2026
Agreed Incoterms /Shipping Terms	CPT, RCF Thal Project Site	Margin	22%
LD Clause	@0.5% of the undelivered portion of Order value	Agent Name & Commission	NA
Ship to Address with GST#:	Rashtriya Chemical and Fertilizers Limited Thal, Thal-Vaishet, Alibag # 402 208 State: Maharashtra, India GSTIN: 27AAACL0140PEZ6	Bill to Address with GST#:	Larsen & Toubro Limited Divison Name: L&T Energy Hydrocarbon 14th Floor, AMN Tower, AMN Tower, Jogeshwari - Vikroli Link Road, Powai Campus, Gate 3, Mumbai Suburban, Maharashtra, 400072

INSTRUMENTATION LIST



☐ Short Delivery	☐ Only for Engg. Activities	Date	29.09.2025
☐ Bought out valves	☐ Others, Please Specify	CORB Reference	DM25066L
Customer / Buyer Name	LARSEN & TOUBRO LIMITED	CEC Quote Number with Date	250702-0400-PA Dtd 28.08.2025
Site / End User Name & Country	Rashtriya Chemicals and Fertilizers Limited, India	Application Engineer SPOC	Karthikeyan Sekar Satheshkumar Eagambaram
Buyer PO Number/Date	LTES/75001-83664/BS Dtd 03.09.2025	Outside Sales Engineer/Manager	Ashokvijay R
Buyer PO/LOI Received Date	12.09.2025	PO Value	INR 9,99,750
Agreed / Approved Payment Terms	Refer PO clause no. 5.0	1)PO Delivery Date 2)Agreed Delivery Date	26.03.2026
Agreed Incoterms /Shipping Terms	CPT, RCF Thal Project Site	Margin	22%
LD Clause	@0.5% of the undelivered portion of Order value	Agent Name & Commission	NA
Ship to Address with GST#:	Rashtriya Chemical and Fertilizers Limited Thal, Thal-Vaishet, Alibag # 402 208 State: Maharashtra, India GSTIN: 27AAACL0140PEZ6	Bill to Address with GST#:	Larsen & Toubro Limited Divison Name: L&T Energy Hydrocarbon 14th Floor, AMN Tower, AMN Tower, Jogeshwari - Vikroli Link Road, Powai Campus, Gate 3, Mumbai Suburban, Maharashtra, 400072

CIRCOR VALVE / DESUPERHEATER GAD



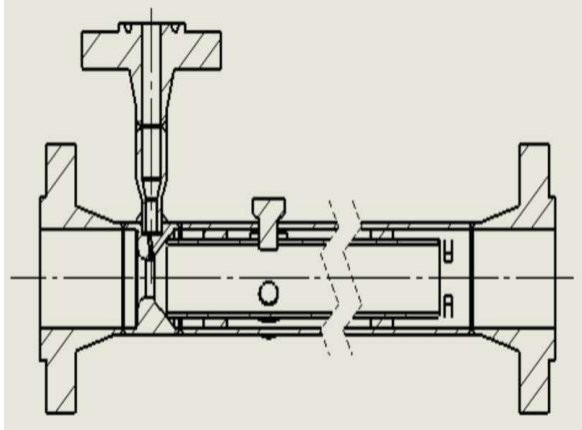
☐ Short Delivery	☐ Only for Engg. Activities	Date	29.09.2025
☐ Bought out valves	☐ Others, Please Specify	CORB Reference	DM25066L
Customer / Buyer Name	LARSEN & TOUBRO LIMITED	CEC Quote Number with Date	250702-0400-PA Dtd 28.08.2025
Site / End User Name & Country	Rashtriya Chemicals and Fertilizers Limited, India	Application Engineer SPOC	Karthikeyan Sekar Satheshkumar Eagambaram
Buyer PO Number/Date	LTES/75001-83664/BS Dtd 03.09.2025	Outside Sales Engineer/Manager	Ashokvijay R
Buyer PO/LOI Received Date	12.09.2025	PO Value	INR 9,99,750
Agreed / Approved Payment Terms	Refer PO clause no. 5.0	1)PO Delivery Date 2)Agreed Delivery Date	26.03.2026
Agreed Incoterms /Shipping Terms	CPT, RCF Thal Project Site	Margin	22%
LD Clause	@0.5% of the undelivered portion of Order value	Agent Name & Commission	NA
Ship to Address with GST#:	Rashtriya Chemical and Fertilizers Limited Thal, Thal-Vaishet, Alibag # 402 208 State: Maharashtra, India GSTIN: 27AAACL0140PEZ6	Bill to Address with GST#:	Larsen & Toubro Limited Divison Name: L&T Energy Hydrocarbon 14th Floor, AMN Tower, AMN Tower, Jogeshwari - Vikroli Link Road, Powai Campus, Gate 3, Mumbai Suburban, Maharashtra, 400072

CIRCOR VALVE / DESUPERHEATER SPECIFICATION & CALCULATION

CIRCOR Flow Technologies India Private Limited

No.15 Naranapuram Village | Thennampalayam to Annur Road | Ponnandampalayam | Coimbatore – 641659 | India

Telephone: - +91 421 232 1600 | www.circor.com

			SPECIFICATION DE-SUPERHEATER CIRCOR Select v15.3.0			SO No./Rev: DM25066L Item Desc.: DSH-Line Item1 Item Rev. / Date: 1 / 29-Sep-2025			
Plant Name: RCF, Ltd Unit No.: Line/PID No.:			Application: De-Superheater Quantity : 1 Primary Line: Tag No.: DS-9801			Customer: Larsen & Toubro Limited PO# / Rev: LTES/75001-83664/BS Project Name: 1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India) End User: Rashtriya Chemicals and Fertilizers Limited			
Plant Site: India									
1		Fluid : Steam		Service :		ShutOff :			
		Sizing Properties Steam:		Units	Max	Norm	Min	Cond4	Cond5
2	SERVICE / CONDITIONS	Inlet Flow Rate		kg/hr	7258	4698	1410		
3		Pressure, P1		kg/cm2(G)	8	8	8		
4		Inlet Temp., T1		°C	416	416	416		
5		Outlet Flow Rate		kg/hr	8878	5747	1725		
6		Outlet Temp., T2		°C	180	180	180		
		Sizing Properties Spray Water (SW):							
7		Flow Rate		kg/hr	1620	1049	315		
8		Pressure		kg/cm2(G)	16	16	16		
9		Temperature		°C	115	115	115		
10	LINE	Steam Size & Sch. Inlet	DN 150 SCH STD	40					
11		Steam Size & Sch. Outlet	DN 150 SCH STD	41					
12		Steam Pipe Mat'l In	ASTM A106 Gr B	42					
13		Steam Pipe Mat'l Outlet	ASTM A106 Gr B	43					
14		Liner Size & Sch.	4 inch SCH STD	44					
15		SW Size & Sch. Inlet	DN 25, SCH 160	46					
16		SW Pipe Mat'l Inlet	ASTM A106 Gr B	47					
17	DSH BODY	Type	Venturi DSH	48					
18		Body Size	6 Inch	49					
19		Pressure Class	300#	50					
20		Steam Dsgn Press/Temp	18.15 / FV /420 °C	51					
21		SW Dsgn Press/Temp	18 / FV / 155	52					
22		Mfr. / Model	CIRCOR	53					
23		Model No.	CIR 3210	54					
24		SW Flange Size	1 Inch	55					
25	POSITIONER	SW Flange Rating	300#	56					
26		Rated Cv	1	57					
27		Characteristic		58					
28		End Connection	RF Flanged	59					
29	MATERIAL	Body/Bonnet Mat'l	ASTM A106 Gr B / A105	60					
30		Stub Material		61					
31		Trim Material		62					
32		Nozzle Material	SS 316 Stellite	63					
33		Trim Type		64					
34	ACC.	Certification	IBR	65					
35		Volume Booster		66					
36		Volume Tank		67					
37	LOC.	Equip. Location	Outdoor	68					
38		Amb. Temp. Min/Max		69					
39		Area Classification	Non-Hazardous	70					
		Min Required Straight Length	6m	71					
		Min Req. Temp Sensors Dist.	12m	72					
		6 inch Steam Pipe A106 Gr. B / Sch Std / WNRF ends with 4 inch Sch40 A106 Grb Liner offered to improve the velocity in minimum flow condition.		73					
		Corrosion Allowance 2 mm		74					
		Steam Pipe PMS 3A1BC		75					
		Spray water Pipe PMS 3A1BC		76					
		5 Deg C more than saturation level considered at DSH outlet to avoid condensate formation			Seat Leakage Other See Factory Notes / QA Plan				
				Press/Temp Tables: ASME B16.34					
Recommended the strainer in the water line (Customer Scope)				Submittal by: Karthikeyan S Company: CIRCOR Flow Technologies India Pvt Ltd Contact Info:					

	DESUPERHEATER ANALYSIS SHEET CIRCOR Select v15.3.0			SO No./Rev: DM25066L / 1 Item Desc.: DSH-Line Item1 Item Rev. / Date: 1 / 29-Sep-2025 Customer: Larsen & Toubro Limited PO# / Rev: LTES/75001-83664/BS Project Name: 1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India) End User: Rashtriya Chemicals and Fertilizers Limited		
Plant Name: RCF, Ltd Unit No.: Line/PID No.: Plant Site: India	Application: Attemperator Spray Water Quantity : 1 Primary Line: Tag No.: DS-9801					
Fluid: Steam						
Sizing Properties Steam:	Units	Max	Norm	Min	Cond4	Cond5
Inlet Flow Rate	kg/hr	7258	4698	1410		
Pressure, P1	kg/cm2(G)	8	8	8		
Inlet Temp., T1	°C	416	416	416		
Saturated Temp	°C	174.69	174.69	174.69		
Inlet Enthalpy	KJ/Kg	3300.17	3300.17	3300.17		
Inlet Specific Volume	m3/kg	0.394	0.394	0.394		
Outlet Flow Rate	kg/hr	8878	5747	1725		
Outlet Temp., T2	°C	180	180	180		
Outlet Enthalpy	KJ/Kg	2786.21	2786.21	2786.21		
Outlet Specific Volume	m3/kg	0.247	0.247	0.247		
Sizing Properties Spray Water (SW):						
Flow Rate	kg/hr	1620	1049	314.73		
Pressure	kg/cm2(G)	16	16	16		
Temperature	°C	115	115	115		
Enthalpy	KJ/Kg	483.63	483.63	483.63		
Rel. Density	Kg/m3	0.95	0.95	0.95		
Steam Inlet Velocity	m/s[Mach No.]	38.44[0.06]	24.88[0.04]	7.47[0.01]		
Steam Outlet Velocity	m/s[Mach No.]	23.99[0.05]	15.53[0.03]	4.66[0.01]		
SW Velocity	m/s					
Velocities		Normal	Normal	Normal		
Required Cv						
Flow Condition		Normal	Normal	Normal		
Shut-Off Pressure						
Steam Dsgn Temp In/Out :						
Minimum Degree of Superheat Required:			5 °C (41 °F)			
Min Required Straight Length (From point of water injection):			6m			
Min Required Temp Sensors Distance (From point of water injection):			12m			
Turn Down Ratio: NA						
Spray Water Quantity Required:						
$Q_w = \frac{H_1 - H_2}{H_2 - H_w} \times Q_s$						
H1 = Enthalpy of Steam at Inlet in KJ/Kg						
H2 = Enthalpy of Steam at Outlet in KJ/Kg						
Hw = Enthalpy of Water in KJ/Kg						
Qs = Steam Flow Rate in Kg/Hr						
$Q_w = \frac{3300.17 - 2786.21}{2786.21 - 1620.08} \times 7258$						
Qw = 1620 kg/hr						
*Invalid/Incomplete.			Submittal by: Company: Contact Info:			

				SPECIFICATION ON-OFF / CONTROL VALVE CIRCOR Select v23.1.0			SO No./Rev: DM25066L / 1 Item Desc.: CV-Line Item2 Item Rev. / Date: 1 / 29-Sep-2025			
Plant Name: RCF, Ltd Unit No.: Line/PID No.:				Application: Attenuator Spray Water Quantity : 1 Primary Line: -			Customer: L&T Energy HydrCarbon PO# LTES/75001-83664/BS Project Name: 1200 MTPD (DAP Basis) NPK Project End User: Rashtriya Chemicals and Fertilizers Limited			
Plant Site: Thal, Maharashtra,INDIA				Tag No.: TCV-A1-9802						
1		Fluid : Water			Critical Pressure : 224.52 kg/cm ²					
		Sizing Properties	Units	Case1	Case2	Case3	Case4	..	Shut-Off	
2	SERVICE / CONDITIONS	Flow Rate	kg/hr	1620	1049	315			---	
3		Inlet Pressure	kg/cm ² [g]	16	16	16			18 kg/cm ²	
4		Outlet Pressure	kg/cm ² [g]	11.47	9.46	8.13			---	
5		Choke Drop	kg/cm ²	12.778	13.15	13.338			---	
6		Inlet Temperature	°C	115	115	115			115	
7		Vapor Pressure	kg/cm ² [g]	0.69	0.69	0.69			---	
8		Rel. Density		0.948	0.948	0.948			---	
9		Absolute Viscosity	cP	0.257	0.257	0.257			---	
10		Required Cv		0.91	0.49	0.13			---	
11		Travel	%	67.9	51.8	15.1			---	
12		Allow. / Pred. SPL, IEC	dBA	85 / 36	85 / 31	85 / 17			---	
13		F_L	5	0.91	0.92	0.93			---	
14	LINE	Pipe Size & Sch. In	DN 25 Sch. 160		53	Type		Pneumatic Spring & Diaphragm		
15		Pipe Size & Sch. Out	DN 25 Sch. 160		54	Mfr. / Model		CIRCOR / CM60		
16		Pipe Insulation	None		55	Size/Eff. Area				
17	BONNET / BODY / VALVE	Type	Cast-Globe		56	Service		Modulating		
18		Body Size	DN25		57	Sprg Actn/Air Fail		Closed		
19		Pressure Class	300 ANSI		58	Spring No.		B6-7G		
20		Dsgn Press	18 kg/cm ² [g]	Dsgn 115 °C	59	Max.Allow. Cyl. Press.		6 bar[g]		
21		Mfr. / Model	CIRCOR / 3100		60	Min. Req'd Press.		4 bar[g]		
22		Model No.			61	Avail. Air Supply				
23		Body/Bonnet Mat'l	A105		62	Max / Min Press		5.5 bar[g] / 4 bar[g]		
24		End Connection	DN25 Raised Face (RF) Flange		63	Bench Range		1.47 - 2.24 kg/cm2		
25		Flg. Face Finish	Std. 125 - 250 rms		64	Orientation		Vertical		
26		End Ext. Mat'l			65	HOD/Override Type		Yes - Side Mounted		
27		Flow Direction	Flow Under Plug		66	Lock Up Valve		No		
28		Type of Bonnet	Standard-UTP		67	Set Press.				
29		Packing Mat'l	Graphite		68	Tubing/Fittings		3/8" NPT SS316		
30	TRIM	Type	Unbalanced - Top Guided		69	Data Plate		Stainless Steel		
31		Cage Type	Contour Plug		70	Mfr. / Model		SIEMENS / 6DR5115-0EN01-0AA3		
32		Characteristic	Equal Percent		71	Input Signal		4-20 mA DC		
33		Travel / Stroke	20 mm		72	On Incr. Signal O/p		Increase		
34		Rated Cv	3.5		73	Type/Protocol		Smart / HART (IP67)		
35		Plug Mat'l	CF8M / SS 316 + Stellite		74	Gauges/By-Pass		Yes / No		
36		Seat Ring Material	CF8M / SS 316 + Stellite		75	Cam Characteristic		Linear		
37		Cage Mat'l	CF8M / SS 316 Cr. Plated		76	Position Feedback		Yes		
38		Stem Mat'l	A479 TY 316		77	Mfr. / Model		SHAVO / 11-4SN-402M-M5KN-NBAO		
39	SPECIAL ACCESSORIES	1/2" NPT SS316 Double Compression Cable Glands offered for positioner (1 no.)			78	Range		0.70 - 7.00 Kg/cm2g		
40					1/2" NPT (F) Adapter offered at AFR inlet w.r.t process connection			79	Filter / Gauge	
41								AIR SET	Set Pressure	
42					SOL	Mfr. / Model				
43						Quantity/Type				
44						Voltage				
45					SWITCHES	Mfr. / Model				
46						Contacts/Rating				
47		Type								
48		TEST	Actuation Points							
49	Hydro Test Press.		77.57 bar[g]							
50	Seat Leakage		ANSI/FCI 70-2 Class V							
51	LOC.	Other		See Factory Notes / QA Plan						
52		Req Stem Thrust:		325 kgf						
					Press/Temp Tables: ASME/ANSI B16.34					
					Submittal by: Karthikeyan Sekar					
					Company: CIRCOR Flow Technologies India Pvt Ltd					
					Contact Info:					

			SPECIFICATION DE-SUPERHEATER CIRCOR Select v15.3.0			SO No./Rev: DM25066L Item Desc.: DSH-Line Item3 Item Rev. / Date: 1 / 29-Sep-2025		
Plant Name: Unit No.: Line/PID No.:			Application: De-Superheater Quantity : 1 Primary Line:			Customer: Larsen & Toubro Limited PO# / Rev: LTES/75001-83664/BS Project Name: 1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India) End User: Rashtriya Chemicals and Fertilizers Limited		
Plant Site: -			Tag No.: DS-9805					

1		Fluid : Steam		Service :			ShutOff :		
		Sizing Properties Steam:		Units	Max	Norm	Min	Cond4	Cond5
2	SERVICE / CONDITIONS	Inlet Flow Rate		kg/hr	5428	4737	1421		
3		Pressure, P1		kg/cm2(G)	3.5	3.5	3.5		
4		Inlet Temp., T1		°C	161.5	161.5	161.5		
5		Outlet Flow Rate		kg/hr	5474	4777	1433		
6		Outlet Temp., T2		°C	153	153	153		
7		Sizing Properties Spray Water (SW):							
8		Flow Rate		kg/hr	46	40	12		
9		Pressure		kg/cm2(G)	16	16	16		
9		Temperature		°C	115	115	115		

10	LINE	Steam Size & Sch. Inlet	DN 150 SCH 40	40	Reference Sketch	
11		Steam Size & Sch. Outlet	DN 150 SCH 40	41		
12		Steam Pipe Mat'l In	ASTM A106 Gr B	42		
13		Steam Pipe Mat'l Outlet	ASTM A106 Gr B	43		
14		Liner Size & Sch.	N/A N/A	44		
15		SW Size & Sch. Inlet	DN 25, SCH 160	45		
16		SW	N/A	46		
17	DSH BODY	Pipe Mat'l Inlet		47	 Fixed Nozzle Desuperheater	
18		Type	Fixed Nozzle - Probe Style	48		
19		Stub Size	3 Inch	49		
20		Pressure Class	150#	50		
21		Steam Dsgn Press/Temp	18.15 / FV / 420 °C	51		
22		SW Dsgn Press/Temp	18 / FV / 155	52		
23		Mfr. / Model	CIRCOR	53		
24		Model No.	CIR 3220	54		
25		SW Flange Size	1 Inch	55		
26		SW Flange Rating	300#	56		
27	Rated Cv	0.04 (FN1 - 1 no.)	57	POSITIONER	Mfr. / Model	
28	Characteristic	RF Flanged	58		Input Signal	
29	End Connection	RF Flanged	59		On Incr. Signal O/p	
30	MATERIAL	Body/Bonnet Mat'l	ASTM A106 Gr B / A105	60	Type/Protocol	
31		Stub Material		61	Gauges/By-Pass	
32		Trim Material		62	Cam Characteristic	
33		Nozzle Material	SS 316 Stellite	63	Position Feedback	
34		Trim Type		64	AIR SET	Mfr. / Model
35	Certification	IBR	65	Range		
36			66	Filter / Gauge		
37	ACC.	Volume Booster		67	Set Pressure	
38		Volume Tank		68		
39				69		
40	LOC.	Equip. Location	Outdoor	70	SOL	Mfr. / Model
41		Amb. Temp. Min/Max		71		Quantity/Type
42		Area Classification	Non-Hazardous	72		Voltage
43		Min Required Straight Length	6m	73	SWITCH	Mfr. / Model
44		Min Req. Temp Sensors Dist.	12m	74		Contacts/Rating
45				75		Type
46		Corrosion Allowance 1.5 mm		76	Actuation Points	
47		Steam Pipe PMS 1A1BC				
48		Spray water Pipe PMS 3A1BC				
49		5 Deg C more than saturation level considered at DSH outlet to avoid condensate formation				

Press/Temp Tables: ASME B16.34			
Recommended the strainer in the water line (Customer Scope)			
Submittal by: Karthikeyan S		Company: CIRCOR Flow Technologies India Pvt Ltd	
Contact Info:			

		DESUPERHEATER ANALYSIS SHEET CIRCOR Select v15.3.0			SO No./Rev: DM25066L / 1 Item Desc.: DSH-Line Item3 Item Rev. / Date: 1 / 29-Sep-2025		
Plant Name: RCF, Ltd Unit No.: Line/PID No.: Plant Site: India		Application: Attemperator Spray Water Quantity : 1 Primary Line: Tag No.: DS-9805			Customer: Larsen & Toubro Limited PO# / Rev: LTES/75001-83664/BS Project Name: 1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India) End User: Rashtriya Chemicals and Fertilizers Limited		
Fluid: Steam							
Sizing Properties Steam:		Units	Max	Norm	Min	Cond4	Cond5
Inlet Flow Rate		kg/hr	5428	4737	1421		
Pressure, P1		kg/cm2(G)	3.5	3.5	3.5		
Inlet Temp., T1		°C	161.5	161.5	161.5		
Saturated Temp		°C	147.46	147.46	147.46		
Inlet Enthalpy		KJ/Kg	2775.14	2775.14	2775.14		
Inlet Specific Volume		m3/kg	0.001	0.001	0.001		
Outlet Flow Rate		kg/hr	5474	4777	1433		
Outlet Temp., T2		°C	153	153	153		
Outlet Enthalpy		KJ/Kg	2755.80	2755.80	2755.80		
Outlet Specific Volume		m3/kg	0.001	0.001	0.001		
Sizing Properties Spray Water (SW):							
Flow Rate		kg/hr	46	40	12.10		
Pressure		kg/cm2(G)	16	16	16		
Temperature		°C	115	115	115		
Enthalpy		KJ/Kg	483.63	483.63	483.63		
Rel. Density		Kg/m3	0.95	0.95	0.95		
Steam Inlet Velocity		m/s[Mach No.]	35.25[0.02]	30.76[0.02]	9.23[0.01]		
Steam Outlet Velocity		m/s[Mach No.]	34.43[0.02]	30.04[0.02]	9.01[0.01]		
SW Velocity		m/s					
Velocities			Normal	Normal	Normal		
Required Cv							
Flow Condition			Normal	Normal	Normal		
Shut-Off Pressure							
Steam Dsgn Temp In/Out :							
Minimum Degree of Superheat Required:				5 °C (41 °F)			
Min Required Straight Length (From point of water injection):				6m			
Min Required Temp Sensors Distance (From point of water injection):				12m			
Contingency:		0 %					
Turn Down Ratio:		NA					
Spray Water Quantity Required:							
$Q_w = \frac{H_1 - H_2}{H_2 - H_w} \times Q_s$							
H1 = Enthalpy of Steam at Inlet in KJ/Kg							
H2 = Enthalpy of Steam at Outlet in KJ/Kg							
Hw = Enthalpy of Water in KJ/Kg							
Qs = Steam Flow Rate in Kg/Hr							
$Q_w = \frac{2775.14 - 2755.8}{2755.8 - 46.21} \times 5428$							
Qw = 46 kg/hr							
*Invalid/Incomplete.				Submittal by: Company: Contact Info:			

				SPECIFICATION ON-OFF / CONTROL VALVE CIRCOR Select v23.1.0				SO No./Rev: DM25066L / 1 Item Desc.: CV-Line Item4 Item Rev. / Date: 1 / 29-Sep-2025	
Plant Name: RCF, Ltd Unit No.: Line/PID No.: Plant Site: Thal, Maharashtra,INDIA				Application: Attemperator Spray Water Quantity : 1 Primary Line: - Tag No.: TCV-A1-9806				Customer: L&T Energy Hydrcarbon PO# LTES/75001-83664/BS Project Name: 1200 MTPD (DAP Basis) NPK Project End User: Rashtriya Chemicals and Fertilizers Limited	
1		Fluid : Water			Critical Pressure : 224.52 kg/cm²				
		Sizing Properties	Units	Case1	Case2	Case3	Case4	..	Shut-Off
2	SERVICE / CONDITIONS	Flow Rate	kg/hr	46	40	12.1			---
3		Inlet Pressure	kg/cm²[g]	16	16	16			18 kg/cm²
4		Outlet Pressure	kg/cm²[g]	5.26	4.84	3.62			---
5		Choke Drop	kg/cm²	15.114	15.114	15.114			---
6		Inlet Temperature	°C	115	115	115			155
7		Vapor Pressure	kg/cm²[g]	0.69	0.69	0.69			---
8		Rel. Density		0.948	0.948	0.948			---
9		Absolute Viscosity	cP	0.257	0.257	0.257			---
10		Required Cv		0.02	0.01	0.004			---
11		Travel	%	65	50	10			---
12		Allow. / Pred. SPL, IEC	dBA	85 / 1	85 / XX	85 / 16			---
13		F_L	5	0.99	0.99	0.99			---
14	LINE	Pipe Size & Sch. In	DN 25 Sch. 160		53	Type Pneumatic Spring & Diaphragm Mfr. / Model CIRCOR / CM60 Size/Eff. Area Service Modulating Sprg Actn/Air Fail Closed Spring No. B6-7G Max.Allow. Cyl. Press. 6 bar[g] Min. Req'd Press. 4 bar[g] Avail. Air Supply Max / Min Press 5.5 bar[g] / 4 bar[g] Bench Range 1.47 - 2.43 kg/cm2 Orientation Vertical HOD/Override Type Yes - Side Mounted Lock Up Valve No Set Press. Tubing/Fittings 3/8" NPT SS316 Data Plate Stainless Steel			
15		Pipe Size & Sch. Out	DN 25 Sch. 160		54				
16		Pipe Insulation	None		55				
17	VALVE / BODY / BONNET	Type	Cast-Globe		56	ACTUATOR			
18		Body Size	DN25		57				
19		Pressure Class	300 ANSI		58				
20		Dsgn Press	18 kg/cm²[g]	Dsgn 155 °C	59				
21		Mfr. / Model	CIRCOR / 3100		60				
22		Model No.			61				
23		Body/Bonnet Mat'l	A105		62				
24		End Connection	DN25 Raised Face (RF) Flange		63				
25		Fig. Face Finish	Std. 125 - 250 rms		64				
26		End Ext. Mat'l			65				
27		Flow Direction	Flow Under Plug		66				
28		Type of Bonnet	Standard-UTP		67				
29		Packing Mat'l	Graphite		68				
30	TRIM	Type	Unbalanced - Top Guided		69	POSITIONER			
31		Cage Type	MF Plug		70				
32		Characteristic	Modified		71				
33		Travel / Stroke	25 mm		72				
34		Rated Cv	0.04		73				
35		Plug Mat'l	SS 440C Hardened		74	AIR SET			
36		Seat Ring Material	SS 440C Hardened		75				
37		Cage Mat'l	Integral with seat		76				
38		Stem Mat'l	A479 TY 316		77				
39	SPECIAL ACCESSORIES	1/2" NPT SS316 Double Compression Cable Glands offered for positioner (1 no.) 1/2" NPT (F) Adapter offered at AFR inlet w.r.t process connection			78	Range 0.70 - 7.00 Kg/cm2g Filter / Gauge Yes (5 Micron) / Yes Set Pressure 4 Barg			
40					79				
41					80	Mfr. / Model Quantity/Type Voltage			
42					81				
43					82	Mfr. / Model Contacts/Rating Type Actuation Points			
44					83				
45					84	Hydro Test Press. 77.57 bar[g] Seat Leakage ANSI/FCI 70-2 Class V Other See Factory Notes / QA Plan Req Stem Thrust: 190 kgf Press/Temp Tables: ASME/ANSI B16.34			
46					85				
47					86				
48					87				
49	LOC.	Equip. Location	Outdoor		88				
50		Amb. Temp. Min/Max			89				
51		Area Classification	Non-Hazardous						
52	Compliance	IBR							
*Equivalent material forms may be used for the trims.					Submittal by: Karthikeyan Sekar Company: CIRCOR Flow Technologies India Pvt Ltd Contact Info:				

Mandatory Spares List						
Valve Tag no.	Actuator Seal Kit	Trim Set	Actuator Set	Soft Parts (Gaskets & Gland Packing)	SMART Poz	AFR
TCV-A1-9802	1 Set	1 Set	1 no.	1 Set	2 no's Offered	2 no's Offered
TCV-A1-9806	1 Set	1 Set	1 no.	1 Set	x	x

x - not offered



☐ Short Delivery	☐ Only for Engg. Activities	Date	29.09.2025
☐ Bought out valves	☐ Others, Please Specify	CORB Reference	DM25066L
Customer / Buyer Name	LARSEN & TOUBRO LIMITED	CEC Quote Number with Date	250702-0400-PA Dtd 28.08.2025
Site / End User Name & Country	Rashtriya Chemicals and Fertilizers Limited, India	Application Engineer SPOC	Karthikeyan Sekar Satheshkumar Eagambaram
Buyer PO Number/Date	LTES/75001-83664/BS Dtd 03.09.2025	Outside Sales Engineer/Manager	Ashokvijay R
Buyer PO/LOI Received Date	12.09.2025	PO Value	INR 9,99,750
Agreed / Approved Payment Terms	Refer PO clause no. 5.0	1)PO Delivery Date 2)Agreed Delivery Date	26.03.2026
Agreed Incoterms /Shipping Terms	CPT, RCF Thal Project Site	Margin	22%
LD Clause	@0.5% of the undelivered portion of Order value	Agent Name & Commission	NA
Ship to Address with GST#:	Rashtriya Chemical and Fertilizers Limited Thal, Thal-Vaishet, Alibag # 402 208 State: Maharashtra, India GSTIN: 27AAACL0140PEZ6	Bill to Address with GST#:	Larsen & Toubro Limited Divison Name: L&T Energy Hydrocarbon 14th Floor, AMN Tower, AMN Tower, Jogeshwari - Vikroli Link Road, Powai Campus, Gate 3, Mumbai Suburban, Maharashtra, 400072

ENQUIRY / RFQ FROM CUSTOMER



Project No. E24003

Rashtriya Chemicals and Fertilizers Limited

1200 MTPD (DAP Basis) NPK Project Thal, Maharashtra (India)

Project Document Title:
Material Requisition for Desuperheater

Project Document Number:
E24003-RCF-IN-MRQ-042

L&T Contract Reference Number:
THAL-NPK-2301 DATED 10TH Oct. 2024

Customer : Rashtriya Chemicals and Fertilizers Limited
PMC : Projects & Development India Limited
LSTK Contractor : Larsen & Toubro Ltd.
Licenser : Incro, S.A.

25-June-25	0	Issued for Review			
			VVC	BRH	SBS
Date	Rev.	Description	Prepared By	Reviewed By	Approved By

Disclaimer:

This document contains information from L&T and its affiliated companies, which is confidential and/or proprietary in nature. No part of this document, data, or information, issued for the captioned project or purpose, shall be used for other than the purpose for which this has been provided and disclose to others or reproduced in any manner, except with the prior written permission of L&T.



1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

Issue and Approval Record

Issued Category					Approval Code				
☒	Engineering	☐	Procurement	☐	Construction	☐	[Code 1] - Approved Consent to proceed		
☐	Information	☐	Enquiry	☐	Construction	☐	[Code 2] - Comments As Noted Work may proceed subject to incorporation of comments		
☒	Review	☐	Purchase	☐	Commissioning	☐	[Code 3] - Rejected Revise and Resubmit. Work shall not proceed		
☐	Approval	☐	Void	☐	As Built	☐	[Code 4] - Information Only Accepted for information only		
Does issue category require approval?						Code:	Date:	Approver:	
☒	Yes	☐	No	☐	NA	Note: This approval does not relieve the document ORIGINATOR any of his obligations and responsibilities under this Contract.			



1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

Revision Record

Sr. No	Rev. No	Page No	Clause /Line No.	Brief Description / Reason of Revision	Remarks

Note:

- A 'vertical line in right-hand margin plus revised text in Blue colour' identifies changes in the latest revision of document.
- Sequential numerical revision philosophy i.e. 0,1,2,3 etc. to identify the document revision's purpose i.e. if issued for review/ approval/ enquiry/ information/ purchase/ construction/ commissioning/ Implementation etc.

Hold List

Hold No.	Hold Description



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

TABLE OF CONTENTS

1.0	Project Brief	5
2.0	Items to be Supplied	5
3.0	Standard Definitions	5
4.0	Codes & standards	6
5.0	Supplier compliance	8
6.0	Certifying Authority	8
7.0	Conflict	8
8.0	Scope of work and supply	8
9.0	Quality System Requirements	11
10.0	Exceptions & Deviations	12
11.0	Proven Track Record	12
12.0	Health Safety and Environment	12
13.0	Preservation, Packing & Shipping	13
14.0	General notes	14
15.0	Technical Requirements	14
16.0	Attachments	18



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

1.0 Project Brief

Rashtriya Chemicals & Fertilizers Limited (RCF) is a public sector undertaking involved in the manufacture and marketing of nitrogenous, phosphatic and potassic fertilizers as well as a variety of industrial chemical products. The company has presently two manufacturing units, one at Trombay and the other at Thal, both located in the state of Maharashtra.

RCF are recognized as one of the most successful integrated agricultural input brands in India. They are ranked amongst the largest fertilizer companies in India. RCF as a business development strategy is exploring for development of phosphatic fertilizer complex in India. In this regard, NPKs have been identified as the prospective fertilizer project by RCF. The outputs DAP/NPK from the fertilizer complex will cater to the local & nearby areas and bridge the demand & production gap in this region.

RCF intends to establish a new 1200 MTPD (DAP Basis) NPK Fertilizer plant along with associated utilities / off-sites facilities in Thal. The plant shall be designed based on DAP (18.46.0) production capacity. In addition to DAP, the plant shall also produce different NP/NPK grades like 20.20.0.13, 10.26.26 and 12.32.16 etc. primarily using Ammonia, Phosphoric Acid, Sulphuric Acid, Muriate of Potash and Urea as required.

M/s Rashtriya Chemicals & Fertilizers Limited (RCF) have entrusted Larsen & Toubro Ltd. for engineering, procurement, and construction work of the DAP project.

Projects & Development India Ltd. (PDIL) has been appointed as CONSULTANT for the aforesaid project.

2.0 Items to be Supplied.

The following items (material and services), required for Rashtriya Chemicals & Fertilizers Limited (RCF) Thal, shall be supplied in accordance with the requirement specified in this requisition and documents attached / referred herewith and mentioned in various attachments.

Sr. No.	Description	Quantity	Remarks
1.	Desuperheater (spray nozzle with Flange)	02	-

3.0 Standard Definitions

Customer	Rashtriya Chemicals and Fertilizers Limited
PMC	Projects & Development India Ltd. (PDIL)
Process Licensor	Incro, S.A.
EPC Contractor	Larsen & Toubro Limited (L&T) is the party which carries out all or part of the design, engineering, procurement, construction, or management of the project.
Purchaser	L&T and its authorized representative assigned for this project by L&T.
Supplier	A party responsible for the supply of equipment, materials or product and related services in accordance with this requisition and its referenced standards and data sheets.
Requisition	This document along with all 'Tables', 'Annexures' and 'Attachments' as attached.
EPC	Engineering Procurement Construction



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

4.0 Codes & standards

- Design, engineering, materials, manufacturing, painting, inspection, testing, preservation, packing, forwarding and documentation shall conform to the codes & standards as mentioned in attached specifications and attachments / data sheets or any other referred documents.
- Wherever edition is not mentioned, latest edition prior to inquiry shall be followed after confirmation from Purchaser. In addition, any requirement as per Governmental Regulations, Local Laws & Regulations & requirement from statutory bodies shall be complied by Supplier. It is assumed that Supplier is in possession of latest edition of applicable codes and standards.
- Where neither code nor standard is specified in Specifications / Attachment, Supplier shall propose applicable codes and standard on which Supplier's design and manufacturing is based. In any case, Supplier must furnish a list of codes and standards other than those specified in this document, which shall be followed by them during engineering for review and approval by Purchaser.

All Valves shall be supplied in accordance with the code requirements mentioned in table below.

API RP 553	Refinery Valves and Accessories for Control and Safety Instrumented Systems
API 609	Butterfly valves: Double Flanged, Lug and Wafer type
API 6D	Pipeline Valves
API 6FA	Specification for Fire Test for Valves
ASME B 1.20.1	Pipe Threads General Purpose (Inch)
ASME B 16.5	Pipe Flanges and Flanged Fittings NPS ½ through NPS 24 Metric/Inch Standard
ASME B 16.20	Metallic Gaskets for pipe Flanges Ring-Joint, Spiral-Wound and Jacketed
ASME B 16.34	Valves-Flanged, Threaded and Welding End
ASME/BPVC-VIII-1	Section VIII, Division 1: Rules for Construction of Pressure Vessels
ASME B 16.10	Face-to-Face and End-to-End Dimensions of valves
ASME B 16.47	Large Diameter Steel Flanges NPS 26 through NPS 60 Metric/Inch Standard
ASME B 40.100	Pressure gauges and Gauge attachments
ANSI FCI 70-2	Control Valve Seat Leakage
IEC 60079-0	Explosive Atmosphere-Part 0: Equipment-General Requirements
IEC 60529	Degree of Protection provided by enclosures (IP code)
IEC 60534-2-1	Industrial Process Control Valves-Part 2-1: Flow Capacity-Sizing Equations for Fluid flow under Installed Conditions
IEC 60534-2-3	Industrial Process Control Valves-Part 2-3: Flow Capacity-Testing Procedures
IEC 60534-2-4	Industrial Process Control Valves-Part 2-4: Flow Capacity-Inherent Flow Characteristics and rangeability
IEC 60534-2-5	Industrial Process Control Valves-Part 2-5: Flow Capacity-Sizing equations for fluid flow through multistage control valve with Inter stage Recovery
IEC 60534.3	Industrial Process Control Valves -Face to Face dimensions (Part 1 to 3)
IEC 60534.8	Industrial Process Control Valves-Noise considerations (Part 1 to 4)
IEC 60801	Electromagnetic compatibility for Industrial Process Measurement and control equipment Part 1: General Introduction
IEC 61000-4	Electromagnetic compatibility (EMC)-Part 4: Testing and Measurement Techniques Set (contains 30 sections)



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

IEC 61000-4-1	Electromagnetic compatibility (EMC)-Part 4-1: Testing and Measurement Techniques-Overview
IEC 61000-4-2	Electromagnetic compatibility (EMC)-Part 4-2: Electrostatic Discharge Immunity Tests
IEC 61000-4-3	Electromagnetic compatibility (EMC)-Part 4-3: Testing and Measurement Techniques-radiated, radio frequency, Electrostatic Field Immunity Tests
IEC 61000-4-4	Electromagnetic compatibility (EMC)-Part 4-4: Testing and Measurement Techniques-Electrical Fast Transient/Burst Immunity Tests
IEC 61000-4-5	Electromagnetic compatibility (EMC)-Part 4-5: Testing and measurement Techniques-Surge Immunity Test
IEC 61508	Functional Safety of electrical / electronic programmable electronic safety related systems (contains 7 sections)
IEC 61511	Functional Safety of Safety Instrumented Systems for the Process Industry Sector (Part 1 to 3)
IS 13947	Degrees of Protection provided by enclosures for low voltage switch gear and control gear
IS 2148	(IEC 60079-1) Specification for Flame proof enclosures of electrical apparatus
ISA 5.1	Instrumentation Symbols & Identification
ISA 7.0.01	Quality Standard for Instrument
ISA 75.01	Industrial Process Control Valves-Part 2-1: Flow Capacity-Sizing Equations for Fluid flow under Installed Conditions
ISA 75.08.01	Face-to-Face Dimensions for Integral Flanged Globe-Style Control valve bodies
ISA 75.08.02	Face-to-Face Dimensions for Flanged and Flangeless Rotary Control Valves (Classes 150, 300, and 600, and PN 10, PN 16, PN 25, PN 40, PN 63 and PN 100)
ISO 5208	Industrial valves-Pressure testing of metallic valves
IEEE C37.90.1	Surge Withstand Capability (SWC) Tests for Relays and Relay Systems associated with electronic power apparatus
PESO	Petroleum and Explosives Safety Organization
RPCB	Rajasthan State Pollution Control Board
OISD-STD-105 to 245	Safety Instrumentation for Process system in Hydrocarbon Industry – allapplicable standards
ISA S7.0	Quality Standard for Instrument Air
ANSI B16.5	Pipe Flanges and Flange Fittings
API 598	Valve Inspection and Testing
API 6FA	Standard for Fire Test for Valves
ISO 10497	Fire Type testing requirements
BS 6 BS 67455 PART2	Testing of Valves – Part 2
ISO 10423	Petroleum and natural gas industries — Drilling and production equipment — Wellhead and Christmas tree equipment
DIN 19234 (NAMUR)	Measurement And Control – Electrical Distance Sensors – Dc Interface For Distance Sensor And Signal Converter



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

5.0 Supplier compliance

Supplier shall submit technical compliance statement, attached with this Requisition, duly filled in along with the offer (Refer **Annexure - 2**).

6.0 Certifying Authority

All approval certificates and Degree of protection Certificate for enclosure shall be provided.

Hardware / Electronic components shall be certified as per relevant international standards like IEC, BS, API etc.

Wherever applicable as per piping class, IBR certification to be considered as indicated in relevant datasheets.

Statutory approval certificate ATEX / IECEx and PESO shall be provided for all electronic items. PESO certification is mandatory and instruments from outside India shall also carry ATEX / IECEx certificate including PESO certification.

Copy of certificate from the competent authority to be provided by the supplier along with the offer.

7.0 Conflict

The requisition shall be read in conjunction with all other specifications and documents attached. During bidding stage, any conflict between the documents, if any, shall be identified in the offer for Purchaser's resolution.

In the event of any conflict actual or implied, among the documents, the most stringent requirements shall be applicable. However, Supplier shall refer back to Purchaser for final decision.

During order execution stage, In the event of conflict, actual or implied, amongst documents relating to this Requisition or enquiry or order, the more stringent requirement(s) shall apply. Supplier shall identify and state such conflict(s) in the course of engineering / manufacturing to obtain Purchaser's concurrence.

Similarly, if there is a conflict between the printed enclosures / catalogues / specification and other statements of Supplier's quotation, superior specification advantageous to the Project shall prevail.

The Project Specifications and Standards are primarily prepared on the basis of Indian regulations, codes and standards. Where Indian codes are not applicable, international American /European/ British /other codes will be utilized accordingly after Purchaser / Customer approval. In any areas of conflict, the order of precedence will be as follows (commencing with the highest):

- Statutory Requirements
- Licensor Documents / Project Specification, Tender Documents, Amendments, TQs
- Instrument datasheet
- This Requisition / Scope
- Standard Specifications / Other specifications (if any)
- Other referred codes and standards

Project specific Engineering Standards/Specifications, as well as Licensor standards, shall be used where these are required to supplement International Standards and Indian National Standards.

8.0 Scope of work and supply

This shall be read with requirement specified in requisition (Including attachments and associated referred documents). Any additional services & materials deemed required from Safety, Operability, and Maintainability aspects should be discussed and agreed with Purchaser before order placement.

It is Supplier's responsibility to verify the selection of type of item, material of construction of each component as per the data mentioned in individual specifications / data sheets / requisition. Supplier to suggest an alternative, in case not found suitable. Supplier shall stand guarantees for all items supplied by them, including their bought-out items. Supplier shall be completely responsible to supply all required materials and services, including that of



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

sub-supplier for satisfying the functional / operational requirements stated in this specifications and Material Requisition.

Items not indicated clearly in the Supplier's offer but required for technical requirements shall be considered as included in the Supplier's scope of supply. Any deviation proposed by any of sub-supplier shall not be considered. Such items shall be completed by Supplier on his full responsibility.

The Supplier's scope of work includes, but not limited to:

Design and Engineering

- (√) Manufacturing
- (√) Supply of materials as specified in the approved requisition
- (√) Acceptance tests and inspection of ex-factory items
- (√) Packing and Forwarding
- (√) Painting, as applicable
- (√) Test Report
- (√) Supply of spares as specified
- (√) Shop Witness Inspection by purchaser / representative
- (√) Pre-commissioning and commissioning spares
- (√) Special tools, if any
- (√) Documentation: The supplier shall supply all the documents as per attached SDR as minimum
- () Start-up, Testing, Pre-Commissioning and Commissioning including checking, etc.

Where,

- (√) means supplier's scope of work / supply
- () excluded from supplier scope of work / supply

Supplier to suggest an alternative, in case the same is not found suitable. Supplier shall stand guarantees for all items supplied by them, including their bought-out items.

Supplier shall note that, in no case their scope of work / supply shall be limited to content mentioned. All agreements as noted in the contractual documentation as mentioned in Purchase order (PO) shall in any case prevail.

8.1. Spare Parts

Spare parts shall be packed for long storage under tropical climatic conditions in suitable cases, clearly marked as to their intended purpose.

8.2. Mandatory Spares (Consumption within 2 years)

Supplier shall supply mandatory spares per attached spare parts list (attachment 12) however, supplier shall recommend and supply Mandatory spares require for smooth start-up and operation of the plant/package system. Mandatory spares list with quantities shall be considered in base scope as separate line item.



1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

8.3. Spare Parts for Erection and Commissioning

Supplier shall supply all spare parts required during erection, construction and commissioning of the item, which shall include any parts requirement replacement, fragile parts, construction allowance for bulk material, consumables including mandatory spares and special tools if any, etc. based on past experience and as per their recommendation.

The list of such recommended spares shall be provided with the offer. Supplier shall include all these spares in the base offer. Supplier shall supply & furnish erection & commissioning spares as per **Annexure-4**.

8.4. List of 2 Years Operational & Maintenance Spares

Supplier is required to submit recommended List of 2 years Operation and Maintenance spares. These spares will not be considered for price evaluation.

8.5. Consumables

Supplier shall submit list of consumables indicating quantities and basis in their offer. If no consumables are envisaged for the items under the requisition, then supplier to mention the same in their offer.

8.6. Special Tools & Tackles

Supplier shall supply the necessary special tool for installation, calibration & maintenance at site that are not commercially available. Details of special tools along with the specific documents / procedure shall be submitted along with base offer. Tool name, Tool number & Quantity shall be indicated with unit price in the Supplier's offer. For special tools & tackles format, refer **Annexure - 3**.

8.7. Inspection and Testing

Supplier shall submit his QA plan of manufacturing, testing and inspection for purchaser's review / approval.

To fulfil the quality assurance requirement, Supplier shall accept free access of Purchaser / Customer /Licensor or their representative at any time until shipment. Shop inspection and tests shall be carried out in accordance with the requirement specified in this Requisition, its attachments, applicable codes and standards.

Supplier shall prepare project specific Inspection Test Plan (ITP) / Quality Assurance Plan (QAP), which shall be in line with project quality assurance plan requirements as specified in these MR & attached specifications, for Purchaser / Customer's approval. Supplier shall also submit related QC procedure like Acceptance Test Procedure for approval by Purchaser/Customer. For reference, Indicative Inspection & test Plan (IITP) is attached with this MR. Supplier shall carry out test and inspection as per approved ITP/QAP. Purchaser's representative / Third Party Inspection Agency / Customer shall join inspection as per approved ITP / QAP.

Inspection and/or witnessing of tests by Purchaser / Purchaser's representative / Third Party Inspection Agency / Customer shall not relieve Supplier of his responsibility for material, quality workmanship and others.

Refer List of attachments: "Quality & Inspection" attachments for detailed QA/QC requirements for this project.

8.8. Supervision during Erection / Installation / Pre-commissioning / Commissioning/Start-up training.

Supplier shall provide services for site supervision for installation, pre-commissioning, start-up and commissioning, site testing, guarantee runs, training etc. as per agreed per diem rates. Supplier shall provide rates for engineer and technician separately.

8.9. Statutory Approval & Certification

Supplier shall be responsible for obtaining all statutory approvals and provide certificate, as specified elsewhere in attached documents of this requisition in line with project specific requirement.



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

Refer List of attachments: "Quality & Inspection" attachments for detailed QA/QC requirements for this project.

8.10. Data & Documentation Requirements

Supplier shall submit data & documents required for post order stage and final documents as per **Annexure-2_Supplier's Compliance Sheet**.

- All bought-out items shall be as per "Approved Supplier / sub-supplier List" attached. Compliance to procurement of material from approved sub-suppliers is mandatory and shall be fully complied with. Acceptance of any deviation from approved Supplier list shall be subject to Purchaser's approval.
- For items not covered in the above list, it is Supplier's responsibility to get such proposed supplier list approved by Purchaser prior to placement of order by Supplier. Supplier shall list down the proposed sub-suppliers for such items and submit the same for Purchaser's approval with necessary documents. Non-acceptance of a particular proposed sub-supplier due to any reasons whatsoever shall not be a cause of schedule and cost implication.
- At any stage, if it comes to the notice of Purchaser that Supplier has procured material / equipment, intentionally or otherwise, from an unapproved sub-supplier and / or items not falling in approved range of sub-supplier, the same shall be rejected forthwith and Supplier shall be liable to replace such material / equipment without any schedule and cost implication to Purchaser.
- It is understood that should the name of sub-suppliers be changed due to change in their Company or Corporate shareholding, Purchaser will accept such Suppliers / sub-suppliers under its new name with prior approval. Any such approval shall, however, not release the Supplier from any of his obligation under the contract; neither shall any such approval signify instruction to use such a sub-supplier. All approved sub-suppliers are deemed to have been freely chosen by the Supplier at his own risk.
- It shall be the responsibility of Supplier to check that the selected models of the item shall meet the performance track record requirements as stated in the Specifications.
- Supplier shall make an independent assessment of capability of all the sub-suppliers for timely deliveries of material / equipment. Any delays in deliveries by sub-supplier(s) shall not be a cause of schedule and cost implication.
- For sub-suppliers included in each category, it will be Supplier's responsibility to ensure that item being ordered on approved sub-supplier has been manufactured and supplied by them (range, thickness, size, capacity, material etc.) in past similar projects or any reputed Fertilizer / Chemical / Hydrocarbon Sector under TPI.

8.11. Guarantee

Guarantee from Supplier shall cover mechanical workmanship and performance of the complete package including their bought-out items. The performance guarantee shall be as specified in the data sheets, project specifications, applicable codes and standards referred in this Requisition or other referred documents.

9.0 Quality System Requirements

- The Supplier shall operate a quality management system in-line with latest version of ISO 9001:2015 or an equally recognized QA Standard, commensurate with the type of goods and scope of services provided. Supplier shall submit valid copy of ISO 9001 certificate or other equivalent standards or quality manual or copy of quality policy signed by Senior Executive along with offer.
- The Purchaser, or purchaser's appointed representative, reserves the right to carry out quality audits at the supplier and sub-supplier's office and workshop, to verify adherence to quality system.



1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

- c. Supplier shall establish suitable control mechanisms to ensure quality in all work phases right from design and development to final delivery. All necessary quality records shall be maintained for evidence of performing the works as per Procedures.
- d. Necessary work procedures, instructions and qualified personnel shall be made available at all work Processes.
- e. Whenever non-conformity is detected by Supplier or by others, Supplier shall ensure to segregate the non-conformity product to prevent unintended use, and suitable corrective action shall be taken without any delay.
- f. Supplier shall nominate Project Quality Manager.

10.0 Exceptions & Deviations

Supplier shall submit a consolidated Exception, Deviations & Clarifications, if any, against Requisition and its attachments in the format (Refer **ANNEXURE –1**) only. Deviation and exception written elsewhere in the offer shall not be considered as deviation. It will be construed that Supplier confirm in totality to the requirement specified in the Requisition. Also, Supplier shall very clearly categorize all Clarification, Exception and Deviation under following categories:

- a. **Category A:** Deviations as per Manufacturer's Standards design and cannot be withdrawn in any circumstances including additional payment or time implication.
- b. **Category B:** Deviations which have been proposed as technical alternative by Supplier. If not accepted by Purchaser / Customer, Supplier shall follow the Requisition requirement / Specifications without any cost and time implications to Purchaser.
- c. **Category C:** Deviations having either 'time' or 'cost' or 'time & cost' implications. If not accepted by Purchaser / Customer, Supplier shall inform implications for compliance to Requisition requirement / Specifications.
- d. **Category D:** Not a deviation against any Requisition requirements. Supplier's clarification or basis for the offer.

11.0 Proven Track Record

Supplier shall select equipment / item very carefully to meet the specified requirement ensuring proven track record of satisfactory performance of selected equipment / item for similar service, meeting proven track record / experience record requirements of project specifications as applicable. Supplier shall not offer prototype equipment / item or special design, which was not used in the past successfully. Supplier shall revise or replace equipment / item at any stage of the project, if Proven Track Record submitted by Supplier does not meet the criteria or is not sufficient.

12.0 Health Safety and Environment

- a. Supplier Products/Supplies/Services shall comply relevant applicable statutory obligations and latest codes & standards and contract obligations. Supplier shall submit the statutory certifications from authorities applicable for the supplies/services covered under this requisition.
- b. Supplier should have a Health, Safety and Environmental management system in-line with ISO 45001:2018 and ISO 14001:2015 or any other equivalent standards. Supplier shall submit valid copy of ISO 45001 and ISO 14001 certificate or other equivalent standards or HSE manual or copy of HSE Policy Statements or Statement that works in accordance with principals of ISO 45001 and ISO 14001 Standard along with offer.



1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

- c. Supplier shall indicate if any specific/potential health, safety, security and environmental aspects & hazards and suggest its mitigation plan related to the equipment installation / operation / maintenance / loading / unloading / handling / removal activities.
- d. Supplier shall submit relevant procedure for preservation and protection during shipping / loading / unloading / construction/commissioning.
- e. The Purchaser, or appointed representative, reserves the right to carry out audits at the supplier and sub-supplier's office and workshop, to verify adherence to Health, Safety and Environmental system.

13.0 Preservation, Packing & Shipping

- a. Supplier shall submit the procedure to protect and preserve the items being quoted during transportation and storage at the construction site. Adequate packing shall be provided to protect against rust, corrosion and damage during transportation and storage.
- b. Supplier shall submit preliminary shipping / packing list indicating Number of boxes, estimated size of package, volume, weight, tag number etc.
- c. Supplier shall take Protection and Preservation measures for the items supplied by them in accordance with their specification considering a storage period of Twelve months from date of shipping.
- d. All moving parts shall be suitably locked to arrest movement during transportation, with proper identification. All ports and openings shall be sealed. Threaded connection shall be plugged with caps or plug. It is Supplier's responsibility to ensure all items are adequately protected during shipment.
- e. Suppliers shall meet all statutory obligations during transportation and storage.
- f. The preparation for shipment of the SUPPLIER shall include, but not be limited to, the following -
 - i. Goods shall not be packed before the issue of purchaser's "Inspection Release Note", authorizing packing/transport or Purchaser's waiver of inspection. Inspection Release Note shall only be issued after issue of Engineering & Quality Clearance form purchaser. Format of same shall be shared after placement of order.
 - ii. Unless otherwise specified, the packing itself shall protect materials/equipment from shocks and weathering, environment, and marine agents, preventing any damage and corrosion of its contents and loss during transport. In addition, the packing and marking must withstand the various tough handling operations, from withdrawal up to the delivery point.
 - iii. In principle, protection of contents and relevant marking must not only be guaranteed for the period of transport, but also for a period of at least one year of outdoor storage under atmospheric conditions at the destination point. Exceptions shall be examined case by case.
 - iv. SUPPLIER shall be responsible for making complete arrangement for transportation including sea-worthy export packing as applicable.
 - v. Supplier shall thoroughly clean and dry all unpainted machinery components, both internally and externally. Dirt, grease, scale, welding spatter and any other foreign matter shall be removed.
 - vi. Metal gaskets, rings, bolts, and nuts shall be greased.
 - vii. Connections and in general all openings shall be adequately protected and plugged to prevent moisture infiltration. Machined / unmachined pipe ends and threaded openings shall be closed and protected by metal plugs of material equal or superior to that of the equipment itself.
 - viii. All packed goods must be properly secured inside the packing to prevent movement during transit.



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

- ix. For material with protective coating, supplier shall indicate the duration of the protection and shall also give instructions for a prolonged period of preservation of the goods.
- x. Materials belonging to the same purchase order, but with different destinations or delivery points, must be packed separately.
- xi. Spare parts shall be packed separately from materials belonging to the main order and per type of equipment, unless otherwise agreed with purchaser.
- xii. All packing material used shall be biologically decomposable. Packing material which endangers or pollutes the environment is prohibited.
- xiii. All individual items included in a single packing unit are to be marked separately.
- xiv. Each package shall be clearly marked in accordance with these requirements.
- xv. Each package mark shall be stencilled with indelible paint on the packing and shall not have any detrimental effect on the parts. Wood/metal plate markings may only be used if authorized by purchaser.
- xvi. Supplier shall include preservation and protection of equipment suitable for long term outdoor storage. Supplier shall identify any special requirements for maintaining the effectiveness of their applied preservation measures.

14.0 General notes

- i. Supplier shall submit all Catalogues / Manuals / Technical literature / a copy of certificates or approvals or data for all the items being supplied including their sub-suppliers.
- ii. Supplier shall specify their local support capability in India. This includes the spare parts inventory and maintenance capabilities.
- iii. Supplier shall comply with the document control list format, the specification format, title block requirements, comment resolution sheet, etc. as per Purchaser's requirements.
- iv. Supplier shall undertake to provide all their outputs in required number of hard copies and as soft copies in the specified software on DVDs and Pen drive. MS Office and AutoCAD is some of the stipulated software. Wherever the output from Supplier is not a generated output, then Supplier shall provide the scanned output in the form of soft file also. The operating manuals and installation manuals shall be provided as soft copies on a DVD. All documentation shall be provided in English language only.

15.0 Technical Requirements

15.1. Control Valve

I. General Requirements

- a. Valve Body Material shall conform to the material specified in the enclosed data sheets. However, it is supplier's responsibility to select the material, which is suitable for pressure, temperature, corrosion, erosion etc. The type of actuators applicable as specified in the individual data sheets. Supply of type of accessories shall be as listed and specified in the individual data sheets.
- b. For all intrinsically safe, explosion proof/flameproof, equipment/ instruments which are manufactured abroad and certified by any statutory authority like ATEX, CMRI, etc. should also have the approval of Petroleum and Explosives Safety Organization (PESO)/ Chief Controller of Explosives (CCOE), Nagpur.



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

- c. Copper, Zinc and their alloys, Fluoroelastomers (FKM) shall not be used for manufacture of any valve internals or parts exposed to atmosphere, especially for Ammonia and Urea services.
- d. All valves specified for Steam, Steam condensate and Boiler Feed Water services shall comply to Indian Boiler Regulation requirements. A certificate to this effect shall from IBR shall be furnished. For valves of imported origin Lloyd's certificate in IBR form III is accepted.
- e. Valve design, body pressure temperature rating and minimum wall thickness shall comply with ASME B16.34.
- f. valve accessories like Filter regulator, Positioner, Solenoid, Booster relay etc. also shall have pneumatic connection 1/2" in size. All accessories shall be mounted on valve body.
- g. Proper canopy shall be provided for all accessories like, Limit Switch, SOV, AFR, Positioner Material of canopy shall be FRP/SS304.
- h. Supplier shall furnish appropriate technical literature, catalogues and drawings showing features, specification, dimensional details, etc.
- i. Valves shall be provided with anti-static devices to achieve electrical continuity between the ball, stem & the valve body
- j. Control valve body, bonnet, bottom flange, line flanges and other pressure containing assemblies shall be of the same material as specified for valve body.
- k. Bonnet flange and bottom flange shall have metallic spiral wound gaskets suitable for the specified service. Gaskets with asbestos bearing fillers shall not be used.
- l. Manual handwheel shall be provided at the side of each actuator wherever specified in the individual specification sheet.

II. Sizing

- a. Sizing of control valves shall be as per ISA-S75-01.
- b. Cv of the valves shall be selected in such a way that the normal maximum flow is attained at about 70% valve opening. The control valve % opening shall be at minimum flow 10-20%, for normal flow 50 to 70%, for maximum flow 75 to 85%.
- c. Where noise level is more than 85 dBA for valve, control valve sizing shall take into account any noise attenuating devices, such as special trim like low db. and diffusers, which may be considered to reduce the noise level within specified limits. The diffuser plates shall be wafering design for easy removal for maintenance. Source treatment for noise shall be preferred over path treatment.
- d. Body size of control valves should be one step lower than pipe size for proper controllability when valves selected for controlling application.
- e. The fluid velocity at the trim outlet shall not exceed 6 m/sec for liquids whereas the velocity of gas or vapour shall not normally exceed 0.3 Mach under operating conditions. In order to reduce the fluid velocity below such limits use can be made of widely-sized valves equipped with reduced trims. Fluid velocity values could exceed the above-mentioned limits in the case of valves provided with a labyrinth plug, cage trim or angle body.

III. Body Material, Connections and Ratings

- a. Valve end connection shall be ASME B16.5. If not specified otherwise, flange connections type shall conform to ASME B 16.5. Face to Face dimension shall be as per ASME B16.10. Flange facing shall be RF with concentrated serrated grooving up to 600 lbs rating and above shall have ring joint facing suitable for octagonal soft metal gaskets as per ASME B16.20.



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

b. Supplier may offer different materials only if they are to a higher specification than those defined on data sheets, subject to Detail Engineering Contractor's / Owner's approval.

IV. Bonnet

Bonnets shall be bolted according to ASME B31.3 with minimum 4 bolts and material shall be the same as the body material. The requirement for an extended bonnet depends on the fluid temperature and the selected packing material. However, it shall be supplied as follows:

- a. Finned/ extended type bonnet shall be provided where service temperature exceeds 200-degree C.
- b. Extended type bonnet shall be provided in general for operating temperature below 0 Degree C.
- c. Standard where operating temperature is between 0°C to 200°C.

The bonnet material shall be same as valve body material,

V. Packing

a. PTFE shall be used as standard packing material for standard bonnet temperatures below 200°C and graphite for higher temperatures. Higher temperatures can be accepted for PTFE if the bonnet is extended. Packing design and material shall be selected carefully for minimum stem friction and live-loading packing boxes shall be considered for PTFE packing.

b. Stem/Shaft material shall be suitable for continuous operation under the process design temperature and process design pressure conditions and shall be compatible with the process fluid.

c. Viton as gasket and packing shall not be used.

d. Gland Packing shall be live loaded high performance / low emission type. Fugitive emissions packing shall be provided for all valves.

e. Bellow seal shall be used where it is required to isolate the packing from the process fluid or where no leakage to atmosphere can be tolerated like toxic, explosive, precious, dangerous fluids. Packing shall not contain asbestos material.

VI. Actuator

a. All actuators shall be pneumatic operated, if not otherwise, specified in the individual technical specification sheets. In general, single acting diaphragm type actuators shall be preferred. However, piston actuators shall be offered only where diaphragm actuators are not suitable.

b. Springs shall be corrosion resistant enclosed type. The compression of the spring shall be adjustable.

c. Actuator diaphragm material shall be in general neoprene rubber.

d. Double-acting piston actuators with volume tank and other accessories ensure correct failure position are acceptable where high thrust is required. Whenever double acting spring-less type of actuator is unavoidable, all accessories like pilot valves, booster relays, non-return valve, pressure gauge, volume tank etc. shall be provided to ensure desired action on air failure. The volume tank shall be sized to stroke the valve fully for two complete cycles. This volume tank shall be supplied with all accessories like PG, safety valve, NRV etc.

e. Actuators shall be sized for a minimum operating air pressure of 4 Kg/ Cm²g and maximum air pressure of 5.5 Kg/ Cm²g.

f. Actuators shall be sized adequately to fully stroke the valve under the maximum differential pressure specified by the process requirements as indicated in instrument datasheet.



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

- g. Open-close operation time of the valve shall be against full shut off pressure.
- h. All the bolts and nuts used in the actuator, bonnet and other parts of the valve body shall be made of SS316.
- i. Internal pneumatic tubing and required tube fitting and accessories of valve provided by valve manufacturer, MOC shall be SS316 as minimum. For large size actuators any higher size of tube like ¾" OD or 1" OD etc. shall be used based on valve manufacturers recommendation to meet the stroke time of valves.
- j. The actuator shall be oversized to minimum 25% sizing.

VII. Air Filter Regulator

- a. The air filter regulators shall be of the reducing-relief valve type, with drainage facility and bolt adjustment provided with a locking facility, e.g. a locking nut, to prevent tampering.
- b. The air filter cartridges shall be of the rigid structure type to resist channeling, rupturing, shrinkage or distortion and shall have a mesh size of 5 micron or better.
- c. Air filter regulator shall be of minimum aluminum material with SS internals and metallic bowl. No PVC part shall be used.
- d. Glass bowl shall not be used. It shall be of manual drain type without any side indicator. The manual drain & regulating knob shall be of metallic construction. Filter regulators shall be mounted in the vertical position.
- E. Air filter regulators shall be fitted with 50mm dial size 316 SS pressure gauges indicating the output air supply pressure to the actuator and/or its components.
- f. Pneumatic connection size ½" NPT(F) for filter regulator shall be provided for customer air supply interface.

VIII. Electro-Pneumatic Smart Valve Positioner (HART Protocol latest revision)

- a. The electro-pneumatic valve positioners shall be intrinsic safe. flame proof for hazardous areas. The certification for hazardous area shall be as per Area classification and execution type specified in the individual specification sheets. The positioners shall be shocked and vibration resistant. The electro-pneumatic positioner shall be suitable for 2/3/4-way field adjustable split-range current input with 100% pneumatic output. Direct/ Reverse action shall be field adjustable. For easy verification of transducer and positioner calibration, transducer output gauge to the positioner shall be preferred. The positioner shall have minimum pressure cut-off feature for positioner protection. All Smart positioners with HART protocol shall have all the capabilities for configuration from remote locations.
- b. The environmental protection for the electro-pneumatic positioner shall be IP67/NEMA 4X. Positioner operating voltage shall be 24 VDC – 2 Wire.
- c. The Pneumatic portion of the electro-pneumatic positioner shall be Forced balance type pneumatic valve positioner with 304SS/ 316SS wetted parts and low air consumption. The positioner shall be suitable for single and double acting actuators. The electro-pneumatic positioner shall be supplied along with the valves and shall be mounted on the valve yoke, fully tubed, complete with input and output gauges and air filter regulator with output gauges. Internal pneumatic tubing of valve provided by valve manufacturer as per manufacturing std with SS 316 tubing shall be considered. This is minimum requirement for Air tubing. For large size actuators any higher size of tube like ¾" OD or 1" OD etc. shall be used based on valve manufacturers recommendation to meet the stroke time of valves. Material of air tubes and fitting shall be minimum SS316. Tube Fitting shall be double compression type of three-piece construction. Tube size in MM and in other sizes are not acceptable.
- d. The position feedback Transmitter shall be provided analog input as standard.
- e. Air lines and fittings shall be in stainless steel. Sizes shall be adequate for the stroking time required. Tubing shall be thin walled with an OD of not less than 6 mm. The larger valves require tubing with a larger diameter



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

- f. Smart positioner should have minimum two pressure gauges and MOC for PG should be minimum SS316. Smart positioner should have local push button and display for configuration, calibration and troubleshooting etc.
- g. Positioner shall be housed in min. Die Case Aluminum enclosure.

IX. Identification and Marking

Each control valve shall be fitted with a SS tag plate and shall contain the following information.

- Tag no.
- Manufacturer's name, serial no. and model
- Body and trim sizes
- Body rating and end connection
- Body and trim material
- Trim characteristics
- Air failure action and spring range
- Air supply pressure
- Actuator Model.
- Cv Value.
- Stem travel in MM
- Flow Direction

16.0 Attachments

A. List of Annexures

Annexure No.	Title	Rev
Annexure-01	Supplier's Exception Deviations & Clarifications	0
Annexure-02	Compliance Sheet to MR	0
Annexure-03	Special Tool & Tackles format	0
Annexure-04	SPIR Format	0
Annexure-04A	List of Commissioning Spares	0
Annexure-05	Sub-supplier List Format	0
Annexure-06	Provent Track Records Format	0
Annexure-07	Supplier Compliance sheet	0
Annexure-08	Desuperheater Dimension Table	0
Annexure-09	Supplier Document Register List	0



1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)

Material Requisition for Desuperheater

E24003-RCF-IN-MRQ-042

Rev.0

25-June-25

B. List of Attachments

Sr. No	Document No	Title	Rev
1	E24003-RCF-IN-DST-042	Datasheet for Desuperheater	0
2	E24003-RCF-PP-SPE-001	Piping Material Specification (PMS)	1
3	E24003-RCF-PP-SPE-002	Valve Material Specification (VMS)	0
4	E24003-RCF-EM-PRO-001	Painting Procedure	0
5	PC193/E-2/P-II/10.0	Spares Parts	4
6	AdVENT-STD-QC-ITP-050	ITP for Desuperheater	01
7	AdVENT-STD-QC-ITP-004	ITP for Control Valve	0
8	E24003-RCF-QC-GENERAL-01	General Requirements for QA/QC	0
9	E24003-RCF-QC-PRO-001	Inspection Methodology & Categorization Plan	2
10	E24003-RCF-QC-SPE-001	General Specification for Positive Material Identification (PMI)	0
11	E24003-RCF-QC-SPE-002	Scope of work for Third Party Inspection Agency	0



Project No. E24003

Rashtriya Chemicals and Fertilizers Limited

1200 MTPD (DAP Basis) NPK Project
Thal, Maharashtra (India)

Project Document Title:
DATASHEET FOR DESUPERHEATER

Project Document Number:
E24003-RCF-IN-DST-042

L&T Contract Reference Number:

THAL-NPK-2301, Dated 10th Oct'2024

Customer : Rashtriya Chemicals and Fertilizers Limited
PMC : Projects & Development India Limited
LSTK Contractor : Larsen & Toubro Limited
Licensor : Incro, S.A.

25-Jun-25	0	Issued for Review			
			VVC	BRH	SBS
Date	Rev.	Description	Prepared By	Checked BY	Approved By

Disclaimer:

This document contains confidential and/or proprietary information of L&T and its affiliated companies. No part of this document, data, or information, issued for the captioned project or purpose, shall be used for other than the purpose for which this has been provided or disclosed to others or reproduced in any manner, except with the prior written permission of L&T.



1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)

DATASHEET FOR DESUPERHEATER

E24003-RCF-IN-DST-042

Rev. 0

25-Jun-25

Issue and Approval Record

Issued Category

☒ Engineering	☐ Procurement	☐ Construction
☐ Information	☐ Enquiry	☐ Construction
☒ Review	☐ Purchase	☐ Commissioning
☐ Approval	☐ Void	☐ As-Built
☐ Design	☐ Other.....	

Approval Code

☐ Code 1 - Approved Consent to proceed
☐ Code 2- Comments As Noted Work may proceed subject to incorporation of comments
☐ Code 3 - Rejected Revise and Resubmit. Work shall not proceed
☐ Code 4 - Information Accepted for information only

Code: **Date** **Approver:**

Is issued Category required approval?

☒ Yes	☐ No	☐ NA
--	------------------------------------	------------------------------------

Note: This approval does not relieve the document ORIGINATOR any of his obligations and responsibilities under this Contract.



1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)

DATASHEET FOR DESUPERHEATER

E24003-RCF-IN-DST-042

Rev. 0

25-Jun-25

Revision Record

Rev.	Date	Prepared By	Checked By	Approved By	Revision Note/ Description
0	25-Jun-25	VVC	BRH	SBS	Issued for Review

Notes:

- **Highlighted cell in yellow colour** identifies changes in the latest revision of document.
- Sequential numerical revision philosophy i.e. 0,1,2,3 etc. to identify the document revision's purpose i.e. if issued for review/ approval/ enquiry/ information/ purchase/ construction/ commissioning/ Implementation etc.

Hold List

Hold No.	Hold Description
1	
2	
3	



1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)

DATASHEET FOR DESUPERHEATER

E24003-RCF-IN-DST-042

Rev. 0

25-Jun-25

S. NO.

PAGE NO.

1.0 GENERAL NOTES

5

2.0 DATASHEET

6



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

DATASHEET FOR DESUPERHEATER

E24003-RCF-IN-DST-042

Rev. 0

25-Jun-25

GENERAL NOTES:

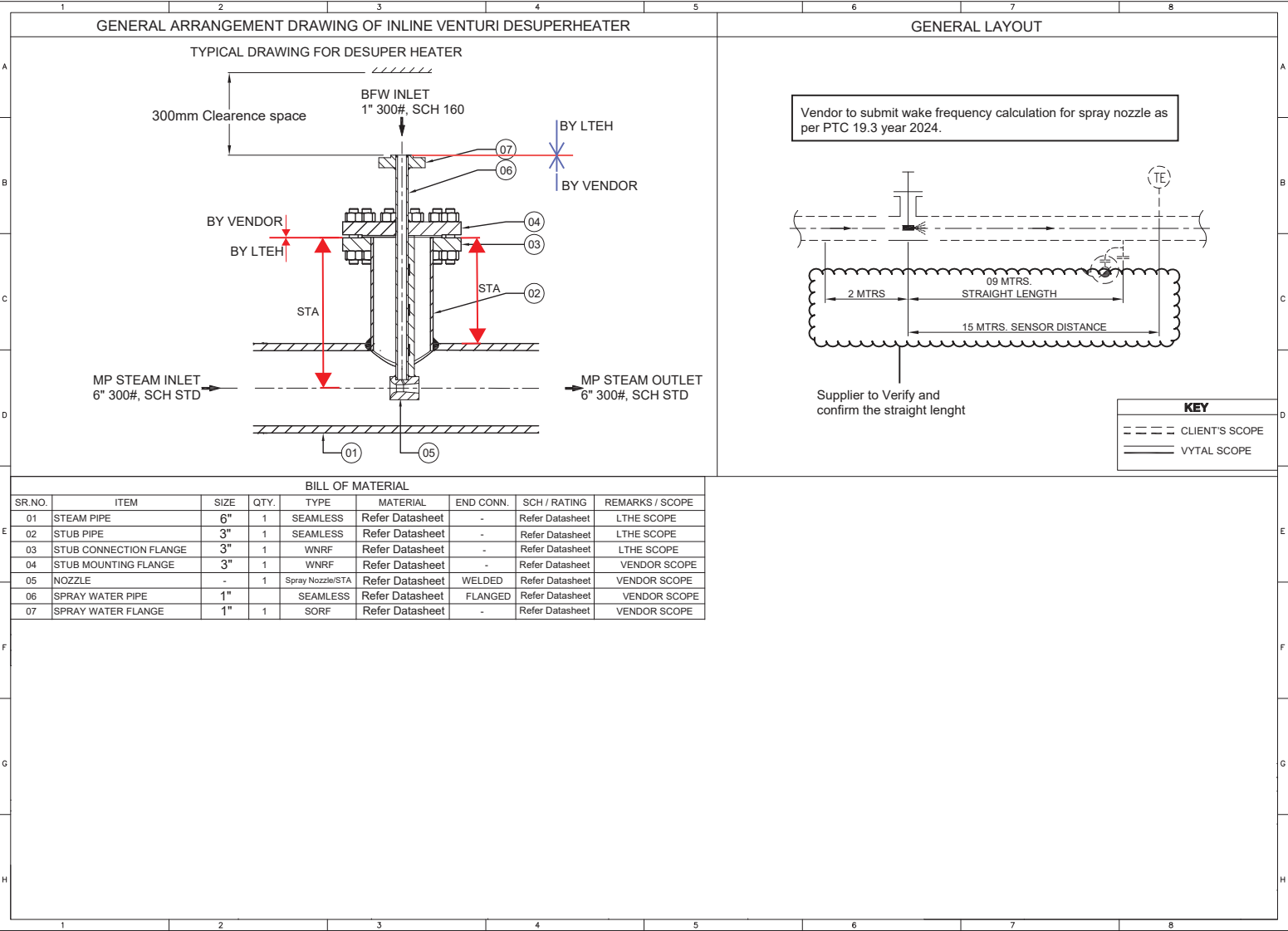
1	Desuperheater datasheet shall be read in conjunction with "MR for Desuperheater, Document Number: "E24003-RCF-IN-MRQ-032" and its all annexures and attachments.
2	Statutory approval certificate ATEX / IECEX and PESO shall be provided for all Electronic items.
3	All valves installed on lines under IBR (Indian Boiler Regulation) shall be certified by IBR and Supplier shall submit IBR Form IIIC.
4	Material offered by Supplier shall be equivalent or superior to the material specified in datasheet, PMS and VMS. It is Supplier's responsibility to offer material which is suitable for project specifications and application.
5	Flange shall be as per ASME B 16.5. Supplier to note that valve flange face finish shall be as per Piping Material specifications, "Document Number: "E24003-RCF-PP-SPE-001"". MOC of Flange shall be as per PMS.
6	Flow direction shall be clearly & permanently marked on the Desuperheater.
7	All required supporting brackets / accessories for the valve / actuator assembly shall be considered by supplier.
8	Accessories such as E/P positioner, Positioner Transmitter and AFR shall be supplied from approved vendor list.
9	Valve along with actuator and accessories shall be supplied assembled, tested and delivered as a single unit. All required supporting brackets / accessories for the valve / actuator assembly shall be considered by Supplier.
10	Lifting lug shall be provided based on size, rating and weight of equipment requirement.
11	Supplier to ensure that there are no dissimilar metal welding. In case of any due mechanical design of the supplier product. Supplier to highlight the same.
12	All Pneumatic Vent Ports (on Actuator and Accessories) shall be plugged with SS304 bug screen.
13	All electronic components shall have EMC immunity conforming to IEC-801.
14	SS 304 tag plate shall be provided on each valve & accessories. Stainless steel nameplates shall have details engraved on it.

GENERAL	1 Tag Number		P&ID No.		DS-9801			E24003-RCF-PC-PID-098						
	2 Service		Process Case		Medium Pressure Steam									
	3 Design Pressure Kg/cm ² g		Min.	Max.	Note-2			Note-2						
	4 Design Temperature °C		Min.	Max.	Note-2			Note-2						
	5 Air Supply Pressure Kg/cm ² g		Min.	Max.	NA			NA						
	6 Elec. Haz. Area Class. STD/Zone/Method/Gr./Temp.		-			-			Safe Area					
PROCESS CONDITIONS	7 Condition		Steam Inlet			Steam Outlet			Quenching Fluid					
	8 Fluid Name		Fluid State		Note-2	Note-2	Note-2	Note-2	Note-2	Note-2				
	9		Min.	Nor.	Max.	Min.	Nor.	Max.	Min.	Nor.	Max.			
	10 Composition													
	11 Flow Rate		Gen. Note 1		Note-2	Note-2	Note-2	Note-2	Note-2	Note-2	Note-2			
	12 Pressure		Kg/cm ² g		Note-2	Note-2	Note-2	Note-2	Note-2	Note-2	Note-2			
	13 Temperature		°C		Note-2	Note-2	Note-2	Note-2	Note-2	Note-2	Note-2			
	14 Mol. Wt. (Gas/Steam)													
	15 Viscosity		cP											
	16 Density (Liq. / Steam)		kg/m ³		Note-2									
	17 Vapour Pressure (Liq.)		Kg/cm ² a											
	18 Cp / Cv (Gas/Steam)		-											
	19 Z (Compressibility) (Gas/Steam)		-											
	20 Shut-off Press. Max.		% Solid											
	21 Critical Press. (Liq.)		% Flash											
	22 Outlet Vap. Mol. Wt.													
	23 Deg. Of Superheat													
	24 Allowable Pressure Loss		Kg/cm ²											
	DESUPERHEATER DATA	27 Cv @ Flow		Min.	Nor.	Max.	STA		STA		STA			
		28 % Opening @ Flow		Min.	Nor.	Max.	STA		STA		STA			
29 Noise @ Flow dBA		Min.	Nor.	Max.	STA		STA		STA					
30 Min Degree of Superheat Required														
31 Turndown Ratio														
32														
BODY & TRIM	33 Type of Desuperheater		Spray Nozzle / STA		LINE	59 Steam Inlet Line No.		150-STM1-A5-7614-3A1BC-H90						
	34 Stub Size		Flange Rating			3" / STA	300#	60 Size		Class	Sch	6"	3A1BC	STD
	35 Body/Bonnet Material		NA			61 Steam Inlet Line I.D.		mm						
	36 Corrosion Allowance		2 mm			62 Steam Outlet Line Number		150-STM1-A5-9801-3A1BC-H80						
	37 Bonnet Type		NA			63 Size		Class	Sch	6"	3A1BC	STD		
	38 Trim		Type	Size		NA	NA	64 Steam Outlet Line I.D.		mm				
	39 Trim Material		NA			65 BFW Inlet Line No.		25-BFW-A5-7621-3A1BC-H50						
	40 Packing Material		NA			66 Size		Class	Sch	1"	3A1BC	S160		
	41 Selected Cv		NA			67 BFW Inlet Line I.D.		mm						
	42 Characteristics		NA			68								
	43 Coolant Inlet Flange Mat.		ASTM A105			69								
	44 Coolant Inlet Pipe Material		ASTM A106 GR.B			70								
	45 Steam Line Flange Mat.		ASTM A105			71								
	46 Steam Line Pipe Material		ASTM A106 GR.B			72								
	47 Nozzle Material		SS316 + Stellite			73								
	48 Nozzle Seat Material		SS316 + Stellite			74								
	49 Soft Seating		NA			75								
	50 Soft Seating Material		NA			76								
	PROCESS CONNECTION	51 TSO		Leakage Class		NA	STA	PURCHASE	77 Manufacturer		STA			
		52							78 Desuperheater Model No.		STA			
53 Conn.		Size	Rating	6" / STA	300#	79 Spray Nozzle Model No.			STA					
54 steam side		Facing	Finish	RF	125-250 AARH	80 Requisition No.			E24003-RCF-IN-MRQ-032					
55 Conn.		Size	Rating	1"	300#	81								
56 water Side		Facing	Finish	RF	125-250 AARH	82								
PROC. NOTE	57 NACE Requirement		NA		FAB. NOTE	83 PWHT Requirement		Yes						
	58 IBR Requirement		Required			84 H2 Service		NA						
Notes: See notes														
					INSTRUMENT DATASHEET Desuperheater			 						
0	Issued for Review		24-Jun-25	VVC	BRH	SBS								
No.	Description		Date	Pred	Chk	App	Code:	Doc. No. E24003-RCF-IN-MRQ-042		Rev.:0				

Tag Number: DS-9801

1. STA: Supplier to Advise
2. Refer Annexure-1
3. BOLT.STUD + 2 NUTS- (HEAVY HEX) : BOLT: A193 GR.B7,NUT: A194 GR.2H.
4. Vendor to submit wake frequency calculation for spray nozzle as per PTC 19.3 year 2024.
5. Supplier to check and suggest that requirement of Thermal Liner where the difference between the coolant teperature and process fluid is greater than 250 degreeC.
6. Supplier to provide following minimum data in sizing calculations. · a. Process flow, pressure & temp. b. Coolant flow, pressure & temp. c. Minimum velocity requirement.
- d. Outlet condition & accuracy of desuperheating (accuracy with which outlet temp. should maintained above set point) e. Actual pressure drop. f. Minimum pipe straight run required for upstream & downstream. g. Minimum distance from desuperheater outlet to install temp. sensor. h. Drain requirement distance from desuperheater outlet.
7. Supplier to ensure that Coolant control valve rangeability shall be higher than PRDS rangeability.

						INSTRUMENT DATASHEET Desuperheater			
0	Issued for Review	24-Jun-25	VVC	BRH	SBS				
No.	Description	Date	Pred	Chk	App	Code:	Doc. No.E24003-RCF-IN-MRQ-042		Rev.:0



Annexure - 01

  			
1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)		Process Datasheet for PRDS - MP Steam	
E24003-RCF-PC-DST-122		Rev. 1	19-Jun-2025

Rev.	1. GENERAL INFORMATION			
	2. ITEM NO.	A1-U-01		
	3. DESCRIPTION	PRDS - IMP Steam		
1	4. P&ID NUMBER	E24003-RCF-PC-PID-098		
	5. LOCATION	OUTDOOR		
	6. SERVICE	STEAM		
	7. QUANTITY	ONE		
	8.			
	9. OPERATING CONDITIONS			
	10. OPERATING CASES	Maximum	Normal	Minimum
	11. INLET STEAM			
	12. OPERATING PRESSURE	kg/cm ² G	14	14
	13. OPERATING TEMPERATURE (NOR / MAX)	°C	420	420
1	14. FLUID DENSITY	kg/m ³	4.69	4.69
1	15. FLOWRATE	kg/hr	7262 (VTA)	4700 (VTA)
1	16. BOILER FEED WATER			1410 (VTA)
1	17. OPERATING PRESSURE (NOTE 6)	kg/cm ² G	16	16
	18. OPERATING TEMPERATURE (NOR / MAX)	°C	115	115
1	19. FLUID DENSITY	kg/m ³	949	949
1	20. FLOWRATE	kg/hr	1670 (VTA)	1081 (VTA)
	21. OUTLET STEAM			324 (VTA)
	22. OPERATING PRESSURE	kg/cm ² G	8	8
	23. OPERATING TEMPERATURE (NOR / MAX)	°C	175.4	175.4
	24. FLUID DENSITY	kg/m ³	4.5	4.5
1	25. FLOWRATE	kg/hr	8932	5781
	26.			1734
	27. DESIGN INFORMATION			
	28. DESIGN PRESSURE (MAX / MIN)	kg/cm ² G	18.15 / FV	
1	29. DESIGN TEMPERATURE	°C	420	
	30. ALLOWABLE PRESSURE DROP (MAX/NOR/MIN)	kg/cm ²	VTA	VTA
	31. INLET STEAM			
	32. LINE SIZE	inches	6"	
	33. RATING	#	300#	
	34. QUENCH DM WATER			
1	35. DESIGN PRESSURE	kg/cm ² G	18 / FV	
	36. DESIGN TEMPERATURE	°C	155	
	37. LINE SIZE	inches	1" (VTA)	
1	38. RATING	#	300#	
	39. OUTLET STEAM			
	40. LINE SIZE	inches	8"	
	41. RATING	#	300#	
	42.			
	43. MATERIAL			
	44. SHELL	CARBON STEEL		
1	45. CORROSION ALLOWANCE	1.5 MM		
	46. INTERNALS	-		
	47. LINING / CLADDING	-		
	48. HEATING COIL / JACKET	-		
	49. FIRE PROOFING	☐ YES ☒ NO		
	50.			
	51. NOTES			
	52.	1) Vendor to specify the type of desuperheating system i.e. mechanical spray nozzle, variable orifice, multi-nozzle spray, steam atomizing or any other type of desuperheater.		
	53.	2) Vendor to provide the following as a minimum		
	54.	- Boiler feed water control valve and its associated bypass valves.		
	55.	- Temperature transmitters & associated piping and fittings.		
	56.	3) Vendor to provide the following		
	57.	- Straight length requirement (upstream and downstream).		
	58.	- Temperature sensing element straight length distance required from quenching point.		
	59.	4) IBR service applicable for piping, fittings and instruments.		
	60.	5) Minimum length of straight pipe before and after spray nozzle shall be provided by vendor.		
	61.	6) Operating pressure values at the upstream of boiler feed water temperature control valve		
	62.			
	63.			
	64.			
	65.			
	66.			
	67.			
	68.			



1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)

Process Datasheet for PRDS - MP Steam

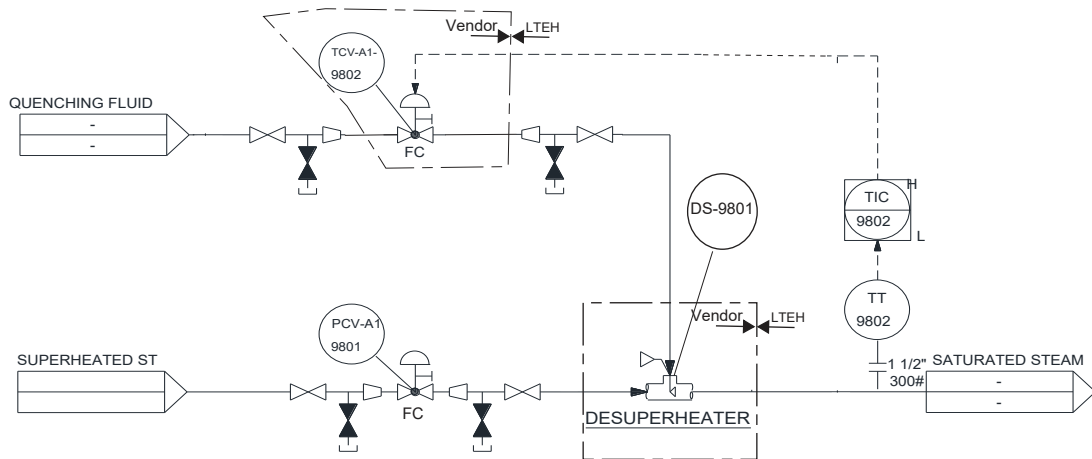
E24003-RCF-PC-DST-122

Rev. 1

19-Jun-202

Rev.

SKETCH



GENERAL	1 Tag Number		P&ID No.		TCV-A1-9802		E24003-RCF-PC-PID-098																																								
	2 Service		Process Case		Boiler Feed Water																																										
	3 Design Pressure Kg/cm ² g		Min.	Max.	FV		18																																								
	4 Design Temperature °C		Min.	Max.	-		155																																								
	5 Air Supply Pressure Kg/cm ² g		Min.	Max.	4		5.5																																								
	6 Elec. Haz. Area Class. STD/Zone/Method/Gr./Temp.				-	-	Safe Area	-																																							
	7 Ex-Certification		POSITIONER	SOV	LIMIT SWITCH	Exd	NA	NA																																							
PROCESS CONDITIONS	8 Fluid Name		Fluid State		Fluid Phase		Boiler Feed Water																																								
	9						Liquid																																								
	10 Flow		Kg/hr		324		1081																																								
	11 Pressure @ Flow		Kg/cm ² g		16		16																																								
	12 Diff. Pressure @ Flow		Kg/cm ²		STA		STA																																								
	13 Temperature		Oper.	Max.	°C		115																																								
	14 Viscosity				cP		-																																								
	15 Density (Liq. / Steam)				kg/m ³		949																																								
	16 Vapour Pressure (Liq.)				Kg/cm ² a		-																																								
	17 Cp / Cv (Gas/Steam)				-		-																																								
	18 Z (Compressibility) (Gas/Steam)				-		-																																								
	19 Shut-off Press. Max. Kg/cm ²		% Solid		18																																										
	20 Critical Press. (Liq.) Kg/cm ² a		% Flash																																												
	21 Mol. Wt. (Gas/Steam)		Outlet Vap. Mol. Wt.																																												
CALCULATED RESULTS	22 Degree of Superheat		Critical Temperature °C																																												
	23 Cv @ Flow		Min.	Nor.	Max.	STA	STA	STA																																							
	24 % Opening @ Flow		Min.	Nor.	Max.	STA	STA	STA																																							
BODY AND TRIM	25 Noise @ Flow dBA		Min.	Nor.	Max.	STA	STA	STA																																							
	26 Body Type		Globe																																												
	27 Body Size Body Rating		STA	300#																																											
	28 Body/Bonnet Material		ASTM A 105																																												
	29 Balanced/Unbalanced		STA																																												
	30 Corrosion Allowance		2 MM																																												
	31 Bonnet Type		Standard (Bolted)																																												
	32 Trim Type Size		STA	STA																																											
	33 Trim (Ball/Plug/Disc) Mtrl		410 Full Hard Faced																																												
	34 Seat Material		STELLITED																																												
	35 Stem/Shaft Material		13% CR.STEEL (NOCASTING)																																												
	36 Guiding type		STA																																												
	37 Rated Trim Travel (mm)		STA																																												
	38 Packing Material Style		Note-2	RENEWABLE																																											
	39 Selected Cv Characteristics		STA	STA																																											
	40 Bolt/Nut Material		Note-3																																												
	41 TSO Leakage Class		NO	IV																																											
	42 Flow Tendency to Valve		Down																																												
	43 Gasket Material		Note-4																																												
	44 Conn. In Size Rating		STA	300#																																											
	45 Conn. In Facing Finish		RF	RF,(125-250 AARH)																																											
	46 Conn. Out Size Rating		STA	300#																																											
	47 Conn. Out Facing Finish		RF	RF,(125-250 AARH)																																											
	ACTUATOR	48 Manufacturer Model		STA	STA																																										
		49 Type		Spring Diaphragm																																											
50 Travel Stop Hand Wheel		-	Required																																												
51 Air Failure Position		FC																																													
52 Open/Close Time Fire Safe		STA	-																																												
POSITIONER	53 Type		SMART 4-20 mA (HART)																																												
	54 Annexure no.		1.1																																												
	55 Action		Direct																																												
	56																																														
	57																																														
	58																																														
	59																																														
PROC. NOTE	60	NACE	IBR	H2	No	Yes	No	FAB. NOTE	95	PWHT Requirement	No																																				
Notes: See notes																																															
<table border="1"> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> <tr><td>0</td><td>Issued for Review</td><td>24-Jun-25</td><td>VVC</td><td>BRH</td><td>SBS</td></tr> <tr><td>No.</td><td>Description</td><td>Date</td><td>Pred</td><td>Chk</td><td>App</td></tr> </table> INSTRUMENT DATASHEET CONTROL VALVE - GLOBE 																																				0	Issued for Review	24-Jun-25	VVC	BRH	SBS	No.	Description	Date	Pred	Chk	App
0	Issued for Review	24-Jun-25	VVC	BRH	SBS																																										
No.	Description	Date	Pred	Chk	App																																										
Code:			Doc. No.E24003-RCF-IN-MRQ-042				Rev.:0																																								

Tag Number: TCV-A1-9802

Notes:

1. STA : Supplier to Advise
2. CORROSION INHIBITED DIE FORMED FLEXIBLE GRAPHITE WITH BRAIDED ANTI EXTRUSION RINGS.
3. ASTM A193 GR B7, ASTM A194 GR 2H
4. SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R.
5. VALVE SHALL BE SUPPLIED WITH IBR CERTIFICATE IN FORM-III C.
6. Side mounted Handwheel (MALLEABLE IRON/CAST ST/FAB.ST/DUCT. IRON) Required.

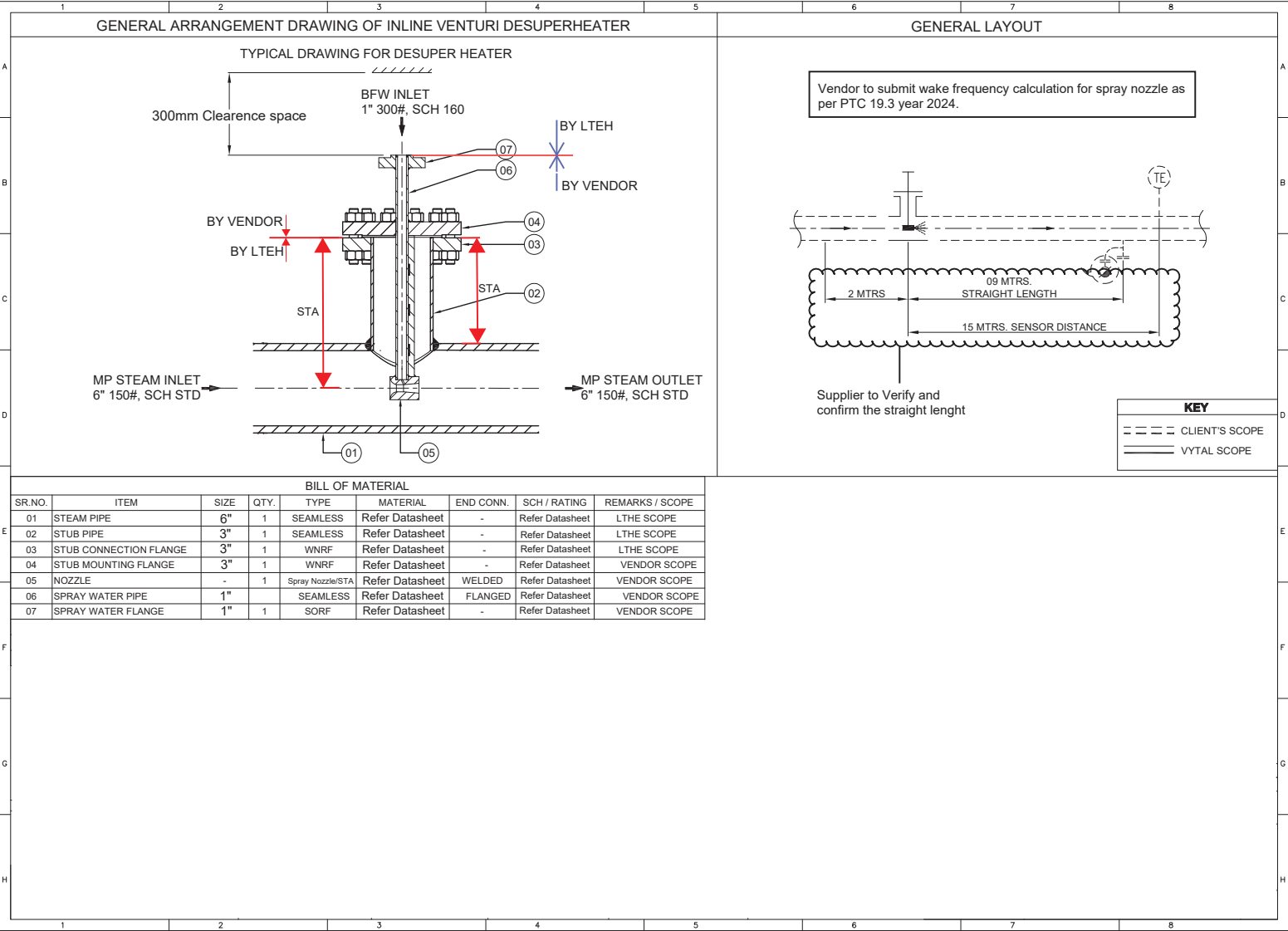
						INSTRUMENT DATASHEET CONTROL VALVE - GLOBE			
0	Issued for Review	24-Jun-25	VVC	BRH	SBS				
No.	Description	Date	Pred	Chk	App	Code:	Doc. No.E24003-RCF-IN-MRQ-042	Rev.:0	

GENERAL	1	Tag Number			P&ID No.			DS-9805			E24003-RCF-PC-PID-098					
	2	Service			Process Case			Medium Pressure Steam								
	3	Design Pressure Kg/cm ² g			Min.	Max.		Note-2			Note-2					
	4	Design Temperature °C			Min.	Max.		Note-2			Note-2					
	5	Air Supply Pressure Kg/cm ² g			Min.	Max.		NA			NA					
	6	Elec. Haz. Area Class. STD/Zone/Method/Gr./Temp.			-		-		-		Safe Area		-			
PROCESS CONDITIONS	7	Condition			Steam Inlet			Steam Outlet			Quenching Fluid					
	8	Fluid Name			Fluid State			Note-2	Note-2		Note-2	Note-2		Note-2	Note-2	
	9				Min.	Nor.	Max.	Min.	Nor.		Max.	Min.	Nor.	Max.		
	10	Composition														
	11	Flow Rate			Gen. Note 1			Note-2	Note-2	Note-2	Note-2	Note-2	Note-2	Note-2	Note-2	
	12	Pressure			Kg/cm ² g			Note-2	Note-2	Note-2	Note-2	Note-2	Note-2	Note-2	Note-2	
	13	Temperature			°C			Note-2	Note-2	Note-2	Note-2	Note-2	Note-2	Note-2	Note-2	
	14	Mol. Wt. (Gas/Steam)														
	15	Viscosity			cP											
	16	Density (Liq. / Steam)			kg/m ³			Note-2								
	17	Vapour Pressure (Liq.)			Kg/cm ² a											
	18	Cp / Cv (Gas/Steam)			-											
	19	Z (Compressibility) (Gas/Steam)			-											
	20	Shut-off Press. Max.			% Solid											
	21	Critical Press. (Liq.)			% Flash											
	22	Outlet Vap. Mol. Wt.														
	23	Deg. Of Superheat														
	24	Allowable Pressure Loss			Kg/cm2											
	25															
	26															
DESUPERHEATER DATA	27	Cv @ Flow			Min.	Nor.	Max.	STA		STA		STA				
	28	% Opening @ Flow			Min.	Nor.	Max.	STA		STA		STA				
	29	Noise @ Flow dBA			Min.	Nor.	Max.	STA		STA		STA				
	30	Min Degree of Superheat Required														
	31	Turndown Ratio														
	32															
BODY & TRIM	33	Type of Desuperheater			Spray Nozzle / STA			LINE	59	Steam Inlet Line No.			150-STM1-A5-9802-1A1BC-H40			
	34	Stub Size	Flange Rating	3" / STA	150#	60	Size		Class	Sch	6"	1A1BC	STD			
	35	Body/Bonnet Material			NA				61	Steam Inlet Line I.D.			mm			
	36	Corrosion Allowance			2 mm				62	Steam Outlet Line Number			250-STL1-A5-9803-1A1BC-H80			
	37	Bonnet Type			NA				63	Size	Class	Sch	6"	1A1BC	STD	
	38	Trim	Type	Size	NA	NA	64		Steam Outlet Line I.D.			mm				
	39	Trim Material			NA				65	BFW Inlet Line No.			25-BFW-A5-9804-3A1BC-H50			
	40	Packing Material			NA				66	Size	Class	Sch	1"	3A1BC	S160	
	41	Selected Cv			NA				67	BFW Inlet Line I.D			mm			
	42	Characteristics			NA				68							
	43	Coolant Inlet Flange Mat.			ASTM A105				69							
	44	Coolant Inlet Pipe Material			ASTM A106 GR.B				70							
	45	Steam Line Flange Mat.			ASTM A105				71							
	46	Steam Line Pipe Material			ASTM A106 GR.B				72							
	47	Nozzle Material			SS316 + Stellite				73							
	48	Nozzle Seat Material			SS316 + Stellite				74							
	49	Soft Seating			NA			75								
	50	Soft Seating Material			NA			76								
	51	TSO	Leakage Class		NA	STA	PURCHAS E	77	Manufacturer			STA				
	52							78	Desuperheater Model No.			STA				
53	Conn.	Size	Rating	6"	150#	79		Spray Nozzle Model No.			STA					
54	steam side	Facing	Finish	RF	125-250 AARH	80		Requisition No.			E24003-RCF-IN-MRQ-032					
PROCESS CONNECTION	55	Conn.	Size	Rating	1"	300#	81									
	56	water Side	Facing	Finish	RF	125-250 AARH	82									
PROC. NOTE	57	NACE Requirement			NA			FAB. NOTE	83	PWHT Requirement			Yes			
	58	IBR Requirement			Required				84	H2 Service			NA			
Notes: See notes																

Tag Number: DS-9805

1. STA: Supplier to Advise
2. Refer Annexure-2
3. BOLT.STUD + 2 NUTS- (HEAVY HEX) : BOLT: A193 GR.B7,NUT: A194 GR.2H.
4. Vendor to submit wake frequency calculation for spray nozzle as per PTC 19.3 year 2024.
5. Supplier to check and suggest that requirement of Thermal Liner where the difference between the coolant teperature and process fluid is greater than 250 degreeC.
6. Supplier to provide following minimum data in sizing calculations. · a. Process flow, pressure & temp. b. Coolant flow, pressure & temp. c. Minimum velocity requirement.
- d. Outlet condition & accuracy of desuperheating (accuracy with which outlet temp. should maintained above set point) e. Actual pressure drop. f. Minimum pipe straight run required for upstream & downstream. g. Minimum distance from desuperheater outlet to install temp. sensor. h. Drain requirement distance from desuperheater outlet.
7. Supplier to ensure that Coolant control valve rangeability shall be higher than PRDS rangeability.

						INSTRUMENT DATASHEET Desuperheater			
0	Issued for Review	24-Jun-25	VVC	BRH	SBS				
No.	Description	Date	Pred	Chk	App	Code:	Doc. No.E24003-RCF-IN-MRQ-042		Rev.:0



Annexure - 02

  			
1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)		Process Datasheet for PRDS - LP Steam	
E24003-RCF-PC-DST-123		Rev. 1	19-Jun-2025

Rev.	GENERAL INFORMATION			
1.	ITEM NO.			
2.	A1-U-02			
3.	DESCRIPTION			
4.	PRDS - LP Steam			
5.	P&ID NUMBER			
6.	E24003-RCF-PC-PID-098			
7.	LOCATION			
8.	OUTDOOR			
9.	SERVICE			
10.	STEAM			
11.	QUANTITY			
12.	ONE			
OPERATING CONDITIONS				
13.	OPERATING CASES	Maximum	Normal	Minimum
14.	INLET STEAM			
15.	OPERATING PRESSURE	kg/cm ² G	8	8
16.	OPERATING TEMPERATURE (NOR / MAX)	°C	175.4	175.4
17.	FLUID DENSITY	kg/m ³	4.50	4.50
18.	FLOWRATE	kg/hr	5432 (VTA)	4740 (VTA)
19.	BOILER FEED WATER			1422 (VTA)
20.	OPERATING PRESSURE (NOTE 6)	kg/cm ² G	16	16
21.	OPERATING TEMPERATURE (NOR / MAX)	°C	115	115
22.	FLUID DENSITY	kg/m ³	949	949
23.	FLOWRATE	kg/hr	68 (VTA)	60 (VTA)
24.	OUTLET STEAM			18 (VTA)
25.	OPERATING PRESSURE	kg/cm ² G	3.5	3.5
26.	OPERATING TEMPERATURE (NOR / MAX)	°C	147.9	147.9
27.	FLUID DENSITY	kg/m ³	2.4	2.4
28.	FLOWRATE	kg/hr	5500	4800
DESIGN INFORMATION				
29.	DESIGN PRESSURE (MAX / MIN)	kg/cm ² G	12 / FV	
30.	DESIGN TEMPERATURE	°C	200	
31.	ALLOWABLE PRESSURE DROP (MAX/NOR/MIN)	kg/cm ²	VTA	VTA
32.	INLET STEAM			
33.	LINE SIZE	inches	6"	
34.	RATING	#	150#	
35.	QUENCH DM WATER			
36.	DESIGN PRESSURE	kg/cm ² G	18 / FV	
37.	DESIGN TEMPERATURE	°C	155	
38.	LINE SIZE	inches	1" (VTA)	
39.	RATING	#	300#	
40.	OUTLET STEAM			
41.	LINE SIZE	inches	10"	
42.	RATING	#	150#	
MATERIAL				
43.	SHELL	CARBON STEEL		
44.	CORROSION ALLOWANCE	1.5 MM		
45.	INTERNAL	-		
46.	LINING / CLADDING	-		
47.	HEATING COIL / JACKET	-		
48.	FIRE PROOFING	☐ YES ☒ NO		
NOTES				
49.	1) Vendor to specify the type of desuperheating system i.e. mechanical spray nozzle, variable orifice, multi-nozzle spray, steam atomizing or any other type of desuperheater.			
50.	2) Vendor to provide the following as a minimum			
51.	- Boiler feed water control valve and its associated bypass valves.			
52.	- Temperature transmitters & associated piping and fittings.			
53.	3) Vendor to provide the following			
54.	- Straight length requirement (upstream and downstream).			
55.	- Temperature sensing element straight length distance required from quenching point.			
56.	4) IBR service applicable for piping, fittings and instruments.			
57.	5) Minimum length of straight pipe before and after spray nozzle shall be provided by vendor.			
58.	6) Operating pressure values at the upstream of boiler feed water temperature control valve			
59.				
60.				
61.				
62.				
63.				
64.				
65.				
66.				
67.				
68.				



1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)

Process Datasheet for PRDS - LP Steam

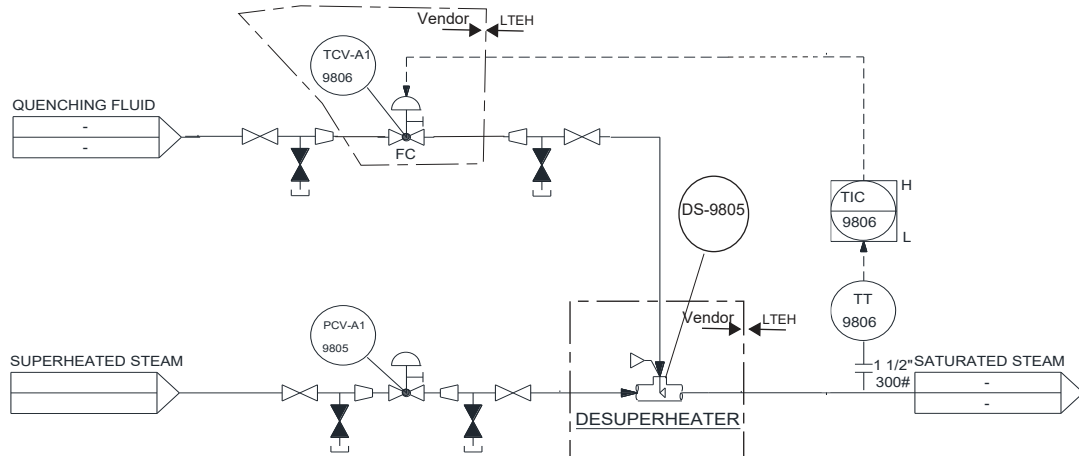
E24003-RCF-PC-DST-123

Rev. 1

19-Jun-2025

Rev.

SKETCH



GENERAL	1	Tag Number			P&ID No.			TCV-A1-9806			E24003-RCF-PC-PID-098							
	2	Service			Process Case			Boiler Feed Water										
	3	Design Pressure Kg/cm ² g			Min.		Max.		FV			18						
	4	Design Temperature °C			Min.		Max.		-			155						
	5	Air Supply Pressure Kg/cm ² g			Min.		Max.		4			5.5						
	6	Elec. Haz. Area Class. STD/Zone/Method/Gr./Temp.							-		-		Safe area - -					
	7	Ex-Certification		POSITIONER		SOV		LIMIT SWITCH		Exd		NA NA						
PROCESS CONDITIONS	8	Fluid Name		Fluid State		Fluid Phase		Boiler Feed Water			Liquid		Single					
	9								Min.		Nor.		Max.					
	10	Flow			Kg/hr			18		60		68						
	11	Pressure @ Flow			Kg/cm ² g			16		16		16						
	12	Diff. Pressure @ Flow			Kg/cm ²			STA		STA		STA						
	13	Temperature		Oper.		Max.		°C		115		115						
	14	Viscosity			cP					-								
	15	Density (Liq. / Steam)			kg/m ³					949								
	16	Vapour Pressure (Liq.)			Kg/cm ² a					-								
	17	Cp / Cv (Gas/Steam)								-								
	18	Z (Compressibility) (Gas/Steam)								-								
	19	Shut-off Press. Max. Kg/cm ²			% Solid			18										
	20	Critical Press. (Liq.) Kg/cm ² a			% Flash													
	21	Mol. Wt. (Gas/Steam)			Outlet Vap. Mol. Wt.													
22	Degree of Superheat			Critical Temperature °C														
CALCULATED RESULTS	23	Cv @ Flow			Min.		Nor.		Max.		STA		STA STA					
	24	% Opening @ Flow			Min.		Nor.		Max.		STA		STA STA					
	25	Noise @ Flow dBA			Min.		Nor.		Max.		STA		STA STA					
BODY AND TRIM	26	Body Type			Globe			SOLENOID VALVE			61		Quantity		-			
	27	Body Size		Body Rating		STA					300#		62		Tag No.A		-	
	28	Body/Bonnet Material			ASTM A 105								63		Tag No.B		-	
	29	Balanced/Unbalanced			STA								64		Annexure no.		-	
	30	Corrosion Allowance			2 MM								65					
	31	Bonnet Type			Standard (Bolted)								66					
	32	Trim		Type		Size					STA STA		67					
	33	Trim (Ball/Plug/Disc) Mtrl			410 Full Hard Faced								68					
	34	Seat Material			STELLITED								69					
	35	Stem/Shaft Material			13% CR.STEEL (NOCASTING)								70					
	36	Guiding type			STA					71								
	37	Rated Trim Travel (mm)			STA					72		Tag No.Open Position		-				
	38	Packing		Material		Style		Note-2 RENEWABLE		73		Tag No.Close Position		-				
	39	Selected Cv		Characteristics		STA		STA		74		Annexure no.		-				
	40	Bolt/Nut Material			Note-3					75								
	41	TSO		Leakage Class		NO		IV		76								
	42	Flow Tendency to Valve			Down					77								
43	Gasket Material			Note-4					78									
ACTUATOR	44	Conn. In		Size		Rating		STA 300#		ACCESSORIES			79		Air Reservoir		-	
	45	Conn. In		Facing		Finish		RF RF,(125-250 AARH)					80		Set Pressure		-	
	46	Conn. Out		Size		Rating		STA 300#					81		Surge Protection		-	
	47	Conn. Out		Facing		Finish		RF RF,(125-250 AARH)					82		Air Lock Device		-	
	48	Manufacturer		Model		STA		STA					83		AFR		Refer Annexure 1.2	
	49	Type			Spring Diaphragm								84					
	50	Travel Stop		Hand Wheel		-		Required					85					
	51	Air Failure Position			FC								86					
	52	Open/Close Time		Fire Safe		STA		Required					87					
	POSITIONER	53	Type			SMART 4-20 mA (HART)			LINE				88		Inlet Line No.		25-BFW-A5-9804-3A1BC-H50	
54		Annexure no.			1.1							89		Size Class Sch		1" 3A1BC S160		
55		Action			Direct							90		Outlet Line No.		25-BFW-A5-9804-3A1BC-H50		
56												91		Size Class Sch		1" 3A1BC S160		
57										PURCHASE			92		Manufacturer		STA	
58													93		Model		STA	
59													94		Requisition No.		E24003-RCF-IN-MRQ-032	
PROC. NOTE	60	NACE	IBR	H2	No	Yes	No	FAB. NOTE			95	PWHT Requirement		No				
Notes: See notes																		
INSTRUMENT DATASHEET CONTROL VALVE - GLOBE  																		
0	Issued for Review			24-Jun-25		VVC	BRH	SBS										
No.	Description			Date		Pred	Chk	App	Code:	Doc. No.E24003-RCF-IN-MRQ-042			Rev.:0					

Tag Number: TCV-A1-9806

Notes:

1. STA : Supplier to Advise
2. CORROSION INHIBITED DIE FORMED FLEXIBLE GRAPHITE WITH BRAIDED ANTI EXTRUSION RINGS.
3. ASTM A193 GR B7, ASTM A194 GR 2H
4. SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R.
5. VALVE SHALL BE SUPPLIED WITH IBR CERTIFICATE IN FORM-III C.
6. Side mounted Handwheel (MALLEABLE IRON/CAST ST/FAB.ST/DUCT. IRON) Required.

						INSTRUMENT DATASHEET CONTROL VALVE - GLOBE		  L&T Energy Hydrocarbon	
0	Issued for Review	24-Jun-25	VVC	BRH	SBS				
No.	Description	Date	Pred	Chk	App	Code:	Doc. No.E24003-RCF-IN-MRQ-042	Rev.:0	



1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)

ANNEXURE 1.1

DATASHEET FOR CONTROL VALVES

Rev. 0

CONTROL VALVE POSITIONER AND POSITION TRANSMITTER

Units : Density- kg/m³, Pressure- kg/cm²g, Temperature- Deg. C Level/Length- mm

1	Type :	Electro-pneumatic Smart Positioners and Positioner Transmitter
2	Safe Area Enclosure Class :	Weatherproof + IP 67
3	Electrical Area Class :	Safe Area
4	Certification :	STA
5	Input Signal :	4-20 mA, HART
6	Air Supply(Max/ Min) :	5.5/ 4 kg/cm ² g
7	Electric Supply	24 VDC Loop Powered
8	Output signal :	3 - 15 psig
9	Configuration Method	Refer Note-2
10	Conduit connections	-
11	Electrical	1/2" NPTF - 2 Nos., Refer Note-5
12	Pneumatic	1/4" NPTF
13	Reverse Polarity protection :	Required
14	Zero / Span Adjustment :	Required
15	Quantity (Nos.)	As per Valve Datasheet
16	Gauges	Required (Supply & Output)
17	CCOE Certification	NA
18	SIL Certification	-
19	Diagnostic Software	Full Diagnostics
20	Manufacturer :	STA
21	Model No. :	STA

NOTES :

- 1) POSITIONERS SHALL HAVE TWO PRESSURE GAUGES (SUPPLY & OUTPUT SIGNALS). MATERIAL OF PRESSURE GAUGES SHALL BE MINIMUM SS.
- 2) VALVE POSITIONER AND POSITIONER TRANSMITTER SHALL BE SMART TYPE AND CAPABLE FOR DIGITAL COMMUNICATION WITH HART MAINTENANCE SYSTEM, DCS AND HART CONFIGURATOR.
- 3) POSITIONER SHALL HAVE SEPARATE COMPARTMENT FOR CABLE TERMINATION.
- 4) PROVISION FOR CANOPY ON VALVE YOKE SHALL BE CONSIDERED BY VENDOR. VENDOR TO PROVIDE SUITABLE CANOPY DESIGN ALONG WITH OFFER.
- 5) ONE CABLE ENTRY SHALL BE USED FOR POSITIONER ANALOG OUTPUT SIGNAL AND OTHER ENTRY SHALL BE PLUGGED SAME APPLICABLE FOR POSITION TRANSMITTER.
- 6) REFER INDIVIDUAL TAG-WISE DATASHEET FOR POSITIONER Ex "d"/ Ex "la" ENCLOSURE REQUIREMENT.



1200 MTPD (DAP Basis) NPK Project, Thal, Maharashtra (India)

ANNEXURE 1.2

DATASHEET FOR CONTROL VALVES

Rev. 0

AIR FILTER REGULATOR

		Units : Density- kg/m3,	Pressure- kg/cm2g,	Temperature- Deg. C	Level/Length- mm
1	GENERAL				
2	Filter Regulator With Gauge		Required		
6	REGULATOR TYPE				
7	Spring & Diaphragm		Required		
8	Intermittent Bleed		-		
9	Other		-		
10	Fluid		Instrument Air		
11	Inlet Pressure - Design	kg/cm2g	8		
12	Inlet Pressure - Minimum	kg/cm2g	4		
13	Outlet Pressure	kg/cm2g	As required for the Valve		
14	Adjustable Range	kg/cm2g	STA		
15	Output Gauge Size / MOC / Connection		2" / SS316 / NPT(M)		
16	Range of the Gauge		0 - 10 kg/cm ²		
17	Range Adjustment		Handwheel with Lock nut		
18	Drain Cock		Required		
19	Connection		1/2" NPT(F)		
20	Construction		Weatherproof to IP 54		
21	REGULATION CHARACTERISTICS				
22	MINIMUM INPUT PRESSURE	kg/cm2g	4		
23	MAXIMUM INPUT PRESSURE	kg/cm2g	5.5		
24	MATERIAL				
25	BODY MATERIAL		Aluminium		
26	BOWL MATERIAL		Aluminium		
27	FILTER		5 Micron Max.		
28	FILTER MATERIAL		Sintered Bronze		
29	DIAPHRAGM		STA		
30	INNER VALVE ASSEMBLY		SS 316		
31	SPRING		STA		
32	ADJUSTABLE SCREW		SS 316		
33	DRAIN PLUG		SS 316		
34	OPTIONS				
35	MOUNTING ACCESSORIES		AFR Shall be mounted on the Valve		
36	HANDWHEEL		Required		
37	QUANTITY		As per Valve Datasheet		
38	MANUFACTURER		STA		
39	MODEL		STA		
40					
41					

NOTES :

- 1 THE RANGE OF THE AIR FILTER REGULATOR SHALL BE BASED ON THE AIR SUPPLY REQUIRED BY VALVE.
- 2 MOUNTING BRACKETS SHALL BE MINIMUM SS 304.



Project No. E24003

Rashtriya Chemicals and Fertilizers Limited

1200 MTPD (DAP Basis) NPK Project Thal, Maharashtra (India)

Project Document Title:
Piping Material Specification (PMS)

Project Document Number:
E24003-RCF-PP-SPE-001

L&T Contract Reference Number:
THAL-NPK-2301, Dated 10th Oct'2024

Customer : Rashtriya Chemicals and Fertilizers Limited
PMC : Projects & Development India Limited
LSTK Contractor : Larsen & Toubro Limited
Licenser : Incro, S.A.

04-Jun-25	1	Issued for Review			
			AS	PSR / RKC	SLK
Date	Rev.	Description	Prepared By	Reviewed By	Approved By

Disclaimer:

This document contains confidential and/or proprietary information of L&T and its affiliated companies. No part of this document, data, or information, issued for the captioned project or purpose, shall be used for other than the purpose for which this has been provided or disclosed to others or reproduced in any manner, except with the prior written permission of L&T.



1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

Issue and Approval Record

Issued Category				Approval Code	
☒	Engineering	☐	Procurement	☐	Construction
☐	Information	☐	Enquiry	☐	Construction
☒	Review	☐	Purchase	☐	Commissioning
☐	Approval	☐	Void	☐	As-Built
☐	Design	☐	Other.....		
Does issue category require approval?					
☒	Yes	☐	No	☐	NA
				☐ [Code 1] - Approved Consent to proceed ☐ [Code 2] - Comments As Noted Work may proceed subject to incorporation of comments ☐ [Code 3] - Rejected Revise and Resubmit. Work shall not proceed ☐ [Code 4] - Information Only Accepted for information only	
				Code: Date: Approver:	
				Note: This approval does not relieve the document ORIGINATOR any of his obligations and responsibilities under this Contract.	



1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

Revision Record

Rev.	Date	Prepared By	Reviewed By	Approved By	Revision Note / Description
0	05-Apr-25	AS	PSR/ RKC	SLK	Issued for review
1	04-Jun-25	AS	PSR/ RKC	SLK	Issued for review

Note:

- A 'vertical line in right-hand margin plus revised text in Blue colour' identifies changes in the latest revision of document.
- Sequential numerical revision philosophy i.e. 0,1,2,3 etc. to identify the document revision's purpose i.e. if issued for review/ approval/ enquiry/ information/ purchase/ construction/ commissioning/ Implementation etc.

Hold List

Hold No.	Hold Description



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

Contents

1.0	Introduction	5
2.0	Scope	5
3.0	Application	5
4.0	Abbreviations	6
5.0	References	8
6.0	Class Designation Code	12
7.0	Attachments	14
	Table-I : Pressure Temperature Rating For Ball, Plug & Butterfly Valves	14
	Annexure-I : Piping Material Class Index	16
	Annexure-II : Piping Material Classes	17



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

1.0 Introduction

Rashtriya Chemicals & Fertilizers Limited (RCF) is a public sector undertaking involved in the manufacture and marketing of nitrogenous, phosphatic and potassic fertilizers as well as a variety of industrial chemical products. The company has presently two manufacturing units, one at Trombay and the other at Thal, both located in the state of Maharashtra.

RCF are recognized as one of the most successful integrated agricultural input brands in India. They are ranked amongst the largest fertilizer companies in India. RCF as a business development strategy is exploring for development of phosphatic fertilizer complex in India. In this regard, NPKs have been identified as the prospective fertilizer project by RCF. The outputs DAP/NPK from the fertilizer complex will cater to the local & nearby areas and bridge the demand & production gap in this region.

RCF intends to establish a new 1200 MTPD (DAP Basis) NPK Fertilizer plant along with associated utilities / off-sites facilities in Thal. The plant shall be designed based on DAP (18.46.0) production capacity. In addition to DAP, the plant shall also produce different NP/NPK grades like 20.20.0.13, 10.26.26 and 12.32.16 etc. primarily using Ammonia, Phosphoric Acid, Sulphuric Acid, Muriate of Potash and Urea as required.

M/s Rashtriya Chemicals & Fertilizers Limited (RCF) have entrusted Larsen & Toubro Ltd. for engineering, procurement, and construction work of the DAP project.

2.0 Scope

This specification covers the various piping classes for Process and Utility Piping (Above Ground & Under Ground). The specification describes the material specifications of the pipes, fittings, flanges, blinds / spacers, valves, bolts / nuts, gaskets, and related components required for various fluids / services of "1200 MTPD (DAP Basis) NPK Project", located at Thal, Maharashtra, India.

This specification has been generated in line with the input received from the Licensor.

3.0 Application

All Piping shall be designed in accordance with the Process Piping Code, ASME B31.3 and IBR latest edition including latest addendum or as agreed upon with Process licensor/client. Once selected shall be followed throughout the job, Particular edition.

Individual piping Class has been generally designed to cover a set of services operating within pressure-temperature combination as per ASME B16.5 / B16.34 or part of it.

For the sake of brevity, the initials of the society to which the codes are referred to may be omitted in the specification. For example, B16.5 is a code referring to ASME; A106 is a standard referring to ASTM.

In any areas of conflict, the order of precedence, commencing with the highest, shall be as follows:

- a) Local Regulatory and Statutory Requirement
- b) Licensor Specifications and Standards
- c) Engineering standards and specifications
- d) International Codes and Standards.

In cases of conflict, interpretation of the most stringent requirement for Owner's approval. In all such cases of conflict, Owner's decision shall be final. Any deviations to the above require prior Owner's approval.



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

The PMS shall be strictly adhered to in the design, requisitioning, purchasing, fabrication, and testing of the piping system. However, deviations of material may occur due to design conditions and/or availability. All substitutions must be duly approved.

All material(s) listed in this specification when supplied / installed shall be brand new and of prime quality and shall conform to the requirements of this specification, Standards / Codes / Specifications referenced herein and shall also be fit for the purpose.

Any exceptions or deviations from this specification along with their merits and justifications shall be brought to the attention of Owner for their review & approval. In case of any dispute or conflict Owner decision will be the final, Decision which will be taken based on leading practices adopted by similar industry for such cases.

4.0 Abbreviations

AARH	Arithmetic Average Roughness Height
ANSI	American National Standards Institute
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing & Materials
AS	Alloy Steel
BE	Bevel End
BBE	Both Bevel End
BHN	Brinell Hardness Number
BS	British Standard
BW	Butt Weld
CA	Corrosion Allowance
CL	Class
CONC	Concentric
CS	Carbon Steel
DP	Dye Penetrant
ECC	Eccentric
E.F.S.W	Electric Fusion Seam Weld
E.R.W	Electric Resistance Weld
FF	Flat Face
FLGD	Flanged
GTAW	Gas Tungsten Arc Welding
GALV	Galvanized



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

HAZ	Heat Affected Zone
HEX	Hexagonal
IBR	Indian Boiler Regulations
ID	Inside Diameter
IGC	Inter Granular Corrosion
I/R	Inner ring
IS	Indian Standard
ISO	International Organization of Standardization
ITP	Inspection & Testing Plan
LAS	Low Alloy Steel
LJ	Lap Joint
LTCS	Low Temperature Carbon Steel
LP	Low Pressure
MFG	Manufacturer's
MAX	Maximum
MIN	Minimum
M/C	Machine
MP	Magnetic Particle
MSS	Manufacturer's Standardization Society
MR	Material Requisition
NACE	National Association of Corrosion Engineers
NB	Nominal Bore
NDT	Non-Destructive Test
NPS	Nominal Pipe Size
NPT (M/F)	Nominal Pipe Thread (Male/Female)
OD	Outside Diameter
O/R	Outer Ring
PE	Plain End
PBE	Plain Both End
PMI	Positive Material Identification
PTFE	Poly Tetra Fluro Ethylene



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

PWHT	Post Weld Heat Treatment
RED	Reducing
RF	Raised Face
RT	Radiography Test
RTJ	Ring Type Joint
SCRD (M/F)	Screwed (Male/Female)
SCH	Schedule
SO	Slip On
SS	Stainless Steel
S.A.W	Submerged Arc Welding
SW	Socket Welded
TBE	Threaded Both End
TEMP	Temperature
TOE	Threaded One End
UT	Ultrasonic Testing
WN	Weld Neck
WAFL	Wafer Lug
WPS	Welding Procedure Specification

5.0 References

Codes, Standards & Specifications

The following latest edition Codes, Standards and Regulations including all mandatory addenda shall be considered as part of this specification. All piping components shall conform to the specification as indicated on the respective piping class listed in this specification.

In case of any conflict only stringent references and acceptances, standards & codes shall be applicable.

DOCUMENT NUMBER	Rev./Ed.	Description
<u>Codes & Standards</u>		
API 6D	2021	Specification for Pipeline & Piping Valves
API 5L	2018	Specification for Line Pipe
API 594	2022	Check Valves: Flanged, Lug, Wafer, and Butt-welding



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

API 598	2023	Valve Inspection & Testing
API 600	2015	Steel Gate Valves—Flanged and Butt-welding Ends, Bolted Bonnets
API 602	2016	Gate, Globe, and Check Valves for Sizes DN 100 (NPS 4) and Smaller for the Petroleum and Natural Gas Industries
API 607	2016	Fire Test for Quarter-turn Valves and Valves Equipped with Non-metallic Seats
API 608	2020	Metal Ball Valves - Flanged, Threaded and Welding Ends
API 609	2016	Butterfly Valves: Double-flanged, Lug- and Wafer-type
API 623	2021	Steel Globe Valves—Flanged and Butt-welding Ends, Bolted Bonnets
API 941	2016	Steels for Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants
ASME B1.1	2022	Unified Inch Screw Threads (Un & UNR Thread forms)
ASME B1.20.1	2013	Pipe Threads, General Purpose (Inch)
ASME B16.5	2020	Pipe Flanges and Flanged Fittings NPS 1/2 through NPS 24 Metric/Inch Standard
ASME B16.9	2018	Factory-Made Wrought Butt-welding Fittings
ASME B16.10	2022	Face-to-Face and End-to-End Dimensions of Valves
ASME B16.11	2021	Forged Fittings, Socket-Welding and Threaded
ASME B16.20	2017	Metallic Gaskets for Pipe Flanges
ASME B16.21	2021	Non-Metallic Flat Gaskets for Pipe Flanges
ASME B16.25	2017	Butt Welding Ends
ASME B16.34	2020	Valve Flanged, Threaded & Welding End.
ASME B16.36	2020	Orifice Flanges
ASME B16.47	2020	Large Diameter Steel Flanges NPS 26 through NPS 60 Metric/Inch Standard
ASME B16.48	2020	Line Blanks



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

ASME B18.31.2	2014	Continuous and Double-End Studs (Inch Series)
ASME B18.2.1	2012	Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)
ASME B18.2.2	2022	Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)
ASME B31.3	2022	Process Piping
ASME B36.10M	2018	Welded and Seamless Wrought Steel Pipe
ASME B36.19M	2018	Stainless Steel Pipe
ASTM A105/ A105M		Standard Specification for Carbon Steel Forgings for Piping Application
ASTM A106		Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
ASTM A153		Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A182/ A182M		Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High Temperature Service
ASTM A193/ A193M		Standard Specification for Alloy-Steel and Stainless-Steel Bolting for High Temperature or High-Pressure Service and Other Special Purpose Applications
ASTM A194/ A194M		Standard Specification for Carbon and Alloy Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both
ASTM A216/ A216M		Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High Temperature Service
ASTM A217		Standard Specification for Steel Castings, Martensitic Stainless and Alloy, for Pressure-Containing Parts, Suitable for High-Temperature Service
ASTM A234/ A234M		Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
ASTM A262		Intergranular corrosion testing
ASTM A312/ A312M		Standard Specification for Seamless, Welded and Heavily Cold Worked Austenitic Stainless-Steel Pipe
ASTM A335		Standard Specification for Seamless Ferritic Alloy-Steel Pipe for High-Temperature Service



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

ASTM A350		Standard Specification for Carbon and Low-Alloy Steel Forgings, Requiring Notch Toughness Testing for Piping Components
ASTM A351/ 351M		Standard Specification for Castings, Austenitic, for Pressure Containing Parts
ASTM A352		Standard Specification for Steel Castings, Ferritic and Martensitic, for Pressure- Containing Parts, Suitable for Low-Temperature Service
ASTM A358		Standard Specification for Electric-Fusion-Welded Austenitic Chromium-Nickel Alloy Steel Pipe for High-Temperature Service
ASTM A370		Standard Test Methods and Definitions for Mechanical Testing of Steel Products
ASTM A388/ A388M		Standard Practice for Ultrasonic Examination of Heavy Steel Forgings
ASTM A403/ A403M		Standard Specification for Wrought Austenitic Stainless Steel Piping Fittings
ASTM A515		Carbon Steel Plate for Intermediate and Higher Temperature Service
ASTM A516/ A516M		Standard Specification for Pressure Vessel Plate, Carbon Steel, for Moderate and Lower Temperature Service
ASTM A671		Standard Specification for Electric-Fusion-Welded Steel Pipe for Atmospheric and Lower Temperatures
ASTM A672		Standard Specification for Electric-Fusion-Welded Steel Pipe for High-Pressure Service at Moderate Temperatures
BS EN ISO 15761	2020	Steel Gate, Globe, and Check Valves for Sizes DN 100 and Smaller, for The Petroleum and Natural Gas Industries
BS 1868	2007	Specification for Steel check valves (flanged and butt-welding ends) for the petroleum, petrochemical and allied industries
BS 1873	2018	Steel Globe and Globe Stop and Check Valves (Flanged and Butt-welding Ends) for Petroleum, Petrochemical and Allied Industries
BS 3799	2017	Steel pipe fitting, screwed and socket welding for the petroleum industry
BS 7531	2006	Specification for Compressed Non-Asbestos Fibre Jointing
IS:778	1999	Copper Alloy Gate, Globe, and Check Valves for Waterworks Purposes
IS 1239	2018	Steel Tubes, Tubulars and Other Wrought Steel Fittings



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

IS 3589	2001	Steel Pipe for Water and Sewage
MSS SP-25	2018	Standard Marking system for Valves, Fittings, Flanges, and Unions
MSS SP-83	2018	Steel Pipe Unions Socket Welding and Threaded
MSS SP-95	2018	Swage(d) Nipples and Bull Plugs
MSS SP-97	2019	Integrally Reinforced Forged Branch Outlet Fittings - Socket Welding, Threaded and Butt-welding Ends
MSS SP-144	2020	Pressure Seal Bonnet Valves
<u>Project Specific Documents</u>		
E24003-RCF-PP-SPE-002	Rev-0	Valve Material Specification (VMS)
E24003-RCF-PP-MS-001	Rev-0	Technical Notes for Valves
E24003-RCF-PP-MS-002	Rev-0	Technical Notes for Piping Bulk Items
E24003-RCF-QC-SPE-001	Rev-0	Specification for NDE requirements of Piping
E24003-RCF-QC-SPE-002	Rev-0	Specification for Marking & Colour Coding
E24003-RCF-QC-SPE-003	Rev-0	Specification for Positive Material Identification
E24003-RCF-PP-SPE-003	Rev-0	Specification for Jacketed Piping

6.0 Class Designation Code

The class designation shall consist of four-letter code "RBMS" plus type of flange face as fifth letter.

3A1SC

3 "R" Code: ANSI Pressure Rating Code

A "B" Code: Base Material

1 "M" Code: Material Standard

S "S" Code: Special Requirements



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

C Type of Flange Face

6.1 The first letter indicates pressure rating: "R" Code

0: Class 125#

1: Class 150#

3: Class 300#

6: Class 600#

6.2 The second letter indicates the base material: "B" Code

A: Carbon Steel

B: Low Alloy Steel

C: Stainless Steel

D: Cast Iron

E: Galvanized Iron

F: Plastic: F1 UHDP, F2 PVC, F3 FRP

G: Lined CS: G1 RLCS, G2 PTFE

6.3 The third letter indicates material standard: "M" Code.

CLASS OF MATERIALS	PIPING MATERIAL	FITTINGS	FORGINGS	CORROSION ALLOWANCE
A1	ASTM A106 GR. B	ASTM A234 GR. WPB	ASTM A105	2.0mm
A2	ASTM A53 B or ASTM A106 GR. B	ASTM A234 GR. WPB	ASTM A105	2.0mm
A3	ASTM A333 GR. 1/ GR. 6	ASTM A420 GR. WPL6	ASTM A350 GR.LF2	2.0mm
C1	ASTM A312 GR.TP304	ASTM A403 WP304	ASTM A182 GR.F304	1.6mm
C2	ASTM A312 GR.TP304L	ASTM A403 WP304L	ASTM A182 GR.F304L	1.6mm
C4	ASTM A312 GR.TP316L	ASTM A403 GR.WPS316L	ASTM A182 GR.F316L	1.6mm
C8	ASTM A312 TP904L(UNS 8904)	ASTM B366 WP 904L	ASTM A182 WP 904L	1.0mm
C9	ASTM A790 UNS 31803	ASTM A815 S 31803	ASTM A182 F51, A815 S 31803	1.6mm
C10	ASTM A790 UNS 32750	ASTM A815 S 32750	STM A182 F53; A815 S 32750	1.6mm
E1	ASTM A106 B, GALVAN.	ASTM A105, GALVAN.	ASTM A105, GALVAN.	1.6mm

6.4 The fourth letter indicates project specific requirement: "S" Code



**1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)**

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

- O: Non Special Requirements
S: Seamless
J: Jacketed
B: IBR
T: Tubing
X: Special service (Brick lined)

6.5 The fifth letter indicates the type of flange face

- A: Flat face
C: Raised face
N: Large Tongue and Groove

7.0 Attachments

- Table-I : Piping Material Class Index
Annexure-I : Piping Material Class Index
Annexure-II : Piping Material Classes

Table 1: Pressure Temperature Rating for Ball, Plug & Butterfly Valves

ITEM	PRESSURE RATING (PIPING CLASS)	SIZE RANGE (NB INCHES)	PRESSURE IN kg / cm ²								
			TEMPERATURE IN °C								
			-29 TO 38	50	65	100	120	150	175	200	>200
BALL (Soft Seated)	150# (AS PER PMS)	Upto 16"	19.98	19.57	19.11	18.04	17.27	16.11	15.09	14.07	DO NOT USE
	300# (AS PER PMS)		52.10	51.08	50.01	47.51	46.9	45.98	45.32	44.66	
	600# (AS PER PMS)		104.11	102.17	100.03	95.03	93.81	91.97	90.65	89.32	
PLUG (Soft Seated)	150# (AS PER PMS)	Upto 8"	19.98	19.57	19.11	18.04	17.27	16.11	15.09	14.07	
	300# (AS PER PMS)		52.10	51.08	50.01	47.51	46.9	45.98	45.32	44.66	
PLUG (Non- Lubricated)	150# (AS PER PMS)	Upto 12"	19.98	19.57	19.11	18.04	17.27	16.11	15.09	14.07	
	300# (AS PER PMS)		52.10	51.08	50.01	47.51	46.9	45.98	45.32	44.66	



1200 MTPD (DAP Basis) NPK Project, Thal,
Maharashtra (India)

Piping Material Specification

E24003-RCF-PP-SPE-001

Rev.1

04-Jun-25

ITEM	PRESSURE RATING (PIPING CLASS)	SIZE RANGE (NB INCHES)	PRESSURE IN kg / cm ²								
			TEMPERATURE IN °C								
			-29 TO 38	50	65	100	120	150	175	200	>200
	600# (AS PER PMS)		104.11	102.17	100.03	95.03	93.81	91.97	90.65	89.32	
Butterfly (Soft Seated)	150# (AS PER PMS)	Upto 24"	19.98	19.57	19.11	18.04	17.27	16.11	15.09	14.07	

NOTE: If the class Pressure / Temperature is lower than this table, class Pressure / Temperature to be followed.

 Rashtriya Chemicals & Fertilizers Ltd.			 Projects and Development India Limited				 L&T Energy Hydrocarbon			
Project Title: 1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL										
Document Title:			PIPING MATERIAL SPECIFICATION (PMS)						REV: 1	
EPCC CONTRACTOR DOCUMENT NO.			E24003-RCF-PP-SPE-001						04-Jun-2025	
ANNEXURE I - PIPING MATERIAL CLASS INDEX										
PIPING CLASSES (ASME B31.3)										
SL. No.	Piping Class	Rating Class - Facing	C.A. (mm)	Size Range	Special Requirement	Basic Material	Service	Limiting Cond.		Remarks
								Pressure (Kg/Sq.cm g)	Temp (°C)	
1	1A1BC	150# - Raised Face	2	1/2" - 14"	IBR	CARBON STEEL	LOW PRESSURE STEAM	19.99 5.6	-29 to +38 425	
2	1A1SC	150# - Raised Face	2	1/2" - 24"	-	CARBON STEEL	PLANT AIR, COATING OIL, COOLING WATER SUPPLY, COOLING WATER RETURN, DEFOAMER, NATURAL GAS, NITROGEN, ABOVE GROUND & UNDER GROUND - FIRE WATER (FW), PROCESS WATER/ RAW WATER, SERVICE WATER, <u>LOW PRESSURE CONDENSATE</u> .	19.99 5.6	-29 to +38 425	
3	1A2SC	150# - Raised Face	2	1/2" - 14"	-	CARBON STEEL	AMMONIA GAS LOW PRESSURE & MEDIUM TEMP.	19.99 5.61	-29 to +38 425	
4	1C1SC	150# - Raised Face	1.6	1/2" - 6"	-	SS304	INSTRUMENT AIR, DEMINERALIZED WATER. (CAT-D)	10	0 to 65	
5	1C2BC	150# - Raised Face	1.6	1/2" - 6"	IBR	SS304L	LP STEAM CONDENSATE	16.21 5.6	-29 to +38 425	
6	1C2SC	150# - Raised Face	1.6	1/2" - 24"	-	SS304L	CHILLED WATER SUPPLY, CHILLE D WATER RETURN. (CAT-D)	6	0 to 65	
7	1C4BC	150# - Raised Face	1.6	1/2" - 8"	-	<u>SS316L</u>	LP STEAM FOR PN SLURRY LINE JACKET	16.21 5.61	-29 to +38 425	JACKET
8	1C4SC	150# - Raised Face	1.6	1/2" - 12"	-	SS316L	SULPHURIC ACID 98% (L.P.), PROCESS WATER TO TAIL GAS SCRUBBER	16.21 4.69	-29 to +38 425	
9	1C8SC	150# - Raised Face	1	1/2" - 16"	-	SS904L	PRENEUTRALIZER SLURRY, PRESCRUBBING PARTIALLY AMMONIATED PHOSPHORIC ACID & PARTIALLY AMMONIATED PHOSPHORIC ACID & PIPE REACTOR FEED.	20.09 12.33	-29 to +38 250	<u>CORE FULLY JACKETED (FOR PRENEUTRALIZER SLURRY ONLY)</u>
10	1C9SC	150# - Raised Face	1.6	1/2" - 24"	-	UNS31803	EFFLUENT RECYCLE, CONCENTRATED PHOSPHORIC ACID, SCRUBBING PARTIALLY AMMONIATED PHOSPHORIC ACID, ACIDULATED WATER FROM TAIL GAS SCRUBBER , WASTE WATER	20.39 16.11	-29 to +38 150	
11	1E1SC	150# - Raised Face	1.6	1/2" - 14"	-	CARBON STEEL (GALVANIZED)	POTABLE WATER, FIRE WATER SPRINKLER SYSTEM AFTER RESTRICTION ORIFICE- A/G	12	0 to 65	
12	3A1BC	300# - Raised Face	2	1/2" - 14"	IBR	CARBON STEEL	MEDIUM PRESSURE STEAM	52.11 29.37	-29 to +38 425	
13	3A2SC	300# - Raised Face	2	1/2" - 14"	-	CARBON STEEL	AMMONIA GAS, <u>NATURAL GAS</u>	52.11 29.37	-29 to +38 425	
14	3A3SC	300# - Raised Face	2	1/2" - 14"	-	LTCS	LIQUID AMMONIA	52.11 29.37	-45 to +38 425	
15	3C4BC	300# - Raised Face	1.6	1/2" - 6"	IBR	SS316L	MEDIUM PRESSURE STEAM, MP STEAM CONDENSATE	42.22 24.37	-29 to +38 425	
16	3C4SC	300# - Raised Face	1.6	1/2" - 6"	-	SS316L	AMMONIA GAS, LIQUID AMMONIA, SULPHURIC ACID 98% (H.P.)	42.22 24.37	-45 to +38 425	
17	3C8SC	300# - Raised Face	1	1/2" - 6"	-	SS904L	LIQUID AMMONIA, PARTIALLY AMMONIATED PHOSPHORIC ACID, SULPHURIC ACID(98%), MEDIUM PRESSURE STEAM	52.31 32.63	-29 to +38 250	
18	15A1BC	1500# - RTJ	2	1/2" - 2"	IBR	CS	BOILER FEED WATER (BFW)	260.33 146.64	-29 to +38 425	

 Rashtriya Chemicals & Fertilizers Ltd.	 Projects and Development India Limited	
Project Title: 1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL		
Document Title:	PIPING MATERIAL SPECIFICATION (PMS)	REV: 1
EPCC CONTRACTOR DOCUMENT NO.	E24003-RCF-PP-SPE-001	04-Jun-2025
ANNEXURE II - PIPING MATERIAL CLASSES		

 Rashtriya Chemicals & Fertilizers Ltd.				 PDIL Projects and Development India Limited				 L&T Energy Hydrocarbon					
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL				PIPING MATERIAL SPECIFICATION (PMS)									
				E24003-RCF-PP-SPE-001				REV: 1		04-Jun-2025			
RATING		BASE MATERIAL		CA		SP. REQUIREMENT		PWHT		PIPE CLASS			
150#		CARBON STEEL		2 MM		IBR		-		1A1BC			
TEMPERATURE (Deg. C)		-29	38	50	100	150	200	250	300	325	350	400	425
PRESSURE (Kg/Sq. cm g)		19.99	19.99	19.58	18.05	16.11	14.07	12.34	10.40	9.48	8.57	6.63	5.61
SERVICE		LOW PRESSURE STEAM.											
SPECIAL NOTES	ITEM	SIZE		DESCRIPTION									
	MAINTENANCE JOINTS	ALL		FLANGED (TO BE KEPT MINIMUM)									
	PIPE JOINTS	1.5" & BELOW		SOCKET WELDED									
		2" & ABOVE		BUTTWELDED									
	TEMP. CONN.			AS PER P&ID									
	PRESS. CONN.			AS PER P&ID									
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001.											
	02	ALL PIPES, FITTINGS, VALVES AND FLANGES SHALL BE TESTED AND TEST CERTIFICATE IN FORM III A FOR PIPE AND FORM III C FOR OTHERS SHALL BE REQD. DULY COUNTERSIGNED BY IBR AUTHORITY OR ITS APPROVED REPRESENTATIVE.											
	03	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.											
	04	REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MSC-002) FOR SPECIAL REQUIREMENTS FOR IBR SERVICE.											
	05	REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MSC-002) FOR DIMENSIONS FOR HANDLE PROJECTION OF SPACER & BLINDS.											



PIPING MATERIAL SPECIFICATION (PMS)

E24003-RCF-PP-SPE-001

REV: 1

04-Jun-2025

RATING

BASE MATERIAL

CA

SP. REQUIREMENT

PWHT

PIPE CLASS

150#

CARBON STEEL

2 MM

IBR

1

1A1BC

BRANCH TABLE

[illegible]

RUN PIPE (SIZE IN INCHES)

LEGEND

H	Half Coupling
R	Pipe to Pipe with Reinforcement Pad
P	Pipe to Pipe
T	Equal/ Reducing Tees
S	Sockolet
W	Weldolet



Rashtriya Chemicals &
Fertilizers Ltd.



Projects and Development
India Limited



L&T Energy
Hydrocarbon

1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL

PIPING MATERIAL SPECIFICATION (PMS)
E24003-RCF-PP-SPE-001

REV: 1

04-Jun-2025

RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	CARBON STEEL	2 MM	IBR	-	1A1BC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	0.75	S160		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	PE	1	1.5	SCH XS		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	2	2	SCH XS		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	3	14	SCH STD		ASME B36.10M	ASTM A106 GR.B		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.10M	ASTM A106 GR.B, L=100mm		
	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.10M	ASTM A106 GR.B, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	14	150	125AARH	ASME B16.5	ASTM A105		
	FLNG. WELD NECK	RF	0.5	14	M / 150	125AARH	ASME B16.5	ASTM A105		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A105		
	FLNG. WELD NECK	RF	2	14	M / 300	125AARH	ASME B16.5	ASTM A105		
	FIG.8	FF	0.5	10	150	125AARH	ASME B16.48	ASTM A105		
	SPACER & BLIND	FF	12	14	150	125AARH	ASME B16.48	ASTM A105		05
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	ELBOW.90	BW	2	14	M		1.5D, ASME B16.9	ASTM A234 GR.WPB-S		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	ELBOW.45	BW	2	14	M		1.5D, ASME B16.9	ASTM A234 GR.WPB-S		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	TEE EQUAL	BW	2	14	M		ASME B16.9	ASTM A234 GR.WPB-S		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM A234 GR.WPB-S		
	REDUC.CONC	BW	2	14	M, M		ASME B16.9	ASTM A234 GR.WPB-S		
	REDUC.ECC	BW	2	14	M, M		ASME B16.9	ASTM A234 GR.WPB-S		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A234 GR.WPB-S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A234 GR.WPB-S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A105		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CAP	BW	2	14	M		ASME B16.9	ASTM A234 GR.WPB-S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A105		
VALVES	GATE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, STEM: 13% CR STEEL; B-16.11, TAG NO.: GAV240SI		
	GATE	FLGD-RF	2	14	150	125AARH	API 600/ ISO 10434	BODY: ASTM A216 GR. WCB, TRIM NO.: 8, STEM: 13% CR STEEL; B-16.5, TAG NO.: GAV244FI		
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, STEM: 13% CR STEEL; B-16.11, TAG NO.: GLV240SI		
	GLOBE	FLGD-RF	2	14	150	125AARH	API 623	BODY: ASTM A216 GR. WCB, TRIM NO.: 8, STEM: 13% CR STEEL, B-16.5; TAG NO.: GLV244FI		
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, B-16.11; TAG NO.: CHV240SI		
	CHECK - SWING	FLGD-RF	2	14	150	125AARH	API 594	BODY: ASTM A216 GR. WCB, TRIM NO.: 8,HINGE: SS316, B-16.5; TAG NO.: CHV244FI		
BOLTING	BOLT.STUD + 2 NUTS-(HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT: A193 GR.B7, NUT: A194 GR.2H		
	BOLT.STUD + 3 NUTS-(HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT: A193 GR.B7, NUT: A194 GR.2H		
GASKETS	GASKET		0.5	14	150		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R; THK=4.5MM		
	GASKET		0.5	14	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 +GRAPHITE FILLER WITH SS304 I/R & CS O/R; THK=4.5MM		

 Rashtriya Chemicals & Fertilizers Ltd.		 Projects and Development India Limited												
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)											
			E24003-RCF-PP-SPE-001		REV: 1									
					04-Jun-2025									
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS									
150#	CARBON STEEL	2 MM	-	-	1A1SC									
TEMPERATURE (Deg. C)	-29	38	50	100	150	200	250	300	325	350	400	425		
PRESSURE (Kg/Sq. cm g)	19.99	19.99	19.58	18.05	16.11	14.07	12.34	10.40	9.48	8.57	6.63	5.61		
SERVICE	PLANT AIR, COATING OIL, COOLING WATER SUPPLY, COOLING WATER RETURN, DEFOAMER, NATURAL GAS, NITROGEN, ABOVE GROUND & UNDER GROUND - FIRE WATER (FW), PROCESS WATER/ RAW WATER, SERVICE WATER, LOW PRESSURE CONDENSATE.													
SPECIAL NOTES	ITEM	SIZE	DESCRIPTION											
	MAINTENANCE JOINTS	ALL	FLANGED (TO BE KEPT MINIMUM)											
	PIPE JOINTS	1.5" & BELOW	SOCKET WELDED											
		2" & ABOVE	BUTTWELDED											
	TEMP. CONN.		AS PER P&ID											
	PRESS. CONN.		AS PER P&ID											
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001.												
	02	FOR RESTRICTION ON USE OF BALL, PLUG & BUTTERFLY VALVES, TABLE-1.												
	03	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.												
	04	REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MSC-002) FOR DIMENSIONS FOR HANDLE PROJECTION OF SPACER & BLINDS.												
	05	FOR UNDERGROUND PIPING, ALL BRANCHES SHALL BE PROVIDED WITH RF-PAD, OF 2D DIAMETER & THICKNESS SIMILAR TO HEADER.												



**1200 MTPD (DAP BASIS) NPK PROJECT,
RCF, THAL**

PIPING MATERIAL SPECIFICATION (PMS)

E24003-RCF-PP-SPE-001

REV: 1

04-Jun-2025

RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	CARBON STEEL	2 MM	-	-	1A1SC

[illegible]

<u>LEGEND</u>	
H	Half Coupling
R	Pipe to Pipe with Reinforcement Pad
P	Pipe to Pipe
T	Equal/ Reducing Tees
S	Sockolet
W	Weldolet

 Rashtriya Chemicals & Fertilizers Ltd.		 पी डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	CARBON STEEL	2 MM	-	-	1A1SC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	0.75	S160		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	PE	1	1.5	SCH XS		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	2	2	SCH XS		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	3	14	SCH STD		ASME B36.10M	ASTM A106 GR.B		
	E.F.S.W	BE	16	24	SCH STD		ASME B36.10M	ASTM A672 GR.B60 CL.12		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.10M	ASTM A106 GR.B, L=100mm		
FLANGES	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.10M	ASTM A106 GR.B, L=100mm		
	FLNG. BLIND	RF	0.5	24	150	125AARH	ASME B16.5	ASTM A105		
	FLNG. WELD NECK	RF	0.5	24	M / 150	125AARH	ASME B16.5	ASTM A105		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A105		
	FLNG. WELD NECK	RF	2	24	M / 300	125AARH	ASME B16.5	ASTM A105		
	FIG.8	FF	0.5	10	150	125AARH	ASME B16.48	ASTM A105		
FITTINGS	SPACER & BLIND	FF	12	24	150	125AARH	ASME B16.48	ASTM A105		04
	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	ELBOW.90	BW	2	14	M		1.5D, ASME B16.9	ASTM A234 GR.WPB-S		
	ELBOW.90	BW	16	24	M		1.5D, ASME B16.9	ASTM A234 GR.WPB-W		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	ELBOW.45	BW	2	14	M		1.5D, ASME B16.9	ASTM A234 GR.WPB-S		
	ELBOW.45	BW	16	24	M		1.5D, ASME B16.9	ASTM A234 GR.WPB-W		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	TEE EQUAL	BW	2	14	M		ASME B16.9	ASTM A234 GR.WPB-S		
	TEE EQUAL	BW	16	24	M		ASME B16.9	ASTM A234 GR.WPB-W		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM A234 GR.WPB-S		
	REDUC.CONC	BW	2	14	M, M		ASME B16.9	ASTM A234 GR.WPB-S		
	REDUC.CONC	BW	16	24	M, M		ASME B16.9	ASTM A234 GR.WPB-W		
	REDUC.ECC	BW	2	14	M, M		ASME B16.9	ASTM A234 GR.WPB-S		
	REDUC.ECC	BW	16	24	M, M		ASME B16.9	ASTM A234 GR.WPB-W		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A234 GR.WPB-S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A234 GR.WPB-S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A105		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CAP	BW	2	24	M		ASME B16.9	ASTM A234 GR.WPB-S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A105		
VALVES	GATE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, STEM: 13% CR STEEL; B-16.11, TAG NO.: GAV240S		
	GATE	FLGD-RF	2	24	150	125AARH	API 600/ ISO 10434	BODY: ASTM A216 GR.WCB, TRIM NO.: 8, STEM: 13% CR STEEL, B-16.5, TAG NO.: GAV244F		
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, STEM: 13% CR STEEL; B-16.11, TAG NO.: GLV240S		
	GLOBE	FLGD-RF	2	24	150	125AARH	API 623	BODY: ASTM A216 GR.WCB, TRIM NO.: 8, STEM: 13% CR STEEL, B-16.5, TAG NO.: GLV244F		
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5; B-16.11, TAG NO.: CHV240S		
	CHECK - SWING	FLGD-RF	2	24	150	125AARH	API 594	BODY: ASTM A216 GR.WCB, TRIM NO 8, HINGE: SS 316, B-16.5, TAG NO.: CHV244F		
	CHECK - DUAL PLATE	WAFL-RF	2	24	150	125AARH	API 594	BODY: ASTM A216 GR.WCB, TRIM NO 8, HINGE: SS 316, B-16.5, TAG NO.: CHV244L		
BUTTERFLY	BUTTERFLY	WAFL-RF	3	24	150	125AARH	API 609 (CAT-A)	BODY: ASTM A216 GR. WCB, TRIM: 13% CR STEEL, B16.5. TAG NO: BUV244L		02

 Rashtriya Chemicals & Fertilizers Ltd.					 PDIL Projects and Development India Limited					 L&T Energy Hydrocarbon				
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL							PIPING MATERIAL SPECIFICATION (PMS)							
							E24003-RCF-PP-SPE-001				REV: 1		04-Jun-2025	
RATING		BASE MATERIAL				CA		SP. REQUIREMENT		PWHT		PIPE CLASS		
150#		CARBON STEEL				2 MM		-		-		1A1SC		
	NEEDLE		SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5; B-16.11, TAG NO.: NDV240S					
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)			0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT: A193 GR.B7, NUT: A194 GR.2H					
	BOLT.STUD + 3 NUTS- (HEAVY HEX)			1	3.25			ASME B18.2.1/ B18.2.2	BOLT: A193 GR.B7, NUT: A194 GR.2H					
GASKETS	GASKET			0.5	24	150		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R; THK=4.5MM					
	GASKET			0.5	24	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R; THK=4.5MM					

 Rashtriya Chemicals & Fertilizers Ltd.				 पी डी आई एल PDIL Projects and Development India Limited										
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL				PIPING MATERIAL SPECIFICATION (PMS)										
				E24003-RCF-PP-SPE-001				REV: 1		04-Jun-2025				
RATING		BASE MATERIAL		CA		SP. REQUIREMENT		PWHT		PIPE CLASS				
150#		CARBON STEEL		2 MM		-		-		1A2SC				
TEMPERATURE (Deg. C)		-29	38	50	100	150	200	250	300	325	350	400	425	
PRESSURE (Kg/Sq. cm g)		19.99	19.99	19.58	18.05	16.11	14.07	12.34	10.40	9.48	8.57	6.63	5.61	
SERVICE		AMMONIA GAS LOW PRESSURE & MEDIUM TEMP.												
SPECIAL NOTES	ITEM		SIZE		DESCRIPTION									
	MAINTENANCE JOINTS		ALL		FLANGED (TO BE KEPT MINIMUM)									
	PIPE JOINTS		1.5" & BELOW		SOCKET WELDED									
			2" & ABOVE		BUTTWELDED									
	TEMP. CONN.				AS PER P&ID									
	PRESS. CONN.				AS PER P&ID									
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .												
	02	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.												
	03	REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MS-C-002) FOR DIMENSIONS FOR HANDLE PROJECTION OF SPACER & BLINDS.												

 Rashtriya Chemicals & Fertilizers Ltd.		 पी डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL		PIPING MATERIAL SPECIFICATION (PMS)			
		E24003-RCF-PP-SPE-001		REV: 1	04-Jun-2025
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	CARBON STEEL	2 MM	-	-	1A2SC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	0.75	S160		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	PE	1	1.5	SCH XS		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	2	2	SCH XS		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	3	14	SCH STD		ASME B36.10M	ASTM A106 GR.B		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.10M	ASTM A106 GR.B, L=100mm		
	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.10M	ASTM A106 GR.B, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	14	150	125AARH	ASME B16.5	ASTM A105		
	FLNG. WELD NECK	RF	0.5	14	M / 150	125AARH	ASME B16.5	ASTM A105		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A105		
	FLNG. WELD NECK	RF	2	14	M / 300	125AARH	ASME B16.5	ASTM A105		
	FIG.8	FF	0.5	10	150	125AARH	ASME B16.48	ASTM A105		
	SPACER & BLIND	FF	12	14	150	125AARH	ASME B16.48	ASTM A105		03
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	ELBOW.90	BW	2	14	M		1.5D, ASME B16.9	ASTM A234 GR.WPB-S		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	ELBOW.45	BW	2	14	M		1.5D, ASME B16.9	ASTM A234 GR.WPB-S		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	TEE EQUAL	BW	2	14	M		ASME B16.9	ASTM A234 GR.WPB-S		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM A234 GR.WPB-S		
	REDUC.CONC	BW	2	14	M, M		ASME B16.9	ASTM A234 GR.WPB-S		
	REDUC.ECC	BW	2	14	M, M		ASME B16.9	ASTM A234 GR.WPB-S		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A234 GR.WPB-S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A234 GR.WPB-S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A105		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CAP	BW	2	14	M		ASME B16.9	ASTM A234 GR.WPB-S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A105		
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, STEM: 13% CR STEEL; B-16.11, TAG NO.: GLV240S		
	GLOBE	FLGD-RF	2	14	150	125AARH	API 623	BODY: ASTM A216 GR. WCB, TRIM NO.: 8, STEM: 13% CR STEEL, B-16.5; TAG NO.: GLV244F		
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, B-16.11; TAG NO.: CHV240S		
	CHECK - SWING	FLGD-RF	2	14	150	125AARH	API 594	BODY: ASTM A216 GR. WCB, TRIM NO.: 8, HINGE: SS316, B-16.5; TAG NO.: CHV244F		
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT: A193 GR.B7, NUT: A194 GR.2H		
	BOLT.STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT: A193 GR.B7, NUT: A194 GR.2H		
GASKETS	GASKET		0.5	14	150		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R; THK=4.5MM		
	GASKET		0.5	14	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R; THK=4.5MM		

 Rashtriya Chemicals & Fertilizers Ltd.		 प्री डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL				PIPING MATERIAL SPECIFICATION (PMS)	
				E24003-RCF-PP-SPE-001	
REV: 1		04-Jun-2025			
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	SS304	1.6 MM	-	-	1C1SC
TEMPERATURE (Deg. C)	0	38	50	65	
PRESSURE (Kg/Sq. cm g)	10.0	10.0	10.0	10.0	
SERVICE		INSTRUMENT AIR, DEMINERALIZED WATER. (CAT-D).			
SPECIAL NOTES	ITEM	SIZE	DESCRIPTION		
	MAINTENANCE JOINTS	ALL	FLANGED (TO BE KEPT MINIMUM)		
	PIPE JOINTS	1.5" & BELOW	SOCKET WELDED		
		2" & ABOVE	BUTTWELDED		
	TEMP. CONN.		AS PER P&ID		
	PRESS. CONN.		AS PER P&ID		
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .			
	02	FOR RESTRICTION ON USE OF BALL, PLUG & BUTTERFLY VALVES, TABLE-1.			
	03	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.			

 Rashtriya Chemicals & Fertilizers Ltd.		 पी डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL		PIPING MATERIAL SPECIFICATION (PMS)			
		E24003-RCF-PP-SPE-001		REV: 1	04-Jun-2025
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	SS304	1.6 MM	-	-	1C1SC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	1.5	80S		ASME B36.19M	ASTM A312 TP304		
	WELDED	BE	2	4	40S		ASME B36.19M	ASTM A312 TP304		
	WELDED	BE	6	6	10S		ASME B36.19M	ASTM A312 TP304		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP304, L=100mm		
	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP034, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	6	150	125AARH	ASME B16.5	ASTM A182 GR.F304		
	FLNG. WELD NECK	RF	0.5	6	M / 150	125AARH	ASME B16.5	ASTM A182 GR.F304		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F304		
	FLNG. WELD NECK	RF	2	6	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F304		
	FIG.8	FF	0.5	6	150	125AARH	ASME B16.48	ASTM A182 GR.F304		
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304		
	ELBOW.90	BW	2	6	M		1.5D, ASME B16.9	ASTM A403 GR.WP304 S		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304		
	ELBOW.45	BW	2	6	M		1.5D, ASME B16.9	ASTM A403 GR.WP304 S		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304		
	TEE EQUAL	BW	2	6	M		ASME B16.9	ASTM A403 GR.WP304 S		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM A403 GR.WP304 S		
	REDUC.CONC	BW	2	6	M, M		ASME B16.9	ASTM A403 GR.WP304 S		
	REDUC.ECC	BW	2	6	M, M		ASME B16.9	ASTM A403 GR.WP304 S		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A403 GR.WP304 S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A403 GR.WP304 S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A182 GR.F304		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304		
	CAP	BW	2	6	M		ASME B16.9	ASTM A403 GR.WP304 S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A182 GR.F304		
VALVES	GATE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 F304, TRIM NO.: 15, STEM: SS 304; B-16.11; TAG NO.: GAV500S		
	GATE	FLGD-RF	2	6	150	125AARH	API 600/ ISO 10434	BODY: ASTM A351 GR. CF8, TRIM NO.: 15, STEM: SS 304; B-16.5; TAG NO.: GAV504F		
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 F304, TRIM NO.: 15, STEM: SS 304; B-16.11; TAG NO.: GLV500S		
	GLOBE	FLGD-RF	2	6	150	125AARH	API 623	BODY: ASTM A351 GR. CF8, TRIM NO.: 15, STEM: SS 304; B-16.5; TAG NO.: GLV504F		
	BALL	FLGD-RF	0.5	6	150	125AARH	API 608/ BS EN ISO 17292	BODY: ASTM A351 GR.CF8, TRIM: SS304, SEAT: RPTFE; B-16.15; TAG NO.: BAV504F		02
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR. F304, TRIM NO.: 15; B-16.11; TAG NO.: CHV500S		
	CHECK - SWING	FLGD-RF	2	6	150	125AARH	API 594	BODY: ASTM A351 GR.CF8, TRIM NO.: 15, HING: SS 304; B-16.5; TAG NO.: CHV504F		
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		
	BOLT.STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		
GASKETS	GASKET		0.5	6	150		ASME B16.21- ASME B16.5	RING, NON ASBESTOS, BS 7531 GR.X, THK. 3MM		
	GASKET		0.5	6	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & SS O/R		

 Rashtriya Chemicals & Fertilizers Ltd.			 Projects and Development India Limited											
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL						PIPING MATERIAL SPECIFICATION (PMS)								
						E24003-RCF-PP-SPE-001				REV: 1	04-Jun-2025			
RATING		BASE MATERIAL		CA	SP. REQUIREMENT		PWHT	PIPE CLASS						
150#		SS304L		1.6 MM	IBR		-	1C2BC						
TEMPERATURE (Deg. C)		-29	38	50	100	150	200	250	300	325	350	400	425	
PRESSURE (Kg/Sq. cm g)		16.21	16.21	15.60	13.56	12.24	11.42	10.71	10.20	9.48	8.57	6.63	5.61	
SERVICE		LP STEAM CONDENSATE.												
SPECIAL NOTES	ITEM		SIZE		DESCRIPTION									
	MAINTENANCE JOINTS		ALL		FLANGED (TO BE KEPT MINIMUM)									
	PIPE JOINTS		1.5" & BELOW		SOCKET WELDED									
			2" & ABOVE		BUTTWELDED									
	TEMP. CONN.				AS PER P&ID									
	PRESS. CONN.				AS PER P&ID									
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .												
	03	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.												
	04	REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MS-002) FOR SPECIAL REQUIREMENTS FOR IBR SERVICE.												

 Rashtriya Chemicals & Fertilizers Ltd.		 पी डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL		PIPING MATERIAL SPECIFICATION (PMS)			
		E24003-RCF-PP-SPE-001		REV: 1	04-Jun-2025
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	SS304L	1.6 MM	IBR	-	1C2BC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	1.5	80S		ASME B36.19M	ASTM A312 GR.TP304L		
	SEAMLESS	BE	2	4	40S		ASME B36.19M	ASTM A312 GR.TP304L		
	SEAMLESS	BE	6	6	10S		ASME B36.19M	ASTM A312 GR.TP304L		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP304L, L=100mm		
	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP034L, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	6	150	125AARH	ASME B16.5	ASTM A182 GR.F304L		
	FLNG. WELD NECK	RF	0.5	6	M / 150	125AARH	ASME B16.5	ASTM A182 GR.F304L		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F304L		
	FLNG. WELD NECK	RF	2	6	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F304L		
	FIG.8	FF	0.5	6	150	125AARH	ASME B16.48	ASTM A182 GR.F304L		
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	ELBOW.90	BW	2	6	M		1.5D, ASME B16.9	ASTM A403 GR.WP304L-S		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	ELBOW.45	BW	2	6	M		1.5D, ASME B16.9	ASTM A403 GR.WP304L-S		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	TEE EQUAL	BW	2	6	M		ASME B16.9	ASTM A403 GR.WP304L-S		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM A403 GR.WP304L-S		
	REDUC.CONC	BW	2	6	M, M		ASME B16.9	ASTM A403 GR.WP304L-S		
	REDUC.ECC	BW	2	6	M, M		ASME B16.9	ASTM A403 GR.WP304L-S		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A403 GR.WP304L-S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A403 GR.WP304L-S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A182 GR.F304L		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	CAP	BW	2	6	M		ASME B16.9	ASTM A403 GR.WP304L-S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A182 GR.F304L		
VALVES	GATE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F304L, TRIM: 15, STEM: SS 316; B-16.11; TAG NO.: GAV400SI		
	GATE	FLGD-RF	2	6	150	125AARH	API 600/ ISO 10434	BODY: ASTM A351 GR.CF3, TRIM: 15, STEM: SS 316; B-16.11; TAG NO.: GAV404FI		
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F304L, TRIM: 15, STEM: SS 316; B-16.11; TAG NO.: GLV400SI		
	GLOBE	FLGD-RF	2	6	150	125AARH	API 623	BODY: ASTM A351 GR.CF3, TRIM: 15, STEM: SS 316; B-16.11; TAG NO.: GLV404FI		
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F304L, TRIM: 15; B-16.11; TAG NO.: CHV400SI		
	CHECK - SWING	FLGD-RF	2	6	150	125AARH	API 594	BODY: ASTM A351 GR.CF3, TRIM: 15, HING: SS 316; B-16.11; TAG NO.: CHV404FI		
BOLTING	BOLT.STUD + 2 NUTS-(HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		
	BOLT.STUD + 3 NUTS-(HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		
GASKETS	GASKET		0.5	6	150		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304L + GRAPHITE FILLER WITH SS304L I/R & SS O/R; THK=4.5MM		
	GASKET		0.5	6	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304L + GRAPHITE FILLER WITH SS304L I/R & SS O/R; THK=4.5MM		

 Rashtriya Chemicals & Fertilizers Ltd.		 पी डी आई एल PDIL Projects and Development India Limited					
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL				PIPING MATERIAL SPECIFICATION (PMS)			
				E24003-RCF-PP-SPE-001		REV: 1	04-Jun-2025
RATING		BASE MATERIAL		CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#		SS304L		1.6 MM	-	-	1C2SC
TEMPERATURE (Deg. C)		0	38	50	65		
PRESSURE (Kg/Sq. cm g)		6.0	6.0	6.0	6.0		
SERVICE		CHILLED WATER SUPPLY, CHILLED WATER RETURN. (CAT-D).					
SPECIAL NOTES	ITEM		SIZE		DESCRIPTION		
	MAINTENANCE JOINTS		ALL		FLANGED (TO BE KEPT MINIMUM)		
	PIPE JOINTS		1.5" & BELOW		SOCKET WELDED		
			2" & ABOVE		BUTTWELDED		
	TEMP. CONN.				AS PER P&ID		
	PRESS. CONN.				AS PER P&ID		
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .					
	02	FOR RESTRICTION ON USE OF BALL, PLUG & BUTTERFLY VALVES, TABLE-1.					
	03	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.					
	04	REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MS-002) FOR DIMENSIONS FOR HANDLE PROJECTION OF SPACER & BLINDS.					

 Rashtriya Chemicals & Fertilizers Ltd.		 पो डी आई एल PDIL Projects and Development India Limited		 L&T Energy Hydrocarbon	
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	SS304L	1.6 MM	-	-	1C2SC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	1.5	80S		ASME B36.19M	ASTM A312 GR.TP304L		
	WELDED	BE	2	4	40S		ASME B36.19M	ASTM A312 GR.TP304L		
	WELDED	BE	6	24	10S		ASME B36.19M	ASTM A312 GR.TP304L		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP304L, L=100mm		
	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP034L, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	24	150	125AARH	ASME B16.5	ASTM A182 GR.F304L		
	FLNG. WELD NECK	RF	0.5	24	M / 150	125AARH	ASME B16.5	ASTM A182 GR.F304L		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M /300	125AARH	ASME B16.5	ASTM A182 GR.F304L		
	FLNG. WELD NECK	RF	2	24	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F304L		
	FIG.8	FF	0.5	10	150	125AARH	ASME B16.48	ASTM A182 GR.F304L		
	SPACER & BLIND	FF	12	24	150	125AARH	ASME B16.48	ASTM A182 GR.F304L		04
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	ELBOW.90	BW	2	6	M		1.5D, ASME B16.9	ASTM A403 GR.WP304L-S		
	ELBOW.90	BW	8	24	M		1.5D, ASME B16.9	ASTM A403 GR.WP304L-W		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	ELBOW.45	BW	2	6	M		1.5D, ASME B16.9	ASTM A403 GR.WP304L-S		
	ELBOW.45	BW	8	24	M		1.5D, ASME B16.9	ASTM A403 GR.WP304L-W		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	TEE EQUAL	BW	2	6	M		ASME B16.9	ASTM A403 GR.WP304L-S		
	TEE EQUAL	BW	8	24	M		ASME B16.9	ASTM A403 GR.WP304L-W		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM A403 GR.WP304L-S		
	REDUC.CONC	BW	2	6	M, M		ASME B16.9	ASTM A403 GR.WP304L-S		
	REDUC.CONC	BW	8	24	M, M		ASME B16.9	ASTM A403 GR.WP304L-W		
	REDUC.ECC	BW	2	6	M, M		ASME B16.9	ASTM A403 GR.WP304L-S		
	REDUC.ECC	BW	8	24	M, M		ASME B16.9	ASTM A403 GR.WP304L-W		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A403 GR.WP304L-S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A403 GR.WP304L-S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A182 GR.F304L		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	CAP	BW	2	24	M		ASME B16.9	ASTM A403 GR.WP304L-S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F304L		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A182 GR.F304L		
VALVES	GATE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F304L, TRIM: 15, STEM: SS 316; B-16.11; TAG NO.: GAV400S		
	GATE	FLGD-RF	2	24	150	125AARH	API 600/ ISO 10434	BODY: ASTM A351 GR.CF3, TRIM: 15, STEM: SS 304; B-16.11; TAG NO.: GAV404F		
	BUTTERFLY	WAFL-RF	3	24	150	125AARH	API 609 (CAT-A)	BODY: ASTM A351 GR. CF3, TRIM: SS304, B16.5; TAG NO: BUV404L		02
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F304L, TRIM: 15; B-16.11; TAG NO.: CHV400S		
	CHECK - DUAL PLATE	WAFL-RF	2	24	150	125AARH	API 594	BODY: ASTM A351 GR.CF3, TRIM: 15, HING: SS 316; B-16.5; TAG NO.: CHV404L		
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		
	BOLT.STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		

 Rashtriya Chemicals & Fertilizers Ltd.					 पी डी आई एल PDIL Projects and Development India Limited					 L&T Energy Hydrocarbon				
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL						PIPING MATERIAL SPECIFICATION (PMS)								
						E24003-RCF-PP-SPE-001				REV: 1		04-Jun-2025		
RATING		BASE MATERIAL				CA		SP. REQUIREMENT		PWHT		PIPE CLASS		
150#		SS304L				1.6 MM		-		-		1C2SC		
GASKETS	GASKET		0.5	24	150		ASME B16.21- ASME B16.5	RING, COMPRESSED ARAMID FIBER WITH NBR BINDER, THK. 3MM						
	GASKET		0.5	24	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304L + GRAPHITE FILLER WITH SS304/ 304L I/R & SS O/R, THK 4.5MM						

 Rashtriya Chemicals & Fertilizers Ltd.				 पी डी आई एल PDIL Projects and Development India Limited								
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL				PIPING MATERIAL SPECIFICATION (PMS)								
				E24003-RCF-PP-SPE-001				REV: 1		04-Jun-2025		
RATING		BASE MATERIAL		CA		SP. REQUIREMENT		PWHT		PIPE CLASS		
150#		SS316L		1.6 MM		-		-		1C4BC		
TEMPERATURE (Deg. C)		-29	38	50	100	150	200	250	300	350	400	425
PRESSURE (Kg/Sq. cm g)		16.21	16.21	15.60	13.56	12.24	11.42	10.71	10.20	8.57	6.63	5.61
SERVICE		LP STEAM FOR PN SLURRY LINE JACKET.										
SPECIAL NOTES	ITEM		SIZE		DESCRIPTION							
	MAINTENANCE JOINTS		ALL		FLANGED (TO BE KEPT MINIMUM)							
	PIPE JOINTS		1.5" & BELOW		SOCKET WELDED							
			2" & ABOVE		BUTTWELDED							
	TEMP. CONN.				AS PER P&ID							
	PRESS. CONN.				AS PER P&ID							
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .										
	02	REFER JACKETING DETAILS FROM SPECIFICATION E24003-RCF-PP-SPE-003.										
	03	FOR CORE PIPING CLASS 1C8SC, JACKETING SHALL BE FULL.										
	04	FOR STEAM INLET & OUTLET JACKET PIPING, JACKET DRAINS AND VENTS, USE 1/2" SW HALF COUPLING SW FOR 2" & SMALLER JACKETS AND 3/4" SW HALF COUPLING FOR 3" & LARGER JACKET.										
	05	MODIFIED CAPS AND SPILT COLLARSSHALL BE USED AS PER SPECIFICATION FOR JACKETING E24003-RCF-PP-SPE-003.										

 Rashtriya Chemicals & Fertilizers Ltd.		 पो डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	SS316L	1.6 MM	-	-	1C4BC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	1.5	80S		ASME B36.19M	ASTM A312 GR.TP316L		
	SEAMLESS	BE	2	4	40S		ASME B36.19M	ASTM A312 GR.TP316L		
	SEAMLESS	BE	6	6	10S		ASME B36.19M	ASTM A312 GR.TP316L		
	E.F.S.W	BE	8	8	10S		ASME B36.19M	ASTM A358 GR. TP316L CL.1		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP316L, L=100mm		
	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP316L, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	8	150	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FLNG. WELD NECK	RF	0.5	8	M / 150	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FLNG. WELD NECK	RF	0.5	8	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F316L		
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	ELBOW.90	BW	2	8	M		1D, ASME B16.9	ASTM A403 GR.WP316L-S		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	ELBOW.45	BW	2	8	M		1D, ASME B16.9	ASTM A403 GR.WP316L-S		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	TEE EQUAL	BW	2	8	M		ASME B16.9	ASTM A403 GR.WP316L-S		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	TEE REDUCING	BW	2	8	M, M		ASME B16.9	ASTM A403 GR.WP316L-S		
	REDUC.CONC	BW	2	8	M, M		ASME B16.9	ASTM A403 GR.WP316L-S		
	REDUC.ECC	BW	2	8	M, M		ASME B16.9	ASTM A403 GR.WP316L-S		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A403 GR.WP316L-S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A403 GR.WP316L-S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A182 GR.F316L		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	CAP	BW	2	8	M		ASME B16.9	ASTM A403 GR.WP316L-S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A182 GR.F316L		
VALVES	GATE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A 182 GR.F316L, TRIM: SS316L HARDFACED, STEM: SS 316L; B-16.11; TAG NO.: GAV420S		
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A 182 GR.F316L, TRIM: SS316L HARDFACED, STEM: SS 316L; B-16.11; TAG NO.: GLV420S		
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A 182 GR.F316L, TRIM: SS316L HARDFACED; B-16.11; TAG NO.: CHV420S		
	PLUG	FLGD-RF	0.5	1.5	150	125AARH	API 599	BODY: ASTM A182 GR.F316L/ A 351 GR.CF3M, TRIM: HARDENED PLUG; B-16.5, TAG NO.: PLV424F		02
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		
	BOLT.STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		
GASKETS	GASKET		0.5	8	150		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS316L + GRAPHITE FILLER WITH SS316 I/R & SS OR		
	GASKET		0.5	8	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS316L + GRAPHITE FILLER WITH SS316 I/R & SS OR		

 Rashtriya Chemicals & Fertilizers Ltd.				 पी डी आई एल PDIL Projects and Development India Limited								
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL				PIPING MATERIAL SPECIFICATION (PMS)								
				E24003-RCF-PP-SPE-001				REV: 1		04-Jun-2025		
RATING		BASE MATERIAL		CA		SP. REQUIREMENT		PWHT		PIPE CLASS		
150#		SS316L		1.6 MM		-		-		1C4SC		
TEMPERATURE (Deg. C)		-29	38	50	100	150	200	250	300	350	400	425
PRESSURE (Kg/Sq. cm g)		16.21	16.21	15.60	13.56	12.24	11.42	10.71	10.20	8.57	6.63	4.69
SERVICE		SULPHURIC ACID 98% (L.P.), PORCESS WATER TO TAIL GAS SCRUBBER.										
SPECIAL NOTES	ITEM		SIZE		DESCRIPTION							
	MAINTENANCE JOINTS		ALL		FLANGED (TO BE KEPT MINIMUM)							
	PIPE JOINTS		1.5" & BELOW		SOCKET WELDED							
			2" & ABOVE		BUTTWELDED							
	TEMP. CONN.				AS PER P&ID							
	PRESS. CONN.				AS PER P&ID							
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .										
	02	FOR RESTRICTION ON USE OF BALL, PLUG & BUTTERFLY VALVES, TABLE-1.										
	03	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.										
	04	FOR SULPHURIC ACID SERVICE USE ONLY THESE VALVES.										
	05	FOR SULPHURIC ACID SERVICE USE ONLY THESE GASKETS.										

 Rashtriya Chemicals & Fertilizers Ltd.		 पो डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	SS316L	1.6 MM	-	-	1C4SC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	1.5	80S		ASME B36.19M	ASTM A312 GR.TP316L		
	SEAMLESS	BE	2	4	40S		ASME B36.19M	ASTM A312 GR.TP316L		
	SEAMLESS	BE	6	6	10S		ASME B36.19M	ASTM A312 GR.TP316L		
	E.F.S.W	BE	8	12	10S		ASME B36.19M	ASTM A358 GR.TP316L CL.1		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP316L, L=100mm		
	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP316L, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	12	150	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FLNG. WELD NECK	RF	0.5	12	M / 150	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FLNG. WELD NECK	RF	2	12	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FIG.8	FF	0.5	10	150	125AARH	ASME B16.48	ASTM A182 GR.F316L		
	SPACER & BLIND	FF	12	12	150	125AARH	ASME B16.48	ASTM A182 GR. F316L		04
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	ELBOW.90	BW	2	6	M		1.5D, ASME B16.9	ASTM A403 GR.WP316L-S		
	ELBOW.90	BW	8	12	M		1.5D, ASME B16.9	ASTM A403 GR.WP316L-W		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	ELBOW.45	BW	2	6	M		1.5D, ASME B16.9	ASTM A403 GR.WP316L-S		
	ELBOW.45	BW	8	12	M		1.5D, ASME B16.9	ASTM A403 GR.WP316L-W		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	TEE EQUAL	BW	2	6	M		ASME B16.9	ASTM A403 GR.WP316L-S		
	TEE EQUAL	BW	8	12	M		ASME B16.9	ASTM A403 GR.WP316L-W		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM A403 GR.WP316L-S		
	REDUC.CONC	BW	2	6	M, M		ASME B16.9	ASTM A403 GR.WP316L-S		
	REDUC.ECC	BW	2	6	M, M		ASME B16.9	ASTM A403 GR.WP316L-S		
	REDUC.ECC	BW	8	12	M, M		ASME B16.9	ASTM A403 GR.WP316L-W		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A403 GR.WP316L-S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A403 GR.WP316L-S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A182 GR.F316L		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	CAP	BW	2	6	M		ASME B16.9	ASTM A403 GR.WP316L-S		
	CAP	BW	8	12	M		ASME B16.9	ASTM A403 GR.WP316L-S		
VALVES	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A182 GR.F316L		
	GATE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A 182 GR.F316L, TRIM: SS316L HARDFACED, STEM: SS 316L; B-16.11; TAG NO.: GAV420S		
	GATE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A 182 GR.F316L, TRIM: ALLOY 20, STEM: SS 316L; B-16.11; TAG NO.: GAV420SS		04
	GATE	FLGD-RF	2	12	150	125AARH	API 600/ ISO 10434	BODY: ASTM A 351 GR.CF3M, TRIM: SS316L HARDFACED, STEM: SS 316L; B-16.11; TAG NO.: GAV424F		
	GATE	FLGD-RF	2	12	150	125AARH	API 600/ ISO 10434	BODY: ASTM A 351 GR.CF3M, TRIM: ALLOY 20, STEM: SS 316L; B-16.11; TAG NO.: GAV424FS		04
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A 182 GR.F316L, TRIM: SS316L HARDFACED, STEM: SS 316L; B-16.11; TAG NO.: GLV420S		
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A 182 GR.F316L, TRIM: ALLOY 20, STEM: SS 316L; B-16.11; TAG NO.: GLV420SS		04
	GLOBE	FLGD-RF	2	12	150	125AARH	API 623	BODY: ASTM A 351 GR.CF3M, TRIM: SS316L HARDFACED, STEM: SS 316L; B-16.11; TAG NO.: GLV424F		
	GLOBE	FLGD-RF	2	12	150	125AARH	API 623	BODY: ASTM A 351 GR.CF3M, TRIM: ALLOY 20, STEM: SS 316L; B-16.11; TAG NO.: GLV424FS		04

 Rashtriya Chemicals & Fertilizers Ltd.					 Projects and Development India Limited					 L&T Energy Hydrocarbon				
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL						PIPING MATERIAL SPECIFICATION (PMS)								
						E24003-RCF-PP-SPE-001					REV: 1		04-Jun-2025	
RATING		BASE MATERIAL			CA		SP. REQUIREMENT		PWHT		PIPE CLASS			
150#		SS316L			1.6 MM		-		-		1C4SC			
VALVES	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A 182 GR.F316L, TRIM: SS316L HARDFACED; B-16.11; TAG NO.: CHV420S						
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A 182 GR.F316L, TRIM: ALLOY 20; B-16.11; TAG NO.: CHV420SS				04		
	CHECK - SWING	FLGD-RF	2	12	150	125AARH	API 594	BODY: ASTM A 351 GR.CF3M, TRIM: SS316L HARDFACED; B-16.11; TAG NO.: CHV424F						
	CHECK - SWING	FLGD-RF	2	12	150	125AARH	API 594	BODY: ASTM A 351 GR.CF3M, TRIM: ALLOY 20; B-16.11; TAG NO.: CHV424FS				04		
	PLUG	FLGD-RF	0.5	6	150	125AARH	API 599	BODY: ASTM A182 GR.F316L/ A 351 GR.CF3M, TRIM: HARDENED PLUG; B-16.5, TAG NO.: PLV424F				02		
	PLUG	FLGD-RF	0.5	6	150	125AARH	API 599	BODY: ASTM A182 GR.F316L/ A 351 GR.CF3M, TRIM: PFA LINED; B-16.5, TAG NO.: PLV424FS				04, 02		
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8						
	BOLT.STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8						
GASKETS	GASKET		0.5	12	150		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS316L + GRAPHITE FILLER WITH SS316 I/R & SS O/R; THK=4.5MM						
	GASKET		0.5	12	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS316L + GRAPHITE FILLER WITH SS316 I/R & SS O/R; THK=4.5MM						
GASKETS	GASKET		0.5	12	150		ASME B16.21- ASME B16.5	RING, 100% RIGID VIRGIN PTFE, 3MM				05		
	GASKET		0.5	12	300		ASME B16.21- ASME B16.5	RING, 100% RIGID VIRGIN PTFE, 3MM				05		

 Rashtriya Chemicals & Fertilizers Ltd.		 PDIL Projects and Development India Limited		 L&T Energy Hydrocarbon	
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
04-Jun-2025					
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	SS904L	1 MM	-	-	1C8SC
TEMPERATURE (Deg. C)	-29	38	50	100	150
	200				
PRESSURE (Kg/Sq. cm g)	20.09	20.09	19.17	16.01	14.68
					13.56
SERVICE		PRENEUTRALIZER SLURRY, PRESCRUBBING PARTIALLY AMMONIATED PHOSPHORIC ACID & PARTIALLY AMMONIATED PHOSPHORIC ACID & PIPE REACTOR FEED.			
SPECIAL NOTES	ITEM	SIZE	DESCRIPTION		
	MAINTENANCE JOINTS	ALL	FLANGED (TO BE KEPT MINIMUM)		
	PIPE JOINTS	1.5" & BELOW	SOCKET WELDED		
		2" & ABOVE	BUTTWELDED		
	TEMP. CONN.		AS PER P&ID		
	PRESS. CONN.		AS PER P&ID		
NOTES	01 NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .				
	02 FOR RESTRICTION ON USE OF BALL, PLUG & BUTTERFLY VALVES, TABLE-1.				
	03 THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.				
	04 REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MSC-002) FOR DIMENSIONS FOR HANDLE PROJECTION OF SPACER & BLINDS.				
	05 EPDM GASKET SHALL ONLY BE USED FOR PHOSPHORIC ACID SERVICE.				
	06 THIS PIPE CLASS COVERS CORE PIPING AND ITS ELEMENTS ONLY. REFER PIPE CLASS 1C4BC FOR JACKET PIPING.				
	07 REFER JACKETING DETAILS FROM SPECIFICATION E24003-RCF-PP-SPE-003.				
	08 FOR ONLY PRENEUTRALIZER SLURRY SERVICE, ALL PIPES, VALVES AND FITTINGS SHALL BE FULLY JACKETED.				
	09 PTFE GASKET SHALL NOT BE USED ABOVE 204°C.				
	10 MODIFIED FLANGES TO CORRESPOND JACKET PIPE, EXCEPT ITS BORE WHICH IS TO SUIT CORE PIPE OD. REFER APPENDIX-I (SPECIFICATION FOR REDUCING SLIP-ON FLANG FOR FULLY JACKETED LINES) OF TECHNICAL NOTES FOR PIPING BULK ITEMS (DOC. NO.: E24003-RCF-PP-MSC-002). CORROSPONDING BLIND FLANGE, FIG. 8, SPACTACLE BLIND SHALL BE DESIGN TO SUIT CONNECTING FLANGES AS PER APPENDIX-I OF PIPING BULK ITEMS (DOC. NO.:E24003-RCF-PP-MSC-002_Rev-0).				
	11 FOR ONLY PRENEUTRALIZER SLURRY SERVICE, VALVES SHALL BE FULLY JACKETED AND EACH VALVE SHALL HAVE FLANGED STEAM SUPPLY AND CONDENSATE RETURN CONNECTIONS.				
	12 TO BE USED FOR SERVICE OTHER THAN PRENEUTRALIZER SLURRY (NON-JACKETED LINE).				
	13 KNIFE GATE VALE SHALL NOT BE USED FOR PRESSURE MORE THAN 10.5 Kg/cm2g FOR SIZE UPTO 24".				
	14 CORE PIPE FOR PRENEUTRALIZER SLURRY LINE SHALL BE USED FOR SIZE UPTO 10" ONLY.				

 Rashtriya Chemicals & Fertilizers Ltd.		 पो डी आई एल PDIL Projects and Development India Limited		 L&T Energy Hydrocarbon	
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	SS904L	1 MM	-	-	1C8SC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	1.5	80S		ASME B36.19M	ASTM A312 GR.TP904L		
	SEAMLESS	BE	2	10	40S		ASME B36.19M	ASTM A312 GR.TP904L		
	SEAMLESS	BE	12	16	10S		ASME B36.19M	ASTM A312 GR.TP904L		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP904L, L=100mm		
	NIPPLE, SEAMLESS	PE/NPTM	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP904L, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	10	150	125AARH	MFG STD	ASTM A182 GR.F904L		10
	FLNG. REDUCING SLIP ON	RF	0.5	10	M / 150	125AARH	APPENDIX-I	ASTM A182 GR.F904L		10
	FLNG. REDUCING SLIP ON	RF	0.5	10	M / 300	125AARH	APPENDIX-I	ASTM A182 GR.F904L		10
	FIG.8	FF	0.5	10	150	125AARH	MFG STD	ASTM A182 GR.F904L		10
	FLNG. BLIND	RF	0.5	16	150	125AARH	ASME B16.5	ASTM A182 GR.F904L		12
	FLNG. WELD NECK	RF	0.5	16	M / 150	125AARH	ASME B16.5	ASTM A182 GR.F904L		12
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F904L		12
	FLNG. WELD NECK	RF	2	16	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F904L		12
	FIG.8	FF	0.5	10	150	125AARH	ASME B16.48	ASTM A182 GR.F904L		12
	SPACER & BLIND	FF	12	16	150	125AARH	ASME B16.48	ASTM A182 GR.F904L		12
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	ELBOW.90	BW	2	16	M		1.5D, ASME B16.9	ASTM B366 GR.WP904L S		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	ELBOW.45	BW	2	16	M		1.5D, ASME B16.9	ASTM B366 GR.WP904L S		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	TEE EQUAL	BW	2	16	M		ASME B16.9	ASTM B366 GR.WP904L S		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	TEE REDUCING	BW	2	16	M, M		ASME B16.9	ASTM B366 GR.WP904L S		
	REDUC.CONC	BW	2	16	M, M		ASME B16.9	ASTM B366 GR.WP904L S		
	REDUC.ECC	BW	2	16	M, M		ASME B16.9	ASTM B366 GR.WP904L S		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM B366 GR.WP904L S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM B366 GR.WP904L S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A182 GR.F904L		12
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	CAP	BW	2	16	M		ASME B16.9	ASTM B366 GR.WP904L S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A182 GR.F904L		
VALVES	KNIFE-GATE	FLGD-RF	2	16	150	125AARH	MSS-SP-81	BODY: ASTM A 182 F904L, TRIM: SS904L, STEM: F904L,B-16.5; TAG NO.: KGV554F		12, 13
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F904L, TRIM: SS904L, B-16.11, TAG NO.: CHV550S		12
	CHECK - SWING	FLGD-RF	2	16	150	125AARH	API 594	BODY-ASTM A182 F904L, TRIM: SS904L, HING: F904L B-16.5; TAG NO.: CHV554F		12
	PLUG	FLGD-RF	0.5	6	150	125AARH	API 598	BODY: ASTM A182 GR.F904L, TRIM: SS904L HARDENED PLUG, B-16.5; TAG NO.: PLV554F		02,12
	PLUG-JACKETED	FLGD-RF	0.5	6	150	125AARH	API 599	BODY: ASTM A182 GR.F904L, TRIM: SS904L HARDENED PLUG, B-16.5; TAG NO.: PLV554FJ		02,11
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8M CL.2, NUT:A194 GR.8M		
	BOLT.STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8M CL.2, NUT:A194 GR.8M		
GASKETS	GASKET		0.5	16	150		ASME B16.21- ASME B16.5	RING, 100% RIGID VIRGIN PTFE, THK. 3MM		
	GASKET		0.5	16	300		ASME B16.21- ASME B16.5	RING, 100% RIGID VIRGIN PTFE, THK. 3MM		

 Rashtriya Chemicals & Fertilizers Ltd.		 PDIL Projects and Development India Limited		 L&T Energy Hydrocarbon	
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
04-Jun-2025					
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	UNS31803	1.6 MM	-	-	1C9SC
TEMPERATURE (Deg. C)	-29	38	50	100	150
PRESSURE (Kg/Sq. cm g)	20.39	20.39	19.88	18.05	16.11
SERVICE		EFFLUENT RECYCLE, CONCENTRATED PHOSPHORIC ACID, SCRUBBING PARTIALLY AMMONIATED PHOSPHORIC ACID, ACIDULATED WATER FROM TAIL GAS SCRUBBER , WASTE WATER.			
SPECIAL NOTES	ITEM	SIZE	DESCRIPTION		
	MAINTENANCE JOINTS	ALL	FLANGED (TO BE KEPT MINIMUM)		
	PIPE JOINTS	1.5" & BELOW	SOCKET WELDED		
		2" & ABOVE	BUTTWELDED		
	TEMP. CONN.		AS PER P&ID		
	PRESS. CONN.		AS PER P&ID		
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .			
	02	FOR RESTRICTION ON USE OF BALL, PLUG & BUTTERFLY VALVES, TABLE-1.			
	03	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.			
	04	REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MSC-002) FOR DIMENSIONS FOR HANDLE PROJECTION OF SPACER & BLINDS.			
	05	KNIFE GATE VALE SHALL NOT BE USED FOR PRESSURE MORE THAN 10.5 Kg/cm2g FOR SIZE UPTO 24".			

 Rashtriya Chemicals & Fertilizers Ltd.		 पो डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	UNS31803	1.6 MM	-	-	1C9SC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	1.5	80S		ASME B36.19M	ASTM A790 UNS 31803		
	SEAMLESS	BE	2	4	40S		ASME B36.19M	ASTM A790 UNS 31803		
	SEAMLESS	BE	6	6	10S		ASME B36.19M	ASTM A790 UNS 31803		
	E.F.S.W	BE	8	24	10S		ASME B36.19M	ASTM A928 UNS 31803 CL.1		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.19M	ASTM A790 UNS 31803, L=100mm		
	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.19M	ASTM A790 UNS 31803, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	24	150	125AARH	ASME B16.5	ASTM A182 GR. F51		
	FLNG. WELD NECK	RF	0.5	24	M / 150	125AARH	ASME B16.5	ASTM A182 GR. F51		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A182 GR. F51		
	FLNG. WELD NECK	RF	2	24	M / 300	125AARH	ASME B16.5	ASTM A182 GR. F51		
	FIG.8	FF	0.5	10	150	125AARH	ASME B16.48	ASTM A182 GR. F51		
	SPACER & BLIND	FF	12	24	150	125AARH	ASME B16.48	ASTM A182 GR. F51		04
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR. F51		
	ELBOW.90	BW	2	6	M		1.5D, ASME B16.9	ASTM A815 S31803 S		
	ELBOW.90	BW	8	24	M		1.5D, ASME B16.9	ASTM A815 S31803 W		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR. F51		
	ELBOW.45	BW	2	6	M		1.5D, ASME B16.9	ASTM A815 S31803 S		
	ELBOW.45	BW	8	24	M		1.5D, ASME B16.9	ASTM A815 S31803 W		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR. F51		
	TEE EQUAL	BW	2	6	M		ASME B16.9	ASTM A815 S31803 S		
	TEE EQUAL	BW	8	24	M		ASME B16.9	ASTM A815 S31803 W		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR. F51		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM A815 S31803 S		
	REDUC.CONC	BW	2	6	M, M		ASME B16.9	ASTM A815 S31803 S		
	REDUC.CONC	BW	8	24	M, M		ASME B16.9	ASTM A815 S31803 W		
	REDUC.ECC	BW	2	6	M, M		ASME B16.9	ASTM A815 S31803 S		
	REDUC.ECC	BW	8	24	M, M		ASME B16.10	ASTM A815 S31803 W		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A815 S31803 S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A815 S31803 S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A182 GR. F51		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A182 GR. F51		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR. F51		
	CAP	BW	2	24	M		ASME B16.9	ASTM A815 S31803 S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR. F51		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR. F51		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A182 GR. F51		
VALVES	KNIFE-GATE	FLGD-RF	2	24	150	125AARH	MSS-SP-81	BODY: ASTM A 995 GR.CD3MN, TRIM: UNS31803, STEM: A479 S31803, B-16.5; TAG NO.: KGV584F		05
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F51, TRIM: UNS31803, B-16.11; TAG NO.: CHV580S		
	CHECK - SWING	FLGD-RF	2	24	150	125AARH	API 594	BODY: ASTM A 995 GR.CD3MN, TRIM: UNS31803, HING: A479 S31803, B-16.5; TAG NO.: CHV584F		
	PLUG	FLGD-RF	0.5	6	150	125AARH	API 599	BODY: ASTM A182 GR.F51/ ASTM A 995 GR.CD3MN, TRIM: UNS31803, HARDENED PLUG, B-16.5; TAG NO.: PLV584F		02
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		
	BOLT.STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		
GASKETS	GASKET		0.5	24	150		ASME B16.21- ASME B16.5	SPIRAL WOUND, SS316L + PTFE FILLER WITH SS316L I/R & SS O/R; THK=4.5MM		
	GASKET		0.5	24	300		ASME B16.21- ASME B16.5	SPIRAL WOUND, SS316L + PTFE FILLER WITH SS316L I/R & SS O/R; THK=4.5MM		

 Rashtriya Chemicals & Fertilizers Ltd.		 PDIL Projects and Development India Limited		 L&T Energy Hydrocarbon	
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
04-Jun-2025					
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	CARBON STEEL	1.6 MM	GALVANISED	-	1E1SC
TEMPERATURE (Deg. C)	0	38	50	65	
PRESSURE (Kg/Sq. cm g)	12	12	12	12	
SERVICE	POTABLE WATER, FIRE WATER SPRINKLER SYSTEM AFTER RESTRICTION ORIFICE- A/G.				
SPECIAL NOTES	ITEM	SIZE	DESCRIPTION		
	MAINTENANCE JOINTS	1.5" & BELOW	UNIONS (SCRF) TO BE KEPT MINIMUM		
		2" & ABOVE	FLANGED (SCREWED) TO BE KEPT MINIMUM		
	PIPE JOINTS	4" & BELOW	SCREWED COUPLING		
		6" & ABOVE	BUTTWELDED		
	TEMP. CONN.		AS PER P&ID		
	PRESS. CONN.		AS PER P&ID		
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .			
	02	ALL NUTS AND BOLTS SHALL BE 'HOT DIP GALVANIZED' AS PER ASTM A 153.			
	03	SURFACES WHERE GALVANISING HAS BEEN BURNT OFF SHALL BE WIRE BRUSHED AND COLD GALVANISED WITH ZINC COATING COMPOUND.			
	04	ALL PIPES SHALL BE 'HOT DIP GALVANIZED' AS PER ASTM A 53 AND OTHER PIPING COMPONENTS AS PER ASTM A153.			
	05	THREADED JOINTS SHALL BE MADE WITH 1" WIDTH PTFE JOINING TAPE.			
	06	PIPE SPOOL REQUIRING WELDING SHALL BE GALVANISED AFTER WELDING.			
	07	THREADED FITTINGS SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.			
	08	FOR BRANCH CONNECTION "X", USE EQUAL TEE/REDUCING TEE WITH NIPPLE AND REDUCING COUPLING.			

 Rashtriya Chemicals & Fertilizers Ltd.		 पो डी आई एल PDIL Projects and Development India Limited		 L&T Energy Hydrocarbon	
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
150#	CARBON STEEL	1.6 MM	GALVANISED	-	1E1SC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	NPTM	0.5	2	XS		ASME B36.10M	ASTM A106 GR.B (GALV.)		
	SEAMLESS	NPTM	3	4	SCH STD		ASME B36.10M	ASTM A106 GR.B (GALV.)		
	SEAMLESS	BE	6	14	SCH STD		ASME B36.10M	ASTM A106 GR.B (GALV.)		
	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.10M	ASTM A106 GR.B (GALV.), L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	14	150	125AARH	ASME B16.5	ASTM A105 (GALV.)		
	FLNG. SCRD	RF	0.5	4	M / 150	125AARH	ASME B16.5	ASTM A105 (GALV.)		
	FLNG. SLIP ON	RF	6	14	M / 150	125AARH	ASME B16.5	ASTM A105 (GALV.)		
	FIG.8	FF	0.5	10	150	125AARH	ASME B16.48	ASTM A105 (GALV.)		
FITTINGS	SPACER & BLIND	FF	12	14	150	125AARH	ASME B16.48	ASTM A105 (GALV.)		
	ELBOW.90	NPTF	0.5	4	3000		ASME B16.11	ASTM A105 (GALV.)		
	ELBOW.90	BW	6	14	M		1.5D, ASME B16.9	ASTM A234 GR.WPB S (GALV.)		
	ELBOW.45	NPTF	0.5	4	3000		ASME B16.11	ASTM A105 (GALV.)		
	ELBOW.45	BW	6	14	M		1.5D, ASME B16.9	ASTM A234 GR.WPB S (GALV.)		
	TEE EQUAL	NPTF	0.5	4	3000		ASME B16.11	ASTM A105 (GALV.)		
	TEE EQUAL	BW	6	14	M		ASME B16.9	ASTM A234 GR.WPB S (GALV.)		
	TEE REDUCING	NPTF	0.5	4	3000		ASME B16.11	ASTM A105 (GALV.)		
	REDUC.CONC	BW	6	14	M, M		ASME B16.9	ASTM A234 GR.WPB S (GALV.)		
	REDUC.ECC	BW	6	14	M, M		ASME B16.9	ASTM A234 GR.WPB S (GALV.)		
	CAP	NPTF	0.5	4	3000		ASME B16.11	ASTM A105 (GALV.)		
	CAP	BW	6	14	M		ASME B16.9	ASTM A234 GR.WPB S (GALV.)		
	CPLNG.FULL	NPTF	0.5	4	3000		ASME B16.11	ASTM A105 (GALV.)		
	CPLNG.RED	NPTF	0.5	4	3000		ASME B16.11	ASTM A105 (GALV.)		
	THREDOLET	NPTF	0.5	1.5	3000		MSS SP 97	ASTM A105 (GALV.)		
	UNION	NPTF	0.5	4	3000		MSS-SP-83	ASTM A105 (GALV.)		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A105 (GALV.)		
VALVES	GATE	NPTF	0.5	1.5	CL.2		IS778 CL.2	BODY: IS 318 GR.LTB 2, TRIM: IS 318 GR.LTB 2, STEM: IS 320 Gr.HT 2, B1.20.1; TAG NO.: GAV9010T		
	GATE	FLGD-RF	2	14	150	125AARH	API 600/ ISO 10434	BODY: ASTM A 216 GR. WCB, TRIM:BRONZE, STEM: BRONZE, B-16.5; TAG NO.: GAV914F		
	GLOBE	NPTF	0.5	1.5	CL.2		IS778 CL.2	BODY: IS 318 GR.LTB 2, TRIM: IS 318 GR.LTB 2, STEM: IS 320 Gr.HT 2, B1.20.1; TAG NO.: GLV9010T		
	GLOBE	FLGD-RF	2	14	150	125AARH	API 623	BODY: ASTM A 216 GR. WCB, TRIM:BRONZE, STEM: BRONZE, B-16.5; TAG NO.: GLV914F		
	CHECK - PISTON LIFT	NPTF	0.5	1.5	CL.2		IS778 CL.2	BODY: IS 318 GR.LTB 2, TRIM: IS 318 GR.LTB 2, STEM: IS 320 Gr.HT 2, B1.20.1., TAG NO: CHV9010T		
	CHECK - SWING	FLGD-RF	2	14	150	125AARH	API 594	BODY: ASTM A 216 GR. WCB, TRIM: BRONZE, B-16.5; TAG NO.: CHV914F		
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B7 (GALV.), NUT:A194 GR.2H (GALV.)		
	BOLT.STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B7 (GALV.), NUT:A194 GR.2H (GALV.)		
GASKETS	GASKET		0.5	14	150		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R; THK=4.5MM		

 Rashtriya Chemicals & Fertilizers Ltd.				 पी डी आई एल PDIL Projects and Development India Limited										
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL				PIPING MATERIAL SPECIFICATION (PMS)										
				E24003-RCF-PP-SPE-001				REV: 1		04-Jun-2025				
RATING		BASE MATERIAL		CA		SP. REQUIREMENT		PWHT		PIPE CLASS				
300#		CARBON STEEL		2 MM		IBR		-		3A1BC				
TEMPERATURE (Deg. C)		-29	38	50	100	150	200	250	300	325	350	400	425	
PRESSURE (Kg/Sq. cm g)		52.11	52.11	51.09	47.52	45.99	44.66	42.73	40.58	39.46	38.34	35.38	29.37	
SERVICE		MEDIUM PRESSURE STEAM.												
SPECIAL NOTES	ITEM		SIZE		DESCRIPTION									
	MAINTENANCE JOINTS		ALL		FLANGED (TO BE KEPT MINIMUM)									
	PIPE JOINTS		1.5" & BELOW		SOCKET WELDED									
			2" & ABOVE		BUTTWELDED									
	TEMP. CONN.				AS PER P&ID									
	PRESS. CONN.				AS PER P&ID									
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .												
	02	FOR RESTRICTION ON USE OF BALL, PLUG & BUTTERFLY VALVES, TABLE-1.												
	03	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.												
	04	REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MS-002) FOR SPECIAL REQUIREMENTS FOR IBR SERVICE.												
	05	REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MS-002) FOR DIMENSIONS FOR HANDLE PROJECTION OF SPACER & BLINDS.												

 Rashtriya Chemicals & Fertilizers Ltd.		 पो डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
300#	CARBON STEEL	2 MM	IBR	-	3A1BC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	1.5	S160		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	2	2	SCH XS		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	3	12	SCH STD		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	14	14	XS		ASME B36.10M	ASTM A106 GR.B		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.10M	ASTM A106 GR.B, L=100mm		
	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.10M	ASTM A106 GR.B, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	14	300	125AARH	ASME B16.5	ASTM A105		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A105		
	FLNG. WELD NECK	RF	2	14	M / 300	125AARH	ASME B16.5	ASTM A105		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 600	125AARH	ASME B16.5	ASTM A105		
	FLNG. WELD NECK	RF	2	14	M / 600	125AARH	ASME B16.5	ASTM A105		
	FIG.8	FF	0.5	10	300	125AARH	ASME B16.48	ASTM A105		
	SPACER & BLIND	FF	12	14	300	125AARH	ASME B16.48	ASTM A105		05
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	ELBOW.90	BW	2	14	M		1.5D, ASME B16.9	ASTM A234 GR.WPB S		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	ELBOW.45	BW	2	14	M		1.5D, ASME B16.9	ASTM A234 GR.WPB S		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	TEE EQUAL	BW	2	14	M		ASME B16.9	ASTM A234 GR.WPB S		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM A234 GR.WPB S		
	REDUC.CONC	BW	2	14	M, M		ASME B16.9	ASTM A234 GR.WPB S		
	REDUC.ECC	BW	2	14	M, M		ASME B16.9	ASTM A234 GR.WPB S		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A234 GR.WPB S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A234 GR.WPB S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A105		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CAP	BW	2	14	M		ASME B16.9	ASTM A234 GR.WPB S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A105		
VALVES	GATE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, STEM: 13% CR STEEL; B-16.11, TAG NO.: GAV240SI		
	GATE	FLGD-RF	2	14	300	125AARH	API 600/ ISO 10434	BODY: ASTM A216 GR. WCB, TRIM NO.: 8, STEM: 13% CR STEEL; B-16.5, TAG NO.: GAV245FI		
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, STEM: 13% CR STEEL; B-16.11, TAG NO.: GLV240SI		
	GLOBE	FLGD-RF	2	14	300	125AARH	API 623	BODY: ASTM A216 GR. WCB, TRIM NO.: 8, STEM: 13% CR STEEL, B-16.5; TAG NO.: GLV245FI		
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, B-16.11; TAG NO.: CHV240SI		
	CHECK - SWING	FLGD-RF	2	14	300	125AARH	API 594	BODY: ASTM A216 GR. WCB, TRIM NO.: 8, HINGE: SS316, B-16.5; TAG NO.: CHV245FI		
	PLUG	FLGD-RF	0.5	6	300	125AARH	API 599	BODY- ASTM A105/ A216 GR.WCB, TRIM-HARDENED PLUG, B-16.5, TAG NO.: PLV245FI		02
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT: A193 GR.B7, NUT: A194 GR.2H		
	BOLT.STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT: A193 GR.B7, NUT: A194 GR.2H		

 Rashtriya Chemicals & Fertilizers Ltd.					 Projects and Development India Limited					 L&T Energy Hydrocarbon				
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL							PIPING MATERIAL SPECIFICATION (PMS)							
							E24003-RCF-PP-SPE-001					REV: 1		04-Jun-2025
RATING		BASE MATERIAL				CA		SP. REQUIREMENT		PWHT		PIPE CLASS		
300#		CARBON STEEL				2 MM		IBR		-		3A1BC		
GASKETS	GASKET		0.5	14	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R; THK=4.5MM						
	GASKET		0.5	14	600		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R; THK=4.5MM						

 Rashtriya Chemicals & Fertilizers Ltd.				 पी डी आई एल PDIL Projects and Development India Limited										
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL				PIPING MATERIAL SPECIFICATION (PMS)										
				E24003-RCF-PP-SPE-001				REV: 1		04-Jun-2025				
RATING		BASE MATERIAL		CA		SP. REQUIREMENT		PWHT		PIPE CLASS				
300#		CARBON STEEL		2 MM		-		-		3A2SC				
TEMPERATURE (Deg. C)		-29	38	50	100	150	200	250	300	325	350	400	425	
PRESSURE (Kg/Sq. cm g)		52.11	52.11	51.09	47.52	45.99	44.66	42.73	40.58	39.46	38.34	35.38	29.37	
SERVICE		AMMONIA GAS, NATURAL GAS .												
SPECIAL NOTES	ITEM		SIZE		DESCRIPTION									
	MAINTENANCE JOINTS		ALL		FLANGED (TO BE KEPT MINIMUM)									
	PIPE JOINTS		1.5" & BELOW		SOCKET WELDED									
			2" & ABOVE		BUTTWELDED									
	TEMP. CONN.				AS PER P&ID									
	PRESS. CONN.				AS PER P&ID									
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .												
	02	FOR RESTRICTION ON USE OF BALL, PLUG & BUTTERFLY VALVES, TABLE-1.												
	03	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.												
	04	REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MSC-002) FOR DIMENSIONS FOR HANDLE PROJECTION OF SPACER & BLINDS.												

 Rashtriya Chemicals & Fertilizers Ltd.		 पो डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
300#	CARBON STEEL	2 MM	-	-	3A2SC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	1.5	S160		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	2	2	SCH XS		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	3	12	SCH STD		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	14	14	XS		ASME B36.10M	ASTM A106 GR.B		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.10M	ASTM A106 GR.B, L=100mm		
	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.10M	ASTM A106 GR.B, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	14	300	125AARH	ASME B16.5	ASTM A105		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A105		
	FLNG. WELD NECK	RF	2	14	M / 300	125AARH	ASME B16.5	ASTM A105		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 600	125AARH	ASME B16.5	ASTM A105		
	FLNG. WELD NECK	RF	2	14	M / 600	125AARH	ASME B16.5	ASTM A105		
	FIG.8	FF	0.5	10	300	125AARH	ASME B16.48	ASTM A105		
	SPACER & BLIND	FF	12	14	300	125AARH	ASME B16.48	ASTM A105		04
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	ELBOW.90	BW	2	14	M		1.5D, ASME B16.9	ASTM A234 GR.WPB S		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	ELBOW.45	BW	2	14	M		1.5D, ASME B16.9	ASTM A234 GR.WPB S		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	TEE EQUAL	BW	2	14	M		ASME B16.9	ASTM A234 GR.WPB S		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM A234 GR.WPB S		
	REDUC.CONC	BW	2	14	M, M		ASME B16.9	ASTM A234 GR.WPB S		
	REDUC.ECC	BW	2	14	M, M		ASME B16.9	ASTM A234 GR.WPB S		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A234 GR.WPB S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A234 GR.WPB S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A105		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CAP	BW	2	14	M		ASME B16.9	ASTM A234 GR.WPB S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A105		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A105		
VALVES	GATE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, STEM: 13% CR STEEL; B-16.11, TAG NO.: GAV240S		
	GATE	FLGD-RF	2	14	300	125AARH	API 600/ ISO 10434	BODY: ASTM A216 GR. WCB, TRIM NO.: 8, STEM: 13% CR STEEL; B-16.5, TAG NO.: GAV245F		
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, STEM: 13% CR STEEL; B-16.11, TAG NO.: GLV240S		
	GLOBE	FLGD-RF	2	14	300	125AARH	API 623	BODY: ASTM A216 GR. WCB, TRIM NO.: 8, STEM: 13% CR STEEL, B-16.5; TAG NO.: GLV245F		
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, B-16.11; TAG NO.: CHV240S		
	CHECK - SWING	FLGD-RF	2	14	300	125AARH	API 594	BODY: ASTM A216 GR. WCB, TRIM NO.: 8, HINGE: SS316, B-16.5; TAG NO.: CHV245F		
	PLUG	FLGD-RF	0.5	6	300	125AARH	API 599	BODY- ASTM A105/ A216 GR.WCB, TRIM- HARDENED PLUG, B-16.5, TAG NO.: PLV245F		02
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT: A193 GR.B7, NUT: A194 GR.2H		
	BOLT.STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT: A193 GR.B7, NUT: A194 GR.2H		

 Rashtriya Chemicals & Fertilizers Ltd.					 PDIL Projects and Development India Limited					 L&T Energy Hydrocarbon				
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL						PIPING MATERIAL SPECIFICATION (PMS)								
						E24003-RCF-PP-SPE-001					REV: 1		04-Jun-2025	
RATING		BASE MATERIAL				CA		SP. REQUIREMENT			PWHT		PIPE CLASS	
300#		CARBON STEEL				2 MM		-			-		3A2SC	
GASKETS	GASKET		0.5	14	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R; THK=4.5MM						
	GASKET		0.5	14	600		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R; THK=4.5MM						

 Rashtriya Chemicals & Fertilizers Ltd.				 PDIL Projects and Development India Limited				 L&T Energy Hydrocarbon					
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL				PIPING MATERIAL SPECIFICATION (PMS)									
				E24003-RCF-PP-SPE-001				REV: 1		04-Jun-2025			
RATING		BASE MATERIAL		CA		SP. REQUIREMENT		PWHT		PIPE CLASS			
300#		LTCS		2 MM		LOW TEMPERATURE SERVICE		-		3A3SC			
TEMPERATURE (Deg. C)		-45	38	50	100	150	200	250	300	325	350	400	425
PRESSURE (Kg/Sq. cm g)		52.11	52.11	51.09	47.52	45.99	44.66	42.73	40.58	39.46	38.34	35.38	29.37
SERVICE		LIQUID AMMONIA.											
SPECIAL NOTES	ITEM		SIZE		DESCRIPTION								
	MAINTENANCE JOINTS		ALL		FLANGED (TO BE KEPT MINIMUM)								
	PIPE JOINTS		1.5" & BELOW		SOCKET WELDED								
			2" & ABOVE		BUTTWELDED								
	TEMP. CONN.				AS PER P&ID								
	PRESS. CONN.				AS PER P&ID								
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .											
	02	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.											
	03	REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MS-002) FOR DIMENSIONS FOR HANDLE PROJECTION OF SPACER & BLINDS.											
	04	IMPACT TESTING REQUIRED FOR ALL WELDED ITEMS. VALUES SHALL BE AS PER ASME B31.3 TABLE 323.35 AND ASTM A 370.											
	05	VALVES SHALL HAVE EXTENDED BONNET AS PER BS 6364 (NON COLD BOX APPLICATION).											

 Rashtriya Chemicals & Fertilizers Ltd.		 पी डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
300#	LTCS	2 MM	LOW TEMPERATURE SERVICE	-	3A3SC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	0.75	S160		ASME B36.10M	ASTM A333 GR.6		
	SEAMLESS	PE	1	1.5	SCH XS		ASME B36.10M	ASTM A333 GR.6		
	SEAMLESS	BE	2	2	SCH XS		ASME B36.10M	ASTM A333 GR.6		
	SEAMLESS	BE	3	12	SCH STD		ASME B36.10M	ASTM A333 GR.6		
	SEAMLESS	BE	14	14	XS		ASME B36.10M	ASTM A333 GR.6		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.10M	ASTM A333 GR.6, L=100mm		
	NIPPLE, SEAMLESS	PE/NPTM	0.5	1.5	M		ASME B36.10M	ASTM A333 GR.6, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	14	300	125AARH	ASME B16.5	ASTM A350 GR.LF2 CL.1		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A350 GR.LF2 CL.1		
	FLNG. WELD NECK	RF	2	14	M / 300	125AARH	ASME B16.5	ASTM A350 GR.LF2 CL.1		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 600	125AARH	ASME B16.5	ASTM A350 GR.LF2 CL.1		
	FLNG. WELD NECK	RF	2	14	M / 600	125AARH	ASME B16.5	ASTM A350 GR.LF2 CL.1		
	FIG.8	FF	0.5	10	300	125AARH	ASME B16.48	ASTM A350 GR.LF2 CL.1		
	SPACER & BLIND	FF	12	14	300	125AARH	ASME B16.48	ASTM A350 GR.LF2 CL.1		03
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A350 GR.LF2 CL.1		
	ELBOW.90	BW	2	14	M	1.5D, ASME B16.9	ASME B16.9	ASTM A420 GR.WPL6-S		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A350 GR.LF2 CL.1		
	ELBOW.45	BW	2	14	M	1.5D, ASME B16.9	ASME B16.9	ASTM A420 GR.WPL6-S		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A350 GR.LF2 CL.1		
	TEE EQUAL	BW	2	14	M		ASME B16.9	ASTM A420 GR.WPL6-S		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A350 GR.LF2 CL.1		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM A420 GR.WPL6-S		
	REDUC.CONC	BW	2	14	M, M		ASME B16.9	ASTM A420 GR.WPL6-S		
	REDUC.ECC	BW	2	14	M, M		ASME B16.9	ASTM A420 GR.WPL6-S		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A420 GR.WPL6-S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A420 GR.WPL6-S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A350 GR.LF2 CL.1		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A350 GR.LF2 CL.1		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A350 GR.LF2 CL.1		
	CAP	BW	2	14	M		ASME B16.9	ASTM A420 GR.WPL6-S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A350 GR.LF2 CL.1		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A350 GR.LF2 CL.1		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A350 GR.LF2 CL.1		
VALVES	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A 350 GR.LF2 CL.1, TRIM NO.:12; B-16.11; TAG NO.: GLV140SE		05
	GLOBE	FLGD-RF	2	14	300	125AARH	API 623	BODY-ASTM A 352 GR.LCB, TRIM NO.:12, B-16.5; TAG NO.: GLV145FE		05
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A 350 GR.LF2 CL.1, TRIM NO.:12; B-16.11; TAG NO.: CHV140S		
	CHECK - SWING	FLGD-RF	2	14	300	125AARH	API 594	BODY-ASTM A 352 GR.LCB, TRIM NO.:12, B-16.5; TAG NO.: CHV145F		
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT:A320 GR.L7, NUT:A194 GR.7		
	BOLT.STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT:A320 GR.L7, NUT:A194 GR.7		
GASKETS	GASKET		0.5	14	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R; THK=4.5MM		
	GASKET		0.5	14	600		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & CS O/R; THK=4.5MM		

 Rashtriya Chemicals & Fertilizers Ltd.				 पी डी आई एल PDIL Projects and Development India Limited								
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL				PIPING MATERIAL SPECIFICATION (PMS)								
				E24003-RCF-PP-SPE-001				REV: 1		04-Jun-2025		
RATING		BASE MATERIAL		CA		SP. REQUIREMENT		PWHT		PIPE CLASS		
300#		SS316L		1.6 MM		IBR		-		3C4BC		
TEMPERATURE (Deg. C)		-29	38	50	100	150	200	250	300	350	400	425
PRESSURE (Kg/Sq. cm g)		42.22	42.22	40.79	35.49	32.02	29.78	28.04	26.61	25.59	24.78	24.37
SERVICE		MEDIUM PRESSURE STEAM, MP STEAM CONDENSATE.										
SPECIAL NOTES	ITEM		SIZE		DESCRIPTION							
	MAINTENANCE JOINTS		ALL		FLANGED (TO BE KEPT MINIMUM)							
	PIPE JOINTS		1.5" & BELOW		SOCKET WELDED							
			2" & ABOVE		BUTTWELDED							
	TEMP. CONN.				AS PER P&ID							
	PRESS. CONN.				AS PER P&ID							
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .										
	02	FOR RESTRICTION ON USE OF BALL, PLUG & BUTTERFLY VALVES, TABLE-1.										
	03	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.										
	04	REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MS-002) FOR SPECIAL REQUIREMENTS FOR IBR SERVICE.										

 Rashtriya Chemicals & Fertilizers Ltd.		 पो डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
300#	SS316L	1.6 MM	IBR	-	3C4BC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	1.5	80S		ASME B36.19M	ASTM A312 GR.TP316L		
	SEAMLESS	BE	2	6	40S		ASME B36.19M	ASTM A312 GR.TP316L		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP316L, L=100mm		
	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP316L, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	6	300	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FLNG. WELD NECK	RF	2	6	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 600	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FLNG. WELD NECK	RF	2	6	M / 600	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FIG.8	FF	0.5	6	300	125AARH	ASME B16.48	ASTM A182 GR.F316L		
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	ELBOW.90	BW	2	6	M		1.5D, ASME B16.9	ASTM A403 GR.WP316L-S		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	ELBOW.45	BW	2	6	M		1.5D, ASME B16.9	ASTM A403 GR.WP316L-S		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	TEE EQUAL	BW	2	6	M		ASME B16.9	ASTM A403 GR.WP316L-S		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM A403 GR.WP316L-S		
	REDUC.CONC	BW	2	6	M, M		ASME B16.9	ASTM A403 GR.WP316L-S		
	REDUC.ECC	BW	2	6	M, M		ASME B16.9	ASTM A403 GR.WP316L-S		
	SWAGE.CONC	BE/PE	0.5	6	M, M		MSS-SP-95	ASTM A403 GR.WP316L-S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A403 GR.WP316L-S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A182 GR.F316L		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	CAP	BW	2	6	M		ASME B16.9	ASTM A403 GR.WP316L-S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A182 GR.F316L		
VALVES	GATE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY-ASTM A182 GR.F316L , TRIM-SS316L HARDFACED, STEM: SS316L; B-16.11; TAG NO.: GAV420SI		
	GATE	FLGD-RF	2	6	300	125AARH	API 600/ ISO 10434	BODY: ASTM A 351 GR.CF3M, TRIM: SS316L HARDFACED, STEM: SS316L; B-16.5, TAG NO.: GAV425FI		
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY-ASTM A182 GR.F316L , TRIM-SS316L HARDFACED, STEM: SS316L; B-16.11; TAG NO.: GLV420SI		
	GLOBE	FLGD-RF	2	6	300	125AARH	API 623	BODY: ASTM A 351 GR.CF3M, TRIM: SS316L HARDFACED, STEM: SS316L; B-16.5, TAG NO.: GLV425FI		
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F316L, TRIM: SS316L HARDFACED; B-16.11, TAG NO.: CHV420SI		
	CHECK - SWING	FLGD-RF	2	6	300	125AARH	API 594	BODY-ASTM A 351 GR.CF3M, TRIM-SS316L HARDFACED, B-16.5; TAG NO.: CHV425FI		
	PLUG	FLGD-RF	0.5	6	300	125AARH	API 599	BODY- ASTM A182 GR.F316L/ A 351 GR.CF3M, TRIM: SS316L & PLUG HARDENED, B-16.5; TAG NO.: PLV425FI		02
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		
	BOLT.STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		
GASKETS	GASKET		0.5	6	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS316L + GRAPHITE FILLER WITH SS316L I/R & SS OR		
	GASKET		0.5	6	600		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS316L + GRAPHITE FILLER WITH SS316L I/R & SS OR		

 Rashtriya Chemicals & Fertilizers Ltd.				 PDIL Projects and Development India Limited				 L&T Energy Hydrocarbon				
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL				PIPING MATERIAL SPECIFICATION (PMS)								
				E24003-RCF-PP-SPE-001				REV: 1		04-Jun-2025		
RATING		BASE MATERIAL		CA		SP. REQUIREMENT		PWHT		PIPE CLASS		
300#		SS316L		1.6 MM		-		-		3C4SC		
TEMPERATURE (Deg. C)		-45	38	50	100	150	200	250	300	350	400	425
PRESSURE (Kg/Sq. cm g)		42.22	42.22	40.79	35.49	32.02	29.78	28.04	26.61	25.59	24.78	24.37
SERVICE		AMMONIA GAS, LIQUID AMMONIA, SULPHURIC ACID 98% (H.P.).										
SPECIAL NOTES	ITEM	SIZE		DESCRIPTION								
	MAINTENANCE JOINTS	ALL		FLANGED (TO BE KEPT MINIMUM)								
	PIPE JOINTS	1.5" & BELOW		SOCKET WELDED								
		2" & ABOVE		BUTTWELDED								
	TEMP. CONN.			AS PER P&ID								
	PRESS. CONN.			AS PER P&ID								
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .										
	02	FOR RESTRICTION ON USE OF BALL, PLUG & BUTTERFLY VALVES, TABLE-1.										
	03	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.										
	04	THESE VALVES & GASKETS SHALL BE USED ONLY FOR SULPHURIC ACID SERVICE.										
	05	FOR LIQUID AMMONIA SERVICE, VALVE SHALL HAVE EXTENDED BONNET AS PER BS6364 (NON COLD BOX APPLICATION).										
	06	THESE VALVES & GASKETS SHALL BE USED ONLY FOR GASEOUS AMMONIA SERVICE.										
	07	PTFE GASKET SHALL NOT BE USED ABOVE 204°C.										
	08	FOR LIQUID AMMONIA SERVICE, IMPACT TESTING IS REQUIRED FOR ALL WELDED ITEMS. VALUES SHALL BE AS PER ASME B31.3 TABLE 323.35 AND ASTM A 370.										

 Rashtriya Chemicals & Fertilizers Ltd.		 Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
300#	SS316L	1.6 MM	-	-	3C4SC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	1.5	80S		ASME B36.19M	ASTM A312 GR.TP316L		
	SEAMLESS	BE	2	6	40S		ASME B36.19M	ASTM A312 GR.TP316L		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP316L, L=100mm		
	NIPPLE, SEAMLESS	NPTM	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP316L, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	6	300	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FLNG. WELD NECK	RF	2	6	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 600	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FLNG. WELD NECK	RF	2	6	M / 600	125AARH	ASME B16.5	ASTM A182 GR.F316L		
	FIG.8	FF	0.5	6	300	125AARH	ASME B16.48	ASTM A182 GR.F316L		
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	ELBOW.90	BW	2	6	M		1.5D, ASME B16.9	ASTM A403 GR.WP316L S		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	ELBOW.45	BW	2	6	M		1.5D, ASME B16.9	ASTM A403 GR.WP316L S		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	TEE EQUAL	BW	2	6	M		ASME B16.9	ASTM A403 GR.WP316L S		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM A403 GR.WP316L S		
	REDUC.CONC	BW	2	6	M, M		ASME B16.9	ASTM A403 GR.WP316L S		
	REDUC.ECC	BW	2	6	M, M		ASME B16.9	ASTM A403 GR.WP316L S		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A403 GR.WP316L S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM A403 GR.WP316L S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A182 GR.F316L		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	CAP	BW	2	6	M		ASME B16.9	ASTM A403 GR.WP316L S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F316L		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A182 GR.F316L		
VALVES	BALL	FLGD-RF	0.5	6	300	125AARH	API 608/ BS EN ISO 17292	BODY: ASTM A 351 GR.CF3M, TRIM: SS316L HARDFACED, B-16.5; TAG NO.: BAV425FE		05, 02
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY-ASTM A182 GR.F316L , TRIM-ALLOY 20, STEM: SS316L; B-16.11; TAG NO.: GLV420SS		04
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY-ASTM A182 GR.F316L , TRIM-SS316L HARDFACED, STEM: SS316L; B-16.11; TAG NO.: GLV420SE		05
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY-ASTM A182 GR.F316L , TRIM-SS316L HARDFACED, STEM: SS316L; B-16.11; TAG NO.: GLV420S		06
	GLOBE	FLGD-RF	2	6	300	125AARH	API 623	BODY: ASTM A 351 GR.CF3M, TRIM: ALLOY 20, STEM: SS316L, B-16.5, TAG NO.: GLV425FS		04
	GLOBE	FLGD-RF	2	6	300	125AARH	API 623	BODY: ASTM A 351 GR.CF3M, TRIM: SS316L HARDFACED, STEM: SS316L, B-16.5, TAG NO.: GLV425FE		05
	GLOBE	FLGD-RF	2	6	300	125AARH	API 623	BODY: ASTM A 351 GR.CF3M, TRIM: SS316L HARDFACED, STEM: SS316L, B-16.5, TAG NO.: GLV425F		06
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F316L, TRIM: ALLOY 20; B-16.11, TAG NO.: CHV420SS		04
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F316L, TRIM: SS316L HARDFACED, B-16.11, TAG NO.: CHV420S		
	CHECK - SWING	FLGD-RF	2	6	300	125AARH	API 594	BODY-ASTM A 351 GR.CF3M, TRIM-ALLOY 20, B-16.5; TAG NO.: CHV425FS		04
	CHECK - SWING	FLGD-RF	2	6	300	125AARH	API 594	BODY-ASTM A 351 GR.CF3M, TRIM-SS316L HARDFACED, B-16.5; TAG NO.: CHV425F		

 Rashtriya Chemicals & Fertilizers Ltd.					 PDIL Projects and Development India Limited					 L&T Energy Hydrocarbon									
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL										PIPING MATERIAL SPECIFICATION (PMS)									
										E24003-RCF-PP-SPE-001					REV: 1		04-Jun-2025		
RATING			BASE MATERIAL				CA		SP. REQUIREMENT			PWHT		PIPE CLASS					
300#			SS316L				1.6 MM		-			-		3C4SC					
VALVES	PLUG		FLGD-RF	0.5	6	300	125AARH	API 599	BODY- ASTM A182 GR.F316L/ A 351 GR.CF3M, TRIM: PFA LINED, B-16.5; TAG NO.: PLV425FS					04, 02					
	PLUG		FLGD-RF	0.5	6	300	125AARH	API 599	BODY- ASTM A182 GR.F316L/ A 351 GR.CF3M, TRIM: SS316L & PLUG HARDENED, B-16.5; TAG NO.: PLV425FE					05, 02					
	PLUG		FLGD-RF	0.5	6	300	125AARH	API 599	BODY- ASTM A182 GR.F316L/ A 351 GR.CF3M, TRIM: SS316L & PLUG HARDENED, B-16.5; TAG NO.: PLV425F					06, 02					
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)			0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8										
	BOLT.STUD + 3 NUTS- (HEAVY HEX)			1	3.25			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8										
GASKETS	GASKET			0.5	6	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS316L + GRAPHITE FILLER WITH SS316L I/R & SS OR					05, 06					
	GASKET			0.5	6	600		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS316L + GRAPHITE FILLER WITH SS316L I/R & SS OR					05, 06					
GASKETS	GASKET			0.5	6	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS316 + PTFE FILLER WITH SS316 I/R & SS OR					04					
	GASKET			0.5	6	600		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS316 + PTFE FILLER WITH SS316 I/R & SS OR					04					

 Rashtriya Chemicals & Fertilizers Ltd.		 PDIL Projects and Development India Limited		 L&T Energy Hydrocarbon																	
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)																		
			E24003-RCF-PP-SPE-001		REV: 1	04-Jun-2025															
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS																
300#	SS904L	1 MM	-	-	3C8SC																
<table border="1"> <tr> <td>TEMPERATURE (Deg. C)</td> <td>-29</td> <td>38</td> <td>50</td> <td>100</td> <td>150</td> <td>200</td> <td>250</td> </tr> <tr> <td>PRESSURE (Kg/Sq. cm g)</td> <td>52.31</td> <td>52.31</td> <td>50.07</td> <td>41.91</td> <td>38.24</td> <td>35.38</td> <td>32.63</td> </tr> </table>						TEMPERATURE (Deg. C)	-29	38	50	100	150	200	250	PRESSURE (Kg/Sq. cm g)	52.31	52.31	50.07	41.91	38.24	35.38	32.63
TEMPERATURE (Deg. C)	-29	38	50	100	150	200	250														
PRESSURE (Kg/Sq. cm g)	52.31	52.31	50.07	41.91	38.24	35.38	32.63														
SERVICE		LIQUID AMMONIA, PARTIALLY AMMONIATED PHOSPHORIC ACID, SULPHURIC ACID(98%).																			
SPECIAL NOTES	ITEM	SIZE	DESCRIPTION																		
	MAINTENANCE JOINTS	ALL	FLANGED (TO BE KEPT MINIMUM)																		
	PIPE JOINTS	1.5" & BELOW	SOCKET WELDED																		
		2" & ABOVE	BUTTWELDED																		
	TEMP. CONN.		AS PER P&ID																		
	PRESS. CONN.		AS PER P&ID																		
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001 .																			
	02	FOR RESTRICTION ON USE OF BALL, PLUG & BUTTERFLY VALVES, TABLE-1.																			
	03	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.																			
	04	FOR SULPHURIC ACID SERVICE THESE VALVES & GASKETS SHALL ONLY BE USED.																			
	05	FOR PHOSPHORIC ACID SERVICE THESE GASKETS SHALL ONLY BE USED.																			
	06	PTFE GASKET SHALL NOT BE USED ABOVE 204°C.																			

**1200 MTPD (DAP BASIS) NPK PROJECT,
RCF, THAL**

PIPING MATERIAL SPECIFICATION (PMS)

E24003-RCF-PP-SPE-001

REV: 1

04-Jun-2025

RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
300#	SS904L	1 MM	-	-	3C8SC

[illegible]

LEGEND

H	Half Coupling
R	Pipe to Pipe with Reinforcement Pad
P	Pipe to Pipe
T	Equal/ Reducing Tees
S	Sockolet
W	Weldolet

 Rashtriya Chemicals & Fertilizers Ltd.		 पी डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL		PIPING MATERIAL SPECIFICATION (PMS)			
		E24003-RCF-PP-SPE-001		REV: 1	04-Jun-2025
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
300#	SS904L	1 MM	-	-	3C8SC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	PE	0.5	1.5	80S		ASME B36.19M	ASTM A312 GR.TP904L		
	SEAMLESS	BE	2	6	40S		ASME B36.19M	ASTM A312 GR.TP904L		
	NIPPLE, SEAMLESS	PBE	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP904L, L=100mm		
	NIPPLE, SEAMLESS	PE/NPTM	0.5	1.5	M		ASME B36.19M	ASTM A312 GR.TP904L, L=100mm		
FLANGES	FLNG. BLIND	RF	0.5	6	300	125AARH	ASME B16.5	ASTM A182 GR.F904L		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F904L		
	FLNG. WELD NECK	RF	2	6	M / 300	125AARH	ASME B16.5	ASTM A182 GR.F904L		
	FLNG. SOCKET WELDED	RF	0.5	1.5	M / 600	125AARH	ASME B16.5	ASTM A182 GR.F904L		
	FLNG. WELD NECK	RF	2	6	M / 600	125AARH	ASME B16.5	ASTM A182 GR.F904L		
	FIG.8	FF	0.5	6	300	125AARH	ASME B16.48	ASTM A182 GR.F904L		
FITTINGS	ELBOW.90	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	ELBOW.90	BW	2	6	M		1.5D, ASME B16.9	ASTM B366 GR.WP904L S		
	ELBOW.45	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	ELBOW.45	BW	2	6	M		1.5D, ASME B16.9	ASTM B366 GR.WP904L S		
	TEE EQUAL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	TEE EQUAL	BW	2	6	M		ASME B16.9	ASTM B366 GR.WP904L S		
	TEE REDUCING	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	TEE REDUCING	BW	2	2	M, M		ASME B16.9	ASTM B366 GR.WP904L S		
	REDUC.CONC	BW	2	6	M, M		ASME B16.9	ASTM B366 GR.WP904L S		
	REDUC.ECC	BW	2	6	M, M		ASME B16.9	ASTM B366 GR.WP904L S		
	SWAGE.CONC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM B366 GR.WP904L S		
	SWAGE.ECC	BE/PE	0.5	3	M, M		MSS-SP-95	ASTM B366 GR.WP904L S		
	SOCKOLET	SW	0.5	1.5	3000		MSS-SP-97	ASTM A182 GR.F904L		
	CAP	NPTF	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	CAP	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	CAP	BW	2	6	M		ASME B16.9	ASTM B366 GR.WP904L S		
	CPLNG.FULL	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	CPLNG.RED	SW	0.5	1.5	3000		ASME B16.11	ASTM A182 GR.F904L		
	HEX. PLUG	NPTM	0.5	0.5	-		ASME B16.11	ASTM A182 GR.F904L		
VALVES	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F904L, TRIM: SS904L, STEM: F904L, B-16.5; TAG NO.: GLV550S		
	GLOBE	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F904L, TRIM: ALLOY20, STEM: F904L, B-16.5; TAG NO.: GLV550SS		04
	GLOBE	FLGD-RF	2	6	300	125AARH	API 623	BODY: ASTM A182 GR.F904L, TRIM: SS904L, STEM: F904L, B-16.5; TAG NO.: GLV555F		
	GLOBE	FLGD-RF	2	6	300	125AARH	API 623	BODY: ASTM A182 GR.F904L, TRIM: ALLOY20, STEM: F904L, B-16.5; TAG NO.: GLV555FS		04
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F904L, TRIM: SS904L, B-16.11; TAG NO.: CHV550S		
	CHECK - PISTON LIFT	SW/ 3000	0.5	1.5	800		API 602/ ISO 15761	BODY: ASTM A182 GR.F904L, TRIM: ALLOY20, B-16.11; TAG NO.: CHV550SS		04
	CHECK - SWING	FLGD-RF	2	6	300	125AARH	API 594	BODY-ASTM A182 GR.F904L, TRIM: SS904L, HING: F904L B-16.5; TAG NO.: CHV555F		
	CHECK - SWING	FLGD-RF	2	6	300	125AARH	API 594	BODY-ASTM A182 GR.F904L, TRIM: ALLOY20, HING: F904L B-16.5; TAG NO.: CHV555FS		04
	PLUG	FLGD-RF	0.5	6.0	300	125AARH	API 599	BODY: ASTM A182 GR.F904L, TRIM: SS904L HARDENED PLUG, B-16.5; TAG NO.: PLV555F		02
	PLUG	FLGD-RF	0.5	6.0	300	125AARH	API 599	BODY: ASTM A182 GR.F904L, TRIM: PFA LINED, B-16.5; TAG NO.: PLV555FS		04, 02



Rashtriya Chemicals &
Fertilizers Ltd.



Projects and Development
India Limited



L&T Energy
Hydrocarbon

**1200 MTPD (DAP BASIS) NPK PROJECT,
RCF, THAL**

PIPING MATERIAL SPECIFICATION (PMS)

E24003-RCF-PP-SPE-001

REV: 1

04-Jun-2025

RATING		BASE MATERIAL			CA	SP. REQUIREMENT		PWHT	PIPE CLASS	
300#		SS904L			1 MM	-		-	3C8SC	
BOLTING	BOLT:STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		
	BOLT:STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT:A193 GR.B8 CL.2, NUT:A194 GR.8		
GASKETS	GASKET		0.5	6	300		ASME B16.21- ASME B16.5	SPIRAL WOUND, SS316 + PTFE FILLER WITH SS316 I/R & SS OR		04, 05
	GASKET		0.5	6	600		ASME B16.21- ASME B16.5	SPIRAL WOUND, SS316 + PTFE FILLER WITH SS316 I/R & SS OR		04, 05
	GASKET		0.5	6	300		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & SS OR		
	GASKET		0.5	6	600		ASME B16.20- ASME B16.5	SPIRAL WOUND, SS304 + GRAPHITE FILLER WITH SS304 I/R & SS OR		

 Rashtriya Chemicals & Fertilizers Ltd.				 पी डी आई एल PDIL Projects and Development India Limited										
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL				PIPING MATERIAL SPECIFICATION (PMS)										
				E24003-RCF-PP-SPE-001				REV: 1		04-Jun-2025				
RATING		BASE MATERIAL		CA		SP. REQUIREMENT		PWHT		PIPE CLASS				
1500#		CARBON STEEL		2 MM		IBR		-		15A1BC				
TEMPERATURE (Deg. C)		-29	38	50	100	150	200	250	300	325	350	400	425	
PRESSURE (Kg/Sq. cm g)		260.33	260.33	255.54	237.59	229.84	223.32	213.83	203.03	197.42	191.50	177.02	146.64	
SERVICE		BOILER FEED WATER (BFW).												
SPECIAL NOTES	ITEM		SIZE		DESCRIPTION									
	MAINTENANCE JOINTS		ALL		FLANGED (TO BE KEPT MINIMUM)									
	PIPE JOINTS		1.5" & BELOW		BUTTWELDED									
			2" & ABOVE		BUTTWELDED									
	TEMP. CONN.				AS PER P&ID									
	PRESS. CONN.				AS PER P&ID									
NOTES	01	NDT REQUIREMENTS AS PER SPECIFICATION E24003-RCF-QC-SPE-001.												
	02	FOR RESTRICTION ON USE OF BALL, PLUG & BUTTERFLY VALVES, TABLE-1.												
	03	THREADED FITTINGS SHALL BE LIMITED TO INSTRUMENT CONNECTIONS, SAMPLE CONNECTIONS, CONNECTIONS TO VENDOR'S EQUIPMENTS & AT ORIFICE FLANGE TAPS AND SHALL BE SEAL WELDED EXCEPT FOR CAPS & SAMPLE TO OPEN SIDE.												
	04	REFER TECHNICAL NOTES FOR PIPING BULK (DOC. NO.: E24003-RCF-PP-MS-C-002) FOR SPECIAL REQUIREMENTS FOR IBR SERVICE.												

 Rashtriya Chemicals & Fertilizers Ltd.		 पो डी आई एल PDIL Projects and Development India Limited			
1200 MTPD (DAP BASIS) NPK PROJECT, RCF, THAL			PIPING MATERIAL SPECIFICATION (PMS)		
			E24003-RCF-PP-SPE-001		REV: 1
RATING	BASE MATERIAL	CA	SP. REQUIREMENT	PWHT	PIPE CLASS
1500#	CARBON STEEL	2 MM	IBR	-	15A1BC

NOTE: M - MATCH SCH AS OF PIPE

ITEMS	TYPE	ENDS	SIZE RANGE (INCH)		SCH /THK (MM) / RATING	FACE FINISH	DIM. STD / DSGN. STD.	MATERIAL DESCRIPTION	CODE	NOTE
			LOWER	UPPER						
PIPES	SEAMLESS	BE	0.5	1.5	160		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	2	3	160		ASME B36.10M	ASTM A106 GR.B		
	SEAMLESS	BE	4	4	XXS		ASME B36.10M	ASTM A106 GR.B		
	NIPPLE, SEAMLESS	BBE	0.5	1.5	M		ASME B36.10M	ASTM A106 GR.B, L=100mm		
FLANGES	FLNG. BLIND	RTJ	0.5	4	1500	63AARH	ASME B16.5	ASTM A105		
	FLNG. WELD NECK	RTJ	0.5	4	M / 1500	63AARH	ASME B16.5	ASTM A105		
	FIG.8	RTJ	0.5	4	M / 1500	63AARH	ASME B16.48	ASTM A105		
	ELBOW.90	BW	0.5	1.5	M		1.5D, ASME B16.9	ASTM A105		
	ELBOW.90	BW	2	4	M		1.5D, ASME B16.9	ASTM A234 GR.WPB S		
	ELBOW.45	BW	0.5	1.5	M		1.5D, ASME B16.9	ASTM A105		
	ELBOW.45	BW	2	4	M		1.5D, ASME B16.9	ASTM A234 GR.WPB S		
	TEE EQUAL	BW	0.5	1.5	M		ASME B16.9	ASTM A105		
	TEE EQUAL	BW	2	4	M		ASME B16.9	ASTM A234 GR.WPB S		
	TEE REDUCING	BW	0.5	1.5	M, M		ASME B16.9	ASTM A105		
	TEE REDUCING	BW	2	4	M, M		ASME B16.10	ASTM A234 GR.WPB S		
	REDUC.CONC	BW	0.5	1.5	M, M		ASME B16.9	ASTM A105		
	REDUC.ECC	BW	2	4	M, M		ASME B16.9	ASTM A234 GR.WPB S		
	WELDOLET	BW	0.5	1.5	S160		MSS-SP-97	ASTM A105		
	CAP	BW	0.5	1.5	M		ASME B16.9	ASTM A105		
VALVES		SW	2	4	M		ASME B16.9	ASTM A234 GR.WPB S		
	GATE	BW	0.5	1.5	1500		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, STEM: 13% CR STEEL; B-16.11, TAG NO.: GAV240BI		
	GLOBE	BW	0.5	1.5	1500		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, STEM: 13% CR STEEL; B-16.11, TAG NO.: GLV240BI		
	CHECK - PISTON LIFT	BW	0.5	1.5	1500		API 602/ ISO 15761	BODY: ASTM A105, TRIM NO.: 5, B-16.11; TAG NO.: CHV240BI		
BOLTING	BOLT.STUD + 2 NUTS- (HEAVY HEX)		0.375	0.875			ASME B18.2.1/ B18.2.2	BOLT: A193 GR.B7, NUT: A194 GR.2H		
	BOLT.STUD + 3 NUTS- (HEAVY HEX)		1	3.25			ASME B18.2.1/ B18.2.2	BOLT: A193 GR.B7, NUT: A194 GR.2H		
GASKETS	GASKET		0.5	4	1500		ASME B16.20- ASME B16.5	OCTAGONAL RING, SOFT IRON (90 BHN MAX.)		



☐ Short Delivery	☐ Only for Engg. Activities	Date	29.09.2025
☐ Bought out valves	☐ Others, Please Specify	CORB Reference	DM25066L
Customer / Buyer Name	LARSEN & TOUBRO LIMITED	CEC Quote Number with Date	250702-0400-PA Dtd 28.08.2025
Site / End User Name & Country	Rashtriya Chemicals and Fertilizers Limited, India	Application Engineer SPOC	Karthikeyan Sekar Satheshkumar Eagambaram
Buyer PO Number/Date	LTES/75001-83664/BS Dtd 03.09.2025	Outside Sales Engineer/Manager	Ashokvijay R
Buyer PO/LOI Received Date	12.09.2025	PO Value	INR 9,99,750
Agreed / Approved Payment Terms	Refer PO clause no. 5.0	1)PO Delivery Date 2)Agreed Delivery Date	26.03.2026
Agreed Incoterms /Shipping Terms	CPT, RCF Thal Project Site	Margin	22%
LD Clause	@0.5% of the undelivered portion of Order value	Agent Name & Commission	NA
Ship to Address with GST#:	Rashtriya Chemical and Fertilizers Limited Thal, Thal-Vaishet, Alibag # 402 208 State: Maharashtra, India GSTIN: 27AAACL0140PEZ6	Bill to Address with GST#:	Larsen & Toubro Limited Divison Name: L&T Energy Hydrocarbon 14th Floor, AMN Tower, AMN Tower, Jogeshwari - Vikroli Link Road, Powai Campus, Gate 3, Mumbai Suburban, Maharashtra, 400072

IMPORTANT CORRESPONDENCES/EMAILS